Algorithmic Trading

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Notation List

$x \in E,\ x \notin E$	membership / non-membership
$E \subseteq F,\ \emptyset$	subset relation / empty set
$X \times Y,\ f : X \to Y$	Cartesian product / function from X to Y
$\Longrightarrow,\ \Longleftrightarrow$	implies / if and only if
$:=$	defined to be equal to
$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	standard number systems
$\mathbb{F}$	field of scalars, usually $\mathbb{R}$ or $\mathbb{C}$
a, b, c, λ	scalars
$\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{0}$	vectors and the zero vector
$\mathcal{U}, \mathcal{V}, \mathcal{W}$	vector spaces or subspaces, depending on context
$\beta = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$	finite ordered list, often a basis
$\operatorname{span}(E),\ \dim(\mathcal{V})$	span of E / dimension of $\mathcal{V}$
$\mathcal{U} + \mathcal{W},\ \mathcal{U} \oplus \mathcal{W},\ \mathcal{V}/\mathcal{U}$	sum, direct sum, quotient space
$T : \mathcal{V} \to \mathcal{W},\ \mathcal{L}(\mathcal{V}, \mathcal{W})$	linear map / space of linear maps
$\ker(T),\ \operatorname{range}(T)$	kernel and range of T
$[\mathbf{v}]_\beta,\ [T]_{\gamma \leftarrow \beta}$	coordinate vector / matrix representation
$A, B, C,\ \operatorname{Mat}(m, n, \mathbb{F})$	matrices / $m \times n$ matrices over $\mathbb{F}$
$A^{-1},\ A^T,\ A^*,\ \operatorname{tr}(A), \det(A)$	standard matrix operations

Declared overrides

Symbols used in a sense other than the guide's house register, with the scope of each override.

σ^2	variance of one period's value shock, not annualised; Grossman–Miller entries
σ	absolute volatility of an arithmetic Brownian midprice, in price units; Avellaneda–Stoikov
$p^{\pm},\ P^{\pm}(\delta)$	arrival and fill probabilities, not prices; static limit-order depth
$\lambda^a,\ \lambda^b$	Poisson execution intensities, not price-impact slopes; Avellaneda–Stoikov
$p,\ \alpha$	probability of the high value, not a price; fraction of informed traders; Glosten–Milgrom half-spreads and market collapse
$A,\ C;\ Q_t,\ q_t$	adverse-selection and transaction-cost spread components (full-spread amounts), not matrices; trade sign, $+1$ for a buy, not a quantity; spread components, spread measures, Roll's estimator and component identification

Preface

This compendium follows the sections of the institutional quantitative finance learning guide, pinned in `guide/`: one chapter per guide section, holding typed, labelled entries that cite their sources at each claim. Notes written before these notes followed the guide's study method are compiled in the final part, Notes awaiting migration, until each is rewritten as entries in its section's chapter.

This content builds upon the foundational works of Rishi K. Narang [s7-narang], Raja Velu [s3-velu], and Marcos López de Prado [alpha-afml], among others, whose insights form the backbone of our discussion.

Chapter 1

Market microstructure and market making

Pathways drawn on: `market-making-lob` (draft entries from the migration: supplemental inventory, information and quoting models, Glosten–Milgrom, a practitioner account of dealer quoting, and textbook accounts of inventory-based quoting; of the pathway's required readings only Avellaneda–Stoikov and Glosten–Milgrom, through its 1983 working paper, are drawn on); and the section overview `execution-market-making` (draft entries from the migration, from Harris, Hasbrouck and Lehalle and Laruelle; none of its readings yet studied). The section beginning with fundamental value covers informed trading, informative prices, market efficiency, inside information and the price formation process, from Harris and Lehalle and Laruelle. The section beginning with spread measures measures and decomposes spreads, from Harris and Hasbrouck (the latter a required reading of this section's overview route, not yet studied); the section beginning with fundamental and transitory volatility covers volatility, returns by time scale, the relation of volatility to spreads, and bubbles and crashes, from Harris, Hasbrouck and Lehalle and Laruelle. Notes for this section that predate the study method are in Chapter 7, in the part Notes awaiting migration.

1.1 Inventory risk and the price of immediacy

Model 1.1 Grossman–Miller market making: the price of immediacy

Statement. Competitive, risk-averse market makers supply immediacy to a liquidity trader who arrives before the trader with the offsetting need, and the price concession they charge compensates exactly for carrying the inventory until that trader arrives [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 21]. In the simplified three-date version, n identical market makers, liquidity trader LT1 holding i units and liquidity trader LT2 holding $-i$ units (arriving at date 2) trade an asset worth $S_3 = \mu + \epsilon_2 + \epsilon_3$ at date 3, and the date-1 equilibrium is [crypto-cartea, same chap., pp. 23–24]

$$S_1 = \mu - \gamma\sigma^2 \frac{i}{n+1}, \qquad q_1^{MM} = q_1^{LT1} = \frac{i}{n+1},$$

so LT1 sells $\frac{n}{n+1}\, i$ at once and each market maker holds $\frac{i}{n+1}$ until date 2, when LT2 arrives, the price is efficient, $S_2 = \mathbb{E}[S_3 \mid \epsilon_2] = \mu + \epsilon_2$, and every agent exits with zero inventory [crypto-cartea, same chap., pp. 22–24]. The liquidity discount is $|S_1 - \mu| = \gamma\sigma^2 \, |i| \, /(n+1)$; when LT1 buys ($i < 0$) the market makers earn the same amount as a premium [crypto-cartea, same chap., p. 24]. Figure 1.1 shows the holdings and trades.

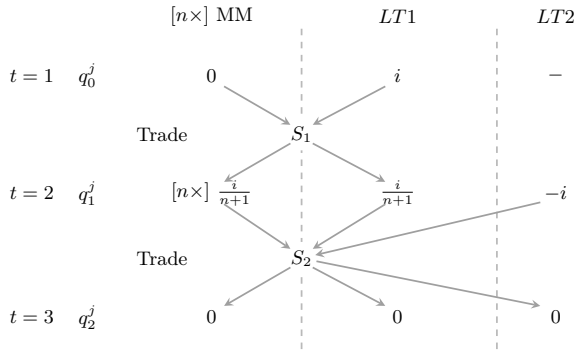


Figure 1.1: Holdings q_t^j and trading in the Grossman–Miller model, after [crypto-cartea, Figure 2.1, p. 23].

Notation. Local override for this entry, the fee result and the liquidity measures: σ^2 is the variance of one period's value shock, in squared price units and not annualised; γ is the coefficient of absolute risk aversion; q_t^j is agent j's holding on leaving date t, and S_t the date-t trade price.

Conditions. Dates $t \in \{1, 2, 3\}$; shocks ϵ_2, ϵ_3 independent $N(0, \sigma^2)$, each announced between dates $t-1$ and t; every agent has CARA utility $U(X) = -\exp(-\gamma X)$; market makers start with no inventory; the arrival of LT2 at date 2 is certain; no trading or holding costs [crypto-cartea, same chap., p. 21]. Interpretation: the concession charges for risk, not for information; no agent is informed, which separates this model from Kyle's. Idea of the proof: solve backwards; with CARA utility and normal shocks each agent's demand is $(\mathbb{E}[\text{next price}] - \text{price})/(\gamma\sigma^2)$, and market clearing at each date gives the prices [crypto-cartea, same chap., pp. 22–23]; the full working is in the derivations artefact listed under *Further reading*.

Practice. The discount rises with the size of the liquidity need $|i|$, risk aversion γ and volatility σ^2, and falls with competition n; as n grows the premium goes to zero, S_1 tends to μ, and LT1's immediate trade tends to i [crypto-cartea, same chap., p. 24]. Competition is itself set by participation costs: in Grossman and Miller's earlier entry stage, the number of market makers falls monotonically as the participation cost c rises, which raises the discount [crypto-cartea, same chap., p. 25]. Per-share fees and two empirical liquidity measures follow from the same equilibrium: see trading fees and price impact and return reversal. Interpretation: the quoted inside spread implied by the model, a bid at $\mu - p$ and an ask at $\mu + p$ with $p = \gamma\sigma^2 |i| / (n+1)$, is $2p$.

Pitfalls. The model solves a single Walrasian clearing rather than a trading mechanism [crypto-cartea, same chap., pp. 26–27], and the offsetting trader arrives for sure; the arrival time and its uncertainty, which drive inventory risk in continuous-time quoting models, are absent. Interpretation: reading every concession as risk compensation overstates inventory costs where order flow is informed.

See also. Kyle single-auction model (contrast: risk-neutral market makers, price impact from information); Avellaneda–Stoikov quoting (generalisation: inventory risk in continuous time with random order arrival); static limit-order depth (the same immediacy supplied through posted orders).

Further reading. [grossman-miller-1988], the original model, including participation costs (not yet read for this entry); own full derivation, from the legacy notes: `notes/artefacts/market-making-lob/derivations.tex`.

Index. Grossman–Miller; immediacy; liquidity premium; liquidity discount; inventory risk; risk transfer; dealer competition; CARA–normal equilibrium

Status. Verification: partly checked 2026-10-14 (every formula and the comparative statics against `crypto-cartea`, pp. 21–27, read from page images of a scanned copy; the original paper not consulted). Stability: draft. Currency: timeless (theory).

Result 1.2 Grossman–Miller with per-share trading fees

Statement. If every trader pays a fee η per share bought or sold, the date-2 price is unchanged, $S_2 = \mu + \epsilon_2$, but for a seller ($i > 0$) the date-1 price falls to [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", pp. 25–26]

$$S_1 = \mu - \gamma\sigma^2 \frac{i}{n+1} - 2\Big(\frac{n}{n+1}\Big)\eta, \qquad q_1^{LT1,*} = \frac{i}{n+1} + 2\Big(\frac{n}{n+1}\Big)\frac{\eta}{\gamma\sigma^2}.$$

The initiating liquidity trader pays his own fee and, through the lower price, the fraction $n/(n+1)$ of the market makers' round-trip fees 2η, and keeps more inventory after date 1 than without fees [crypto-cartea, same chap., p. 26].

Notation. As in the Grossman–Miller model.

Conditions. The model of the Grossman–Miller entry, with fees proportional to shares rather than to dollar volume, positions remaining after date 2 liquidated at date 3, and desired trades large enough that every participant prefers trading to not trading [crypto-cartea, same chap., p. 25]. Idea of the proof: fees cancel at

date 2 because remaining positions are sold either then or at date 3; at date 1 they reduce the market makers' demand by the round-trip fee, $q_1^{MM} = (\mu - S_1 - 2\eta)/(\gamma\sigma^2)$, and clearing $i = nq_1^{MM} + q_1^{LT1}$ gives the price [crypto-cartea, same chap., pp. 25–26].

Practice. Liquidity traders whose desired trade is small relative to the fee do not trade at all [crypto-cartea, same chap., p. 25]. Participation costs and fees act differently: participation costs work through the number of market makers, while fees work mainly through the discount and the immediacy obtained, since each market maker pays 2η per trade but recovers $2n\eta/(n+1)$ through the discount [crypto-cartea, same chap., p. 26].

Pitfalls. Real exchange fees are often proportional to dollar volume and differ by side and by maker or taker [crypto-cartea, same chap., p. 25]; the closed form assumes the same per-share fee for everyone.

See also. Grossman–Miller model (the base case, $\eta = 0$).

Index. trading fees; exchange fees; fee incidence; participation cost; no-trade region; liquidity discount

Status. Verification: partly checked 2026-10-14 (the formulas and fee incidence against `crypto-cartea`, pp. 25–26, from page images of a scanned copy in which several signs are faint; the printed results agree with the clearing condition on the same page). Stability: draft. Currency: timeless (theory).

Result 1.3 Grossman–Miller liquidity measures: linear price impact and negative autocovariance

Statement. Read as trading in a limit-order book, where LT1's market orders walk the book posted by the market makers, the Grossman–Miller date-1 price is linear in the quantity LT1 trades [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", pp. 26–27]:

$$S_1 = \mu + \lambda\, q^{LT1}, \qquad \lambda = -\frac{1}{n}\,\gamma\sigma^2, \qquad q^{LT1} = i\,\frac{n}{n+1}.$$

With public news ϵ_1 before LT1 trades and efficient price $\mu_t = \mathbb{E}[S_3 \mid \epsilon_1, \dots, \epsilon_t]$, the price changes $\Delta_1 = S_1 - S_0$ and $\Delta_2 = S_2 - S_1$ satisfy [crypto-cartea, same chap., pp. 27–28]

$$\operatorname{Cov}[\Delta_1, \Delta_2] = -\lambda^2 \operatorname{Var}[q^{LT1}] < 0.$$

Notation. As in the Grossman–Miller model; here q^{LT1} is the quantity LT1 trades at date 1, not a holding, and λ is the slope of price on that quantity.

Conditions. The model of the Grossman–Miller entry, with shocks $\epsilon_1, \epsilon_2, \epsilon_3$ i.i.d. normal with variance σ^2, and liquidity demand independent of news [crypto-cartea, same chap., pp. 27–28]. Idea of the proof: $S_1 = \mu_1 + \lambda q^{LT1}$ and $S_2 = \mu_2$, so the order-pressure term enters Δ_1 and reverses in Δ_2 [crypto-cartea, same chap., p. 28].

Practice. A more liquid market has a smaller $|\lambda|$, from more competition, less risk aversion or lower volatility [crypto-cartea, same chap., p. 27]. As λ goes to zero, so does the autocovariance, and the price process converges to the efficient-price martingale [crypto-cartea, same chap., p. 28]. The two measures coincide here only because the model is essentially static; in dynamic models they capture different things, permanent effects of order flow on the efficient price as against the temporary effect in each trader's execution price [crypto-cartea, same chap., p. 28].

Pitfalls. Interpretation: an empirical regression of price changes on signed order flow, or a negative return autocovariance, mixes this inventory effect with information and bid–ask bounce; this model alone cannot attribute it. Compare the information slope of Kyle's model.

See also. Grossman–Miller model (source of the equilibrium); Kyle single-auction model (contrast: a price-impact slope from information).

Index. price impact; Kyle's lambda (contrast); liquidity measure; autocovariance of price changes; return reversal; temporary impact; transitory price pressure

Status. Verification: partly checked 2026-10-14 (both formulas and the comparative statics against `crypto-cartea`, pp. 26–28, from page images of a scanned copy). Stability: draft. Currency: timeless (theory).

1.2 Quoting with limit orders

Model 1.4 Static limit-order depth with exponential fill probability

Statement. A small, risk-neutral market maker with costless inventory and infinite patience chooses depths $\delta^{\pm}$ from the midprice S_t for a sell order at $S_t + \delta^+$ and a buy order at $S_t - \delta^-$; a buy (sell) market order arrives with probability p^+ (p^-) and reaches depth δ with probability $e^{-\kappa^{\pm}\delta}$, and expected profit per trade is maximised by [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 29, eq. (2.3)]

$$\max_{\delta^+,\delta^-} \left\{ p^+ e^{-\kappa^+\delta^+}\delta^+ + p^- e^{-\kappa^-\delta^-}\delta^- \right\}, \qquad \delta^{\pm,*} = \frac{1}{\kappa^{\pm}}.$$

The optimal depth equals the mean depth of the book on that side [crypto-cartea, same chap., p. 30]. This is a static simplification set up in place of the dynamic Ho–Stoll dealer model, which the authors give as the usual first reference [crypto-cartea, same chap., p. 28]. Figure 1.2 shows the depths and fill probabilities.

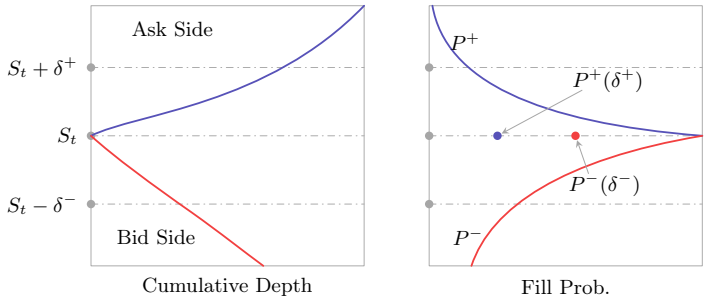


Figure 1.2: Limit-order depths and probability of execution, after [crypto-cartea, Figure 2.2, p. 30].

Notation. Local override for this entry: $p^{\pm}$ are probabilities of market-order arrival, not prices, and $P^{\pm}(\delta)$ is the probability that an arriving market order reaches depth δ, not a bond or put price.

Conditions. Everyone agrees on the value S_t (no information asymmetry); other market makers' orders form a fixed book that does not react; the market maker has no inventory or trading costs and liquidates at the midprice [crypto-cartea, same chap., pp. 28–29]. Idea of the proof: the problem separates by side, and the first-order condition $p^{\pm}e^{-\kappa^{\pm}\delta^{\pm}}(1-\kappa^{\pm}\delta^{\pm}) = 0$ gives the depth; the arrival probabilities scale profit but not the optimal depth (Interpretation, own derivation).

Practice. The model isolates the trade-off between margin per trade and probability of execution [crypto-cartea, same chap., p. 30]; quoting deeper earns more per fill and fills less often.

Pitfalls. The authors list what makes it unrealistic: the arrival and fill probabilities are special, constant and exogenous; the market maker's decision does not affect other traders; and the objective is static [crypto-cartea, same chap., p. 30]. Interpretation: there is no inventory penalty and no adverse selection, so the depth does not respond to position or to informed flow; Avellaneda–Stoikov adds inventory.

See also. Avellaneda–Stoikov quoting (generalisation: dynamic, with inventory and the same exponential intensity); Grossman–Miller model (the immediacy this quoting supplies).

Further reading. [ho-stoll-1981], the dynamic dealer model this simplifies (not yet read for this entry; see the pathway backlog); own full derivation, from the legacy notes: `notes/artefacts/market-making-lob/derivations.tex`.

Index. limit-order placement; quote depth; fill probability; probability of execution; exponential order-book depth; static market making; Ho–Stoll (static simplification)

Status. Verification: partly checked 2026-10-14 (the problem, the solution and the stated limitations against `crypto-cartea`, pp. 28–30, from page images of a scanned copy; the Ho–Stoll paper not consulted). Stability: draft. Currency: timeless (theory).

Model 1.5 Avellaneda–Stoikov inventory-aware quotes

Statement. A dealer with CARA utility, inventory q and horizon T quotes around a reservation (indifference) price rather than the midprice s [s6-stoikov, sec. 2.2, eq. (8), p. 219]:

$$r(s, q, t) = s - q\gamma\sigma^2(T - t).$$

With symmetric exponential execution intensities $\lambda^a(\delta) = \lambda^b(\delta) = Ae^{-k\delta}$, an expansion of the value function in q with a first-order approximation of the order-arrival terms gives the same reservation price and a bid–ask spread, independent of inventory, of [s6-stoikov, sec. 3.2, eqs. (29)–(30), p. 221]

$$\delta^a + \delta^b = \gamma\sigma^2(T - t) + \frac{2}{\gamma}\ln\left(1 + \frac{\gamma}{k}\right).$$

Notation. Local override for this entry: the midprice follows $dS_u = \sigma\, dW_u$, so σ is an absolute volatility in price units per square root of the model's time unit, not the annualised volatility of returns; s is the current midprice; $\delta^a = p^a - s$ and $\delta^b = s - p^b$ are the quote distances; λ^a, λ^b are Poisson execution intensities, not price-impact slopes [s6-stoikov, secs. 2.1 and 2.4, pp. 218–219].

Conditions. Arithmetic Brownian midprice with no drift and a zero interest rate; exponential utility of terminal wealth plus inventory marked at the midprice; one-share limit orders, updated continuously at no cost, filled at Poisson rates that decrease with distance from the midprice [s6-stoikov, secs. 2.1–2.4, pp. 218–219]. The exponential intensity follows from a constant market-order frequency, a power-law order-size distribution

and a logarithmic temporary impact, with $A = \Lambda/\alpha$ and $k = \alpha K$; a power-law impact gives a power-law intensity instead [s6-stoikov, sec. 2.5, eq. (12), p. 220]. Exact quotes solve the HJB equation and two implicit equations for the distances [s6-stoikov, sec. 3.1, eqs. (15)–(19), pp. 220–221]; the closed form above is an approximation [s6-stoikov, sec. 3.2, p. 221]. A long position ($q > 0$) puts the reservation price below the midprice, so both quotes shift down to sell [s6-stoikov, sec. 2.2, p. 219].

Practice. In 1000 simulations with $s = 100$, $T = 1$, $\sigma = 2$, $\gamma = 0.1$, $k = 1.5$, $A = 140$ and $dt = 0.005$, the inventory strategy earned a mean profit of 65.0 with standard deviation 6.6, against 68.4 and 12.7 for a strategy quoting the same average spread of 1.49 symmetrically around the midprice; final inventory had standard deviation 2.9 against 8.4 [s6-stoikov, sec. 3.3, Table 1, p. 222]. The symmetric strategy earns slightly more because it trades more, but its P&L is much more dispersed [s6-stoikov, sec. 3.3, p. 222].

Pitfalls. Interpretation: the closed form is a control template, not a production model; real books add queue position, tick size, discrete inventory limits, latency, rebates, hedging instruments and fill probabilities that change with regime. The approximate spread does not depend on inventory only because the intensities are exponential [s6-stoikov, sec. 3.2, p. 221]. Interpretation: the formula has no inventory limit; a hard limit needs the bounded-inventory solution read in this pathway (`wide-gueant-lehalle-tapia`, not yet read).

See also. Static limit-order depth (special case: no inventory risk, static); Grossman–Miller model (the same inventory-risk premium in a three-date equilibrium).

Further reading. Own working from the legacy notes: `notes/artefacts/market-making-lob/derivations.tex`.

Index. Avellaneda–Stoikov; reservation price; indifference price; inventory skew; optimal bid–ask spread; market making with limit orders; exponential fill intensity; HJB market making

Status. Verification: partly checked 2026-10-14 (the reservation price, intensity, spread and simulation results against the author-hosted journal PDF of `s6-stoikov`, pp. 217–222; the HJB equation and expansion steps read but not re-derived). Stability: draft. Currency: as published in 2008; review annually with the catalogue record.

1.3 Informed trading and informative prices

Definition 1.6 Fundamental value, market value and noise: what informative prices are informative about

Statement. The *market value* of an instrument is the price at which traders can buy or sell it. Its *fundamental value* (intrinsic value) is the expected present value of all present and future benefits and costs of holding it: the value everyone would agree on if all knew everything known about the instrument, analysed it properly and perceived its benefits and costs alike [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.1, pp. 222–223]. The difference between the two is *noise*; prices are informative when they are near fundamental values and completely informative when they equal them [alpha-harris, same sec., p. 223]. Writing V_t for fundamental value and P_t for the price, noise is $\epsilon_t = P_t - V_t$. Fundamental values use only information available now, so they are the best estimates of perfect-foresight values, not those values; and their changes are unpredictable, since a predictable change would mean they did not yet reflect the information behind the prediction [alpha-harris, same sec., p. 223]. Relative to a price, information is of three kinds:

- *public*: available to any trader, such as public news and past prices and volumes [alpha-harris, same chap., pp. 223 and 240];
- *private*: held only by some traders, whether learnt first or produced by their own research [alpha-harris, same chap., pp. 223 and 228];
- *stale*: already in the price, meaning it cannot be used to forecast future price changes; traders who act on it are *pseudo-informed*, and in fact uninformed [alpha-harris, same chap., sec. 10.5.2.1, p. 229].

Notation. V_t fundamental value, P_t market price, $\epsilon_t = P_t - V_t$ noise; the sign is a convention, since Harris defines noise only as the difference.

Conditions. Fundamental values are not observed, so whether a price is informative or noisy cannot easily be determined [alpha-harris, same chap., p. 223]. Prices become informative in two ways: when values are common knowledge no one trades at other prices; otherwise informed traders' trading moves prices towards their estimates, which is the more important route because values are rarely common knowledge [alpha-harris, same chap., sec. 10.6.1, pp. 236–237]. Information gets into the price when all traders understand its significance or when informed traders push the price towards their estimates [alpha-harris, same chap., p. 229].

Practice. Whether information is stale can rarely be checked directly. Harris's guide: information that is unique and that others could not have anticipated is probably not stale; information acted on slowly, widely known, cheap to obtain, or reasonably anticipated probably is. His example: Occidental Petroleum rose 10 percent the day after its long-ill chairman died and fell back the next day, because his death was expected and

its date was not material [alpha-harris, same chap., p. 229 and box, p. 230]. Fischer Black would call stock prices informative if they lay between half and twice their fundamental values [alpha-harris, same chap., box, p. 223].

Pitfalls. Two conditioning sets are in use. Harris's fundamental value conditions on everything known to anyone, private information included. Hasbrouck calls the expectation of the payoff conditional on *all public information* the fundamental value or efficient price [s6-hasbrouck, sec. 3.2, p. 26]: the m_t of the random-walk decomposition. Harris's noise is therefore not Hasbrouck's pricing error $s_t = p_t - m_t$. A price equal to the public-information expectation has a zero pricing error but nonzero noise whenever private information has not yet reached it (Interpretation). The legacy notes defined fundamental value as "implied by all payoff-relevant information", which reads as the perfect-foresight value that Harris distinguishes from it. They also described private information as "costly or restricted"; private information can be produced lawfully from public data by research [alpha-harris, chap. "Insider Trading", p. 585].

See also. fundamental and transitory volatility (the statistical version, conditioned on public information); informed trading and aggregation (how prices become informative); market efficiency; inside information; outside spread (value traders' estimates in use).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. fundamental value; intrinsic value; market value; noise; informative prices; perfect-foresight value; random walk; public information; private information; stale information; pseudo-informed traders; efficient price

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 222–230, 236–237, 240 and 585, and `s6-hasbrouck`, 2007 Oxford printing, p. 26). Stability: draft. Currency: timeless (concepts).

Result 1.7 Informed trading makes prices informative: the price as an aggregate of value estimates

Statement. Informed traders buy when the price is below their estimate of fundamental value and sell when it is above. Their buying pushes prices up and their selling pushes them down, so their trading moves prices towards their estimates [alpha-harris, chap. "Informed Traders and Market Efficiency", secs. 10.2–10.3, pp. 223–224]. Harris's algebraic illustration [alpha-harris, same chap., box, p. 225]: N traders hold unbiased forecasts $f_i = V + e_i$ of the value V, each wants the position $D_i = a(f_i - P)$, and the security is in zero net supply. Market clearing, $\sum_i a(f_i - P) = 0$, gives

$$P = \frac{1}{N}\sum_{i=1}^{N} f_i = V + e_M, \qquad e_M = \frac{1}{N}\sum_{i=1}^{N} e_i.$$

With independent errors, $e_M \to 0$ as N grows, by the law of large numbers: prices are most informative when many informed traders collect information independently. Interpretation: if the errors are also independent of V with common variance σ^2, then $\mathbb{E}[P \mid V] = V$ and $\mathrm{Var}(P - V) = \sigma^2/N$.

Conditions. When forecasts differ in accuracy, the best combination is a weighted average that gives more weight to the more accurate. Well-informed traders tend to be wealthier and take larger positions, so prices approximate such an optimally weighted average: the market is a statistical calculator [alpha-harris, same chap., p. 224 and box, p. 226]. Interpretation: with trader-specific slopes a_i the same clearing condition gives $P = \sum_i a_i f_i / \sum_i a_i$. Many mistaken traders, or one large one, make prices less informative, until their losses drive them out [alpha-harris, same chap., p. 224]. Informed traders do not trade to make prices informative: price impact is their cost, and they want prices to adjust only after they have built their positions [alpha-harris, same chap., p. 224]. They profit only from uninformed traders; competition among them pushes prices closer to values and lowers their profits until the least successful just cover their costs [alpha-harris, same chap., secs. 10.6.3 and 10.6.5, pp. 237–239].

Practice. Trade aggressively when the information will soon be common knowledge or many others will act on it; trade slowly (stealth trading) when the advantage will last [alpha-harris, same chap., sec. 10.4, pp. 225–226]. The most profitable traders have estimates that are both precise and *orthogonal*, uncorrelated with other traders' [alpha-harris, same chap., sec. 10.6.2.1, p. 237].

Pitfalls. Aggregation removes independent errors, not shared ones. Interpretation: if the errors have common variance σ^2 and pairwise correlation ρ, then $\mathrm{Var}(e_M) = \sigma^2[\rho + (1-\rho)/N] \to \rho\sigma^2$, so a shared data source, model or stale signal leaves an error that no number of traders removes; Harris notes that traders who use the same information and methods have highly correlated estimates [alpha-harris, same chap., p. 237]. Harris also says the market's forecast error is smaller than the average individual error whenever the individual errors are not identical [alpha-harris, same chap., box, p. 225]. That holds for squared errors, since $\left(\frac{1}{N}\sum_i e_i\right)^2 \le \frac{1}{N}\sum_i e_i^2$ with equality only when all e_i are equal. For absolute errors the two are equal whenever all errors have the same sign (Interpretation). The legacy proposition assumed the price is the average forecast; in the box this follows from linear demands with a common slope and zero net supply, an assumption Harris says simplifies

the arithmetic without changing the qualitative results. The legacy mechanism had informed traders buy when $P_t < V_t$: they act on estimates, and lose when their estimates are poor [alpha-harris, same chap., p. 223].

See also. fundamental value; market efficiency (why prices cannot be fully informative); Kyle single-auction model and Glosten–Milgrom model (equilibrium models of informed trading); price formation process.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. informed traders; informative prices; information aggregation; forecast averaging; law of large numbers; correlated errors; orthogonal estimates; stealth trading; statistical calculator; price impact as a cost

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 222–226 and 236–239, the box on p. 225 included; no page images). Stability: draft. Currency: timeless (theory).

Definition 1.8 Market efficiency: weak, semistrong and strong forms, and the definition with information costs

Statement. In an *efficient market*, prices reflect all information that traders can acquire and profitably trade upon. The definition builds in the costs of acquiring information and of acting on it, and the impact of informed trading on prices [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.6.6, p. 240]. Harris sets it beside the three traditional forms [alpha-harris, same sec., p. 240]:

- *weak form*: prices reflect all information in past prices, so no one can predict price changes from past prices alone;
- *semistrong form*: prices reflect all public information, which includes all public news and past prices and volumes in all securities and contracts;
- *strong form*: prices reflect all public and private information as soon as it is known; since informed traders could never profit, only markets in instruments whose values are common knowledge are strong-form efficient.

Conditions. Prices never fully reflect all the information traders could collect: information that is too expensive or too inconsequential is not acquired [alpha-harris, same sec., p. 240]. Harris's market paradox: if prices were fully informative, informed trading would not pay, so no one would make them informative. Prices and values change constantly, so gaps open and informed trading closes them: prices are informative but not always [alpha-harris, same chap., sec. 10.6.4, pp. 238–239]. Lehalle and Laruelle make the same point with Grossman and Stiglitz (1980), not consulted: market dynamics must contain enough inefficiency to reward those who improve the information in prices [alpha-laruelle, Introduction, p. 4]. How informative prices are depends on the costs of information and of trading, and on there being uninformed traders to lose to informed ones [alpha-harris, same chap., secs. 10.6.3 and 10.7, pp. 238 and 243–244].

Practice. Informative prices are not free. What uninformed traders lose to informed traders pays for much of the information in prices, so policies that frustrate informed traders can make prices less informative and markets more liquid; restrictions on insider trading are an example [alpha-harris, same chap., sec. 10.6.7, p. 241]. Harris concludes that these restrictions narrow spreads and make prices less informative in the short run [alpha-harris, chap. "Insider Trading", sec. 29.4, pp. 597–598]. The trade-off is not universal: publishing material fundamental information raises both liquidity and informativeness [alpha-harris, chap. "Informed Traders and Market Efficiency", sec. 10.6.8, pp. 241–242]. Trading halts before material releases protect uninformed traders, especially those who offer liquidity, at the cost of a few minutes' delay in price formation, a trade Harris judges worth making [alpha-harris, same chap., box, p. 241]. Most published studies find markets weak-form efficient, and semistrong-form efficient for public information that is easy to obtain and interpret [alpha-harris, same chap., p. 240].

Pitfalls. Comparing market structures needs the change in benefits, not their totals, and the benefits of informative prices and of liquidity are both hard to measure [alpha-harris, same chap., sec. 10.6.7, p. 241]. The legacy notes put volumes in the weak-form information set; Harris's weak form uses past prices, and his semistrong set adds volumes. They also called the cost-sensitive definition *economic* efficiency, a name Harris does not use. Their trade-off said informed trading transfers wealth from "slower" liquidity traders and that efficiency-raising policies reduce "displayed" liquidity. Harris speaks of uninformed traders and liquidity suppliers, and of liquidity in general.

See also. fundamental value; informed trading and aggregation; inside information (the main example of the trade-off); spread determinants (asymmetric information widens spreads).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`. Grossman and Stiglitz (1980), cited by Lehalle and Laruelle, not consulted.

Index. market efficiency; efficient markets; weak-form efficiency; semistrong-form efficiency; strong-form efficiency; information costs; market paradox; Grossman–Stiglitz paradox; liquidity and informative prices; public information; trading halts

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 237–244 and 597–598, and `alpha-laruelle`, 2018 World Scientific printing, p. 4). Stability: draft. Currency: timeless (concepts); the empirical summary as of 2002.

Definition 1.9 Inside information: material nonpublic information and who may not trade on it (United States, as summarised in 2002)

Statement. For economic analysis, Harris defines *inside information* as material information about the value of a security that is not available to public traders, and *material* information as information that would cause prices to change if it were widely known; *insider trading* is trading on it [alpha-harris, chap. "Insider Trading", sec. 29.1, p. 584]. Inside information keeps its status however many people pass it on, and loses it only when it is publicly available, through a broadly distributed press release or a public filing [alpha-harris, same sec., p. 585]. The legal definitions come from statutes, regulations and case law, are complex, and call for an attorney [alpha-harris, same sec., p. 584]. The SEC's summary, as quoted by Harris from its web page of January 2002, covers trading in breach of a fiduciary duty or other relationship of trust and confidence while in possession of material nonpublic information, and tipping, trading by those tipped, and misappropriation [alpha-harris, same sec., box, p. 585]. Those who may not trade on such information include:

- *corporate insiders*: senior managers, directors, large shareholders, key employees, and anyone involved in corporate activities that will materially affect the firm's value [alpha-harris, sec. 29.1.1, p. 587];
- employees of law, banking, brokerage and printing firms given the information to serve the corporation, and government employees who learn it at work [alpha-harris, sec. 29.1, box, p. 585];
- *tippees*: friends, family and business associates of insiders, and the friends of their friends [alpha-harris, same sec., pp. 584–585];
- *misappropriators*, who take confidential information from an employer or other source. Chiarella's conviction was reversed in 1980 because he owed no fiduciary duty to the firms whose stock he traded; Harris says that under the later law he would be convicted for misappropriation [alpha-harris, same sec., box, p. 586]. Winans was convicted under the misappropriation doctrine [alpha-harris, same sec., box, p. 587].

Analysts whose private information comes only from analysing public information are not trading on inside information [alpha-harris, same sec., p. 585].

Conditions. Jurisdiction and date: United States law as Harris summarises it in 2002. Most countries prohibit insider trading, but only a few, mainly the United States, Canada and Great Britain, regularly and seriously enforce their laws [alpha-harris, chap. "Insider Trading", p. 584]. Regulation FD (2000) requires an issuer that discloses material nonpublic information to anyone to disclose it publicly [alpha-harris, same chap., box, p. 586].

Practice. Corporations restrict their employees' trading: many require corporate approval before a trade, may refuse it before a release, or allow trading only in a window after quarterly earnings [alpha-harris, same chap., sec. 29.1.1, p. 588]. Exchange surveillance looks for large price changes on large volume before significant news [alpha-harris, same chap., sec. 29.2, p. 588]. Insider trading is hard to prove: a sophisticated speculator can claim independent analysis of public data, and trading on information that is not revealed in a discrete event is essentially undetectable from prices and volumes [alpha-harris, same chap., pp. 589–590]. Like all informed trading, insider trading hurts liquidity suppliers and raises spreads for uninformed traders [alpha-harris, same chap., sec. 29.3.1.2, pp. 591–592].

Pitfalls. Harris's categories are legal categories summarised for economists, not a legal test; check the law in force. The reporting deadlines, prearranged-plan rules and penalties he gives [alpha-harris, same chap., pp. 587–591] are as of 2002, are not repeated here, and were not checked against current law. The legacy notes used "temporary insider" and a "relationship, benefit or confidence" test for tipping. Neither term appears in the passages checked. They also named auditors, consultants and exchange staff, whom the SEC summary does not list. A legacy principle, *signal tradability requires permission*, said that data provenance, confidentiality duties, firm restricted lists, compliance approvals and execution capacity decide whether a predictive signal may be used. The passages checked support confidentiality duties (misappropriation), employer approval of employees' own trades (p. 588), and the dependence of informed trading profits on liquidity [alpha-harris, chap. "Informed Traders and Market Efficiency", box, p. 236]. Firm restricted lists, compliance approval of signals and the provenance of purchased data are not in them.

See also. fundamental value (private and public information); market efficiency (insider-trading restrictions in the information–liquidity trade-off); spread determinants (enforced insider-trading rules as a secondary determinant).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. inside information; insider trading; material nonpublic information; MNPI; materiality; corporate insiders; tippees; misappropriation; Chiarella; Regulation FD; trading windows; preclearance; market surveillance; signal permission

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 236 and 584–592; the SEC web page Harris quotes was not consulted). Stability: draft. Currency: dated, United States law as summarised in 2002; not checked against current law or other jurisdictions.

Definition 1.10 Price formation process: immediate impact and its decay

Statement. The *price formation process* (PFP) is how trades and information become prices. Lehalle and Laruelle report a consensus that it consists of the impact of each trade combined with the variation of the asset's "fair price", after Bouchaud et al. (2002), not consulted [alpha-laruelle, Introduction, p. 27]. They split it into two phases [alpha-laruelle, same, Figure 8, pp. 27–29]:

1. *immediate impact*: trades move the price through the balance of buyers and sellers, and of liquidity-providing and liquidity-consuming orders, together with moves of the spread;
2. *decay*, or the *resilience* of the book: liquidity returns, or does not, according to the distance between the best quotes and investors' views of the fair price.

Pressure inconsistent with participants' consensus generates only temporary impact; pressure consistent with it is not pushed back, and the impact is permanent [alpha-laruelle, same, pp. 3 and 28]. The PFP is mainly driven by information [alpha-laruelle, same, p. 7].

Conditions. The more investors with views of the fair price take part, the faster impact decays. Market makers with inventory constraints contribute to the decay without any view of the fair price, but only if their inventory exceeds the turnover of a price jump [alpha-laruelle, Introduction, p. 28]. Post-trade transparency speeds the decay by warning traders that the last price is out of line with their views [alpha-laruelle, same, p. 28]. A design tension runs through it: the more information traders have, the better the PFP should be, but each trader fears information leakage, front-running and excessive impact [alpha-laruelle, same, pp. 19–20].

Practice. Market makers buy from early sellers and sell to later buyers, so that most temporary oscillations shrink to a spread related to the asset's risk [alpha-laruelle, Introduction, pp. 3–4]. Volatility is the random, unexpected component of the PFP, and the more efficient the microstructure, the lower intraday volatility [alpha-laruelle, chap. 2, sec. 2.5.1, pp. 171–172]. In a fragmented market, a consolidated tape organises how information is shared [alpha-laruelle, Introduction, p. 7]; the legacy fragmentation and outage blocks are assigned to `market-structure-execution-quality`. The authors see high-frequency traders' growing role as a possible threat to the decay, since they need not hold a view of the fair price and have small inventory limits [alpha-laruelle, same, p. 28]; that question belongs to `hft-market-design`.

Pitfalls. The impact of a single large order depends on what others are doing. If many others buy at the same time, the impact of a large buy is permanent; if most sell, it can be almost invisible. Impact is measured only by averaging over many market configurations [alpha-laruelle, Introduction, p. 3]. "Clean" and "noisy" PFP, the legacy terms, are not defined by the authors, although they speak of a "less noisy PFP" [alpha-laruelle, chap. 2, sec. 2.5.1, p. 177]. Interpretation: the noise is the transitory component of price changes.

See also. informed trading and aggregation (the information side); component identification (permanent and transitory price effects of trades); Kyle single-auction model (permanent impact from information); Grossman–Miller model (temporary concession for immediacy); returns by time scale.

Further reading. The legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`. Bouchaud et al. (2002), cited for the decomposition, not consulted.

Index. price formation process; PFP; fair price; immediate price impact; impact decay; resilience; temporary impact; permanent impact; post-trade transparency; information leakage; consolidated tape

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, Introduction, pp. 3–7, 19–20 and 27–29, and chap. 2, pp. 171–172 and 177; Figure 8 not inspected). Stability: draft. Currency: timeless (concept); the regulatory context is European, 2007–2018.

1.4 Information and price impact

Model 1.11 Kyle single-auction model: informed trading and linear price impact

Statement. One risk-neutral insider who knows the asset value $v \sim N(\mu, \sigma_v^2)$, price-insensitive noise traders with net order $u \sim N(0, \sigma_u^2)$ independent of v, and competitive risk-neutral market makers who see only the net order flow $y = x + u$ and set $S(y) = \mathbb{E}[v \mid y]$ [crypto-cartea, chap. "A Primer on the Microstructure

of Financial Markets", pp. 31–32] have a linear equilibrium [catalogue-0496, chap. "Strategic Trader Models I: Informed Traders", p. 93, eqs. (4.3)–(4.5)]:

$$x(v) = \beta(v - \mu), \qquad S(y) = \mu + \lambda y, \qquad \beta = \frac{\sigma_u}{\sigma_v}, \qquad \lambda = \frac{\sigma_v}{2\sigma_u}.$$

The insider's ex ante expected profit is $\frac{1}{2}\sigma_v\sigma_u$ [catalogue-0496, same chap., p. 97, eq. (4.17)], and the posterior variance of v after the auction is exactly half the prior, whatever the order flow: price reveals half of the insider's information [catalogue-0496, same chap., p. 98, eq. (4.20)].

Notation. Source notation: `catalogue-0496` writes p_0 for μ, Σ_0 for σ_v^2, σ_μ for σ_u and P for the price; `crypto-cartea` writes σ for σ_v. Here λ is a price-impact slope, in price per share.

Conditions. A single auction; the insider does not observe u before trading; normality of v and u; market makers earn zero expected profit given the order flow, so price is semi-strong efficient [catalogue-0496, same chap., pp. 91–92]; [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 32]. There is no other linear equilibrium, but nonlinear equilibria are not excluded [catalogue-0496, chap. "Strategic Trader Models I: Informed Traders", p. 97]. Idea of the proof: given a linear price rule the insider's expected profit $x(v - \mu) - \lambda x^2$ is quadratic, so $\beta = 1/(2\lambda)$; given a linear strategy, v and y are jointly normal and the projection $\mathbb{E}[v \mid y]$ is linear in y; the two conditions fix β and λ [catalogue-0496, same chap., pp. 93–96].

Practice. Doubling the noise-trader standard deviation doubles the insider's order and expected profit and leaves the ex ante price unchanged [catalogue-0496, same chap., p. 97]. Market makers are risk-neutral, so any premium compensates their informational disadvantage and is borne by the price-insensitive liquidity traders [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 31]. Interpretation (own derivation): the insider trades only part of the mispricing, because a larger order reveals more through y; λ rises with σ_v and falls with σ_u.

Pitfalls. A linear price rule implies a negative price for a large enough sell imbalance, so it is at best an approximation of actual price behaviour [catalogue-0496, chap. "Strategic Trader Models I: Informed Traders", p. 96]. The joint-normality assumption drives the predictions on trade size and volume [catalogue-0496, same chap., p. 98]. The textbook account in `crypto-cartea` prints the equilibrium price as $\mu + 2(x + u)\sigma_u/\sigma$, so $\lambda = 2\sigma_u/\sigma$ [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 33]; this is the reciprocal of the value in `catalogue-0496`, eq. (4.5), and of the value implied by its own preceding step $\beta = (2\lambda)^{-1}$. The authors' errata correct the price to $\mu + (x + u)\sigma/(2\sigma_u)$, so $\lambda = \sigma/(2\sigma_u)$, in agreement with `catalogue-0496` [crypto-cartea-errata, entries for p. 33]. Do not copy λ from the printed page.

See also. Grossman–Miller model (contrast: a premium for risk, not for information); Grossman–Miller liquidity measures (contrast: an inventory price-impact slope).

Further reading. [s6-kyle], the original model and its dynamic version (not yet read for this entry); own full derivation, from the legacy notes: `notes/artefacts/market-making-lob/derivations.tex`.

Index. Kyle model; Kyle's lambda; price impact; informed trading; insider; noise traders; adverse selection; semi-strong efficiency; information revelation; toxic order flow

Status. Verification: partly checked 2026-10-14 (the equilibrium, profit and posterior variance against `catalogue-0496`, pp. 91–98, from page images of a scanned 1998 printing; the setup against `crypto-cartea`, pp. 31–33, whose printed λ conflicts as noted under *Pitfalls*; Kyle's paper not consulted; the p. 33 misprint confirmed against the authors' errata page 2026-10-15). Stability: draft. Currency: timeless (theory).

1.5 Adverse selection and the bid–ask spread

Model 1.12 Glosten–Milgrom sequential-trade model: spreads from adverse selection

Statement. In a pure dealership market, investors arrive one at a time, randomly and anonymously, see the specialist's bid b_t and ask a_t, and buy one unit at the ask, sell one unit at the bid, or leave; some arrivals are informed about the asset's future value V, the rest trade for liquidity reasons [glosten-milgrom-1983, sec. II, pp. 3–4]. An arrival with information $\mathcal{F}_t$ has reservation price $Z_t = \rho_t\,\mathbb{E}[V \mid \mathcal{F}_t]$ and buys if $Z_t > a_t$, sells if $Z_t < b_t$; competitive, risk-neutral specialists with information $\mathcal{S}_t$ earn zero expected profit on each purchase and each sale, so the quotes solve [glosten-milgrom-1983, p. 6]

$$a_t = \mathbb{E}[V \mid \mathcal{S}_t,\ Z_t > a_t], \qquad b_t = \mathbb{E}[V \mid \mathcal{S}_t,\ Z_t < b_t].$$

Then $a_t \geq \mathbb{E}_t[V] \geq b_t$, with strict inequalities whenever informed buying above and informed selling below $\mathbb{E}_t[V]$ have positive probability: adverse selection alone produces a spread, with no inventory cost, risk aversion or monopoly power [glosten-milgrom-1983, Prop. 1, pp. 7–9]. Each transaction price is the specialist's updated expectation of V, so transaction prices form a martingale with respect to the specialist's and the public information [glosten-milgrom-1983, Prop. 2, p. 10].

Notation. $V > 0$ is the asset's value, realised at a future date, with finite variance; ρ_t is the arriving investor's liquidity (time-preference) parameter, with $\rho = 1$ for the specialist; $\mathcal{F}_t$ and $\mathcal{S}_t$ are the information of the arrival and of the specialist; $\mathbb{E}_t$ is expectation given the information both share [glosten-milgrom-1983, pp. 4–7]. The paper writes A_t, B_t for a_t, b_t, F_t, S_t for $\mathcal{F}_t, \mathcal{S}_t$.

Conditions. Every participant is risk-neutral with utility $\rho x V + c$ over shares x and consumption c; ρ is independent across traders and carries no information about V [glosten-milgrom-1983, pp. 4–5]. Uninformed participation is exogenous; trades are single units; short positions in cash or stock cost nothing; zero profit on each side is motivated by Bertrand competition between equally informed specialists, and existence of the quotes is not shown in general [glosten-milgrom-1983, pp. 4–6]. If the specialist sees only public information, the ask is the smallest a with $a \geq \mathbb{E}[V \mid \mathcal{H}_t, Z_t > a]$ and the bid the largest b with $b \leq \mathbb{E}[V \mid \mathcal{H}_t, Z_t < b]$ [glosten-milgrom-1983, p. 7]. Idea of the proof of Proposition 1: conditional on a buy, the ask equals the expectation of V given the buy; since buying selects arrivals whose own expectation exceeds a threshold and ρ is uninformative, that conditional expectation is at least $\mathbb{E}_t[V]$ [glosten-milgrom-1983, p. 8].

Practice. The spread is larger, for given information, when insiders' information is finer, when the ratio of informed to uninformed arrivals rises, or when the dispersion of the uninformed liquidity parameter around its median falls [glosten-milgrom-1983, Prop. 5, pp. 15–16]. If buy and sell probabilities stay above $\theta_a, \theta_b > 0$, prices converge and the root-mean-square spread over the first k trades is at most $\sqrt{\mathrm{Var}(V)/(\theta_a \theta_b k)}$ [glosten-milgrom-1983, Prop. 3, p. 11]; with Poisson arrivals the average spread is then of the order of one over the square root of volume, because trades reveal information [glosten-milgrom-1983, pp. 18–19]. The specialist and the arrivals come to agree on the expected value of V [glosten-milgrom-1983, Prop. 4, p. 13]. The market maker need not know which traders are informed: since well-informed traders rarely reveal themselves, dealers treat every order as possibly informed and revise prices with the side of each trade [alpha-harris, chap. "Dealers", pp. 288–289]. The two-point version with closed-form half-spreads is a separate result.

Pitfalls. If the liquidity motive is weak relative to the insiders' advantage, the zero-profit quotes may preclude trade, and the market shuts down until enough public information neutralises the insiders' advantage [glosten-milgrom-1983, p. 9]; if the bounded-probability condition fails, the spread need not shrink: the market may close, or every trade may occur at the bid [glosten-milgrom-1983, p. 12]. Proposition 5 compares spreads for given information; it does not say that stocks with more insiders have wider spreads on average, since more insiders also reveal information faster [glosten-milgrom-1983, p. 17]. The authors list unit lots, the absence of risk aversion and inventory costs, and a fixed underlying value as limitations [glosten-milgrom-1983, sec. IV, pp. 24–25]. The locators are to the 1983 discussion paper, not the 1985 journal article that the catalogue names.

See also. Glosten–Milgrom half-spreads (special case: two values, closed form); market collapse (special case: price-sensitive liquidity traders); adverse-selection spread component (special case: a textbook example); Kyle single-auction model (contrast: batch order flow and a strategic insider); Grossman–Miller model (contrast: a spread for inventory risk, without information).

Further reading. [s6-glosten], the published version (not consulted for this entry); the legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. Glosten–Milgrom; sequential trade model; adverse selection; bid–ask spread; informed trading; competitive specialist; zero-profit quotes; martingale prices; semi-strong efficiency; market breakdown; toxic order flow

Status. Verification: partly checked 2026-10-15 (every statement against the scanned 1983 discussion paper, pp. 1–25, read from page images with OCR for running text; the 1985 journal version not consulted; the Harris passage from the text layer of a local copy). Stability: draft. Currency: timeless (theory).

Result 1.13 Glosten–Milgrom half-spreads with two values

Statement. Let the asset's value be V_H with probability p and $V_L < V_H$ otherwise, let a fraction α of traders be informed, and let competitive risk-neutral market makers quote $a = \mu + \Delta_a$ and $b = \mu - \Delta_b$ around $\mu = \mathbb{E}[V \mid \mathcal{F}]$. Zero expected profit on each side gives the half-spreads [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 35]

$$\Delta_a = \frac{\alpha p}{\alpha p + (1-\alpha)/2}\,(V_H - \mu) = \frac{V_H - \mu}{1 + \frac{1-\alpha}{\alpha}\,\frac{1/2}{p}}, \qquad \Delta_b = \frac{\mu - V_L}{1 + \frac{1-\alpha}{\alpha}\,\frac{1/2}{1-p}}.$$

With trading over time, $p_t = \mathbb{P}(V = V_H \mid \mathcal{F}_t)$ and $\mu_t = \mathbb{E}[V \mid \mathcal{F}_t]$ are updated by Bayes' rule from the order flow, the same formulas hold with p_t, μ_t, and each execution price equals the expected value given the history and the side of the trade, so the price at execution times is a martingale [crypto-cartea, same chap., p. 36].

Notation. Local override for this entry and market collapse: p is the probability of the high value, not a price; α is the fraction of informed traders; Δ_a, Δ_b are the ask and bid half-spreads; $\mathcal{F}$ is public information, including the order flow. Source notation: `crypto-cartea` writes v for V.

Conditions. A simplified, essentially static version of the model: one-unit orders; informed traders know V and buy one unit if $a < V_H$ and $V = V_H$, sell one unit if $b > V_L$ and $V = V_L$; liquidity traders are price-

insensitive and buy or sell with probability $\frac{1}{2}$ each, whatever the value; market makers are risk-neutral, carry inventory at no cost and compete to zero expected profit, using only public information [crypto-cartea, same chap., pp. 34–35]. Idea of the proof: a buy comes from a liquidity trader with weight $(1-\alpha)/2$, who pays Δ_a, and from an informed trader with weight αp, who costs $V_H - \mu - \Delta_a$; setting the weighted profit to zero gives Δ_a, and the bid is symmetric [crypto-cartea, same chap., p. 35].

Practice. Reading asymmetric information as toxicity, α is its prevalence, and p with $V_H - \mu$, or $1-p$ with $\mu - V_L$, its magnitude on the buy or sell side; market makers widen each half-spread as the prevalence and magnitude of toxicity on that side rise [crypto-cartea, same chap., pp. 35–36]. Interpretation (own derivation): as $\alpha \to 0$ both half-spreads vanish, and as $\alpha \to 1$ the quotes reach V_H and V_L, where only informed traders would trade.

Pitfalls. This is a textbook simplification: the paper allows a general value distribution and price-sensitive uninformed traders [glosten-milgrom-1983, sec. II, pp. 3–6]. With price-insensitive liquidity traders a monopolist would quote $a = V_H$, $b = V_L$; the formulas rely on competition between market makers [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 35].

See also. Glosten–Milgrom model (generalisation); market collapse (extension: price-sensitive liquidity traders); adverse-selection spread component (contrast: a parametrisation by the dealer's pricing error).

Index. Glosten–Milgrom half-spread; ask half-spread; bid half-spread; toxicity; prevalence of informed trading; Bayesian quote updating; zero-profit market making

Status. Verification: partly checked 2026-10-15 (the setup, both formulas, the dynamic version and the interpretation against `crypto-cartea`, pp. 34–36, read from page images of a scanned copy; the zero-profit algebra re-derived). Stability: draft. Currency: timeless (theory).

Result 1.14 Glosten–Milgrom with price-sensitive liquidity traders: market collapse

Statement. If liquidity trader i trades only when the half-spread is at most his urgency c_i, with c_i distributed with cumulative distribution function F across liquidity traders, the zero-profit half-spreads are defined implicitly by [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 37]; [crypto-cartea-errata, entry for p. 37]

$$\Delta_a = \frac{V_H - \mu}{1 + \frac{1-\alpha}{\alpha}\,\frac{(1-F(\Delta_a))/2}{p}}, \qquad \Delta_b = \frac{\mu - V_L}{1 + \frac{1-\alpha}{\alpha}\,\frac{(1-F(\Delta_b))/2}{1-p}}.$$

Widening a half-spread drives out liquidity traders; if urgency is small, the only solutions may be the extreme ones, $\Delta_a = V_H - \mu$ and $\Delta_b = \mu - V_L$: the market collapses, only informed traders, who are then indifferent, might trade, and any trade reveals the value [crypto-cartea, chap. "A Primer on the Microstructure of Financial Markets", p. 37].

Notation. As in the two-value half-spreads; c_i is the cash-equivalent gain to liquidity trader i from completing the desired trade.

Conditions. The two-value model of the half-spreads entry, with only the fraction $1 - F(\Delta)$ of would-be liquidity buyers or sellers trading at half-spread Δ [crypto-cartea, same chap., p. 37]. "Small" urgency means that no liquidity trader's urgency exceeds the expected value of the asset: $F(\mu - \epsilon) = 1$ for some $\epsilon > 0$ [crypto-cartea, same chap., p. 37, n. 5].

Practice. The paper gives the same mechanism: when the proportion of insiders is high or the liquidity motive of the uninformed weak, the zero-profit quotes can preclude trade, a lemons problem as in Akerlof, and the market reopens when public information neutralises the insiders' advantage [glosten-milgrom-1983, pp. 9 and 17].

Pitfalls. The book prints $F(\Delta_a)$ in the bid formula; the authors' errata correct it to $F(\Delta_b)$, as given above [crypto-cartea-errata, entry for p. 37]. Do not copy the bid formula from the printed page.

See also. Glosten–Milgrom half-spreads (the price-insensitive case, $F \equiv 0$); Glosten–Milgrom model (generalisation, with market shutdown).

Index. market collapse; market breakdown; market shutdown; lemons problem; price-sensitive liquidity traders; urgency; liquidity dry-up; adverse selection

Status. Verification: partly checked 2026-10-15 (the formulas, the collapse condition and its footnote against `crypto-cartea`, p. 37, from page images of a scanned copy; the correction against the authors' errata page, read from the equation source of the page; the shutdown passages against the 1983 discussion paper). Stability: draft. Currency: timeless (theory).

Result 1.15 Spread components: the adverse-selection component, its two views, and the total spread

Statement. The bid–ask spread is split, for analysis, into a *transaction-cost* (transitory) component C, which pays the dealer's normal costs of doing business, and an *adverse-selection* (permanent) component A, which recovers from uninformed traders what the dealer loses to informed ones [alpha-harris, chap. "Bid/Ask Spreads", pp. 298–299]. A dealer first estimates the value V_0 from all current information, then the values

V^B and V^S she would estimate if the next trader were a buyer or a seller; $A = V^B - V^S$, and she quotes [alpha-harris, same chap., Figure 14-1, p. 301]

$$a = V^B + \tfrac{1}{2}C, \qquad b = V^S - \tfrac{1}{2}C, \qquad a - b = A + C.$$

If the dealer learns nothing more from the next trade than its side, the conditional estimate becomes her new value: after a buy both quotes rise by $A/2$ [alpha-harris, same chap., Figure 14-2, pp. 301–302]. In a simple example based on the Glosten–Milgrom model, a dealer believes that the security is worth V unconditionally, that the next trader is equally likely to buy or sell and is informed with probability α, and that it is worth $V+e$ if an informed trader buys and $V-e$ if an informed trader sells [alpha-harris, chap. "Bid/Ask Spreads", p. 320]. Information view: conditional on a buy the dealer's value estimate is $(1-\alpha)V + \alpha(V+e) = V + \alpha e$, conditional on a sell $V - \alpha e$, so the adverse-selection component is $2\alpha e$ [alpha-harris, same chap., p. 320]. Accounting view: expected profit on the next trade is $\frac{1}{2}(a-b) - \alpha e$, so breaking even requires $a - b = 2\alpha e$ [alpha-harris, same chap., pp. 320–321]. Both views give the same component [alpha-harris, same chap., p. 300]; in the example $A = 2\alpha e$ and $C = 0$.

Notation. A and C are amounts of the full spread, in price units; a, b are the ask and bid. Source notation: Harris names the components in words and writes V_0^B, V_0^S for V^B, V^S (Figure 14-1); in his appendix he writes P for α, E for e (the dealer's pricing error when the trader is informed), and A, B for the ask and bid a, b, so his A there is not the component.

Conditions. Normal costs of doing business are ignored; the example assumes that dealers know the value when informed traders trade, which the result does not need [alpha-harris, same chap., pp. 300 and 320]. The two views agree when the loss from trading with an informed trader equals the pricing error, as when the dealer cannot restore inventory before prices adjust; if the dealer can rebalance but does not recognise the informed trade, informed traders keep trading with him until the price adjusts, and his cumulative loss is still the original pricing error [alpha-harris, same chap., p. 302]. Low barriers to entry and competition from public limit orders, which undercut wide dealer quotes, limit monopoly spreads [alpha-harris, same chap., p. 298]; in a competitive dealer market free entry and exit drive dealers to normal profits, so the total spread is this component plus a transaction-cost component that covers operating costs, financing, any monopoly profit and any premium for inventory risk [alpha-harris, same chap., pp. 298–299]. V^B and V^S lie equally far from V_0, at $V_0 \pm A/2$, when informed buyers and sellers are equally likely and their expected pricing errors equal [alpha-harris, same chap., p. 301]. A three-way split (order processing, inventory, adverse selection) separates the inventory-risk premium out of C; Hasbrouck likewise notes that a spread impounds noninformational costs and inventory effects as well as asymmetric information [s6-hasbrouck, sec. 5.5, p. 53]. In the structural sequential-trade version with a cost c per trade, $a = \mathbb{E}[V \mid \text{buy}] + c$ and $b = \mathbb{E}[V \mid \text{sell}] - c$; the bid and ask sequences are martingales but trade prices are not [s6-hasbrouck, sec. 5.4.2, p. 49]. There c is a per-side amount, so Harris's C corresponds to $2c$ (Interpretation).

Practice. The component rises with the probability that the next trader is informed, with the materiality of the information and with trade size, because dealers believe large traders are often well informed [alpha-harris, chap. "Dealers", pp. 289–291 and 295]. Price changes from the adverse-selection component are permanent, while the transaction-cost component causes bid–ask bounce [alpha-harris, chap. "Bid/Ask Spreads", pp. 299–300]. Volatile instruments have wider spreads, both through inventory risk for risk-averse dealers and as a proxy for asymmetric information [alpha-harris, same chap., pp. 312–313]; the other determinants are in spread determinants. Uninformed traders lose to informed ones whichever order they use: their limit orders fill quickly when informed traders want the other side and fail to fill when informed traders compete on the same side, and their market orders pay spreads widened by adverse selection, so that the component is in effect a fee dealers charge market-order traders for bearing adverse-selection risk [alpha-harris, same chap., sec. 14.3, pp. 303–304, Table 14-1].

Pitfalls. The legacy notes said that A is twice the dealer's mark-to-market loss against a correctly signed informed trade. In the example that holds only if $\alpha = 1$: against an informed buyer the dealer's loss relative to V is the whole pricing error e, while $A/2 = \alpha e$ is the expected revision given a buy, that is, the expected loss per buy, informed and uninformed buyers together, of a dealer whose ask is at V (Interpretation: own derivation from the example, pp. 300–302 and 320). The two views equate a probability-weighted loss with a conditional revision, not A with the loss on one informed trade. This is a textbook example, not the Glosten–Milgrom paper, whose quotes are expectations conditional on a trade at the quote (the model); the legacy note that attributed $a = V + \alpha e$, $b = V - \alpha e$ to Glosten and Milgrom followed this example. The accounting view fixes only the width $a - b$; the level $V \pm \alpha e$ comes from the information view (Interpretation). Harris prints the dealer's profit against an informed buyer as $A - (V - E)$; his stated result requires $A - (V + E)$ [alpha-harris, chap. "Bid/Ask Spreads", p. 321].

See also. Glosten–Milgrom model (generalisation); Glosten–Milgrom half-spreads (contrast: a parametrisation by the value distribution); dealer quote adjustment (practice: how dealers set and revise quotes); spread measures (effective and realised spreads, which measure the components after the fact); component identification

(estimating A and C from prices and trade signs).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. adverse selection spread component; permanent spread component; transaction cost spread component; transitory spread component; order-processing cost; total spread; dealer quote decomposition; Glosten–Milgrom theorem (Harris); information view; accounting view; spread decomposition; bid–ask bounce; uninformed traders lose

Status. Verification: partly checked 2026-10-15 (the example, both derivations, the conditions and the spread determinants against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 289–304, 312–313 and 320–321; p. 321 and Figure 14-1, p. 301, also from page images; the structural version against the text layer of a local copy of `s6-hasbrouck`, 2007 Oxford printing, pp. 49 and 53). Stability: draft. Currency: timeless (theory).

1.6 Dealer inventory and quote adjustment

Definition 1.16 Dealer inventory imbalance and the two inventory risks

Statement. A dealer's *target inventory* is the position it wants to hold, and its *inventory imbalance* is actual inventory minus target [alpha-harris, chap. "Dealers", p. 283]. Dealers face two inventory risks, distinguished by whether future price changes are correlated with the imbalance [alpha-harris, same chap., p. 285]:

- *diversifiable inventory risk*: price changes independent of the imbalance; gains and losses are equally likely, they do not cause losses in the long run, and dealing in many instruments reduces their effect on total profit [alpha-harris, same chap., pp. 285–286];
- *adverse-selection risk*: price changes inversely correlated with the imbalance, because informed traders take the side their information favours, make order flow one-sided and push the dealer's inventory away from target before prices move [alpha-harris, same chap., p. 286].

Conditions. If long and short positions are equally costly, a dealer that does not also speculate, hedge or invest has a target of zero; where shorting costs more, targets are positive; speculating dealers set targets by their views [alpha-harris, same chap., pp. 283–284]. Large imbalances are costly to finance and expose the dealer to losses if prices move, and an imbalance beyond the dealer's capital forces liquidation [alpha-harris, same chap., p. 284]; a dealer can also hedge an imbalance with a correlated instrument [alpha-harris, same chap., p. 285, Table 13-2]. Interpretation: the costs of carrying inventory thus group into price risk, the financing of long and the borrowing of short positions, the use of capital, and the basis risk and costs of hedges; recall risk and risk-limit use, which the legacy notes also listed, are not in the passages checked.

Practice. Dealers avoid both risks by keeping inventory under control, so most attend to *market values*, the prices that produce two-sided order flow, more than to fundamental values, which they discover as a by-product [alpha-harris, same chap., pp. 286–287]. Adverse selection is still the most important determinant of dealer profitability [alpha-harris, same chap., p. 286]. Inventory risk does not need informed traders: diversifiable inventory risk is present when no one is better informed, risk-averse dealers require a premium for it, mainly in highly volatile instruments, and that premium is part of the transaction-cost spread component [alpha-harris, chap. "Bid/Ask Spreads", pp. 299 and 313]; see spread components.

Pitfalls. "Diversifiable" here means unrelated to the dealer's imbalance; within one instrument it is still a risk that a risk-averse dealer is paid to bear, as in Grossman–Miller (Interpretation). A falling inventory after buys does not by itself say which risk is present; separating them needs the subsequent price path (Interpretation).

See also. dealer quote adjustment (how dealers control the imbalance); Grossman–Miller model (a model of the first risk); Glosten–Milgrom model (a model of the second); inventory-based quotes (models of quotes as functions of the imbalance).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. inventory imbalance; target inventory; inventory risk; diversifiable inventory risk; adverse selection risk; inventory spread component; inventory costs; financing cost; market value; fundamental value; dealer risk management

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 283–287, 299 and 313). Stability: draft. Currency: timeless (concepts).

Method 1.17 Dealer quote adjustment for inventory and suspected informed trading

Statement. Dealers control inventory and adverse selection with four levers: the level of both quotes, their skew, the width of the spread and the sizes shown.

1. *Inventory.* To reduce inventory, lower bid and ask, decrease the bid size and raise the ask size; to increase it, do the opposite; an impatient dealer can take another trader's offer or hit another trader's bid, or hedge with a correlated instrument [alpha-harris, chap. "Dealers", pp. 284–285, Table 13-2].

2. *After a suspected informed trade.* After selling to a suspected well-informed buyer, raise the ask and lower its size to discourage further informed buying, raise the bid and its size to rebuild inventory, or buy at another trader's ask or buy a correlated instrument; after buying from a suspected informed seller, do the reverse [alpha-harris, same chap., pp. 287–288, Table 13-3].

3. *Before the next trade.* Set the ask at the value expected if the next trader buys and the bid at the value expected if the next trader sells, adjusting more when informed trading is more likely and the information more material [alpha-harris, same chap., pp. 288–290]; see the adverse-selection component.

4. *Size.* Avoid displaying large size, so that traders must ask for it and reveal themselves [alpha-harris, same chap., p. 291].

Conditions. Dealers rarely know who is informed: informed traders hide behind brokers and claim to be uninformed, and rules of thumb (large, active, impatient traders are often informed; small retail traders rarely) are useful but unreliable, so every order is treated as possibly informed [alpha-harris, same chap., pp. 288–289]. Spreads widen with asymmetric information, volatility and weak utilitarian trading interest [alpha-harris, chap. "Bid/Ask Spreads", pp. 312–313] (spread determinants).

Practice. Searching for prices that bring two-sided order flow is price discovery: it finds market values [alpha-harris, chap. "Dealers", p. 284]. Demanding liquidity to rebalance gives a negative realised spread [alpha-harris, same chap., p. 284]. After a buy, raising quotes is right if the buyer was informed and a mistake if not, when value traders sell to the dealer and realised spreads fall; doing nothing is the reverse; buying at another trader's ask may limit losses against an informed buyer but is a big mistake against an uninformed one [alpha-harris, same chap., pp. 291–292, Table 13-4]. Poorly informed dealers more often mistake uninformed for informed clients, so they turn inventory quickly and earn smaller realised spreads; well-informed dealers can hold larger positions longer and earn larger realised spreads [alpha-harris, same chap., pp. 292–294]. Value traders trade when prices diverge from fundamental value, often after such mistakes or after risk-averse dealers obtain large concessions for taking big positions, and in doing so restore dealers' inventories [alpha-harris, same chap., pp. 291 and 294]. Some dealers avoid informed traders by trading only with retail or known clients, refusing anonymous size, or slicing large orders over time, which lets them offer more liquidity to the flow they accept [alpha-harris, same chap., pp. 290–291].

Pitfalls. Informed traders can bluff, sending small orders on one side to move a dealer before trading on the other; against traders they cannot identify, dealers should adjust prices at the same rate per quantity whatever the side, speed, size or sequence of orders [alpha-harris, same chap., pp. 292 and 294]. Interpretation: refusing flow protects the dealer but reduces the immediacy available to others; one-sided flow turns dealing into warehousing risk. These are heuristics without an objective function; Avellaneda–Stoikov derives the inventory skew from one.

See also. dealer inventory risk (prerequisite); adverse-selection spread component (the spread set before the next trade); Avellaneda–Stoikov quoting (a model of the inventory skew); Glosten–Milgrom model (a model of the revision after each trade); inventory-based quotes (textbook models of quotes as functions of inventory); spread determinants (why spreads differ across instruments).

Further reading. The legacy text it replaces, tables included, is kept verbatim in `notes/artefacts/market-making-lob/derivations.tex`.

Index. inventory control; quote skew; quote shading; quote size; dealer tactics; price discovery; market value; realised spread; informed-trader identification; refusing flow; value traders; bluffing; toxic flow

Status. Verification: partly checked 2026-10-15 (every step and table summary against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 283–294 and 312–313). Stability: draft. Currency: timeless (practice description from 2002; venue examples dated).

Model 1.18 Inventory-based dealer quotes: reservation prices linear in inventory (Stoll; Ho–Stoll as reported by O'Hara)

Statement. In Hasbrouck's version of Stoll's one-period model, a dealer with CARA utility $U(W) = -e^{-\gamma W}$ holds n shares of a security with payoff $X \sim N(\mu_X, \sigma_X^2)$, borrowing and lending at a zero rate, and posts a bid B for one share that leaves him indifferent to its execution; equating certainty equivalents gives [s6-hasbrouck, sec. 11.2, eq. (11.2), p. 110]

$$B = \mu_X - (2n+1)\,\frac{\gamma\sigma_X^2}{2}.$$

If the dealer could first trade at a notional price P, his subjective valuation, his optimal holding would be $n^* = (\mu_X - P)/(\gamma\sigma_X^2)$, and at that optimum $B = P - \gamma\sigma_X^2/2$ [s6-hasbrouck, same sec., pp. 110–111]. The

same indifference for a one-share sale gives $A = \mu_X - (2n-1)\gamma\sigma_X^2/2$ (Interpretation: own derivation; the source gives only the bid), so the quotes are

$$B = r - \tfrac{1}{2}\gamma\sigma_X^2, \qquad A = r + \tfrac{1}{2}\gamma\sigma_X^2, \qquad r = \mu_X - \gamma\sigma_X^2\, n = P - \gamma\sigma_X^2\,(n - n^*)\,:$$

the midpoint of the quotes falls linearly as inventory rises above target, and the spread $\gamma\sigma_X^2$ does not depend on inventory.

Notation. n is the dealer's inventory and n^* his target; r is the reservation (indifference) midpoint; A, B are the dealer's ask and bid. The source writes α for the risk aversion γ, and μ_x, σ_x^2 in eq. (11.2). The legacy notes wrote $r_t = m_t - \kappa(q_t - q^\star)$ with spread s_t; here $\kappa = s = \gamma\sigma_X^2$, $q^\star = n^*$ and $m = P$.

Conditions. One period, a normally distributed payoff, exponential utility, one-share trades, and no information asymmetry: the quotes compensate only for being pulled off the optimal portfolio [s6-hasbrouck, same sec., p. 110]. Idea of the proof: with CARA utility and normal wealth the certainty equivalent is $\mu_W - \gamma\sigma_W^2/2$; set it equal with and without the trade, and solve for the price [s6-hasbrouck, same sec., p. 110]. With two correlated securities the bid for the first is $B_1 = \mu_1 - \frac{1}{2}\gamma\sigma_1\big[(1+2n_1)\sigma_1 + 2\rho n_2\sigma_2\big]$, lower when the securities are positively correlated, but at the optimum it is again $P_1 - \gamma\sigma_1^2/2$ and the correlation drops out, a consequence of CARA utility [s6-hasbrouck, same sec., eq. (11.3), p. 111].

Practice. O'Hara's account of the multiperiod model of Ho and Stoll (1981): a monopolistic, risk-averse dealer facing Poisson buy and sell arrivals whose rates are linear in his ask and bid adjustments around a fixed true price p, quoting $p + a$ and $p - b$ to maximise expected utility of terminal wealth over a finite horizon; the problem has no known closed form, and Ho and Stoll approximate it with a first-order expansion and quadratic utility [catalogue-0496, sec. 2.3, pp. 29–35]. As O'Hara reports them, three properties follow: the spread is the risk-neutral monopoly spread, set by the slopes of the arrival functions, plus a risk adjustment that rises with relative risk aversion, transaction size, the stock's variance and the time remaining; transaction uncertainty by itself does not affect the spread; and the spread does not depend on inventory, the dealer instead shifting both quotes (with true price 50, quotes 48 and 52 move to 47 and 51 after a purchase at the bid) by an amount set by risk aversion, the stock's risk and his wealth [catalogue-0496, same sec., pp. 35–36]. Garman's fixed-quote dealer is ruined with probability one, within days for realistic parameters, which is why quotes must depend on inventory; in Amihud and Mendelson's risk-neutral model with inventory bounds both quotes fall as inventory rises, but the spread *widens* with distance from the preferred position and the quotes are not symmetric about the true value [s6-hasbrouck, secs. 11.1.1–11.1.2, pp. 107–109].

Pitfalls. Neither original paper was consulted: the model is Hasbrouck's version of Stoll (1978), and the Ho–Stoll properties are O'Hara's report, not checked against the paper, `ho-stoll-1981`, which the route's `catalogue-0527` does not identify by title. A spread independent of inventory is a feature of these models, not a general result (Amihud and Mendelson above). O'Hara notes that a fixed true price is fundamental to the Ho–Stoll analysis, and that its finite horizon, with liquidation certain at the end, implies spreads narrowing predictably through the day, which seems unrealistic [catalogue-0496, sec. 2.3, p. 36]. Empirically, quotes rising after a customer purchase fit inventory control and adverse selection alike; NYSE specialist quote changes are explained mostly by trades, with inventories adding little, and dealers manage positions more through non-public quotes, interdealer trading and discretion over participation than through posted quotes [s6-hasbrouck, secs. 11.4–11.5, pp. 115–117]; see component identification.

See also. Avellaneda–Stoikov quoting (generalisation: continuous time, with the spread also independent of inventory in its approximation); Grossman–Miller model (contrast: risk compensation set by market clearing, not by a quoting dealer); dealer inventory risk (prerequisite); dealer quote adjustment (practice).

Further reading. [ho-stoll-1981], the multiperiod model (not consulted: no accessible copy); the legacy block it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. inventory-based quoting; inventory skew; reservation price; Stoll dealer model; Ho–Stoll; Amihud–Mendelson; Garman; dealer risk aversion; CARA dealer; inventory control; preferred inventory

Status. Verification: partly checked 2026-10-15 (the Stoll version, eqs. (11.2)–(11.3), and the inventory evidence against the text layer of a local copy of `s6-hasbrouck`, 2007 Oxford printing, pp. 106–117; the Ho–Stoll account against page images of `catalogue-0496`, 1998 Wiley printing, pp. 29–36; the ask and the reservation price derived here; neither Stoll (1978) nor Ho and Stoll (1981) consulted). Stability: draft. Currency: timeless (theory); the evidence is from 1990s NYSE data.

1.7 Measuring and decomposing the spread

Definition 1.19 Spread measures: quoted, inside, relative, effective and realised spreads

Statement. With best bid b_t, best ask a_t and midpoint $m_t = \frac{1}{2}(a_t + b_t)$:

- the *quoted spread* is $a_t - b_t$; the *inside spread* is the highest bid's distance from the lowest ask across all dealers, and is no wider than the narrowest dealer's spread [alpha-harris, chap. "Dealers", p. 280];

- the *relative spread* is the spread as a fraction of price, the only basis on which spreads of different instruments compare [alpha-harris, chap. "Bid/Ask Spreads", p. 312];
- for a trade at p_t with sign $Q_t = +1$ if buyer-initiated and -1 if seller-initiated, the *effective spread* is twice the signed difference from the midpoint at the time of the trade, and the *realised spread* twice the signed difference from the midpoint a set interval Δ later, commonly 5, 10, 15 or 60 minutes [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", pp. 424–425]:

$$ES_t = 2Q_t(p_t - m_t), \qquad RS_t = 2Q_t(p_t - m_{t+\Delta}), \qquad ES_t = RS_t + 2Q_t(m_{t+\Delta} - m_t).$$

The last term, the signed midpoint revision after the trade, is sometimes used as an estimate of the trade's price impact [s6-hasbrouck, sec. 14.2.2, eq. (14.2), p. 148].

Notation. Q_t is the trade sign. Harris's effective and realised spreads are doubled, full-spread amounts; Hasbrouck's *effective cost* $p_t - m_t$ and *realised cost* $p_t - m_{t+5}$ are per-side amounts written for a buy order, with the midpoint five minutes after the order's receipt, as in the SEC's Rule 605 [s6-hasbrouck, same sec., p. 148]. Harris calls the undoubled signed difference the *liquidity premium* [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 425]. The legacy notes wrote ϵ_t for Q_t, s_t for the quoted spread and PI_t for the impact term, and measured the relative spread against m_t.

Conditions. The effective spread equals the quoted spread when every purchase is at the ask and every sale at the bid; it is smaller with price improvement and larger when large orders fill outside the quotes. Realised and effective spreads are equal if the midpoint does not move over the interval; usually it moves with the trade, so realised spreads are smaller [alpha-harris, same chap., p. 425]. When the analyst did not take part in the trades, signs are inferred, usually by the Lee–Ready rule, which overstates costs for traders who do not only take liquidity and cannot see orders filled in several trades [alpha-harris, same chap., p. 423]. Delayed trade reports bias effective costs down, and hidden orders and market-maker improvement distort the inferred sign [s6-hasbrouck, sec. 14.3.3, pp. 150–151].

Practice. Quoted minus effective spread measures the price improvement dealers give; effective minus realised spread measures dealers' losses to well-informed traders [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", pp. 425–426]. The realised cost is the revenue of a dealer who sold to the customer at p_t and covered at the later midpoint [s6-hasbrouck, sec. 14.2.2, p. 148]. The effective spread is the least noisy benchmark estimator, because its benchmark is contemporaneous; the realised spread grows noisier the longer the interval [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 427]. The quoted spread measures the cost of immediacy for small orders only; depth and resiliency are other aspects of liquidity [alpha-harris, chap. "Bid/Ask Spreads", p. 312].

Pitfalls. Harris also uses *realised spread* for the difference between the prices at which a dealer actually sells and buys on a round trip, which can be negative when quotes move against the inventory [alpha-harris, chap. "Dealers", p. 280]; that is a dealer's outcome, not the midpoint-based estimator above. Mixing doubled and per-side conventions changes figures by a factor of two. Interpretation: effective minus realised spread is the permanent component only if the interval is long enough for prices to adjust and short enough to exclude unrelated news, and quote moves for inventory that have not yet reversed fall into it too; see component identification.

See also. spread components (what the measures estimate); Roll's estimator (alternative: a spread from trade prices alone); outside spread (contrast: an undisplayed willingness to trade).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. quoted spread; inside spread; relative spread; half-spread; effective spread; realised spread; realized spread; effective cost; realized cost; liquidity premium; price impact; price improvement; Rule 605; Lee–Ready; trade signing; transaction cost measurement

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 280–281, 312 and 423–427, and `s6-hasbrouck`, 2007 Oxford printing, pp. 148 and 150–151). Stability: draft. Currency: as of the 2002 and 2007 printings for the reporting rules (Rule 605 named by Hasbrouck); definitions timeless.

Result 1.20 Roll's serial-covariance spread estimator

Statement. Let trade prices be $P_t = V_t + \frac{1}{2}SQ_t$, where the fundamental value V_t is a random walk with independent innovations ε_t of mean zero and variance σ^2, S is the spread, and the trade sign $Q_t = \pm 1$ is a buy with probability one-half, independently of earlier signs and of ε_t and ε_{t+1}. Then [alpha-harris, chap. "Volatility", box, p. 415]

$$\operatorname{Var}(\Delta P_t) = \sigma^2 + \tfrac{1}{2}S^2, \qquad \operatorname{Cov}(\Delta P_t, \Delta P_{t-1}) = -\tfrac{1}{4}S^2, \qquad S = 2\sqrt{-\operatorname{Cov}(\Delta P_t, \Delta P_{t-1})},$$

and all autocovariances of order two or more are zero [s6-hasbrouck, sec. 3.4, p. 30]. Roll's estimator replaces the covariance by its sample value [alpha-harris, chap. "Volatility", p. 415].

Notation. S is the full spread, in price units; changes are between successive trades. Hasbrouck writes $p_t = m_t + cq_t$ with half-spread $c = S/2$, efficient-price innovation u_t of variance σ_u^2, and γ_0, γ_1 for the variance and first autocovariance of price changes, so $\gamma_0 = 2c^2 + \sigma_u^2$, $\gamma_1 = -c^2$, $c = \sqrt{-\gamma_1}$ and $\sigma_u^2 = \gamma_0 + 2\gamma_1$ [s6-hasbrouck, sec. 3.4, eqs. (3.3)–(3.5), pp. 29–30]. Harris calls the estimate an effective spread [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 434].

Conditions. The spread is constant and noninformational: competitive dealers quote $m_t \pm c$ to cover a cost c per trade [s6-hasbrouck, sec. 3.4, p. 29]. Idea of the proof: $\Delta P_t = \varepsilon_t + \frac{1}{2}S(Q_t - Q_{t-1})$; every cross-product has zero expectation except $\mathbb{E}Q_{t-1}^2 = 1$, which gives $-\frac{1}{4}S^2$ for the first autocovariance [alpha-harris, chap. "Volatility", p. 415]. The variance splits into a fundamental part σ^2 and a transitory part from bid–ask bounce [alpha-harris, same chap., pp. 415–416].

Practice. The estimator is used where quotes are unavailable [s6-hasbrouck, sec. 3.4, p. 30]; Harris advises using it only then [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 434]. For one NYSE stock (PCO, October 2003), $\hat{\gamma}_1 = -0.0000294$ gives $c = \$0.017$ and a spread of \$0.034, against a time-weighted average quoted spread of \$0.032 [s6-hasbrouck, sec. 3.4, p. 30].

Pitfalls. In that sample the assumptions fail: the spread ranged from \$0.01 to \$0.49, successive trade signs had correlation about 0.34, and midpoint changes were positively correlated with the latest trade sign [s6-hasbrouck, sec. 3.4, p. 30]. The biases go both ways. Positively autocorrelated signs, $\mathrm{Corr}(q_t, q_{t-1}) = \rho > 0$, give $\gamma_1 = -c^2(1 - 2\rho)$ and bias $\hat{c}$ down; signs positively correlated with the efficient-price innovation, as with informed trading, give $\gamma_1 = -c(c + \rho\sigma_u)$ and bias $\hat{c}$ up [s6-hasbrouck, sec. 4.4, Exercises 4.2–4.3, pp. 38–39]. A positive sample autocovariance is not uncommon, and estimation error alone can produce one when the model is correct; the square root is then undefined [s6-hasbrouck, sec. 4.5, p. 39]. Harris says violations typically make it underestimate the spread [alpha-harris, chap. "Liquidity and Transaction Cost Measurement", p. 434], and that reversals lasting beyond one trade make Roll's variance components understate transitory volatility [alpha-harris, chap. "Volatility", p. 416]. The legacy notes listed informed trading among causes of a less negative covariance; under Hasbrouck's Exercise 4.3 it makes the covariance more negative. Price discreteness and hidden liquidity, also listed there, are not discussed in the passages checked.

See also. component identification (generalisation: the generalized Roll model, with an information component); fundamental and transitory volatility (generalisation: random-walk decompositions with a general pricing error); spread measures (when quotes are available); Grossman–Miller liquidity measures (contrast: negative autocovariance from inventory risk).

Further reading. The legacy blocks it replaces, with the full derivation, verbatim: `notes/artefacts/market-making-lob/derivations.tex`. Roll's 1984 paper was not consulted.

Index. Roll model; Roll estimator; serial covariance spread estimator; bid–ask bounce; negative autocovariance; transitory volatility; implicit spread; effective spread estimate

Status. Verification: partly checked 2026-10-15 (the model and derivation against the page image of `alpha-harris`, p. 415, and its text layer, pp. 416 and 434; the parametrisation, example, biases and estimation against the text layer of `s6-hasbrouck`, pp. 29–30 and 38–39; the original paper not consulted). Stability: draft. Currency: timeless (method); the example is from 2003 data.

Result 1.21 Separating transitory and permanent spread components: bounce, revisions and identification

Statement. The transaction-cost component makes prices bounce between bid and ask, so its price changes reverse; the adverse-selection component moves the dealers' value estimates, so its price changes are permanent and unpredictable; both depend on whether a buyer or a seller initiated the trade, and the usual decomposition explains current and future price changes with the signs of trades [alpha-harris, chap. "Bid/Ask Spreads", pp. 299–300 and 302–303]. In Hasbrouck's generalized Roll model the efficient price moves with the trade and with public news, and trades occur at a cost c either side of it [s6-hasbrouck, sec. 8.2, eq. (8.1), pp. 67–68]:

$$m_t = m_{t-1} + \lambda q_t + u_t, \qquad p_t = m_t + cq_t, \qquad \text{ask} - \text{bid} = 2(c + \lambda),$$

$$\gamma_0 = c^2 + (c + \lambda)^2 + \sigma_u^2, \qquad \gamma_1 = -c(c + \lambda),$$

with all higher autocovariances zero [s6-hasbrouck, sec. 8.3, eq. (8.3), p. 68]. Two moments cannot identify the three parameters λ, c, σ_u^2: trade prices alone do not separate the components. Only the variance of efficient-price changes, $\lambda^2 + \sigma_u^2 = \gamma_0 + 2\gamma_1$, is identified without restrictions [s6-hasbrouck, same sec., pp. 68–69].

Notation. $q_t = \pm 1$ is the trade sign; λ is the adverse-selection (information) cost and c the noninformational cost, both per side; u_t is nontrade public information, uncorrelated with q_t; γ_0, γ_1 are the variance and first autocovariance of Δp_t. In Harris's terms the adverse-selection component is 2λ and the transaction-cost component $2c$ (Interpretation).

Conditions. Timing: after the trade at $t - 1$ the efficient price is m_{t-1}; public news u_t arrives; dealers set bid and ask symmetrically about $m_{t-1} + u_t$; the trade q_t arrives and the efficient price becomes m_t [s6-hasbrouck, sec. 8.2, p. 68]. Setting $\lambda = 0$ gives Roll's model; setting $\sigma_u^2 = 0$ or $c = 0$ identifies the rest, but

such restrictions are economically strong [s6-hasbrouck, sec. 8.3, p. 69].

Practice. With trade signs, analyses indicate that in most markets the adverse-selection component is larger than the transaction-cost component [alpha-harris, chap. "Bid/Ask Spreads", p. 303]. Inventory control predicts the same short-run quote response as adverse selection: after a customer buys, the dealer's quotes rise either way. The two separate in the long run, since information effects are permanent while inventory effects vanish once offsetting trades restore the dealer's position; for NYSE specialists, most quote dynamics are attributable to trades, with inventories adding little explanatory power [s6-hasbrouck, sec. 11.4, p. 115]. Identification therefore needs more than prints: trade signs and subsequent quote revisions and, to separate inventory, dealer positions.

Pitfalls. An estimated permanent component includes any inventory-driven quote change that has not reversed within the horizon studied, and the transitory component includes any mean reversion from uninformed order imbalances (Interpretation). Where the dealer is counterparty to every one-unit trade, $q_t = -\Delta I_t$ for inventory I_t [s6-hasbrouck, sec. 11.3.2, p. 112], and a regression on current and lagged trades and inventories is then collinear; the NYSE data avoid this because the specialist does not take part in every trade [s6-hasbrouck, sec. 11.4, p. 115].

See also. Roll's estimator (special case: $\lambda = 0$); spread components (the theory being estimated); spread measures (a model-free split by effective and realised spreads); inventory-based quotes (the inventory alternative).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. spread decomposition; transitory component; permanent component; bid–ask bounce; generalized Roll model; identification; trade signs; inventory versus information; price revision; adverse selection estimation

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, pp. 299–303, and `s6-hasbrouck`, pp. 67–69, 112 and 115). Stability: draft. Currency: timeless (method); the NYSE evidence is from 1990s data.

Result 1.22 Determinants of bid–ask spreads: primary factors, order-driven equilibrium and observable proxies

Statement. Three primary factors set spreads [alpha-harris, chap. "Bid/Ask Spreads", sec. 14.6.1, pp. 311–313]:

- *asymmetric information* widens them, most when informed traders know material information;
- *volatility* widens them: it raises the value of the timing option that standing limit orders and dealer quotes give away when they cannot be adjusted at once, makes diversifiable inventory risk costly to risk-averse dealers, and proxies for asymmetric information;
- *utilitarian trading interest* narrows them: active markets spread dealers' fixed costs over more trades, make inventory easier to lay off, draw public limit orders, and dilute informed flow.

In a continuous order-driven market the equilibrium spread makes the marginal trader indifferent between limit and market orders; it would be zero if traders were identical and equally informed, could cancel instantly and free, and cared neither for time nor risk [alpha-harris, same chap., secs. 14.4.1–14.4.2, pp. 304–310]. Relaxing those assumptions gives Harris's Table 14-2 [alpha-harris, same chap., p. 311]: information asymmetry (increases spreads; great importance), time to cancel limit orders (increases; high), volatility (increases; high), limit-order management costs (increases; moderate), the value of traders' time (increases; moderate), the commission difference between limit and market orders (one for one), and risk aversion (increases; low).

Conditions. With costly limit-order management, the equilibrium spread equals the expected cost of managing limit orders plus twice the value of the timing option [alpha-harris, same chap., p. 307]; with only a commission difference, it equals that difference [alpha-harris, same chap., p. 306]. If all traders were equally uninformed, uncertainty would narrow spreads, because value estimates are less volatile than values; uncertainty widens spreads through asymmetry [alpha-harris, same chap., p. 310]. The primary factors interact: asymmetry widens spreads, which deters uninformed traders and lowers volume, which widens them further [alpha-harris, same chap., p. 312]. Public limit-order traders without dealers' costs can undercut dealers, who survive by exploiting their view of the order flow [alpha-harris, same chap., sec. 14.5, pp. 310–311].

Practice. Observable *secondary* determinants stand in for the primary ones [alpha-harris, same chap., sec. 14.6.2, pp. 314–317]. For asymmetric information: disclosure rules, market-condition reports, analyst and press coverage, economy-wide commodity contracts, diversified portfolios and diversified firms, established rather than emerging industries, firm age, enforced insider-trading rules, and expected material news such as earnings. For volatility: financial and operating leverage and uncertain growth for stocks; perishability, storage costs, low inventories and weather for commodities; political, inflation and interest-rate uncertainty for currencies; inflation, rates and credit quality for bonds. For utilitarian interest: trading activity (volume,

number of trades), firm size, debt issue size and on-the-run status, and contracts that replicate risks many traders want to hedge. When asymmetric information is severe or utilitarian interest small, spreads can be so wide that the market fails; small firms then finance through banks and venture capitalists, whose due diligence, restrictions and supervision rights need a private relationship [alpha-harris, same chap., sec. 14.6.3, p. 317]. Empirically, daily spreads rise with daily volatility, the spread rewarding market makers for the risk they bear, and across small-tick instruments spreads follow $\psi_i = a + b\,\sigma_i/\sqrt{N_i}$, with N_i the daily number of trades [alpha-laruelle, chap. 2, secs. 2.2 and 2.3.1, pp. 144–145 and 150–151]; see spread–volatility regression. When the spread is stuck at one tick, liquidity providers respond by changing size at the best quotes rather than the spread [alpha-laruelle, same chap., sec. 2.2, p. 148]. In crashes liquidity suppliers withdraw: in 1987 buyers waited until they believed all sellers were satisfied, and some Nasdaq dealers took their phones off the hook [alpha-harris, chap. "Bubbles, Crashes, and Circuit Breakers", pp. 561 and 563].

Pitfalls. Secondary factors are correlated with more than one primary factor and with each other, so an observed effect can have the opposite sign to the prediction (emerging industries attract volume), and tests need controls [alpha-harris, same chap., p. 314]. Volatility has a second, opposite effect through gambling and hedging interest, which raise activity [alpha-harris, same chap., pp. 316–317]. Trading activity is the best observable predictor of spreads, but Harris keeps it secondary to preserve the distinction between theoretical and observable determinants [alpha-harris, same chap., p. 312]. These are qualitative predictions; the importance ratings in Table 14-2 are Harris's judgement, not estimates. The legacy notes' "availability of close risk substitutes" is not Harris's wording: his factor is risk replication for hedgers. A legacy principle, *volatility is a liquidity price*, listed smaller displayed size and faster cancellations among liquidity providers' responses to volatility. The passages checked support wider spreads and, in crashes, withdrawal. Lehalle and Laruelle report book size moving inversely to the spread over the day [alpha-laruelle, chap. 2, sec. 2.2, p. 148]; that is co-movement, not a measured response to volatility.

See also. spread components (the components these factors move); market collapse (a model of market failure); outside spread (the spread of value traders); dealer quote adjustment; spread–volatility regression (an empirical test of the volatility determinant); fundamental and transitory volatility.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. spread determinants; asymmetric information; volatility; utilitarian traders; trading activity; equilibrium spread; order-driven market; timing option; limit-order management cost; commission difference; secondary determinants; spread proxies; market failure; liquidity and capital structure; volatility as a liquidity price; liquidity withdrawal; tick size and depth

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 303–318, Table 14-2 included, and pp. 561 and 563; and of `alpha-laruelle`, 2018 World Scientific printing, pp. 144–151; no page images). Stability: draft. Currency: timeless (theory); institutional examples from 2002; the European evidence from 2008–2016.

Definition 1.23 Outside spread: the prices at which value traders will trade

Statement. A value trader's *outside spread* is the interval between the prices at which he is willing to buy and to sell; value traders rarely quote these prices, so as not to reveal their value estimates or give free trading options [alpha-harris, chap. "Value Traders", sec. 16.2, p. 340]. Writing $\widehat{V}$ for his value estimate, he buys only below b^o and sells only above a^o, and $s^o = a^o - b^o$. Interpretation: if every cost and risk of a one-unit trade could be summarised in an all-in half-cost $c > 0$, the expected surplus would be $\widehat{V} - P - c$ from buying and $P - \widehat{V} - c$ from selling. He would trade only outside the no-trade band $[\widehat{V} - c, \widehat{V} + c]$, so $b^o = \widehat{V} - c$ and $a^o = \widehat{V} + c$. The spread widens with [alpha-harris, same chap., pp. 340–344]:

- adverse selection from news traders who know of value changes he has not yet learnt;
- the winner's curse: the risk that his estimate is wrong, larger when values are hard to estimate and when many others estimate them, with losses from the market impact of trading on a wrong estimate;
- the costs of research, which must be recovered also on the instruments analysed that offer no trade.

Conditions. Value trading is profitable only when price differs from fundamental value, which happens in two ways. New information changes the value: news traders, by definition the first to learn it, profit and push the price to the new value. Or uninformed traders push the price away from value: value traders profit and push it back. Value traders therefore profit only by trading against uninformed traders' demands for liquidity [alpha-harris, same chap., sec. 16.1, pp. 338–339]. They do so directly, through limit orders or blocks, or indirectly through dealers whose inventories the uninformed demand has unbalanced [alpha-harris, same chap., secs. 16.1.1–16.1.2, pp. 339–340]. Outside spreads are much wider than dealer spreads: value traders trade when order flow is one-sided and cannot learn their mistakes from it, take larger positions held until prices revert, so bear more inventory risk and financing cost, trade less volume over which to spread fixed costs,

and earn at best half the outside spread when price returns to value, while a dealer often earns the whole spread on a round trip [alpha-harris, same chap., sec. 16.3, pp. 344–345].

Practice. Value traders buy when dealers lower their quotes to shed inventory taken from uninformed sellers, and so supply depth to the market and resiliency against uninformed price moves; dealers take larger positions where value traders follow the market closely [alpha-harris, same chap., pp. 340, 345–346]. When their value estimates differ enough, value traders' outside spreads overlap and they trade with each other [alpha-harris, same chap., p. 343]. Not buying does not oblige a trader to sell: uncertain estimates mean he should speculate only when he believes an instrument is very mispriced, and the larger the apparent mispricing, the more likely it is a mistake [alpha-harris, same chap., box, p. 343]. In continuous markets the winner's curse bites because traders trade at the prices they quote, and the market aggregates estimates sequentially; in a two-sided single-price auction it matters little [alpha-harris, same chap., pp. 342–343].

Pitfalls. The outside spread is not observed in quotes or trades (Interpretation: it is inferred from where value traders act). The legacy notes listed leverage and borrowing among its financing costs; Harris names the cost of financing larger positions, without that detail [alpha-harris, same chap., p. 344]. A single half-cost c understates the problem: adverse selection and the winner's curse are not per-unit costs but depend on what a trade at a given price reveals and on how many others are estimating the value, which is why Harris treats them as risks that widen the spread (Interpretation). The legacy notes gave value traders three sources of profit: information changes, uninformed liquidity demand and dealer inventory pressure. In Harris, information changes are where news traders profit; value traders gain there only by correcting news traders' under- or overreaction [alpha-harris, chap. "Informed Traders and Market Efficiency", Table 10-1, p. 230, and p. 239]. Dealer inventory pressure is the indirect channel of uninformed demand [alpha-harris, chap. "Value Traders", pp. 339–340].

See also. spread measures (contrast: displayed and transaction spreads); spread determinants; dealer quote adjustment (value traders restore dealers' inventories); fundamental value; informed trading and aggregation.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. outside spread; value traders; winner's curse; research costs; resiliency; depth; liquidity supplier of last resort; news traders; no-trade band; value-trading profits; uninformed liquidity demanders

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 338–346, and pp. 230 and 239 for the profit sources). Stability: draft. Currency: timeless (concepts).

1.8 Volatility, liquidity and crashes

Definition 1.24 Fundamental and transitory volatility: the random-walk and transitory components of prices

Statement. *Volatility* is the tendency of prices to change unexpectedly. *Fundamental volatility* comes from unanticipated changes in instrument values; *transitory volatility* comes from trading by impatient uninformed traders, which pushes prices away from values until they revert. Total volatility is the sum of the two [alpha-harris, chap. "Volatility", pp. 410, 413–414]. In statistical form, the trade price is a random-walk efficient price plus a stationary pricing error [s6-hasbrouck, sec. 8.6, p. 72]:

$$p_t = m_t + s_t, \qquad m_t = m_{t-1} + w_t, \qquad \Delta p_t = w_t + \Delta s_t,$$

where s_t has mean zero, is covariance-stationary, and may be serially correlated and correlated with w_t. Fundamental volatility is the variance σ_w^2 of w_t. If the price changes have the moving-average representation $\Delta p_t = \theta(L)\varepsilon_t$, then $\sigma_w^2 = \theta(1)^2\sigma_\varepsilon^2$, the same for every structural model consistent with the data, while the variance of the pricing error has only a lower bound [s6-hasbrouck, same sec., eqs. (8.10)–(8.13), pp. 73–74]. Over long horizons $\sigma_w^2 \approx \mathrm{Var}(p_t - p_{t-k})/k$, because microstructure effects matter less as the interval grows [s6-hasbrouck, sec. 8.3, p. 69].

Notation. p_t is the trade price, m_t the efficient (fundamental) price, $s_t = p_t - m_t$ the pricing error, w_t the efficient-price innovation; $\theta(1)$ is the sum of the moving-average coefficients. Harris writes V_t for value and ε_t for its change. The legacy notes wrote $P_t = M_t + u_t$; Hasbrouck's u_t is public news in the generalized Roll model, so this entry uses s_t.

Conditions. Fundamental volatility comes only from *unexpected* changes in the factors that set values. Predictable price changes that pay holders for carrying costs and risk, such as a zero-coupon bond's creep towards par, are not volatility; an unexpected change in rates is [alpha-harris, chap. "Volatility", pp. 411–412]. When news is common knowledge prices can change without trading; when few know it, prices move on high volume as the informed trade; and the price revisions that make up the adverse-selection spread component contribute to fundamental volatility [alpha-harris, same chap., p. 411]. Transitory volatility includes bid–

ask bounce, produced by the transaction-cost (transitory) spread component, and the price effects of large orders and cumulative uninformed imbalances that reverse when value traders and arbitrageurs act; it includes both the price change and its reversal. Value traders, arbitrageurs and dealers do not cause it but resolve it [alpha-harris, same chap., pp. 413–414]. Roll's model is the special case $s_t = \frac{1}{2}SQ_t$ with trade signs independent of w_t, which gives the two-term variance $\sigma^2 + \frac{1}{2}S^2$ (see Roll's estimator).

Practice. Fundamental factors differ by instrument: supply and demand, interest rates and storage costs for commodities; rates and credit quality for bonds; management, resources and technologies, product and input markets and rates for stocks; inflation, macroeconomic policy and trade and capital flows for currencies. Volatility is higher for commodities that are costly to store or perishable, especially with long production pipelines, variable demand or inelastic demand; for firms whose value depends on unresolved uncertainties, such as research outcomes or the growth behind high price–earnings ratios; for instruments exposed to political risk; and for the equity of highly leveraged firms [alpha-harris, same chap., pp. 411–413]. Because transitory price changes revert, negative serial correlation of price changes is a strong indicator of transitory volatility: over successive trades for bid–ask bounce, and over minutes to months for uninformed order imbalances [alpha-harris, same chap., p. 414]. A variance ratio $V_{M,N} = [\mathrm{Var}(p_t - p_{t-M})/M]/[\mathrm{Var}(p_t - p_{t-N})/N]$ equals one for a random walk; when microstructure effects dominate short horizons, $V_{M,N} > 1$ for $M < N$ [s6-hasbrouck, sec. 8.7, eq. (8.14), pp. 74–75]. Transitory volatility moves with uninformed traders' transaction costs and is small in liquid markets [alpha-harris, chap. "Volatility", p. 414]. Regulators can have no lasting effect on fundamental volatility but can change transitory volatility, in either direction [alpha-harris, same chap., p. 410]. Circuit breakers slow fundamental price changes; their net effect on transitory volatility is indeterminate, lowering it when they restrain uninformed traders, order anticipators and bluffers and raising it when they restrain informed traders, dealers and arbitrageurs, and empirical studies have been inconclusive. Margins and transaction taxes cannot discriminate between these trader types [alpha-harris, chap. "Bubbles, Crashes, and Circuit Breakers", secs. 28.3.2–28.3.4, pp. 575–577].

Pitfalls. Negative serial correlation is not proof of transitory volatility: successive spot prices of a perishable commodity are prices of different items, for different delivery dates, so their negative correlation can be fundamental [alpha-harris, chap. "Volatility", box, p. 413]. Its absence is not proof of a random walk either: lagged price adjustment makes price changes positively autocorrelated [s6-hasbrouck, sec. 8.6, Exercise 8.2, p. 74], and negative autocorrelation at some lags can offset positive autocorrelation at others to give a variance ratio near one [s6-hasbrouck, sec. 8.7, p. 75]. Roll's variance components understate transitory volatility when reversals take longer than one trade [alpha-harris, chap. "Volatility", p. 416]. The split of $\mathrm{Var}(\Delta p_t) = \sigma_w^2 + \mathrm{Var}(\Delta s_t) + 2\,\mathrm{Cov}(w_t, \Delta s_t)$ beyond σ_w^2 is not identified; assigning the covariance term to transitory volatility, as the legacy notes did, is a convention, and Harris's "sum of the two" holds when the components are uncorrelated (Interpretation). The legacy notes also listed *episodic* and *continuing* volatility as further sources. Harris uses episodic volatility for episodes of large price changes in short intervals [alpha-harris, same chap., p. 410], a description rather than a third source; continuing volatility does not appear in the passages checked.

See also. Roll's estimator (special case and estimator); component identification (the same split for spreads: the generalized Roll model); Garman–Klass volatility (measuring total volatility from ranges); bubbles and crashes (episodes mixing both components); spread determinants (volatility as a determinant); fundamental value (Harris's value, conditioned on all information, not only public); returns by time scale.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. volatility; fundamental volatility; transitory volatility; episodic volatility; total volatility; random-walk decomposition; efficient price; pricing error; permanent and transitory components; variance ratio; negative serial correlation; bid–ask bounce; storage costs; perishable commodities; political risk; leverage; circuit breakers; regulating volatility

Status. Verification: partly checked 2026-10-15 (against the text layers of local copies of `alpha-harris`, 2002 Oxford printing, pp. 410–417 and 575–577, and `s6-hasbrouck`, 2007 Oxford printing, pp. 69 and 72–75). Stability: draft. Currency: timeless (concepts and method).

Method 1.25 Garman–Klass volatility from open, high, low and close prices

Statement. Lehalle and Laruelle estimate intraday volatility over a fixed interval from its open, high, low and close prices p^O, p^H, p^L, p^C and its average price $\bar{p}$ [alpha-laruelle, chap. 2, sec. 2.5.1, p. 172]:

$$\mathrm{Vol}_{GK} = \frac{\sqrt{\frac{1}{2}\left(p^H - p^L\right)^2 - (2\log 2 - 1)\left(p^C - p^O\right)^2}}{\bar{p}}.$$

1. Choose the interval. Lehalle and Laruelle use 5 minutes to show intraday seasonality and 15 minutes to compare markets [alpha-laruelle, same chap., secs. 2.2 and 2.5.1, pp. 146 and 172].

2. Record the open, high, low and close of each interval and apply the formula.

3. Average across intervals or days; the comparison of FTSE 100 and IBEX 35 components used a 20-day moving average of daily averages [alpha-laruelle, same chap., sec. 2.5.1, p. 172].

Notation. The superscripts O, H, L, C avoid a clash with the transaction-cost component C. The legacy notes used log ratios, $\sigma_{GK} = \sqrt{\frac{1}{2}(\log H/L)^2 - (2\log 2 - 1)(\log C/O)^2}$; since $\log(p^H/p^L) \approx (p^H - p^L)/\bar{p}$ for small ranges, the two forms agree to first order (Interpretation).

Conditions. Volatility is not observed: each estimate presumes a model. Under an arithmetic Brownian price the natural estimator sums squared trade-to-trade changes, but on high-frequency data that estimator is polluted by bid–ask bounce, and with jumps they must be removed first. Garman–Klass is a measure not polluted too much by the bounce. Implied volatility measures something else, the uncertainty priced into options [alpha-laruelle, same chap., sec. 2.2, pp. 146–147].

Practice. Lehalle and Laruelle use it for the intraday seasonality of volatility, with volume, spread and book size, on Royal Dutch Shell, 2011–2016 (Figure 2.12, pp. 146–148), and to compare the intraday volatility of FTSE 100 and IBEX 35 components after MiFID, finding no significant effect of the UK's fragmentation on intraday volatility [alpha-laruelle, same chap., sec. 2.5.1, pp. 172–174].

Pitfalls. The coefficient $2\log 2 - 1$ and the estimator's properties come from Garman and Klass (1980), which was not consulted; Lehalle and Laruelle give the formula without derivation or assumptions. The estimate is per interval and relative to price; scaling it to a daily or annual figure needs an assumption about how variance adds across intervals (Interpretation). When the high and low are trade prices they include part of the spread, so on short intervals the range still carries some bounce (Interpretation).

See also. fundamental and transitory volatility (what the total contains); spread–volatility regression (a use of interval volatilities); Roll's estimator (the bounce that trade-based estimators pick up); returns by time scale (open-to-close and overnight returns).

Further reading. The original estimator [garman-klass-1980], not consulted. The legacy block, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. Garman–Klass; range-based volatility; open high low close; OHLC; intraday volatility; realised volatility; bid–ask bounce; volatility estimation; intraday seasonality

Status. Verification: partly checked 2026-10-15 (the formula against the page image of `alpha-laruelle`, 2018 World Scientific printing, p. 172; its use against the text layer, pp. 146–148 and 172–174; the original paper not consulted). Stability: draft. Currency: timeless (method); the applications use 2007–2016 European data.

Definition 1.26 Intraday, overnight and close-to-close returns: volatility and correlation by time scale

Statement. For day t with opening price p_t^O and closing price p_t^C, Lehalle and Laruelle define the *intraday* and *overnight* returns [alpha-laruelle, chap. 2, sec. 2.5.1, eqs. (2.2)–(2.3), p. 175]:

$$r_t^{\text{intra}} = \log p_t^C - \log p_t^O, \qquad r_t^{\text{on}} = \log p_t^O - \log p_{t-1}^C,$$

and compute the correlation between two stocks separately on each series, over a six-month sliding window, averaging the pairwise correlations across an index's components [alpha-laruelle, same sec., pp. 174–175]. Interpretation: the close-to-close (*extraday*) return is exactly their sum, $r_t^{\text{cc}} = r_t^{\text{on}} + r_t^{\text{intra}}$. Volatility likewise has an intraday and an extraday scale: investors with extraday constraints bear extraday volatility, and intraday volatility only through its effect on that scale [alpha-laruelle, same chap., sec. 2.5, p. 169].

Notation. Superscripts O and C for open and close, as in Garman–Klass, avoid a clash with the transaction-cost component C.

Conditions. The two returns come from different market mechanisms: overnight the market is organised as an auction fixing, during the day as a continuous double auction. The authors hold that the opening fixing is less contaminated by microstructure noise than the closing fixing, and so better reflects investors' consensus on the price [alpha-laruelle, same chap., sec. 2.5.1, p. 177]. Interpretation: $\text{Var}(r^{\text{cc}}) = \text{Var}(r^{\text{on}}) + \text{Var}(r^{\text{intra}}) + 2\,\text{Cov}(r^{\text{on}}, r^{\text{intra}})$, so extraday risk mixes the two scales and their covariance.

Practice. On the components of the CLES 60, DAX, CAC 40 and IBEX 35, 1991–2012, correlation rises mainly on the overnight side, and overnight correlation explains well the spikes in extraday correlation, to which it is closer; intraday correlation acts as a baseline that overnight correlation keeps bouncing above [alpha-laruelle, same sec., Figures 2.27–2.28, pp. 174–177]. Intraday correlation spiked in summer 2010 and in August 2011, and for three of the indices was in October 2010 at its highest level since 1991 [alpha-laruelle, same sec., pp. 174 and 176]. The authors read intraday correlation as an intermediate marker in a contagion process by which microscopic causes have macroscopic effects [alpha-laruelle, same sec., p. 174].

Pitfalls. The authors attribute part of the March–October 2010 rise in extraday correlation to rising intraday correlation, the consequence of a less noisy PFP, and offer growing high-frequency activity as one possible explanation [alpha-laruelle, same sec., p. 177]: a conjecture, not a test. The legacy notes said daily correlation spikes come from different mechanisms at different horizons; the authors find the spikes mainly overnight. The legacy notes tied overnight correlation to auction and news revaluation and overnight volatility

to information arrival. The auction is in the source; news arriving overnight is not stated in the passages checked (Interpretation). A legacy principle said intraday frictions matter for daily risk when they change the closing or auction price or the path used by benchmark investors. The source asks whether changes in intraday price formation show at the extraday scale [alpha-laruelle, same chap., sec. 2.5, p. 169], and compares the noise in the two fixings (p. 177); benchmark paths are not discussed there. Interpretation: the returns depend on which prices are used, auction prints or last trades, and the open and close of a fragmented market may differ by venue.

See also. Garman–Klass volatility (volatility within intraday intervals; the FTSE 100 and IBEX 35 comparison of the same section); fundamental and transitory volatility (variance across horizons); price formation process. The legacy Flash Crash and systemic-risk blocks of the same section of the source are assigned to `hft-market-design` and `market-structure-execution-quality`.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. intraday return; overnight return; close-to-close return; extraday volatility; intraday volatility; overnight correlation; intraday correlation; correlation spikes; opening auction; closing auction; contagion; microstructure and macro prices

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, pp. 169 and 174–177, figure captions included; no page images). Stability: draft. Currency: as of the 1991–2012 data; definitions timeless.

Method 1.27 Spread–volatility regression in levels and logs

Statement. To test how the spread moves with volatility, Lehalle and Laruelle proceed in five steps [alpha-laruelle, app. A, sec. A.10, pp. 292–298]:

1. Compute the spread ψ_i and the volatility σ_i over the same intervals; their example uses 15-minute intervals of Rio Tinto from January 2012.
2. Fit $\psi_i = a_0 + a_1\sigma_i + \epsilon_i$ by ordinary least squares: $\hat{a}_1 = \sum_i(\sigma_i - \bar{\sigma})(\psi_i - \bar{\psi}) / \sum_i(\sigma_i - \bar{\sigma})^2$, $\hat{a}_0 = \bar{\psi} - \hat{a}_1\bar{\sigma}$.
3. Inspect the distributions: both variables looked log-normal.
4. Fit $\log\psi_i = a_0 + a_1\log\sigma_i + \epsilon_i$, which in levels is the multiplicative model $\psi = A_0\sigma^{a_1}e^{\epsilon}$, $A_0 = e^{a_0}$.
5. If $\epsilon \sim N(0, \sigma_\epsilon^2)$, then $\mathbb{E}[\psi] = A_0\sigma^{a_1}e^{\sigma_\epsilon^2/2}$, so predict levels with $\hat{\psi} = \hat{A}_0\sigma^{\hat{a}_1}e^{\hat{\sigma}_\epsilon^2/2}$, which they call the unbiased predictor.

A positive $\hat{a}_1$ is consistent with the spread rewarding market makers for the risk they take: the higher the volatility, the more they ask [alpha-laruelle, chap. 2, sec. 2.3.1, p. 150].

Notation. ψ is the spread, σ the volatility over the interval, a_0, a_1 the intercept and slope, σ_ϵ^2 the residual variance. A_0 is the level intercept, not the adverse-selection component A.

Conditions. The least-squares assumptions are those of the simple linear model: the regressor measured without error; errors with mean zero, constant variance and no correlation with each other or with the regressor; normal errors for the t tests [alpha-laruelle, app. A, sec. A.10.1, pp. 293–295]. Across small-tick instruments, Lehalle and Laruelle report the relation $\psi_i = a + b\,\sigma_i/\sqrt{N_i}$, with N_i the average number of trades per day and a, b shared within an asset class; for large-tick instruments, $\psi_i = a + (b_1\sigma_i + b_2\,\eta\,\text{tick})/\sqrt{N_i}$, where $\eta = N_c/(2N_a)$ is half the ratio of price continuations to alternations, smaller the more the price bounces [alpha-laruelle, chap. 2, sec. 2.2, pp. 144–145].

Practice. For Rio Tinto, the level fit gave $\hat{a}_0 = 2.4894$, $\hat{a}_1 = 0.0485$, $R^2 = 0.2404$; the log fit $\hat{a}_0 = 0.5609$, $\hat{a}_1 = 0.2252$, $R^2 = 0.1553$; the back-transformed predictor, with $\hat{\sigma}_\epsilon = 0.2509$, explains 0.1097 of the variance of the spread in levels [alpha-laruelle, app. A, Figures A.14, A.17 and A.18, pp. 296–298]. Daily spreads rise with daily volatility on BNP Paribas and BP, 2008–2012; points off the line in August 2011 coincide with the short-selling ban, when market makers could not meet their commitments in financial stocks [alpha-laruelle, chap. 2, sec. 2.3.1, Figure 2.13, pp. 150–151].

Pitfalls. The fit is descriptive: in the example volatility explains a quarter or less of the spread's variance. Unless the comparison controls for them, other variables drive both series. Tick size: when the spread is stuck at one tick, liquidity providers add size at the best quotes instead, so in large-tick assets volatility may show in depth rather than in the spread [alpha-laruelle, chap. 2, sec. 2.2, p. 148]. Time of day: spread, volatility, volume and book size share an intraday seasonality, high after the open, so pooling intervals across the day mixes the seasonal pattern with the relation [alpha-laruelle, same sec., pp. 145–148]. Trading activity: the cross-sectional relation scales volatility by $1/\sqrt{N_i}$ [alpha-laruelle, same sec., pp. 144–145]. Harris's general warning applies: observable determinants are correlated with each other and with more than one primary factor, so tests need controls [alpha-harris, chap. "Bid/Ask Spreads", p. 314]. The legacy notes also listed venue and depth as controls; the passages checked do not discuss them as regression controls. Causation runs both ways,

since a wider spread adds bounce to trade-based volatility (Interpretation). The predictor is unbiased only with known a_0, a_1 and σ_ϵ^2; with estimates plugged in it is approximately unbiased (Interpretation).

See also. spread determinants (the theory being tested: volatility as a primary determinant); Garman–Klass volatility (an interval volatility); spread measures (which spread to regress).

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`.

Index. spread and volatility; spread–volatility regression; log-log regression; log-normal; retransformation bias; unbiased predictor; ordinary least squares; tick size; number of trades; bid–ask bounce ratio; intraday seasonality; market depth

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-laruelle`, 2018 World Scientific printing, pp. 144–151 and 292–298, figure captions included; `alpha-harris`, p. 314). Stability: draft. Currency: as of the 2008–2016 data used; method timeless.

Mechanism 1.28 Bubbles and crashes: how mispricing builds and unwinds

Statement. A *bubble* is a rise of prices substantially above fundamental values, quickly or over a long period; a *crash* is a very fast fall, often but not always after a bubble. Since fundamental values are not common knowledge, bubbles and crashes can occur at all [alpha-harris, chap. "Bubbles, Crashes, and Circuit Breakers", sec. 28.1, p. 556]. The typical sequence [alpha-harris, same chap., sec. 28.1.1, pp. 556–557]:

1. Buyers grow overly optimistic about new technologies or markets, often without recognising what prices already reflect, and push prices up.

2. Momentum traders and order anticipators buy and accelerate the rise; early buyers grow more confident and early sellers buy back to stop their losses.

3. Value traders and arbitrageurs cannot sell enough (no position to sell, too little capital to carry a short, short selling hard) or will not (they expect uninformed buying to continue, or doubt their own value estimates).

4. Sellers start trading aggressively: long-term holders, early buyers taking gains, contrarians, value traders, arbitrageurs. Momentum buyers lose interest and optimists lose confidence.

5. Late buyers sell to stop losses, margin calls and stop orders force more selling, and order anticipators sell ahead of them: prices fall quickly. Panic can take prices below value, and a bounce follows.

The price accelerator (Harris, after a manuscript by Treynor): rising prices move wealth from pessimists with short positions to optimists with long ones; the richer optimists may buy more, and the losing pessimists may buy to cover margin calls, so prices rise further; the sellers are mild optimists and mild pessimists [alpha-harris, same chap., box, p. 556].

Conditions. Most bubbles and crashes involve both fundamental and transitory volatility. A bubble usually starts on good news, a fundamental change; pseudo-informed traders then overreact, the transitory part. It usually bursts on bad news, but the volume and price response is out of proportion to the news because of the overpricing that preceded it; the reversion towards value counts as transitory volatility, since it occurs only because uninformed traders inflated prices, and panic below value adds more [alpha-harris, same chap., sec. 28.1.2, pp. 557–558].

Practice. Harris's 1987 account shows the mechanisms together: portfolio insurance traded like stop orders, and uncertainty about how much of it was waiting kept buyers out; NYSE order printers could not keep up, so traders did not know whether their orders had filled; index futures led the cash market down and traded at a discount of more than 10 percent once index arbitrageurs stopped; and buy-side traders could not reach Nasdaq dealers [alpha-harris, same chap., sec. 28.2.2, pp. 559–563]. Crashes rarely have a single cause: traders become confused about risk, attend to the wrong issues and, when events move fast, panic [alpha-harris, same chap., sec. 28.2.8, p. 571].

Pitfalls. A large fall without news of commensurate importance does not show that the market failed. Most crashes correct earlier pricing errors; if they were failures, prices would rebound fully once traders saw them below value, but after most crashes prices rebound little [alpha-harris, same chap., box, p. 559]. Attribution therefore looks at the reversal after the event, at what preceded it, and at overloaded systems, forced selling and broken links between markets, as in 1987 (Interpretation, from the cases on pp. 557–563). Bubbles are hard to identify in real time, and analysts generally understand the causes of extreme volatility better after the fact [alpha-harris, same chap., p. 555]. The legacy list named policy among the stories that start a bubble; Harris's general account names technology and new markets, and his Japanese example turns on loose monetary policy [alpha-harris, same chap., sec. 28.2.7, pp. 570–571].

See also. fundamental and transitory volatility (the two components in each phase); outside spread (why value traders wait); spread determinants. The legacy flash-crash, liquidity-spiral and shock-propagation blocks are assigned to `hft-market-design` and remain in Chapter 7.

Further reading. The legacy blocks it replaces, verbatim: `notes/artefacts/market-making-lob/derivations.tex`. Treynor's manuscript "The Canonical Bubble" was not consulted.

Index. bubble; crash; market break; meltdown; price accelerator; momentum traders; order anticipators; limits to arbitrage; margin calls; stop-loss orders; dead cat bounce; portfolio insurance; 1987 crash; crash attribution; market failure

Status. Verification: partly checked 2026-10-15 (against the text layer of a local copy of `alpha-harris`, 2002 Oxford printing, pp. 555–563 and 570–571). Stability: draft. Currency: timeless (mechanism); the case material is historical, to 2002.

Notes awaiting migration

Chapter 2

ML and alternative data

2.1 Exploratory Data Analysis

2.1.1 Data Taxonomy

A brief overview of the types of data used in systematic trading.

Four essential types of financial data

Fundamental Data	Market Data	Analytics	Alternative Data
Assets	Price/Yield/IV	Analyst Recommendation	Satellite/CCTV
Liabilities	Volume	Credit Ratings	Google Searches
Sales	Dividend/Coupons	Earnings Expectations	Twitter/Chats
Costs/Earnings	Open Interest	News Sentiment	Metadata
Macro Variables	Quotes/Cancellations	...	...
...	Aggressor Side		
	...		

Remark 2.1.1. *Fundamental Data Characteristics*

i. Data published is indexed by last date included in report, which precedes date of release.

ii. Data is often backfilled or re-instated, and data vendor may overwrite initial values with corrections.

iii. Data is extremely regularised and low frequency.

Remark 2.1.2. *Market Data Characteristics*

i. Raw feed contains unstructured information, such as FIX messages (allow full construction of trading book), or full collection of BWIC (bids wanted in competition) responses.

ii. FIX data is not trivial to process, $\sim$ 10TB generated on daily basis

Remark 2.1.3. *Analytics Data Characteristics*

i. Derivative data as processed based on original source. Signal already extracted from the original source.

ii. Costly, methodology used in production may be biased or opaque.

Remark 2.1.4. *Alternative Data Characteristics*

i. Produced by individuals, business processes, and sensors.

ii. Primary information that has not made it to other sources.

iii. Cost and privacy concerns. May be useful if it annoys data infrastructure team.

Definition 2.1.5. *Reference Data*

i. Trading Universe: evolving daily to incorporate new listings, de-listings etc. Knowing when a particular stock no longer trades is important to avoid survivor bias.

ii. Symbology Mapping: ISIN, SEDOL, RIC, Bloomberg Tickers etc. Data is not static, symbols may change, complicating historical data merges. Mapping needs to persist as point-in-time data and allow for historical 'as-of-date' usage, require implementation of bi-temporal data structure.

iii. Ticker Changes: for reasons described in symbology mapping. To maintain historical table of ticker changes to seamlessly go up and down time series data.

iv. Corporate Actions Calendars: contain stock and cash dividends (announcement, execution date), stock splits, reverse splits, rights offer, mergers and acquisitions, spin off, free float or shares outstanding adjustments, quotation suspensions etc.
For dividends, announcements may coincide with more volatility, jumps in price time series. Allow building of strategies that look to benefit from the added volatility.
For stock splits, reverse splits, rights offers, all historical data need to be adjusted backward to reflect the split (both volume and price).
For M&A, spin-offs, to account for changes in valuation, hence used in Merger Arbitrage strategies.
Suspensions result in gaps in data, may impact backtesting.

v. Static Data: country, sector, primary exchange, currency, and quote factor. May be used to group instruments based on fundamental similarities (hence for pairs trading). Maintaining a table of quotation currency per instrument necessary to aggregate positions at portfolio level.

vi. Exchange Specific Data: individual exchanges have variety of differences to be accounted for when designing trading strategies. First group of information concerns the hours and dates of operations:

1. Holiday Calendar: Strategies trading simultaneously in several markets and leveraging correlation may not perform as well if one market is closed and another is open.
2. Exchange Sessions Hours: Different sessions (Pre-Market, Continuous Core, After-Hour etc.); auction times and respective cutoff times for oder submission; lunch break restricting intraday trading and auctions before/after lunch; settlement times for futures market. Daylight Saving Time (DST) adjustments; length of trading hours during course of the year; different trading hours by venue.
3. Disrupted Days: Exchange outages or trading disruptions, market data issues. To be recorded so they can be filtered out when building or testing strategies.

Second group of information governing the mechanics of trading:

1. Tick Size: Minimum eligible price increment; may vary by instrument and as a function of price.
2. Trade and Quote Lots: Minimum size increment for quotes or trades.
3. Limit-Up and Limit-Down Constraints: Maximum daily fluctuations of securities, and whether trading is paused or can only be traded at better prices than the threshold.
4. Short Sell Restrictions: Restrict shot sells not to trade at price worse than last price, or not to create a new quote that will be lower than the lowest prevailing quote. Impact ability to source liquidity.

vii. Market Data Condition Codes: vary per exchange and asset class, and each market event may be attributed to several codes at once. To build mapping table of condition codes and what they mean (i.e., auction trade, lit or dark trade, cancelled or corrected trade, regular trade, off-exchange trade reporting, block-size trade, trade originating from multi-leg order such as option spread trade etc.). To access liquidity for trading algorithm, trades published for reporting purposes must be excluded and not be used to update some of the aggregated daily data used in construction of trading strategies.

viii. Special Day Calendars: days wth distinct liquidity characteristics to be accounted for in both execution strategies and in alpha generation process. These (non-exhaustive) irregular events may be:

1. Half trading days preceding Christmas and following thanksgiving in US
2. Ramandan even in Turkey
3. Taiwan market opening on weekend to make up for lost trading days during holiday periods
4. Korean market changing trading hours on day of nationwide university entrance exam
5. Brazilian market opening late on day following the Carnival
6. Last trading days of months and quarters (investors rebalance portfolios)
7. Index rebalancing dates, where intraday volume distribution is significantly skewed toward EODs
8. Options and futures expiry dates (quarterly/monthly expiry, Triple Witching in US, Special Quotations in Japan) where excess trading volume and different intraday patterns result from hedging activity and portfolio adjustments.

Model normal days first. Special days are modelled either independently, or using normal days as baseline.

ix. Futures-Specific Reference Data: to know which contract was live at any point of time by using expiry calendar, and the most liquid contract. Equity index futures are most liquid for first contract available (front month), energy futures such as oil are more liquid for second contract. Hence to know which contract carry the most significant price formation characteristics, and what is true liquidity available.
Note there is no real standardised expiry frequency that applies across markets. When computing rolling-window metrics, to account for potential roll dates (due to investors rolling forward positions) that may

have happened during the time span. May blend volume time series prior to roll date and after roll date. Futures market also have different market phases during the day with significantly different liquidity characteristics. Various market data metrics (volume profile, average spread, average bid-ask sizes etc) should be computed separately for each market phase by maintaining a table of start and end times of each session for each contract.

x. Options-Specific Reference Data (Options Chain): expiry date and strike price combination (option chain). Map of equity tickers to option tickers with strike and expiry dates allow for design for more complex investment and hedging strategies (i.e., distance to strike, change in open interest of puts and calls).

xi. Market-Moving News Releases: macroeconomic announcements. To maintain calendar of dates and times of their occurrences to assess their impact on strategies. Central bank announcements or meeting minutes releases about major economies (FED/FOMC, ECB, BOE, BOJ, SNB), Non-Farm Payrolls, Purchasing Managers' Index, Manufacturing Index, Crude Oil Inventories etc.
Stock-specific releases such as earnings calendars, specialised sector events (for healthcare, biotech etc).

xii. Related Tickers: tickers that are related to each other as they fundamentally represent the same underlying asset. Allows efficient opportunity exploitation. Primary tickers to composite tickers mapping (for markets with fragmented liquidity), dual listed/fungible securities in US and Canada, ADR or GDR, local and foreign boards in Thailand etc.

xiii. Composite Assets: ETFs, Indexes, Mutual Funds etc. May be used to achieve desired exposures, or as cheap hedging instruments, and provide arbitrage opportunities when they deviate from NAV. To maintain information such as time series of their constituents and value of any cash component, divisor used to translate NAV into quoted price, constituent weights.

xiv. Latency tables: for higher frequency trading strategies. Contains distribution of latency between different data centres for more efficient order routing, and reordering data that are recorded in different locations.

Definition 2.1.6. *Market Data Feed*

i. Level I Data (Trade and BBO Quotes): trades and top of book quotes. Enough to reconstruct Best Bid and Offer (BBO). Also contains information in form of trade status (cancelled, reported late etc), trade and quote qualifiers (odd lot, normal trade, auction trade, Intermarket Sweep, average price reporting, on which exchange etc). May be used to analyse sequence of events and decide if a given print should be used to update the last price and total volume traded at a point in time.

ii. Level II Data (Market Depth): addition of quote depth data, displays all lit limit order book updates (price changes, addition or removal of shares quoted) at any level in the book, for all of the lit venues in fragmented markets.

iii. Level III Data (Full Order View): message data. Each order arriving is attributed a unique ID for tracking over time, and is precisely identified when it is executed, cancelled, or amended. Possible to build a full (with national depth) book at any moment intraday. Example from US market:

1. Timestamp: number of milliseconds after midnight
2. Ticker: equity symbol (up to 8 char)
3. Order: Unique order ID
4. T: message type. 'B' is add buy order; 'S' is add sell order'; 'E' is execute outstanding order in part; 'C' is cancel outstanding order in part; 'F' is execute outstanding order in full; 'D' is delete outstanding order in full; 'X' is bulk volume for cross event; 'T' is execute non-displayed order
5. Shares: order quantity for 'B', 'S', 'E', 'X', 'C', 'T' messages. Zero for 'F', 'D' messages
6. Price: order price for 'B', 'S', 'X', 'T' messages. Zero for cancellation and executions. Last 4 digits are decimal, padded to right with zeroes. Divide by 1000 to convert to currency value.
7. MPID: Market Participant ID associated with transaction (4 char)
8. MCID: Market Centre Code for originating exchange (1 char)

A few special types of orders worth mentioning are:

1. Order subject to price sliding: execution price may be one cent worse than display price at NASDAQ; ranked at locking price as hidden order, displayed at the price one minimum price variation inferior to locking price. New order ID will be used if order is replaced as a display order.
2. Pegged order: based on NBBO, not routable, new timestamp given upon repricing; display rule vary over exchanges
3. Mid-point peg order: non-displayed, may result in half-penny execution

4. Reserve order: displayed size is ranked as displayed limit order; reserve size is behind non-displayed orders and pegged orders in priority.
   Minimum display quantity is 100, amount replenished from reserve size when it falls below 100 shares; New timestamp created, displayed size re-ranked upon replenishment.
5. Discretionary order: displayed at one price while passively trading a more aggressive discretionary price. Order becomes active when shares are available within discretionary price range. Order ranked last in priority. Execution price may be worse than display price.
6. Intermarket sweep order: can be executed without need for checking prevailing NBBO.

Using these data, we may model: the pattern of inter-arrival times of various events; arrival and cancellation rates as a function of distance from nearest touch price; arrival and cancellation rates as a function of other information, such as in the queue on either side of the book, order book imbalance etc.
Once modelled, we may analyse: the impact of market order on limit order book; chances for limit order to move up the queue from given entry position; probability of earning the spread; expected direction of price movement over a short horizon.

Definition 2.1.7. *Binned Data*

i. Open, High, Low, Close (OHLC) and Previous Close Price: indication on trading activity and intraday volatility. Distance traveled between lowest and highest points is indication of market sentiment. Previous close has to be adjusted for corporate actions and dividends.

ii. Last Trade before Close (Price/Size/Time): how much the close price may have jumped in final moments of trading; how stable it is as a reference value for next day.

iii. Volume: trading activity indicator, especially when level jumps from long term average. Collect volume breakdown between lit and dark venues for execution strategies.

iv. Auctions Volume and Price: price discovery event when significant volume prints occur.

v. VWAP: indication of trading activity on the day. Easier to algorithmically execute large orders with VWAP than a single print.

vi. Short Interest/Days-to-Cover/Utilisation: good proxy for investor position. Short pressure an indication of upcoming short term moves: large short interest is bearish view from institutional investors. Utilisation level of available securities to borrow gives indication of how much room is left for future shorting. Days-to-Cover to assess magnitude of potential short squeeze (if sellers unwind position, fraction of available daily liquidity needed); larger value indicates larger potential of sudden upswing on heavily shorted securities.

vii. Futures Data: insight into activity or large investors through open interest data. Offer arbitrage opportunities if their basis exhibits mis-pricing compared to one's dividend estimates.

viii. Index-Level Data: source of relative measures for instrument specific features (index OHLC, volatility). Normalised features identify individual instruments deviating from their benchmarks.

ix. Options Data: information on position of traders through open interest and Greeks.

x. Asset Class Specific: yield/benchmark rates (repo, 2y, 10y, 30y), CDS spreads, US Dollar Index

Definition 2.1.8. *Granular Intraday Microstructure Activity*

i. Number and Frequency of Trades: proxy for activity level, and how continuous it is. Low number of trades mean harder execution, and may be more volatile

ii. Number and Frequency of Quote Updates: similar proxy for activity level

iii. Top of Book Size; proxy for liquidity of instruments (larger top of book size makes it possible to trade larger order quasi immediately)

iv. Depth of Book (price and size): similar proxy for liquidity

v. Spread Size (average, median, time weighted average): proxy for cost of trading. Parametrised distribution used to identify opportunities if they are cheap or expensive

vi. Trade size (average, median): to identify intraday liquidity opportunities when examining volume available in the order book.

vii. Ticking time (average, median): representation of how often one should expect changes in the order book first level. For execution algorithms for which the frequency of updates (adding/cancelling child orders, re-evaluating decisions etc.) should commensurate with characteristics of the traded instrument.

Daily distribution can be used as start of day estimates and updated intraday with online Bayesian updates.
Last group of daily data is derived from previous two groups but stored pre-computed to save time during research phase, or to be used as normalising values:

i. X-day Average Daily Volume (ADV) / Average Auction Volume

ii. X-day Volatility (close-to-close, open-to-close etc)

iii. Beta with respect to index or sector (plain beta, or asymmetric up-days/down-days beta)

iv. Correlation matrix

When binning data, this may be grouped into bins ranging from a few seconds to 30 minutes. Minute bar data are used for volume and spread profiles to prevent introducing excess noise due to market friction.

Definition 2.1.9. *Fundamental Data and Other Data*

i. Key Ratios: Earnings Per Share (EPS), Price-to-Earning (P/E), Price-to-Book (P/B), etc.

ii. Analyst Recommendations: aggregated values given consensus valuation

iii. Earnings data: estimations by research analysts provide quarterly earning estimates which can be used as indication of performance of stock before actual value is published

iv. Holders: sudden changes in ownership indicate changes in sentiment by sophisticated investor

v. Insiders Purchase/Sale: indicator of future stock price moves from group of people who have access to best possible information about the company

vi. Credit Ratings: credit downgrades resulting in higher funding costs have negative impact on equity prices

2.1.2 Financial Data Structures

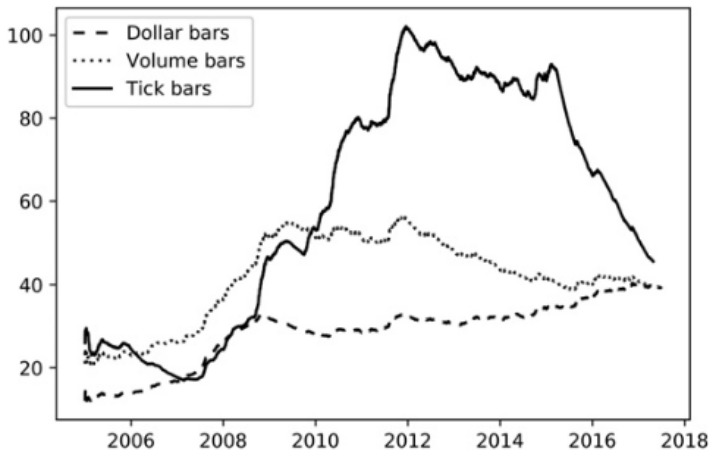


Figure 2.1: Average daily frequency of tick, volume, and dollar bars

Remark 2.1.10. *Standard BARS*
Method to transform a series of observations arriving at irregular frequency into a homogeneous series derived from regular sampling.

i. Time Bars: obtained by sampling information at fixed intervals. Information collected includes timestamp, volume-weighted average price (VWAP), open price, close price, high price, low price, volume etc.
To be avoided as markets do not process information at constant time interval. Time bars oversample information in low-activity periods and under-sample information in high-activity periods. Time bars exhibit poor statistical properties, i.e., serial correlation, heteroscedasticity, non-normality of returns.

ii. Tick Bars: sample variables extracted each time a pre-defined number of transactions take place. Allows synchronisation of sampling with a proxy of information arrival.
Sampling as a function of trading activity creates returns closer to IID Normal [ane-geman-2000]. When constructing tick bars, to be aware of outliers, as many exchanges carry out auction at open and at close; order book accumulates bids and offers without matching. Order fragmentation introduces some arbitrariness in number of ticks. Matching engine protocols may split one fill into multiple artificial partial fills as a matter of operational convenience.

iii. Volume Bars: samples every time a pre-defined amount of security's units that have been exchanged.
Achieves better statistical properties than sampling tick bars.
Convenient artefact for studying market microstructure theories.

iv. Dollar Bars: samples an observation every time a pre-defined market value is exchanged. Used when the analysis involves significant price fluctuations. Robust against corporate actions such as splits, reverse splits, issuance of new shares, buying back existing shares.
Bar size could be dynamically adjusted as a function of free-floating market cap of a company or outstanding amount of issued debt.

Remark 2.1.11. *Information-Driven Bars*
Method to sample more frequently when new (micro-structural) information arrives to the market.

i. Tick Imbalance Bars: sample bars whenever tick imbalance exceeds expectations. To determine tick index T such that accumulation of signed ticks exceeds a given threshold.
Let $\{(p_t, v_t)\}_{t=1,\dots,T}$ be sequence of ticks where p_t and v_t is the price and volume associated with tick t.
Let tick rule define a sequence $\{b_t\}_{t=1,\dots,T}$ where

$$b_t = \begin{cases} b_{t-1} & \text{if } \Delta p_t = 0 \\ \frac{|\Delta p_t|}{\Delta p_t} & \text{if } \Delta p_t \neq 0 \end{cases}$$

The tick imbalance at time T is defined as

$$\theta_T = \sum_{t=1}^{T} b_t$$

Compute expected value of θ_T at beginning of the bar,

$$E_0[\theta_T] = E_0[T](P[b_t = 1] - P[b_t = -1]) = E_0[T](2P[b_t = 1] - 1)$$

where $E_0[T]$ is expected size of tick bar, $P[b_t = 1]$ and $P[b_t = -1]$ is unconditional probability that a tick is classified as a buy and sell. In practice, $E_0[T]$ and $(2P[b_t = 1]-1)$ may be estimated as an exponentially weighted moving average of T and b_t values from prior bars.
Define the tick imbalance bar (TIB) as a T^* contiguous subset of ticks such that

$$T^* = \arg\min_T \{|\theta_T| \geq E_0[T] \mid 2P[b_t = 1] - 1\}$$

where the size of expected imbalance is implied by $|2P[b_t = 1] - 1|$.
When θ_T is more imbalanced than expected, a low T will satisfy the conditions.
TIBs are produced more frequently under presence of informed trading (asymmetric information that triggers one-side trading). TIBs are buckets of trades containing equal amounts of information.

ii. Volume/Dollar Imbalance Bars: sample bars when volume or dollar imbalances diverge from expectations.
First, define imbalance at time T as

$$\theta_T = \sum_{t=1}^{T} b_t v_t$$

where v_t may represent ether number of securities traded (VIB) or dollar amount traded (DIB).
The expected value of θ_T at the beginning of the bar is then computed as

$$\begin{aligned} E_0[\theta_T] &= E_0\left[\sum_{t|b_t=1}^{T} v_t\right] - E_0\left[\sum_{t|b_t=-1}^{T} v_t\right] \\ &= E_0[T](P[b_t = 1]E_0[v_t|b_t = 1] - P[b_t = -1]E_0[v_t|b_t = -1]) \\ &= E_0[T](v^+ - v^-) \end{aligned}$$

where the initial expectation of v_t is decomposed into component contributed by buys and sells. Then

$$E_0[\theta_T] = E_0[T](2v^+ - E_0[v_t])$$

In practice, $E_0[T]$ and $(2v^+ - E_0[v_t])$ may be estimated as exponentially weighted moving average of T and $b_t v_t$ values from prior bars. Next, define VIB or DIB as a T^*-contiguous subset of ticks such that

$$T^* = \arg\min_T \{|\theta_T| \geq E_0[T] \mid 2v^+ - E_0[v_t]\}$$

where the size of expected imbalance is implied by $|2v^+ - E_0[v_t]|$.
When θ_T is more imbalanced then expected, a low T will satisfy the conditions.

VIB and DIB addresses concerns on tick fragmentation and outliers, and also addresses the issues of corporate actions, as the bar size is adjusted dynamically.

iii. Tick Runs Bars: sample bars when the sequence of buys in overall volume diverges from expectations. For the case when large traders sweep order book, use iceberg orders, or slice parent orders into multiple children, all leaving a trace of runs in the $\{b_t\}_{t=1,\dots,T}$ sequence. Define length of current run as

$$\theta_T = \max\left\{\sum_{t|b_t=1}^{T} b_t - \sum_{t|b_t=-1}^{T} b_t\right\}$$

The expected value of θ_T at beginning of bar is computed as

$$E_0[\theta_T] = E_0[T]\max\{P[b_t = 1], 1 - P[b_t = 1]\}$$

In practice, $E_0[T]$ and $P[b_t = 1]$ may be estimated as exponentially weighted moving average of T and proportion of buy ticks from prior bars. Next, define TRB as T^*-contiguous subset of ticks such that

$$T^* = \arg\min_T\{\theta_T \geq E_0[T]\max\{P[b_t = 1], 1 - P[b_t = 1]\}\}$$

where the expected count of ticks from runs is implied by $\max\{P[b_t = 1], 1 - P[b_t = 1]\}$.
When θ_T exhibits more runs than expected, a low T will satisfy these conditions.
Instead of measuring length of longest sequence, count number of ticks of each side without offsetting.

iv. Volume/Dollar Runs Bars: sample bars when volume or dollars traded by one side exceed expectation for a bar. First, define volume or dollars associated with a run as

$$\theta_T = \max\left\{\sum_{t|b_t=1}^{T} b_t v_t - \sum_{t|b_t=-1}^{T} b_t v_t\right\}$$

where v_t may either represent volume (VRB) or dollar amount exchanged (DRB). The expected value of θ_T at beginning of the bar is then

$$E_0[\theta_T] = E_0[T]\max\{P[b_t = 1]E_0[v_t|b_t = 1], (1 - P[b_t = 1])E_0[v_t|b_t = -1]\}$$

In practice, $E_0[T], P[b_t = 1], E_0[v_t|b_t = 1], E_0[v_t|b_t = -1]$ may be estimated as exponentially weighted moving average of T, proportion of buy ticks, buy volumes, and sell volumes from prior bars. Next, define a volume runs bar (VR) as T^*-contiguous subset of ticks such that

$$T^* = \arg\min_T\{\theta_T \geq E_0[T]\max\{P[b_t = 1]E_0[v_t|b_t = 1], (1 - P[b_t = 1])E_0[v_t|b_t = -1]\}\}$$

expected volume from runs is implied by $\max\{P[b_t = 1]E_0[v_t|b_t = 1], (1 - P[b_t = 1])E_0[v_t|b_t = -1]\}\}$.
When θ_T exhibits more runs than expected, volume from runs is greater than expected, a low T will satisfy these conditions.

Definition 2.1.12. *Multi-Product Series: ETF Trick*
To model a basket of securities as if it was a single cash product. To transform any complex multi-product dataset into a single dataset that resembles a total return ETF.

Method 2.1.13. *ETF Trick*
Produce a time series that reflects the value of \$1 invested. Changes in the series will reflect changes in PnL, series will be strictly positive, and implementation shortfall will be taken into account. The bars contain:

i. Raw open price of instrument $i = 1, \dots, I$ at bar $t = 1, \dots, T$: $o_{i,t}$
ii. Raw close price of instrument $i = 1, \dots, I$ at bar $t = 1, \dots, T$: $p_{i,t}$
iii. USD value of one point of instrument $i = 1, \dots, I$ at bar $t = 1, \dots, T$: $\varphi_{i,t}$. This includes forex rate.
iv. Volume of instrument $i = 1, \dots, I$ at bar $t = 1, \dots, T$: $v_{i,t}$
v. Carry, dividend, or coupon paid by instrument i at bar t: $d_{i,t}$. Variable can also be used to charge margin costs or costs of funding.

All instruments $i = 1, \dots, I$ were tradable at bar $t = 1, \dots, T$. Even if some instruments were not tradable over entirety of time interval $[t - 1, t]$, at least they were tradable at times $t - 1$ and t.

For basket of securities with allocation vector ω_t rebalanced (or rolled) on bars $B \subseteq \{1, \ldots, T\}$, the \$1 investment value $\{K_t\}$ is derived as

$$h_{i,t} = \begin{cases} \frac{\omega_{i,t} K_t}{o_{i,i+1}\varphi_{i,t} \sum_{i=1}^{I} |\omega_{i,t}|} & \text{if } t \in B \\ h_{i,t-1} & \text{otherwise} \end{cases}$$

$$\delta_{i,t} = \begin{cases} p_{i,t} - o_{i,t} & \text{if } (t-1) \in B \\ \Delta p_{i,t} & \text{otherwise} \end{cases}$$

$$K_t = K_{t-1} + \sum_{i=1}^{I} h_{i,t-1}\varphi_{i,t}(\delta_{i,t} + d_{i,t})$$

where $K_0 = 1$ is the initial AUM. Variable $h_{i,t}$ is the holdings of instrument i at time t, $\delta_{i,t}$ is change of market value between $t-1$ and t for instrument i. Note profits or losses are being reinvested whenever $t \in B$, hence preventing negative prices. Dividends $d_{i,t}$ are already embedded in K_t.
The purpose of $\omega_{i,t} \left(\sum_{i=1}^{I} |\omega_{i,t}| \right)^{-1}$ is to de-lever the allocations.
Let τ_i be transaction cost associated with trading \$1 of the instrument. Three additional variables that the strategy needs to know for every observed bar t are:

i. Rebalance Costs: variable cost $\{c_t\}$ associated with allocation rebalance is

$$c_t = \sum_{i=1}^{I} (|h_{i,t-1}|\, p_{i,t} + |h_{i,t}|\, o_{i,t+1})\varphi_{i,t}\tau_i \quad \forall t \in B$$

Note c_t is not embedded in K_t, as shorting will generate fictitious proceeds when allocation is rebalanced. In code, $\{c_t\}$ is treated as a (negative) dividend.

ii. Bid-Ask Spread: the cost $\{\tilde{c}_t\}$ of buying or selling one unit of this ETF,

$$\tilde{c}_t = \sum_{i=1}^{I} |h_{i,t-1}|\, p_{i,t}\varphi_{i,t}\tau_i$$

When a unit is bought or sold, strategy must charge this cost $\tilde{c}_t$.

iii. Volume: volume traded $\{v_t\}$ is determined by least active member in the basket. Let $v_{i,t}$ be volume traded by instrument i over bar t. The number of tradable basket units is

$$v_t = \min_i \left\{ \frac{v_{i,t}}{|h_{i,t-1}|} \right\}$$

Transaction costs functions may not be linear, and can be simulated by the strategy.

Method 2.1.14. *ETF Trick: Computation of Allocation Vector with PCA*
Consider an IID multivariate Gaussian process with means vector μ of size $N \times 1$, and covariance matrix V of size $N \times N$. First, perform spectral decomposition $VW = W\Lambda$, where columns in W are reordered so that elements of Λ diagonal are sorted in descending order. Second, given allocations vector ω, portfolio risk is

$$\sigma^2 = \omega' V \omega = \omega' W \Lambda W' \omega = \beta' \Lambda \beta = (\Lambda^{1/2}\beta)'(\Lambda^{1/2}\beta)'$$

where β is projection of ω on orthogonal basis. Third, Λ is a diagonal matrix, thus

$$\sigma^2 = \sum_{n=1}^{N} \beta_n^2 \Lambda_{n,n}$$

The risk attributed to the nth component is

$$R_n = \beta_n^2 \Lambda_{n,n} \sigma^{-2} = [W'n]_n^2 \Lambda_{n,n} \sigma^{-2}$$

with $R' 1_N = 1$, and 1_N is a vector of N ones.
Note $\{R_n\}_{n=1,\ldots,N}$ is distribution of risks across orthogonal components.

Next, compute vector ω which delivers user-defined risk distribution R. Note from earlier,

$$\beta = \left\{ \sigma \sqrt{\frac{R_n}{\Lambda_{n,n}}} \right\}_{n=1,\dots,N}$$

which represents allocation in new (orthogonal basis).
The allocation in old basis is $\omega = W\beta$. Rescaling ω re-scales σ, hence keeping risk distribution constant.

Method 2.1.15. *ETF Trick: Single Futures Roll*
To work with non-negative rolled series, derive price series of \$1 investment as follows:

i. Compute time series of rolled futures prices
ii. Compute return r as rolled price change divided by previous roll price
iii. Form a price series using these returns

These methods allow us to produce a continuous, homogeneous, and structured dataset from collection of unstructured financial data. Note however, that several ML algorithms do not scale well with sample size. ML algorithms achieve higher accuracy when they attempt to learn from relevant examples.

Method 2.1.16. *Sampling for Reduction*
To reduce the amount of data used to fit ML algorithm, downsampling could be used.

i. Sequential sampling at constant step size (linspace sampling)
ii. Sampling randomly using uniform distribution (uniform sampling)

Note both samples do not necessarily contain subset of most relevant observations.

Method 2.1.17. *Event-Based Sampling: CUMSUM Filter*
Bets are often placed after some event takes place, hence to let ML algorithm learn whether there is an accurate prediction function under these circumstances, CUSUM filter could be used.
This is a quality-control method, to detect shift in mean value of measured quantity away from a target value.
Let $\{y_t\}_{t=1,\dots,T}$ be IID observations arising from a locally stationary process. The cumulative sums are

$$S_t = \max\{0, S_{t-1} + y_t - E_{t-1}[y_t]\}, \quad S_0 = 0$$

An action will be recommended at the first t satisfying $S_t \geq h$ for some threshold h (filter size).
Note $S_t = 0$ whenever $y_t = E_{t-1}[y_t] - S_{t-1}$, The zero floor means some downward deviations will be skipped.
The filter is set up to identify a sequence of upside divergences from any reset level zero.
The threshold is activated when

$$S_t \geq h \Leftrightarrow \exists \tau \in [1, t] \mid \sum_{i=\tau}^{t} (y_i - E_{i-1}[y_t]) \geq h$$

This concept of run-ups can be extended to include run-downs, giving symmetric CUSUM filter.

$$\begin{aligned} S_t^+ &= \max\{0, S_{t-1}^+ + y_t - E_{t-1}[y_t]\}, \quad S_0^+ = 0 \\ S_t^- &= \min\{0, S_{t-1}^- + y_t - E_{t-1}[y_t]\}, \quad S_0^- = 0 \\ S_t &= \max\{S_t^+, -S_t^-\} \end{aligned}$$

2.1.3 Data Labelling Techniques

Method 2.1.18. *Labelling with Fixed-Time Horizon Method*
Given features matrix X with I rows, $\{X_i\}_{i=1,\dots,I}$ drawn from some bards with index $t = 1, \dots, T$, where $I \leq T$, let an observation X_i be assigned a label $y_i \in \{-1, 0, 1\}$,

$$y_i = \begin{cases} -1 & \text{if } r_{t_{i,0}, t_{i,0}+h} < -\tau \\ 0 & \text{if } \left| r_{t_{i,0}, t_{i,0}+h} \right| \leq \tau \\ 1 & \text{if } r_{t_{i,0}, t_{i,0}+h} > \tau \end{cases}$$

$$r_{t_{i,0}, t_{i,0}+h} = \frac{p_{t_{i,0}+h}}{p_{t_{i,0}}} - 1$$

where τ is a pre-defined constant threshold, $t_{i,0}$ is index of bar immediately after X_i takes place, $t_{i,0} + h$ is index of h-th bar after $t_{i,0}$, and $r_{t_{i,0}, t_{i,0}+h}$ is price return over bar horizon h.

Remark 2.1.19. *Limitations of Fixed-Time Horizon Method*

i. Time bars do not exhibit good statistical properties (as seen earlier)

ii. The same threshold τ is applied regardless of observed volatility.
Compute daily volatility at intraday estimation points, applying span of n days t an exponentially weighted moving standard deviation.

Method 2.1.20. *Labelling with Triple-Barrier Method*
Labels an observation according to first barrier touched out of three barriers.

i. Set two horizontal barriers and one vertical barrier. Horizontal barriers are defined by profit-taking and stop-loss limits, which are a dynamic function of estimated volatility (realised or implied). Third barrier is the number of bars elapsed since the position was taken (expiration limit).

ii. If upper barrier is touched first, label observation as 1. If lower barrier is touched first, label observation as -1. If vertical barrier is touched first, either label by sign of the return or with 0.

Note that the method is path-dependent. To label an observation, need to account for entire path spanning $[t_{i,0}, t_{i,0} + h]$ where h defines the vertical barrier (expiration limit). Let $t_{i,1}$ be the time of first barrier touch with return as $r_{t_{i,0},t_{i,1}}$. The horizontal barriers may not be symmetric.

Remark 2.1.21. *Triple-Barrier Method Configurations*
Denote a barrier configuration by triplet $[pt, sl, t1]$ which are the upper barrier, lower barrier, physical barrier. Set value as 0 if barrier is inactive, and 1 if barrier is active.
The three useful configurations are:

i. $[1, 1, 1]$: to realise profit, but have set a maximum tolerance for losses and a holding period.

ii. $[0, 1, 1]$: to exit after a number of bars, unless stopped-out.

iii. $[1, 1, 0]$: take profit as long as not stopped-out.

The three less realistic configurations are:

i. $[0, 0, 1]$: equivalent to fixed-time horizon method.

ii. $[1, 0, 1]$: position held until a profit is made or maximum holding period is exceeded, without regard for immediate unrealised losses

iii. $[1, 0, 0]$: position is held until a profit is made. Could lock in loose position for years.

The two illogical configurations are:

i. $[0, 1, 0]$: aimless. Hold position until stopped-out.

ii. $[0, 0, 0]$: no barriers. Position locked forever, no label generated.

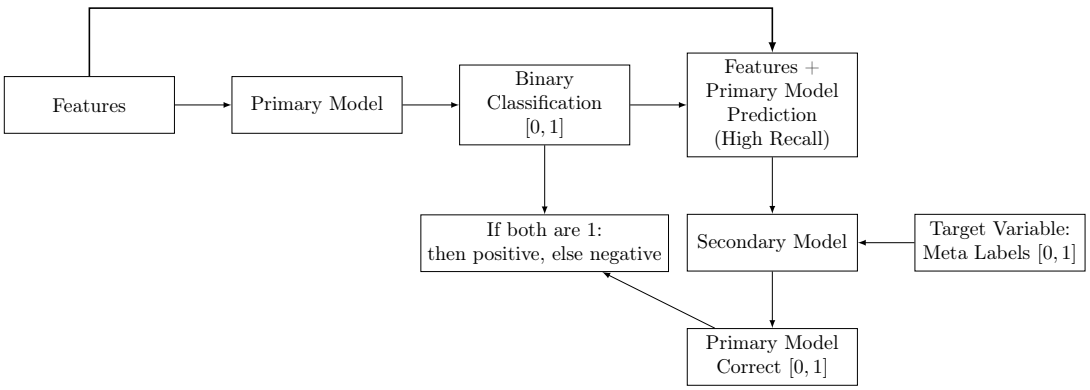


Figure 2.2: Meta-Labelling Process

Method 2.1.22. *Meta-Labelling*
The technique is particularly helpful to achieve higher F1-scores.
First, build a model that achieves high recall, even if precision is not particularly high. Second, correct for low precision by applying meta-labelling to positives predicted by primary model.
Meta-labelling will filter out false positives, where majority of positives have been identified by primary model. The second model's purpose is to determine if the positive from primary model is true or false.

i. Train a primary model (binary classification)

ii. A threshold level is determined at which the primary model has a high recall, ROC curves could be used to help determine a good level.

iii. Typical features of second model are as follows:

   i. Primary model features concatenated with predictions from first model.
   ii. Market state
   iii. Features indicative of false positives
   iv. Distribution related
   v. Recent model performance

   Meta Labels are used as target variable in second model. Fit the second model

iv. Prediction from the secondary model is combined with the prediction from the primary model and only where both are true, is your final prediction true.

Remark 2.1.23. *Limitations of Meta-Labelling*

i. If model has overfit the data, meta-labelling will not add much value

ii. If every trade is not treated as an independent observation, the meta-model is forced to determine day-to-day exposures, which is the wrong way to apply the technique

iii. Technique trades recall for precision. Require a large number of trades to train on, while being happy with reduction in trade frequency

2.1.4 Data Sample Weights

Note that most of ML literature is based on IID assumption, and ML applications usually fail in finance as these assumptions are unrealistic in the case of financial time series.

Remark 2.1.24. *Overlapping Outcomes*
Let label y_i be assigned to an observed feature X_i, where $y_i = f([t_{i,0}, t_{i,1}])$ is a function over the interval. When $t_{i,1} > t_{j,0}$ and $i < j$, then y_j will depend on common return $r_{t_{j,0},\min\{t_{i,1},t_{j,1}\}}$ (over interval $[t_{j,0}, \min\{t_{i,1}, t_{j,1}\}]$). The series of labels $\{y_i\}_{i-1,\ldots,J}$ are not IID whenever there is overlap between any two consecutive outcomes, i.e., $\exists i \mid t_{i,1} > t_{i+1,0}$. If this is resolved by restricting bet horizon to $t_{i,1} \leq t_{i+1,0}$, there is no overlap, but this will lead to coarse models where features sampling frequency is limited by horizon used to determine outcome. To investigate outcomes that lasted a different duration, samples have to be resampled with different frequency. In addition, if path-dependent labelling technique is to be applied, the sampling frequency will be subordinated to first barrier's touch. Hence, to use $t_{i,1} > t_{i+1,0}$, leading to overlapping outcomes.

Method 2.1.25. *Estimating Uniqueness of Label*
Let two labels y_i and y_j be concurrent at time t, both a function of at least one common return $r_{t-1,t} = \frac{p_t}{p_{t-1}} = 1$. To compute the number of labels that are a function of given return $r_{t-1,t}$:

i. For each $t = 1, \ldots, T$, form a binary array $\{1_{t,i}\}_{i=1,\ldots,I}$ where $1_{t,i} \in \{0, 1\}$.
Variable $1_{t,i} = 1$ if and only if $[t_{i,0}, t_{i,1}]$ overlaps with $[t-1, t]$ and $1_{t,i} = 0$ otherwise.

ii. Compute the number of labels concurrent at t, $c_t = \sum_{i=1}^{I} 1_{t,i}$

Method 2.1.26. *Average Uniqueness of Label*
To estimate label's uniqueness (non-overlap) across its lifespan.

i. Uniqueness of label i at time t is $u_{t,i} = 1_{t,i} c_t^{-1}$.

ii. Average uniqueness of label i is average $u_{t,i}$ over label's lifespan, $\overline{u}_i = (\sum_{t=1}^{T} u_{t,i})(\sum_{t=1}^{T} 1_{t,i})^{-1}$.

Note that $\{\overline{u}_i\}_{i=1,\ldots,I}$ are not used for forecasting the label, hence there is no information leakage.

Remark 2.1.27. *IID and Oversampling*
Probability of not selecting item i after I draws with replacement on set of I items is $(1 - I^{-1})^I$. As $I \to \infty$, note that $(1 - I^{-1})^I \to e^{-1}$. Number of unique observations drawn to be expected is $(1 - e^{-1}) \approx \frac{2}{3}$.
If maximum number of overlapping outcomes is $K \leq I$, probability of not selecting a particular item i after I draws with replacement on set of I items is $(1-K^{-1})^I$. As sample size increase, probability can be approximated as $(1 - I^{-1})^{I\frac{K}{I}} \approx e^{-\frac{K}{I}}$. Implication is that incorrectly assuming IID draws lead to oversampling.

Method 2.1.28. *Sampling with Bootstrap, Redundancy*
Sampling with bootstrapping on observations where $I^{-1}\sum_{i=1}^{I}\overline{u}_i \ll 1$, in-bag observations will increasingly be redundant to each other, and very similar to out-of-bag observations. Two solutions may be:

i. Drop overlapping outcomes before performing bootstrap.
As overlaps are not perfect, dropping an observation due to overlap will lead to extreme loss in information.

ii. Utilise the average uniqueness $I^{-1}\sum_{i=1}^{I}\overline{u}_i$ to reduce undue influence of outcomes that contain redundant information. Ensure in-bag observations are not sampled at frequency much higher than uniqueness.

Method 2.1.29. *Sequential Bootstrap*
Draws made according to changing probability that controls for redundancy.

i. Observation X_i is drawn from uniform distribution, $i \sim U[1, I]$.
Probability of drawing any value i is $\delta_i^{(1)} = I^{-1}$.

ii. Second draw, to reduce probability of drawing observation X_j with highly overlapping outcome.
Let φ be sequence of draws (may include repetitions), where $\{\varphi^{(1)}\} = \{i\}$.
Uniqueness of j at time t is $u_{t,j}^{(2)} = 1_{t,j}(1+\sum_{k\in\varphi^{(1)}} 1_{t,k})^{-1}$, which is the uniqueness from adding alternative j's to existing sequence of draws $\varphi^{(1)}$.
Average uniqueness of j is average $u_{t,j}^{(2)}$ over j's lifespan, $\overline{u}_j^{(2)} = (\sum_{t=1}^{T} u_{t,j})(\sum_{t=1}^{T} 1_{t,j})^{-1}$.
A second draw can be made based on updated probabilities $\{\delta_j^{(2)}\}_{j=1,\ldots,I}$:

$$\delta_j^{(2)} = \overline{u}_j^{(2)}\left(\sum_{k=1}^{I}\overline{u}_j^{(2)}\right)^{-1}$$

where $\sum_{j=1}^{I}\delta_j^{(2)} = 1$. Do a second draw, update $\varphi^{(2)}$, and re-evaluate $\{\delta_j^{(3)}\}_{j=1,\ldots,I}$.

iii. Process is repeated until I draws have taken place.

Process draws samples much close to IID, verified by increase in $I^{-1}\sum_{i=1}^{I}\overline{u}_i$.

Method 2.1.30. *Weighting Observations by Uniqueness and Absolute Return*
Let labels be a function for return sign ($\{-1, 1\}$ for standard label, $\{0, 1\}$ for meta-label). The sample weights can be defined in terms of sum of attributed returns over event's life-span, $[t_{i,0}, t_{i,1}]$,

$$\tilde{w}_i = \left|\sum_{t=t_{i,0}}^{t_{i,1}}\frac{r_{t-1,t}}{c_t}\right|, \quad w_t = \tilde{w}_i\left(\sum_{j=1}^{I}\tilde{w}_j\right)^{-1}$$

where $\sum_{i=1}^{I} w_i = I$. The method weigh an observation as a function of absolute log returns that can be attributed uniquely to it. Lower returns should be assigned higher weights.

Method 2.1.31. *Time Decay Weighting*
To let sample weights decay as new observations arrive.
Let $d[x] \geq 0\ \forall x \in [0, \sum_{i=0}^{I}\overline{u}_i]$ be time-decay factors multiplying sample weights from earlier.
The final weight has no decay, $d[\sum_{i=1}^{I}\overline{u}_i] = 1$, and all other weights will adjust relative to that.
Let $c \in (-1, 1]$ be user-defined parameters that determines decay function as follows:

i. If $c \in [0, 1]$, then $d[1] = c$ with linear decay

ii. If $c \in (-1, 0)$, then $d[-c\sum_{i=1}^{I}\overline{u}_i] = 0$, with linear decay between $[-c\sum_{i=1}^{I}\overline{u}_i, \sum_{i=1}^{I}\overline{u}_i]$, and $d[x]\ \forall x \leq -c\sum_{i=1}^{I}\overline{u}_i$.

If given linear piecewise function $d = \max\{0, a + bx\}$, requirements are met by following boundary conditions:

i. $d = a + b\sum_{i=1}^{I}\overline{u}_i = 1 \Rightarrow a = 1 - b\sum_{i=1}^{I}\overline{u}_i$

ii. Contingent on c:

1. $d = a + b0 = c \Rightarrow b = (1 - c)(\sum_{i=1}^{I}\overline{u}_i)^{-1} \quad \forall c \in [0, 1]$
2. $d = a - bc\sum_{i=1}^{I}\overline{u}_i = 0 \Rightarrow b = [(c + 1)\sum_{i=1}^{I}\overline{u}_i]^{-1}\ \forall c \in (-1, 0)$

In the implementation, decay takes place according to cumulative uniqueness. Note that

i. $c = 1$ means there is no time decay

ii. $0 < c < 1$ means weights decay linearly over time, but every observation still receives strictly positive weight, regardless of age

iii. $c = 0$ means weights converge linearly to zero over time

iv. $c < 0$ means oldest portion cT of observations receive zero weight (erased from memory)

Method 2.1.32. *Class Weighting*
Weights for underrepresented labels. Critical in classification problems where the most important classes have rare occurrences. To assign higher weights to samples associated with those rare labels.

2.1.5 Fractionally Differentiated Features

Standard stationarity transformations (i.e. integer differentiation) reduce signal by removing memory. Although stationarity is necessary for inferential purposes, it is rarely the case that we want all memory to be erased. Fractionally differentiated processes exhibit long-term persistence and anti-persistence, hence enhancing the forecasting power compared to standard ARIMA approach.

Definition 2.1.33. *BackShift Operator*
Let B be the backshift operator applied to a matrix of real-valued features $\{X_t\}$, where $B^k X_t = X_{t-k}$ for any integer $k \geq 0$. By binomial expansion, we then have

$$(1-B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k = \sum_{k=0}^{\infty} \prod_{i=0}^{k-1} (d-i) \frac{(-B)^k}{k!} = \sum_{k=0}^{\infty} (-B)^k \prod_{i=0}^{k-1} \frac{d-i}{k-i}$$
$$= 1 - dB + \frac{d(d-1)}{2!} B^2 - \frac{d(d-1)(d-2)}{3!} B^3 + \cdots$$

Remark 2.1.34. *Properties of Fractionally Differentiated Features*
Let d be a real (non-integer) positive number. The arithmetic series consists of dot product

$$\tilde{X}_t = \sum_{k=0}^{\infty} \omega_k X_{t-k}$$
$$\omega = \left\{ 1, -d, \frac{d(d-1)}{2!}, -\frac{d(d-1)(d-2)}{3!}, \ldots, (-1)^k \prod_{i=0}^{k-1} \frac{d-1}{k!}, \ldots \right\}$$
$$X = \{X_t, X_{t-1}, \ldots, X_{t-k}, \ldots\}$$

where ω are the weights, X are the values. Properties of these features are:

i. Long memory: if d is a positive integer number, then

$$\prod_{i=0}^{k-1} \frac{d-i}{k!} = 0 \quad \forall > d$$

and memory beyond that point is cancelled.

ii. Iterative weight generation: given sequence of weights ω, for $k = 0, \ldots, \infty$, the weights are

$$\omega_k = -\omega_{k-1} \frac{d-k+1}{k}, \quad \omega_0 = 1$$

iii. Convergence: For $k > d$, if $\omega_{k-1} \neq 0$, then

$$\left| \frac{\omega_k}{\omega_{k-1}} \right| = \left| \frac{d-k+1}{k} \right| < 1$$

and $\omega_k = 0$ otherwise. Hence weights converge asymptotically to zero.
For positive d and $k < d+1$, then $\frac{d-k+1}{k} \geq 0$, which makes initial weights alternate in sign.
For non-integer d, once $k \geq d+1$, ω_k will be negative if $\text{int}[d]$ is even, and positive otherwise.
In summary, $\lim_{k\to\infty} = 0^-$ when $\text{int}[d]$ is even, and $\lim_{k\to\infty} = 0^+$ when $\text{int}[d]$ is odd.
In special case $d \in (0,1)$, that $-1 < \omega_k < 0 \; \forall k > 0$. Alternate weight signs makes $\{\tilde{X}_t\}_{t=1,\ldots,T}$ stationary, as memory wanes or is offset over the long run.

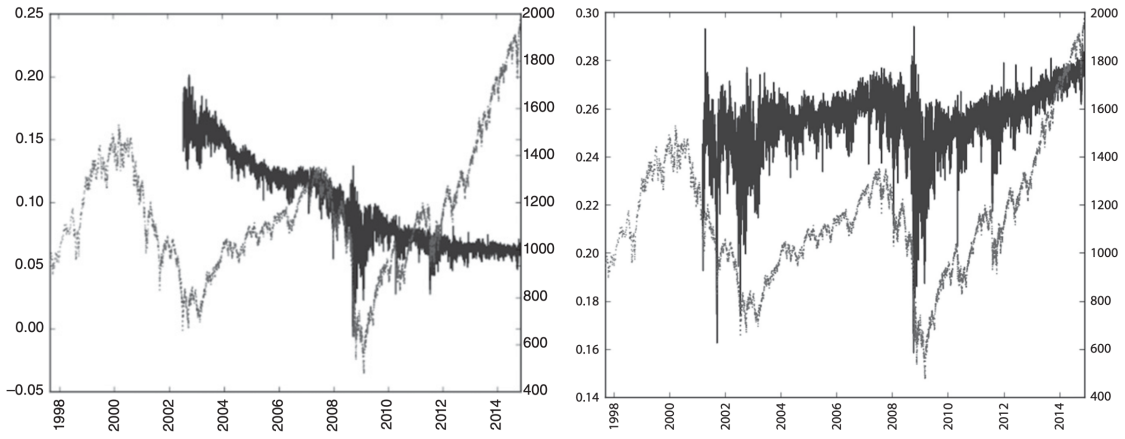


Figure 2.3: Fractional differentiation controlling for weight loss with expanding and fixed-width window

Method 2.1.35. *Expanding Window*
Given time series T with real observations $\{X_t\}_{t=1,\dots,T}$, for each l, the relative weight loss is defined as

$$\lambda_l = \sum_{j=T-l}^{T} |\omega_j| \Big/ \sum_{i=0}^{T-1} |\omega_i|$$

Given tolerance level $\tau \in [0,1]$, determine value l^* such that $\lambda_{l^*} \leq \tau$ and $\lambda_{l^*+1} > \tau$. This value l^* corresponds to the first results $\{\tilde{X}_t\}_{t=1,\dots,l^*}$, where weight-loss is beyond acceptable threshold $\lambda_t > \tau$.
From Remark 2.1.34, it is clear λ_{l^*} depends on convergence speed of $\{\omega_k\}$, which in turn depends on $d \in [0,1]$. For $d = 1, \omega_k = 0\ \forall k > 1$, and $\lambda_l = 0\ \forall l > 1$, hence it suffices to drop $\tilde{X}_1$.
As $d \to 0^+$, l^* increases, and larger portion of initial $\{\tilde{X}_t\}_{t=1,\dots,l^*}$ needs to be dropped to keep the weight loss $\lambda_{l^*} < \tau$. Note that there will be negative drift caused by negative weights added to initial observations as window is expanded. By controlling for weight loss, negative drift is still substantial as $\{\tilde{X}_t\}_{t=l^*+1,\dots,T}$ are computed on an expanding window.

Method 2.1.36. *Fixed-Width Window*
Drop weights after their modulus $|\omega_k|$ decreases below a given threshold τ. This is equivalent to finding the first l^* such that $|\omega_{l^*}| \geq \tau$ and $|\omega_{l^*+1}| \leq \tau$, setting a new variable $\tilde{\omega}_k$:

$$\tilde{\omega_k} = \begin{cases} \omega_k & \text{if } k \leq l^* \\ 0 & \text{if } k > l^* \end{cases}, \qquad \tilde{X}_t = \sum_{k=0}^{l^*} \tilde{\omega}_k X_{t-k} \quad \text{for } t = T - l^* + 1, \dots, T$$

Note that the same vector of weights is used across all estimates of $\{\tilde{X}_t\}_{t=l^*,\dots,T}$, hence avoiding negative drift caused by expanding window's added weights.
Distribution has skewness and excess kurtosis from memory, but it is stationary.

Chapter 3

Rates: yield curves, term premia and derivatives

3.1 Time Value of Money

Definition 3.1.1. *Time Value of Money Formulas*

Concept	Formula	Notes
Expected Annual Rate	$\text{EAR} = (1 + r_p)^m - 1$	r_p periodic rate, m periods/year
	$\text{EAR} = e^r - 1$	Continuous
Future Value (Continuous)	$FV_N = PVe^{r_s N}$	r_s stated rate, N years
Present Value (Frequent)	$PV = FV_N(1 + \frac{r}{m})^{-mN}$	m periods/year
Ordinary Annuity	$FV_N = A\left[\frac{(1+r)^N - 1}{r}\right]$	First CF one period from now
Annuity Due	$FV_N = A\left[\frac{(1+r)^N - 1}{r}\right](1 + r)$	First CF today
Perpetuity	$PV = \frac{A}{r}$	Infinite cash flows

Definition 3.1.2. *Return Metrics*

Metric	Formula / Description	Notes
Internal Rate of Return	$\text{NPV} = 0$	PV inflows = PV outflows
Money-Weighted Return	IRR on portfolio	Timing and size of CFs
Time-Weighted Return	$r = \sqrt[1/n]{\prod_{i=1}^{n}(1 + HPR_i)} - 1$	Geometric mean
Annualised Return	$r = (1 + HPR)^{365/\text{Days}} - 1$	HPR scaled to annual
Continuously Compounded	$r_{\text{CC}} = \ln(1 + HPR)$	Log return
Gross Return	Before fees	Pre-fees
Net Return	After fees	Post-fees
Pre-Tax Nominal	Before taxes	Pre-tax
After-Tax Nominal	After taxes	Post-tax
Real Return	$(1 + r_{\text{Real}}) = \frac{(1+r_{\text{Nom}})(1+r_{\text{Risk}})}{(1+\pi)}$	Inflation-adjusted
Leveraged Return	$r_L = \frac{r(V_0+V_B)-r_B V_B}{V_0}$	V_B borrowed, r_B borrow cost

Chapter 4

Option pricing, volatility models and hedging

Based on the classic by John C. Hull [s7-hulloptions].

4.1 Fundamentals of the Market

The derivatives market is much larger than the stock market in terms of underlying assets. Derivatives may be used for hedging, speculation, or arbitrage; and also transfer a wide range of risks from one entity to another.

Definition 4.1.1. A *derivative* involves two parties agreeing to a future transaction, with value depending on the values of other underlying variables.

A derivatives exchange is a market where individuals and companies trade standardised contracts as defined by the exchange. Once two traders have agreed to trade a product offered by the exchange, it is handled by the exchange clearing house, which takes care of the credit risk by requiring each trade to deposit margin.

If the trade is taken over-the-counter (OTC), participants may present it to a central counterparty (CCP) or clear the trade bilaterally. With the 2008 financial crisis, OTC market is forced to become more like the exchange-traded market, with changes as follows:

i. Standardised OTC derivatives between two financial institutions in US must, whenever possible, be traded on a swap execution facility (SEF), where market participants can post bid and ask quotes, and can trade by accepting the quotes of other market participants.

ii. Require that a CCP be used for most standardised derivatives transactions between financial institutions.

iii. All trades must be reported to a central repository.

4.1.1 Forward, Futures, and Options

Definition 4.1.2. A *spot contract* is an agreement to buy or sell an asset almost immediately. A *forward contract* is an agreement or buy or sell an asset at a certain future time for a certain price.

Let S_T be the spot price of asset at maturity, K is delivery price.
The payoff from a long position in a forward contract is $S_T - K$.
The payoff from a short position in a forward contract is $K - S_T$.

Futures contract are traded on an exchange, unlike forwards which are traded OTC. Majority of futures contract do not lead to delivery, as positions are closed prior to delivery period by entering an opposite trade to the original one.

Party in short position may file notice of intention to deliver with the exchange when they are ready to deliver. If the asset is a commodity, the grade of commodity are specified. The contract size specifies the amount of asset that has to be delivered. The place for delivery must also be specified, as commodities may involve significant transportation costs. The delivery month of the commodity may also be specified, and are chosen by the exchange to meet the needs of market participants. Trading typically ceases a few days before the last day on which delivery can be made. Daily price movement limits are also specified by exchange to

prevent speculative excess causing large price movements; in this case trading ceases for the day.

As the delivery price for a futures contract is approached, the futures price converges to the spot price of the underlying asset.

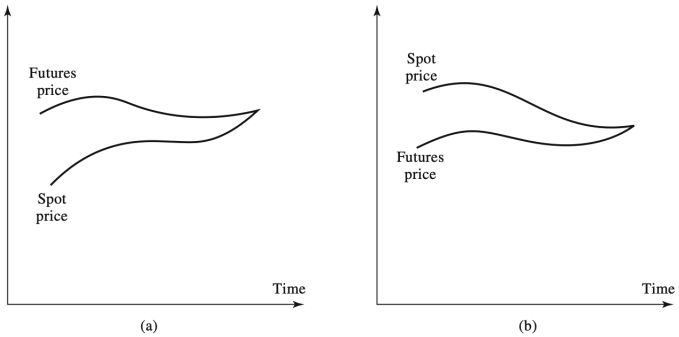


Figure 4.1: Convergence of futures price to spot price.

Suppose futures price is above spot price during delivery period. Traders have clear arbitrage opportunity: short futures, long asset, make delivery. The futures price will then fall.
If futures price is below spot price during delivery period, traders will long futures, wait for delivery, and the futures price will then rise.

Options are traded both on exchanges and OTC.

Definition 4.1.3. A *call option* (*put option*) gives the holder the right to sell (buy) the underlying asset by a certain date for a certain price.

Definition 4.1.4. An *American option* can be exercised at any time up to expiration date. An *European option* can only be exercised on expiration date itself.

4.1.2 Clearing House

Margin accounts are used by exchanges to organise trading so that contract defaults are avoided.
Trader has to keep funds in a margin account; the amount to be deposited at the time the contract is entered into is the *initial margin*. At each of trading day, margin account is adjusted to reflect trader's P&L (*daily settlement*, *marking to market*). Daily settlement leads to funds flowing daily between traders with long positions and traders with short positions; this daily flow of funds to reflect P&L is the *variation margin*. Trading via brokers requires a *maintenance margin*, which is lower than initial margin; if balance in margin account falls below maintenance margin, trader receives *margin call* and is required to top up to initial margin level within short period of time, or else broker closes out the position. Trader is also entitled to withdraw any balance in margin account that is in excess of the initial margin. Brokers pay interest on balance in margin account.
A forward contract is settled at end of life, while futures contract is settled daily. Minimum levels for initial and maintenance margins are set by exchange clearing house. Minimum margin levels are determined by variability of the price of underlying asset, revised when necessary.

The clearing house acts as intermediary in futures transactions, and keep track of all daily transactions for calculating net positions of each of its members. Members must provide initial margin reflecting the total number of contracts cleared. The maintenance margin is set equal to initial margin. In determining margin requirements, the number of contracts outstanding is calculated on a net basis rather than a gross basis. Members are required to contribute to a guaranty fund, which is used in the event that a member defaults, and the member's margin is insufficient to cover losses.

4.1.3 OTC Markets

OTC markets use central counterparties (CCPs), which perform the same role as exchange clearing houses. Members of CCP provide both initial margin and daily variation margin, and contribute to a guaranty fund. If an OTC derivative transaction has been agreed upon between parties A and B, and CCP accepts the transaction, they become the counterparty to both A and B. CCP hence takes on credit risk of both A and B. Transactions are valued daily, and there are daily variation margin payments between members.

For bilaterally cleared OTC, two companies enter a master agreement covering all their trades (ISDA). The agreement includes a credit support annex (CSA), requiring both parties to provide collateral. Collateral agreements in CSAs usually require transactions to be valued daily. Since 2016, regulations require both initial margin and variation margin between financial institutions.

Definition 4.1.5. *Credit spread* is the difference between the interest rate and risk free rate.

(Treasury Rates) The rates on Treasury bills and Treasury bonds. The Treasury rates of developed countries are regarded as risk-free rates, as it is assumed that there is no chance the government will default.

(Overnight Rates) Borrowing and lending overnight by financial institutions to match asset and liabilities requirements for reserves; the overnight rate is the *federal funds rate*. The weighted average of rates in brokered transactions is the *effective federal funds rate*. The Federal Reserve may intervene with its own transactions to raise or lower the rates.

(Repo Rates) Secured borrowing rates; the difference between the price at which securities are sold and then repurchased. If structured carefully, involves very little credit risk. Most common type of repo is an *overnight repo*. In longer term arrangements, *term repos* are used.

The terminal value of an investment A invested at interest rate of R per annum, compounded m times per annum, is $A(1+\frac{R}{m})^{mn}$. If $m=1$, the rate is the *equivalent annual interest rate*. If continuous compounded is used, then the terminal value at the end of a year is Ae^R.

The *n-year spot-rate* is the interest rate earned on a zero-coupon bond. The *Bond Price* is the present value of all cash flows that will be received by owner of the bond, with different spot rate for each cash flow. The *Bond Yield* is the discount rate that, when applied to all cash flows, gives a bond price equal to its market price. The *Par Yield* for a certain bond maturity is the coupon rate that causes the bond price to equal its par value.

Definition 4.1.6. *Forward Rates* are rates implied by current spot rates for periods of time in the future.

Given R_1, R_2 the spot rates for maturities T_1, T_2 respectively, and R_F the forward rate between T_1 and T_2, then

$$R_F = \frac{R_2T_2 - R_1T_1}{T_2 - T_1} = R_2 + (R_2 - R_1)\frac{T_1}{T_2 - T_1}$$

Given the spot rate R for maturity T, the *instantaneous forward rate* for maturity of T is then

$$R_F = R + T\frac{\partial R}{\partial T}$$

If $P(0,T) = e^{-RT}$ is the price of zero-coupon bond maturity at time T, the equation is then

$$R_F = -\frac{\partial}{\partial T}\ln P(0,T)$$

Definition 4.1.7. *Forward Rate Agreement (FRA)* is an agreement to exchange a predetermined rate for a reference rate that will be observed in the market at a future time.

Let R_K be the fixed rate, R_F be the current forward rate for the reference rate, τ be the period of time to which the rates apply, L be the principal in the contract.
For the party that receives the fix rate, the FRA has a present value of

$$\tau(R_K - R_F)L$$

For the party that pays the fix rate, the FRA has a present value of

$$\tau(R_F - R_K)L$$

Definition 4.1.8. The *Duration* of a bond is a measure of how long the holder of the bond has to wait before receiving the present value of the cash payments.

Let c_i be cash flow at time t_i $(1 \le i \le n)$. Bond price B, yield y (continuously compounded) are related by

$$B = \sum_{i=1}^{n} c_i e^{-yt_i}$$

The Duration D of the bond is then

$$D = \sum_{i=1}^{n} t_i \left[\frac{c_i e^{-yt_i}}{B} \right]$$

where the term in square brackets is ratio of present value of cash flow at t_i to bond price. Duration is hence a time-weighted average of the times when payments are made.
The relationship between duration and yield is as follows:

$$\Delta B = \frac{dB}{dy} \Delta y$$
$$\frac{\Delta B}{B} = -D \Delta y$$

If y is expressed with compounding frequency of m times a year, then the relationship becomes

$$\Delta B = -\frac{BD\Delta y}{1 + y/m}$$

Hence, the *modified duration* is

$$D^* = \frac{D}{1 + y/m}$$

The duration relationship is then

$$\Delta B = -BD^* \Delta y$$

When duration is used for bond portfolios, it is assumed that the yields of all bonds will change by approximately the same amount, i.e., a parallel shift in the spot yield curve. The portfolio may still be exposed to shifts that are either large or non-parallel.
Convexity may be used to improve the relationship in the equation. Convexity is defined as

$$C = \frac{1}{B} \frac{d^2 B}{dy^2}$$

By Taylor series, we then have the relationship

$$\frac{\Delta B}{B} = -D\Delta y + \frac{1}{2} C (\Delta y)^2$$

To determine the underlying reasons for the shape of the spot curve, theories on term structure of interest rates are referred to.

(Expectations Theory) A forward interest rate corresponding to a certain future period is equal to the expected future zero interest rate for that period.

(Market Segmentation Theory) There exists no relationship between short, medium, and long-term interest rates, as investors do not readily switch from one maturity to another.

(Liquidity Preference Theory) Investors prefer to preserve their liquidity and invest funds for short periods of time. Borrowers prefer to borrow at fixed rates for long periods of time. Hence forward rates are greater than expected future spot rates.

4.2 Forwards and Futures

Based on the book by John Hull [s7-hulloptions].

4.2.1 Pricing

To examine how forward prices and futures prices are related to spot price, we assume the following are true for some market participants:

i. No transaction costs
ii. Same tax rate on all net trading profits
iii. Money borrowed and lend are at the same risk-free rate
iv. Arbitrage opportunities are taken advantaged of as they occur

Let T be time until delivery date (in years), S_0 be price of underlying asset today, F_0 be price of forward or futures today, r be zero-coupon risk-free rate per annum in continuous compounding (maturing in T years).

Consider a forward contract on underlying asset with spot price S_0 that provides no income. Then

$$F_0 = S_0 e^{rT}$$

If $F_0 > S_0 e^{rT}$, long asset and short forward. If $F_0 < S_0 e^{rT}$, short asset and long forward.
If short sales are not possible, and arbitrage opportunities exist, then if $F_0 > S_0 e^{rT}$, investor may:

1. Borrow S_0 in cash at an interest rate r for T years
2. Buy 1 unit of asset
3. Enter forward contract to sell 1 unit of asset

At time T, asset is sold for F_0, investor to repay $S_0 e^{rT}$ loan, making profit $F_0 - S_0 e^{rT}$.
If $F_0 < S_0 e^{rT}$, investor may:

1. Sell asset for S_0
2. Invest proceeds at interest rate r for time T
3. Enter into forward contract to buy 1 unit of asset

At time T, cash has grown to $S_0 e^{rT}$. Investor repurchase asset for F_0, makes profit of $S_0 e^{rT} - F_0$.

If the underlying asset provide income with present value of I during life of forward, then

$$F_0 = (S_0 - I)e^{rT}$$

If $F_0 > (S_0 - I)e^{rT}$, investor may long asset and short forward. If $F_0 < (S_0 - I)e^{rT}$, investor may short asset and long forward. If short sales are not possible, investors owning the asset will sell the asset and long forward.

If the underlying asset provides a known yield rather than income, with q as the average yield, then the following strategy must generate zero profit to prevent arbitrage:

1. Borrow S_0 to buy one unit of asset at time 0
2. Enter into forward to sell e^{qT} units of asset at time T for F_0
3. Close the forward by selling e^{qT} units of the asset at time T

Hence we have

$$S_0 e^{rT} = e^{qT} F_0$$

or

$$F_0 = S_0 e^{(r-q)T}$$

When a forward contract is first entered, the value is close to zero. Let K be delivery price negotiated some time ago, with T years delivery date, r is T-year risk-free interest rate, F_0 is forward price if contract is negotiated today. Let f be value of forward contract today. Then

$$f = (F_0 - K)e^{-rT}$$

In the case of stock indices, if $F_0 > S_0 e^{(r-q)T}$, profit can be made by buying stocks underlying the index at spot price and shorting the index futures contract. If $F_0 < S_0 e^{(r-q)T}$, short the stocks and long futures. This is known as *index arbitrage*.

In the case of currencies, let r_f be foreign risk-free rate in a foreign-denominated bond, S_0 be spot price of local currency in foreign currency, F_0 be forward or future price of local currency in foreign currency. Then the *interest rate parity* relation persists in the form

$$F_0 = S_0 e^{(r-r_f)T}$$

For commodities, let U be present value of storage costs net of income during life of forward. Then

$$F_0 = (S_0 + U)e^{rT}$$

If storage costs are proportional to price of commodity, they can be treated as negative yield. Let u be storage costs per annum as proportion of spot price net of any yield earned on asset. Then

$$F_0 = S_0 e^{(r+u)T}$$

If $F_0 > (S_0 + U)e^{rT}$, then investor may take advantage of arbitrage:

1. Borrow $S_0 + U$ at risk-free rate, purchase one unit of commodity and pay storage costs
2. Short futures on one unit of commodity

If $F_0 < (S_0 + U)e^{rT}$, then investor may take advantage of arbitrage:

1. Sell commodity, save storage costs, invest proceeds at risk-free interest rate
2. Long futures contract

Benefits from holding physical assets are *convenience yields*, such as by crude oil manufacturers. Let y be the convenience yield, then

$$F_0 e^{yT} = (S_0 + U)e^{rT}$$

If storage costs per unit are a constant proportion of spot price, then y is defined such that

$$F_0 e^{yT} = S_0 e^{(r+u)T}$$

Convenience yield reflects market expectation on future availability of the commodity. The greater the possibility that shortages will occur, the higher the commodity yield.

The *cost of carry* measures storage cost plus interest paid to finance the asset less income earned:

i. Non-dividend paying stock: r, as no storage and income is earned
ii. Stock index: $r - q$, as income is earned at rate q on asset
iii. Currency: $r - r_f$
iv. Commodity: $r - q + u$, where it provides income at rate q and requires storage costs at rate u

4.2.2 Hedging with Futures

The fundamentals of hedging with futures are *hedge-and-forget* strategies, where no changes is made to adjust the hedge once it has been put in place.

Definition 4.2.1. *(Basic Principles of Futures Hedging)*
The objective is to take a position that neutralises the risk as far as possible.

i. *Short Hedge*: short position on futures.
Used when hedger already owns an asset and will sell the asset at some time in the future; or when asset is not owned right now but will be owned and ready for sale sometime in the future.
ii. *Long Hedge*: long position on futures.
Used when hedger will purchase an asset in the future and wants to lock in the price now.

	Short Hedge	Long Hedge
May 15	Spot: 50 Futures: 49	Spot: 50 Futures: 49
August 15 Scenario 1	Spot: 45 Gain from hedge: 4	Spot: 45 Loss from hedge: 4
August 15 Scenario 2	Spot: 55 Loss from hedge: 6	Spot: 55 Gain from hedge: 6

In practice, hedging is not perfect due to factors as follows:

1. Asset being hedged is not exactly the same as the asset underlying the futures contract.
2. Uncertainty as to exact date in which the asset will be bought or sold.
3. Hedge may require the futures contract to be closed out before its delivery month.

These lead to *basis risk*.

Definition 4.2.2. The *basis* in a hedging situation is defined as

$$\text{Basis} = \text{Spot Price} - \text{Futures Price}$$

An increase/decrease in basis is a strengthening/weakening of the basis.

Definition 4.2.3. Let S_i be spot price at time t_i, F_i be futures price at time t_i, b_i be basis price at time t_i. Assume hedge is placed at time t_1, closed at time t_2. Price realised for asset is S_2, profit from futures position is $F_1 - F_2$. Effective price obtained for asset hedging is therefore $S_2 + F_1 - F_2 = F_1 + b_2$.
If b_2 is known, perfect hedge will result. The *basis risk* is the hedging risk from uncertainty associated with b_2.

Definition 4.2.4. *Cross Hedging* occurs when the asset underlying the futures contract is not the same as the asset whose price is being hedged.

Cross hedging is often used when futures of the original asset being hedged are not actively traded on the market, and the hedger seeks an alternative asset to hedge the original asset.

Definition 4.2.5. *Hedge Ratio* is the ratio of size of position taken in futures contract to the size of exposure.

Assuming no daily settlement of futures contracts, hedger seeks a hedge ratio that minimises variance of hedged position value. Let ΔS be change in spot price, ΔF change in futures price. Assuming linear relationship,

$$\Delta S = a + b\Delta F + \epsilon$$

where a, b are constants, ϵ is an error term. Suppose hedge ratio is h. Change in value of position per unit of exposure to S is:

$$\Delta S - h\Delta F = a + (b - h)\Delta F + \epsilon$$

Standard deviation is minimised by setting $h = b$. Let minimum variance hedge ratio be h^*. Then

$$h^* = \rho\frac{\sigma_S}{\sigma_F}$$

where σ_S, σ_F is standard deviation of $\Delta S, \Delta F$ respectively, ρ is coefficient of correlation.

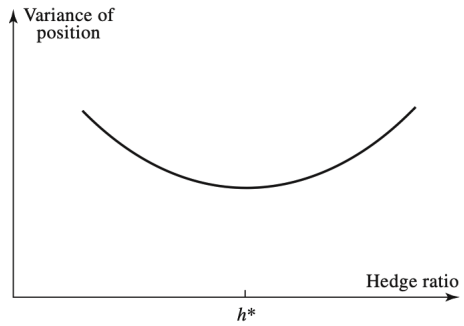


Figure 4.2: Dependence of variance of position on hedge ratio.

Hedge effectiveness is the proportion of variance eliminated by hedging. This is R^2 from regression of ΔS against ΔF, and equals ρ^2. Parameters ρ, σ_S, σ_F are estimated from historical data on ΔS and ΔF.

The optimal number of futures to be used in hedging is

$$N^* = \frac{h^* Q_A}{Q_F}$$

where Q_A is size of potion being hedged (units), Q_F is size of one futures contract (units). The futures contract should be on $h^* Q_A$ units of the asset.

If daily settlement is used, there are a series of one-day hedges, and thus let $\hat{\sigma}_S, \hat{\sigma}_F$ be standard deviation of percentage one-day changes in spot and future price respectively, $\hat{\rho}$ be correlation between percentage one-day changes in spot and future prices. The optimal one day hedge is then

$$h^* = \hat{\rho}\frac{\hat{\sigma}_S S}{\hat{\sigma}_F F}$$

and the optimal number of futures to be used is then

$$N^* = \hat{\rho}\frac{\hat{\sigma}_S S Q_A}{\hat{\sigma}_F F Q_F}$$

If an interest $r\%$ per annum is earned or paid over the remaining life of the hedge, then the optimal number of futures is $N^*/(1 + 0.01r)$; this is *tailing the hedge*.

Stock index futures may be used to hedge a well diversified equity portfolio. Let V_A, V_F be the current value of portfolio and one futures contract respectively.
If portfolio mirrors the index, the optimal hedge ratio is then 1.0, and number of futures contracts to be shorted is then $N^* = \frac{V_A}{V_F}$. If portfolio do not mirror the index, then capital asset pricing model (CAPM) should be used to determine beta (β), and the number of futures contracts to be shorted is then $N^* = \beta\frac{V_A}{V_F}$, assuming maturity of futures contract is close to the maturity of the hedge.

If instead, the hedger wishes to change the beta of portfolio where $\beta > \beta^*$, a short position $(\beta - \beta^*)\frac{V_A}{V_F}$ is required. If $\beta < \beta^*$, then a long position $(\beta^* - \beta)\frac{V_A}{V_F}$ is required.

Stock index hedging is typically used when the portfolio manager is uncertain about performance of market, but is confident that the stocks in the portfolio will outperform the market. The hedger may also be planning to hold a portfolio for a long period of time and requires short-term protection in an uncertain market situation.

If expiration date of hedge is later than delivery dates of all futures contracts that may be used, then the hedger may *stack and roll* by closing out one futures contract and taking the same position in a futures contract with a later delivery date.

4.2.3 Interest Rate Futures

The *day count* defines the way in which interest accrues, and is expressed as X/Y, where X is the way in which number of days between two dates is calculated, Y is total number of days in reference period.

$$\text{interest earned between two dates} = \frac{X}{Y} \times \text{interest earned in reference period}$$

The three common day count conventions used in United States are:

i. Actual/Actual (in period): for Treasury Bond
ii. 30/360: for corporate and municipal bonds, assumes 30 days a month and 360 days a year
iii. Actual/360: for money market instruments

Prices of money market instruments are quoted using discount rate, which is interest earned as a percentage of final face value. Let P be quoted price, Y be cash price, n remaining life of Treasury bill in calendar days:

$$P = \frac{360}{n}(100 - Y)$$

Prices of treasury bonds are quoted in dollars and thirty-seconds of a dollar, i.e., 120-05 or $120\frac{5}{32}$, which is a bond value of $\$120,156.25$. The quoted price is *clean price*, and the cash paid by purchaser is *dirty price*:

$$\text{cash price} = \text{quoted price} + \text{accrued interest since last coupon date}$$

Chapter 5

Multi-asset and systematic macro

Global macro is the strategy of identifying economic regime shifts early and expressing them through the most liquid instruments. The primary objective is surviving and profiting from crises, rather than maximising returns in stable periods [s7-gliner].

5.1 Regime and World State

5.1.1 Survey of Global Macro Landscape

Definition 5.1.1. *Global Macro*
Global macro is a top-down investment strategy that trades economic regimes, not securities.
The core idea is simple but powerful: macro forces dominate asset prices when they shift, and those forces operate across countries, asset classes, and policy domains. Global macro traders

i. Form views on growth, inflation, monetary policy, and geopolitics
ii. Express those views via equities, rates, FX, and commodities
iii. Are unconstrained by geography, asset class, or direction (long/short)

Global macro is often considered the most flexible and opportunistic hedge fund strategy, due to the scope of traded products and the number of markets it covers. Its aim is to preserve capital, using stringent risk management to limit drawdowns.

Remark 5.1.2. *Existence of Global Macro as a Strategy*
Macro exists because economic regimes are unstable, but markets price them slowly and asymmetrically.
Three structural reasons:

i. Policy is discrete, markets are continuous: Central banks and governments move in jumps (rate decisions, QE, capital controls). Markets must reprice entire distributions after the fact.
ii. Capital is sticky: Large allocators (pensions, insurers, sovereign funds) cannot reposition quickly. Macro traders exploit this inertia.
iii. Cross-asset linkages are under-arbitraged: Many investors specialise in a single asset class. Macro traders operate between silos.

Hence macro is a strategy of structural inefficiency, not informational advantage.

Remark 5.1.3. *Types of Global Macro Strategies*
The four broad approaches are:

i. Discretionary Macro
Relies on trader experience, intelligence, and knowledge to take subjective bets on global markets. Requires serious organization and data processing skills. Uses top-down analysis of risks and opportunities across industries, sectors, countries, and the macroeconomic situation.
Discretionary traders can execute *directional trades* (betting on asset direction) or *relative value trades* (pairing assets to capture value differentials). Strong in crises and regime breaks where human judgment outperforms models.
ii. Systematic Macro
Model-driven, employing quantitative factors to produce trading positions that remove human emotion. Prides itself on stringent process, strong back-tests, and ability to operate solely on quantitative analysis.

Over long periods, systematic funds produce more consistent returns than discretionary strategies; however, in periods of high volatility, they tend to underperform (e.g., 2008).
The ability to trade multiple liquid asset classes allows these funds to scale to very large AUM.

iii. High-Frequency Macro
Uses sophisticated technology to trade very short-term (millisecond to hours) dislocations around macro events. Processing speed is paramount. Limited scalability compared to discretionary and systematic.

iv. CTAs (Commodity Trading Advisors)
Primarily futures-based, momentum-driven trend-following strategies. Position sizing methodology originated with the Turtle Traders. Perform well over longer periods but subject to large drawdowns.

It is wise to allocate to both discretionary and systematic macro in a balanced manner. Discretionary is negatively correlated during stress, while systematic provides predictable allocation with back-tested drawdowns.

Remark 5.1.4. *Trade Types in Macro*
Discretionary macro traders express views through two primary trade structures:

i. Directional Trades: Betting on an asset moving in a particular direction. Example: going long copper on a bullish commodities view.

ii. Relative Value Trades: Pairing or grouping assets to capture the relative value differential between them. Example: during European crisis, short Italian 5-year BTPs vs. long German Bobls, betting Italy will see yields rise relative to Germany.

iii. Thematic Trades: Longer-term positions based on structural macro views.
Examples: "fiat money debasement", hence long gold; "Japan debt unsustainability", hence short JGBs.
Thematic trades should be sized smaller than short-term trades because they can take years to play out and may incur negative carry.

Remark 5.1.5. *Return Profile Across Cycles*
Global macro's defining feature is return asymmetry across market regimes. Empirical characteristics:

i. Lower average volatility than equities

ii. Comparable or higher long-run returns

iii. Low or negative correlation to equities during crises

iv. Higher Sharpe ratio than most hedge fund strategies

Macro tends to underperform in bull markets (when staying long works), but outperform during dislocations, recessions, and policy shifts. This asymmetry exists because macro traders can go short, can hold cash, can trade volatility, rates, and FX directly (flexibility that equity-focused strategies lack).

Remark 5.1.6. *Behaviour in Economic Crisis*
In stress regimes:

i. Equity managers are trapped long (mandates prevent shorting or holding cash)

ii. Credit liquidity vanishes (bids disappear, spreads gap wider)

iii. Correlations converge toward 1 (diversification fails)

Macro funds, by contrast, can:

i. Reduce gross exposure rapidly

ii. Short risk assets directly

iii. Express views via rates, FX, or volatility instead of equities

iv. Profit from the dislocation rather than merely survive it

Remark 5.1.7. *Global Macro Correlation Properties*
Global macro's real value is conditional correlation:

i. In normal times: low correlation to equities

ii. In crises: negative correlation to equities

This makes macro an effective portfolio diversifier and tail-risk hedge. Combined with high liquidity (trading the most liquid markets globally), macro is the most popular hedge fund allocation by pension funds.

Remark 5.1.8. *Liquidity Advantage*
One of most liquid hedge fund strategies because it trades the most liquid underlying markets:

i. G10 FX and major EM currencies
ii. Government bond futures (Treasuries, Bunds, JGBs, Gilts)
iii. Equity index futures (S&P 500, Euro Stoxx, Nikkei, etc.)
iv. Major commodity futures (crude oil, gold, copper, etc.)

This liquidity allows macro funds to:

i. Scale to very large AUM without market impact
ii. Enter and exit positions quickly during stress
iii. Offer better redemption terms than illiquid strategies

5.1.2 Role of Central Banks in Global Macro

Remark 5.1.9. *Monetary Policy Regimes*
Central banks define the macro regime more than any other actor. Markets do not trade rates, they trade policy intent under constraints. Regimes fall into following types:

i. Conventional Policy Regime: Policy rate is the main tool. Yield curve responds normally. Transmission is through rates to credit, to growth, then to inflation.
ii. Unconventional Policy Regime: Policy rate loses effectiveness (zero lower bound). Balance sheet becomes the primary tool. Transmission shifts to asset prices, expectations, and FX.
iii. Crisis/Emergency Regime: Stability is more important than inflation targeting. Rules are suspended. Liquidity provision dominates all other objectives.

Remark 5.1.10. *Objectives of Central Banks*
Central banks typically have mandates that include:

i. Price Stability: Most central banks target inflation around 2%
ii. Full Employment: Explicit in Fed's dual mandate; implicit elsewhere
iii. Financial System Stability: Lender of last resort function; systemic risk monitoring
iv. Currency Stability: Explicit for some (SNB, PBOC); implicit for others

Under stress, these objectives may conflict. Central banks must choose which objective to sacrifice. Understanding these trade-offs is critical for macro trading.

Remark 5.1.11. *Conventional Central Bank Tools*

i. Policy Rate: Set by changing overnight rate at which banks lend and borrow to meet reserve requirements. The primary tool in normal times.
ii. Reserve Ratio: Raising reserve ratio decreases money multiplier and contracts money supply.
iii. Open Market Operations (OMO): Buying and selling government bonds in local currency. Repos are short-term agreements where the central bank agrees to repurchase securities within a specified time.
iv. Currency Intervention:
    1. Sterilised intervention: Trading own currency via FX reserves without changing monetary base
    2. Unsterilised intervention: Allows for changes in monetary base

Remark 5.1.12. *Taylor's Rule*
Maps macroeconomic deviations to policy stance:

$$i_t = r^* + \pi_t + \alpha(\pi_t - \pi^*) + \beta(y_t - y_t^*)$$

where r^* is the neutral real rate (unobservable, typically estimated 2–3%), π_t is current inflation, π^* is the inflation target, $(y_t - y_t^*)$ is the output gap, and α, β are response coefficients (typically 0.5).
The rule states that the nominal policy rate must rise more than one-to-one with inflation (Taylor Principle); otherwise real rates fall, inflation becomes self-reinforcing, and expectations de-anchor.
Central banks treat Taylor's rule as a benchmark, communication tool, and political shield.

Remark 5.1.13. *The Impossible Trinity*
A country cannot simultaneously maintain:

i. Fixed exchange rate

ii. Free capital flows

iii. Independent monetary policy

This constraint explains currency crises (ERM 1992, Asian crisis 1997), forced rate hikes or cuts, FX pegs breaking, and why EM central banks behave "irrationally" under stress. Countries must choose two of three.

Remark 5.1.14. *Central Bank Balance Sheet*

i. Monetary Base: Sum of currency in circulation and reserves held at the central bank. Can only be controlled via open market operations.

ii. Money Supply: Changes in money supply are leading predictors of GDP growth and inflation. The Quantity Theory of Money:
$$M \times V = P \times Y$$
where M is money supply, V is velocity of money (turnover rate), P is price level, Y is real output.

Post-2008, central bank balance sheets expanded dramatically. The Fed releases weekly H.4.1 report showing sources and uses of reserves.

Remark 5.1.15. *Types of Money Supply*

i. $M0$: Physical currency in circulation (notes and coins)

ii. $M1$: $M0$ + demand deposits, checking accounts, traveler's checks

iii. $M2$: $M1$ + savings deposits, money market accounts, small time deposits, retail money market funds

iv. $M3$: $M2$ + large time deposits, institutional money market funds, repos, eurodollars

Central banks primarily control $M0$; broader measures depend on banking system credit creation.

Remark 5.1.16. *Zero Lower Bound and the Liquidity Trap*
When central bank rate is at or near zero:

i. Rate cuts lose marginal effectiveness

ii. Expectations dominate outcomes

iii. Forward guidance becomes critical

iv. Fiscal policy starts to matter more

Yield curves flatten naturally; carry trades distort; volatility is suppressed artificially.
Some central banks have gone to negative rates, testing the effective lower bound.

Remark 5.1.17. *Quantitative Easing (QE)*
Central banks purchase financial assets (government bonds, MBS, corporate bonds), creating electronic reserves to expand the balance sheet. QE works through multiple channels:

i. Portfolio rebalancing (investors pushed into riskier assets)

ii. Risk premia compression (term premium, credit spreads)

iii. Signalling effects (commitment to low rates)

iv. Exchange rate channel (currency weakening)

Bank lending does not mechanically follow reserve creation. Transmission depends on credit demand and bank capital. Effectiveness diminishes with successive rounds.
Fed research shows QE1 lowered 10-year yields by 100bp; QE2 by only 14bp.

Remark 5.1.18. *Hawks and Doves*
Central bankers have individual biases that shape their policy views:

i. Hawks: Wary of inflation, favour tighter policy, defend currency value

ii. Doves: Favour growth and employment, tolerate higher inflation, support looser policy

When a known hawk makes dovish statements (or vice versa), it signals a significant shift and moves markets more than statements consistent with known bias.
Tracking individual central banker positions is essential for anticipating policy shifts.

Remark 5.1.19. *Central Bank Communication*
Central bank communication is a policy tool in itself:

i. Statements: Released after each meeting; market moves on hawkish/dovish tone changes
ii. Minutes: Released with lag (2 to 3 weeks); reveal vote splits and internal debates
iii. Press Conferences: Chair/President Q&A; tone and body language matter
iv. Projections: Dot plots (Fed), staff forecasts; reveal expected rate path
v. Speeches: Individual members signalling; watch for coordinated messaging

Example: BOE June 2012 minutes revealed 5–4 vote against QE (vs 8–1 in May), correctly predicting QE expansion in July.

Remark 5.1.20. *Major Central Banks*

i. Federal Reserve (Fed): Dual mandate (price stability, full employment). FOMC meets 8 times/year with 12 voting members (7 governors, NY Fed, 4 rotating).
Most important central bank globally due to USD reserve status.
ii. European Central Bank (ECB): Price stability mandate (HICP $< 2\%$). Governing Council of 25 members. Key rates: Main Refinancing Operations (MRO), deposit facility rate. Tools include LTRO, TLTRO, OMT, APP.
iii. Bank of Japan (BOJ): Targets 2% inflation (chronically undershoots). Policy Board of 9 members. Pioneer of QE, yield curve control (YCC), negative rates.
iv. Bank of England (BOE): 2% inflation target. MPC of 9 members with individual accountability. First major central bank to publish inflation forecasts.
v. Swiss National Bank (SNB): Inflation target $< 2\%$. Notable for 2011–2015 EUR/CHF floor at 1.20 and subsequent abandonment. Massive FX reserve accumulation.
vi. People's Bank of China (PBOC): Multiple objectives including growth, stability, FX management. Manages CNY within trading band. Uses reserve requirements actively.

Remark 5.1.21. *ECB-Specific Tools*

i. LTRO/TLTRO: Long-Term Refinancing Operations provide multi-year funding to banks against collateral. 3-year LTROs in 2011 to 2012 injected over EUR1 trillion.
ii. OMT (Outright Monetary Transactions): Announced September 2012. Unlimited bond purchases for countries in EFSF/ESM programs. Never actually used but stabilised spreads.
iii. APP (Asset Purchase Programme): QE-equivalent. Purchases of government bonds, covered bonds, corporate bonds, ABS.
iv. TARGET2: Payment system showing intra-Eurozone imbalances. German TARGET2 claims rose to EUR700B+ during crisis, reflecting capital flight from periphery.

Remark 5.1.22. *Fed Crisis-Era Programmes*
During and after 2008, the Fed deployed unprecedented tools:

i. Central Bank Swap Lines: USD liquidity to foreign central banks to prevent LIBOR spikes
ii. TAF (Term Auction Facility): 28–84 day unsecured lending without discount window stigma
iii. TSLF (Term Securities Lending Facility): Treasury lending against lower-quality collateral
iv. CPFF (Commercial Paper Funding Facility): Commercial paper purchases to prevent market freeze
v. TALF (Term ABS Loan Facility): Lending against consumer ABS to support credit markets
vi. Agency MBS Purchases: \$1.25 trillion in QE1 alone

Understanding these tools helps anticipate future crisis responses.

5.1.3 Economic Data Releases and Demographics

Remark 5.1.23. *Data Releases and Market Impact*
Data releases are central bank inputs, not just statistics. Markets price expectations before the release; the move comes from the surprise, not the level. Key principles:

i. Surprise Direction: Above/below consensus drives immediate price action. Magnitude of surprise matters more than absolute level.

ii. Revision Risk: Prior releases are frequently revised. Initial release moves markets; revisions move markets again if material.

iii. Hierarchy: Not all data is equal. NFP and CPI dominate in the US; PMIs dominate globally due to frequency and forward-looking nature.

iv. Regime Dependence: What data matters depends on the regime. In low-inflation regimes, growth dominates. In high-inflation regimes, inflation dominates everything.

Back-testing asset responses to past data releases (queries) can inform trade sizing and direction. However, false positives are common. If wrong, cut quickly.

Remark 5.1.24. *Growth Indicators*
Growth data sets the directional bias for risk assets, but markets trade changes, not levels.

i. GDP (Headline, QoQ, YoY): Measures total production in current or constant prices (nominal, real GDP). Released quarterly with preliminary, revised, and final prints. Backward-looking but sets narrative.

ii. Purchasing Managers' Indices (PMIs): Survey business managers on employment, new orders, inventories, production, supplier deliveries, prices, backlogs, imports, and exports on scale of better/same/worse vs prior month. Above 50 = expansion; below 50 = contraction. Most important leading indicator.

iii. Industrial Production: Measures growth in manufacturing, utilities, and mining. More cyclical than services. Released monthly. Leading indicator for GDP forecasting.

iv. Trade Volumes, Baltic Dry Index (BDI): Shipping costs for dry bulk cargo. Leading indicator on global trade activity since it cannot be speculated (no futures market).

PMIs and surveys move markets more than GDP because they are higher frequency and forward-looking.
Growth surprises affect equities (earnings expectations), commodities (demand), FX (via growth differentials).

Remark 5.1.25. *Inflation*
Inflation data is primarily a policy variable. It constrains or enables central bank action.

i. Consumer Price Index (CPI)/Core CPI: Prices paid by consumers for basket of goods and services. Core excludes food and energy (volatile). Central banks target core.

ii. Producer Price Index (PPI): Average change in selling prices received by domestic producers. Leads CPI by 1–3 months as input costs pass through.

iii. PCE (Personal Consumption Expenditures): Fed's preferred inflation measure. Differs from CPI in weights and methodology.

iv. Inflation Expectations: Market-based (breakevens from TIPS) or survey-based (Michigan, NY Fed). Central banks watch expectations closely. De-anchoring is a regime change.

Inflation surprises directly affect real yields, curve shape, FX via real rate differentials.
Inflation matters most when central banks are near constraints (ZLB, credibility questioned).

Remark 5.1.26. *Employment Data*
Labor data bridges growth, inflation, and central bank reaction function.

i. Non-Farm Payrolls (NFP): Number of workers in US excluding farms, private households, nonprofits, self-employed, and military. Released first Friday of each month. Among highest-volatility releases for rates, FX, equities.

ii. Unemployment Rate: Unemployed as percentage of labor force. Headline number; can be distorted by participation changes.

iii. Participation Rate: Labor force as percentage of civilian population. Structural decline (aging) vs cyclical decline (discouraged workers) matters.

iv. Wage Growth (Average Hourly Earnings): More important than job count for inflation outlook.

v. JOLTS (Job Openings and Labor Turnover Survey): Shows job openings, hires, quits. Quits rate indicates labor market tightness and worker confidence.

Wage growth drives services inflation, which is stickier than goods inflation.
Fed's dual mandate makes employment data directly policy-relevant.

Remark 5.1.27. *External Accounts*
External balances define structural currency pressure and financing needs.

i. Current Account: Net exports + net foreign income + transfer payments. Deficit countries need capital inflows or currency depreciation; surplus accumulate reserves and have structurally stronger currencies.

ii. Capital Account/Financial Account: Portfolio flows, FDI, banking flows. Can finance current account deficits but creates vulnerabilities (sudden stop risk).

iii. Trade Balance: Exports minus imports. Incremental changes in exports or imports can have big impact on trade balance as percentage of GDP.

iv. Terms of Trade (ToT): Export prices divided by import prices. Improves when export prices rise relative to imports. Currency depreciation typically worsens ToT.

v. FX Reserves: Central bank holdings of foreign currency and gold. Countries with large reserves can intervene; watching reserve changes indicates intervention.

Persistent current-account deficits require higher yields to attract capital or eventual currency adjustment.

Remark 5.1.28. *Government/Fiscal Indicators*
Fiscal data determines long-term sustainability, not short-term price action.

i. Budget Balance: Revenues minus expenditures. Deficit = borrowing required. Deficit/GDP ratio matters more than absolute level.

ii. Debt-to-GDP: Stock measure of accumulated borrowing. Matters more for EM and Euro periphery than for reserve currency issuers.

iii. Primary Balance: Budget balance excluding interest payments. Shows underlying fiscal stance; positive primary balance means debt/GDP can stabilise if growth exceeds interest rate.

iv. Debt Sustainability: Interest payments as % of revenue. Rising share means deteriorating sustainability.

Fiscal deterioration matters when markets doubt financing ability or central bank independence.
Sovereign risk appears first in bond spreads, CDS, FX (especially EM).

Remark 5.1.29. *Consumption Indicators*
Consumption drives developed-market growth (70%+ of US GDP).

i. Retail Sales: Consumer demand for finished goods. Durable vs non-durable breakdown matters.

ii. Consumer Confidence: Survey of expected financial situation. Conference Board and Michigan surveys in US. Leading indicator, confidence leads spending, but noisy.

iii. Personal Income and Spending: Income growth enables spending growth. Savings rate = income minus spending.

iv. Savings Rate: Higher savings = lower current consumption but more sustainable. US savings rate has declined structurally; Asian savings rates remain high.

Consumption data is most relevant for equity indices, cyclical sectors, domestic currencies.
US consumer accounts for $\sim 15\%$ of global GDP directly through consumption.

Remark 5.1.30. *Housing Indicators*
Housing is cyclically important and credit-sensitive.

i. Building Permits: Lead housing starts since permits are required before construction. Most forward-looking housing indicator.

ii. Housing Starts: Counted when construction begins. Gauge of housing demand and construction activity.

iii. Existing/New Home Sales: Volume of transactions. Prices vs volumes can diverge.

iv. Home Price Indices: Case-Shiller, FHFA. Lag actual market conditions.

v. Household Formations: Structural driver of housing demand. Post-crisis decline reduced demand.

Housing matters for rate sensitivity, consumer wealth effects, and financial stability.

Remark 5.1.31. *Business and Industry Indicators*
These indicators show where growth is coming from and where it's heading.

i. Manufacturing vs Services PMI: Manufacturing is more cyclical and globally linked. Services dominate GDP but move slower.

ii. New Orders: Most forward-looking PMI subcomponent. Rising new orders = future production.

iii. Inventories: High inventories relative to sales = future production cuts. Inventory cycles matter for commodities and industrial equities.

iv. Capacity Utilisation: Actual output relative to potential. Over 100% = overheating, inflation risk. Well under 100% = slack, deflation risk.

v. Durable Goods Orders: Capital expenditure intentions. Core capital goods orders (ex-aircraft, ex-defence) is key business investment indicator.

ISM New Orders minus Inventories spread is a reliable leading indicator of manufacturing cycles.

Remark 5.1.32. *Demographics*
Slowest-moving macro variable, but the hardest constraint. Demographics cannot be QE'd.

i. Working-Age Population (15–64): Drives potential growth. Many developed countries have peaked or will peak soon.

ii. Prime Saving Age (35–69): Peak saving years. Countries past this peak see declining savings and current accounts.

iii. Dependency Ratio: Non-working population (young + old) divided by working population. Higher ratio = more fiscal pressure, lower savings.

iv. Median Age: Summary statistic for demographic structure. Europe/Japan: 40s; US: 38; EM: 20s–30s.

v. Retirement Age vs Life Expectancy: Gap determines pension system sustainability. Life expectancy rising; retirement ages barely moving.

Demographics affect long-term growth, savings rates, current accounts, equilibrium real rates.

Remark 5.1.33. *Demographic Implications by Country*
Peak working-age populations determine structural trajectories:

i. Already peaked: Germany (2006), Japan (2016), Netherlands (1989), UK (2009), US (2007)

ii. Peaking soon: China (2032), Korea (2023), Russia (2025), Spain (2022)

iii. Long runway: India (2050), Indonesia (2050), Nigeria (2050), Philippines (2050)

Countries past peak will see declining current accounts, rising dependency ratios, pressure on fiscal positions. Jean-Claude Trichet: "Current account balance is an important summary indicator that signals losses of competitiveness and emerging imbalances."

Remark 5.1.34. *Building Queries for Data Events*
Queries back-test asset behaviour around past occurrences of specific events:

i. Identify the event (e.g., BOE QE announcement, Fed rate hike, NFP beat)

ii. Find historical dates with similar conditions

iii. Calculate asset returns +1 day, +2 days, +1 week, +1 month after event

iv. Compute hit rates (percentage of times asset moved in expected direction)

v. Assess risk/reward: average return vs standard deviation

Queries are powerful for short-term event trading and understanding portfolio exposure to upcoming releases. Caveat: False positives are common. If position moves against you immediately after event, cut quickly.

Remark 5.1.35. *Key Data Release Calendar (US)*
Monthly cadence of major releases:

i. First Friday: Employment Report (NFP, unemployment rate, wages)

ii. Mid-month: CPI, Retail Sales, Industrial Production

iii. Third week: Housing Starts, Building Permits

iv. Late month: GDP (quarterly), PCE, Durable Goods

v. First business day: ISM Manufacturing PMI

vi. Third business day: ISM Services PMI

FOMC meetings (8 per year) and ECB meetings (6 per year) often dominate all other releases.
Economic calendars (Bloomberg ECO, Reuters, Trading Economics) are essential tools.

5.2 Transmission and Cross-Asset Mapping

This section services as a primer for four basic asset classes and their usage in global macro trading.

5.2.1 Building Blocks of Global Macro

Remark 5.2.1. *Macro Strategy Product Groups*
The four core product groups in global macro are:

i. Fixed Income (FI) (Policy and Growth Sensor): encodes growth expectations, inflation expectations, central bank credibility. Yield curves summarise the entire macro narrative; steepening means growth/reflation; flattening or inversion means slowdown/recession. Fixed income tells what the market thinks will happen before it happens.

ii. Foreign Exchange (FX) (Adjustment Mechanism): FX reflects relative, not absolute, macro conditions. Sensitive to rate differentials, capital flows, balance of payments, risk-on/risk-off regimes. FX absorbs and redistributes macro shocks across countries.

iii. Equities (Growth Expression): equities are long growth, long liquidity, short uncertainty. Equity indices aggregate earnings expectations, discount rates, risk appetite. Volatility (VIX) is a second-order macro signal, not just an equity signal. Equities express the confidence (or fear) embedded in the macro outlook.

iv. Commodities (Inflation and Real-Economy Link): driven by physical supply and demand, inventories, geopolitics. Highly sensitive to growth cycles, inflation shocks, currency moves (especially USD). Commodities anchor macro views in real economic constraints.

These are the most liquid markets globally, which allows macro strategies to scale to large AUM while maintaining execution quality.

Remark 5.2.2. *Relationship Between Assets*
These relationships are regime-dependent, not permanent laws:

i. Growth Shocks: rates rise, equities rally, cyclicals outperform, commodities rise. FX favours growth-linked currencies (AUD, CAD, EM FX).

ii. Recession Shocks: rates fall, curves flatten/invert, equities sell off, USD strengthens (flight to safety), commodities weaken (except gold).

iii. Inflation Shocks: nominal rates up, real rates determine equity impact. Commodities outperform. FX punishes low-credibility central banks.

iv. Liquidity Shocks: USD strengthens sharply, credit spreads widen, correlations spike toward 1, volatility explodes. Cross-currency basis blows out.

A single macro thesis should have expressions across multiple asset classes, each with different risk profiles and convexity characteristics.

Remark 5.2.3. *Asset Correlation Dynamics*
Asset correlations change under stress. Diversification fails when needed most, where assets that appear uncorrelated in calm periods may collapse together in crises.
Macro traders profit when they anticipate correlation breakdowns and position for nonlinear regime shifts. This is why macro portfolios use convex instruments (options), size conservatively, and focus on liquidity.
Traders must know what assets they are trading in aggregate. If entire portfolio has assets with high correlation to one another, they can be stopped out of all positions more easily if the market turns against them.

Remark 5.2.4. *Position Sizing Framework*
Position sizing is arguably more important than knowing what to buy or sell. The first rule of investing is to preserve capital and avoid loss.
Position sizing aims to adjust each position for volatility. This normalises all positions such that losses are more predictable and have the same probability of occurring:

i. Calculate N = 20-day exponential moving average of True Range

ii. True Range = max(High − Low, High − Previous Close, Previous Close − Low)

iii. Use $2N$ for stop-loss calculation (two standard deviations) to avoid being stopped out by normal volatility

iv. With $2N$, there is approximately 2% chance of being stopped out on any given day (vs. 16% with $1N$)

This methodology, developed by the Turtle Traders, allows traders to take positions in a broad range of markets while monitoring risk consistently.

Remark 5.2.5. *Risk/Reward Discipline*
Given human error and transaction costs, traders can only hope to be right about 50% of the time. Having good risk/reward on each trade will make or break long-term success.
As a baseline, always strive for a 1:3 risk-to-reward ratio or higher:

i. For every trade risking 25 basis points, expect to make at least 75 basis points
ii. Set targets before the trade is executed to avoid biases of irrational human decision making
iii. Log reasons for each trade in a trade journal
iv. If original thesis no longer holds true, take trade off, but do not create excuses for taking trades off early

Remark 5.2.6. *Gap Risk*
Gap risk is one of the biggest risks when sizing positions. Gap risk occurs when prices change without any trading in between (e.g., overnight, over weekends). This risk is especially acute in:

i. EM currencies and assets (less liquid, more event-driven)
ii. Positions held over weekends or holidays
iii. Periods of elevated geopolitical risk
iv. Earnings/data release windows

Remark 5.2.7. *Risk Utilisation*
Risk-taking is nonlinear. Even with 50% conviction, should not run 50% of risk utilisation.

i. Base case: set a maximum VaR limit (e.g., 5% daily VaR at 95% confidence)
ii. If up 5% on the year, allow VaR limit to increase to 6%
iii. If down 2.5% on the year, decrease VaR limit to 4%
iv. At the start of each new year, reset to 0% return regardless of prior year performance

The intention is to create steady returns while preserving capital and avoiding blow-up risk.

Remark 5.2.8. *Stress Testing*
Stress testing aims to go beyond VaR to find potential worst-case scenarios:

i. Run historical back-test on P&L for previous 500 trading days ($\approx$ 2 years)
ii. Sort from best to worst days; examine the average of the 5 worst days (worst 1%)
iii. In periods of low volatility, also test against extreme historical events: October 1987 crash, 1998 LTCM/Russian default, 2008 financial crisis, 2011 Japan earthquake, 2015 CNY devaluation, 2020 COVID crash

Knowing maximum loss outside of VaR helps traders understand exposure to tail events.

Remark 5.2.9. *Performance Metrics*
Key metrics for evaluating macro trading performance:

i. Sharpe Ratio: $\frac{E[R]-R_f}{\sigma}$, risk-adjusted return; higher is better. Assumes normal distribution.
ii. Sortino Ratio: Sharpe with only downside volatility, differentiating between good and bad volatility.
iii. Maximum Drawdown: peak-to-trough decline; critical for understanding worst-case capital loss.
iv. Value at Risk (VaR): 95% or 99% daily VaR (expected loss exceeded on 1 in 20 or 100 days).

When evaluating strategies, test Sharpe ratios rather than raw returns. Break down P&L by asset class, duration, and conviction level to identify strengths and weaknesses.

Remark 5.2.10. *Macro View Thesis*
A macro view expressed in only one asset class is incomplete. A single macro thesis should have multiple expressions, each with different risk profiles:

i. Growth view: rates (duration), equities (beta), FX (pro-cyclical currencies), commodities (cyclicals)
ii. Inflation view: commodities (energy, agriculture), breakevens, FX (commodity currencies), TIPS
iii. Crisis/Deflation view: bonds (duration), USD/JPY/CHF (havens), volatility (VIX, swaptions), gold
iv. Policy divergence view: FX (rate differentials), yield curve spreads, relative equity indices

Having multiple expressions allows for confirmation across asset classes and provides natural hedging if one leg underperforms.

5.2.2 Foreign Exchange in Global Macro

FX plays three special roles simultaneously:

i. Macro shock absorber

ii. Macro transmission channel

iii. Macro expression instrument

Remark 5.2.11. *FX Market Structure*
The FX market is the world's largest and most liquid market, and operates 24 hours per day, 5.5 days per week across global financial centers (London, New York, Tokyo, Singapore, Hong Kong).
Key participants include: central banks (intervention, reserve management), commercial banks (market-making, proprietary trading), corporations (hedging trade flows), asset managers (currency overlay, alpha generation), hedge funds (macro speculation, carry trades), and retail traders.
The market is predominantly OTC, with spot, forwards, and swaps as the main instruments. FX futures trade on exchanges (CME) but represent a small fraction of total volume.

Remark 5.2.12. *Role of the US Dollar*
Global macro is de facto USD-centric. USD is the global reserve currency, most commodities are USD-priced, global funding markets clear in USD, and cross-border leverage is USD-denominated.
The USD is involved in approximately 85% of all FX transactions globally. Many "local" macro trades are actually USD liquidity trades in disguise.
Stress often appears first in US Dollar Index (DXY), cross-currency basis, EM FX before local assets.
The DXY Index measures USD against a basket of six major currencies: EUR (57.6%), JPY (13.6%), GBP (11.9%), CAD (9.1%), SEK (4.2%), CHF (3.6%). The DXY is not trade-weighted and over-represents EUR.

Remark 5.2.13. *Special Drawing Rights (SDRs)*
SDRs are international reserve assets created by the IMF to supplement member countries' official reserves. The SDR basket is reviewed every five years. SDRs serve as a unit of account for IMF transactions and provide insight into official reserve diversification trends.

Remark 5.2.14. *Currency Regimes*
Currency regimes matter more than models. FX does not behave uniformly. Regime dominates signal.

i. Free-floating: price discovery happens in FX

ii. Managed/pegged: pressure builds elsewhere such as in rates, reserves, capital controls

iii. Hard pegs/Currency boards: FX looks stable until it breaks violently

FX valuation only works within regime constraints. Ignore the regime and "cheap" can stay cheap forever.

Remark 5.2.15. *Central Bank Intervention*
Central banks intervene in FX markets for multiple objectives:

i. Defend a peg or managed float band

ii. Smooth excessive volatility

iii. Counter speculative attacks

iv. Accumulate or deploy reserves

Intervention effectiveness depends on: reserve adequacy, policy credibility, coordination with monetary policy, and market liquidity conditions. Intervention against fundamental pressures typically fails. Watch for reserve depletion as a leading indicator of forced adjustment.

Remark 5.2.16. *Valuation Techniques*

i. Market Implied and Risk-Appetite Signals

   1. Equity Index Price Performance: strong local equities (vs peers) attract inflows and support currency; sharp equity underperformance foreshadows FX weakness via risk-off deleveraging and foreign selling.
   2. Credit/Sovereign Risk: widening sovereign spreads/CDS signals default/policy credibility/funding stress, weakening FX or raising devaluation risk premia. Also a regime detector for EM FX.
   3. Sentiment/Positioning: call-put skew on three-month risk reversals. Acts as a multiplier. Can create overshoots/undershoots around fair value in crowded carry or crisis episodes.

ii. Trade and Competitiveness Fundamentals

1. Trade Balance/External Account: structurally strong trade position often supports FX; persistent deficits can make the currency reliant on financing.
2. Trade-Weighted Index (TWI): reflects currency's effective competitiveness against trading partners; the currency basket that matters for the real economy.
3. Economic Activity: strong activity can support FX via higher expected returns and inflows, but it can also worsen the trade balance.
4. Export Partner Growth: mapping who buys exports and whether those buyers are expanding/contracting. Export partner growth leads exporter's FX through expected trade receipts.

iii. Macro Sustainability Indicators

1. Debt-to-GDP: high public debt with FX liabilities or weak domestic savings can increase risk premia and currency vulnerability, as it tightens the policy constraint set (inflation vs austerity vs default).
2. Current Account Balance: persistent deficits require capital inflows; sudden stops trigger sharp FX adjustment.
3. FX Reserves: adequacy (months of import cover, short-term debt coverage) determines policy space.

iv. Long-Term Valuation Anchors

1. Purchasing Power Parity (PPP): the idea that exchange rates should adjust so that identical goods cost the same in different countries. In practice, PPP convergence can take years or decades.
2. GDP Per Capita: structural development/productivity proxy. Richer, higher productivity economies tend to sustain stronger real exchange rates over time (Balassa-Samuelson effect).

v. Carry Indicators

1. OIS Differential: compares respective policy rates of two countries. Carry is the interest rate differential between two currencies.
2. LIBOR/SOFR Differential: reflects unsecured interbank rates. Can be decomposed into risk-free rate + credit spread component. Using relative spreads gives a sense of direction; interest rate differential drives demand for a currency.
3. Risk-Adjusted Carry: adjusts carry for volatility. Outperforms standard carry trades and is useful in cross-currency analysis:

$$\text{Carry-to-Risk Ratio} = \frac{\text{3M Carry (Rate Differential)}}{\text{30D Realized Volatility}}$$

Within high-yielding currency pairs, one would ideally want the highest carry-to-risk ratio. Volatility and carry components are always in flux, so adjustments should be made frequently.

Remark 5.2.17. *Carry Trade Mechanics*
Carry strategies are typically executed by borrowing in lower-yielding currencies and buying higher-yielding currencies. The carry trade is a crowded trade; in large risk-off markets, the strategy suffers significant losses as positions unwind simultaneously.
Carry is the return earned on holding a currency, assuming the exchange rate is held constant. In practice, exchange rate moves often dominate carry returns. Historically, carry strategies exhibit positive returns on average but with significant negative skewness (small gains, occasional large losses).

Remark 5.2.18. *Carry Trade Risk Factors*
Key risks to carry trades include:

i. Crowding Risk: carry trades are popular, leading to violent unwinds during risk-off episodes.
ii. Gap Risk: EM currencies can gap significantly on weekends or during crises, bypassing stop-losses.
iii. Regime Shifts: central bank policy changes can rapidly alter rate differentials.
iv. Correlation Breakdown: in stress, all carry currencies tend to sell off together against funding currencies.
v. Liquidity Risk: EM FX can become illiquid precisely when exits are needed most.

Finding the carry-to-risk ratio is useful, as is comparing all currency pairs on a relative basis. Measuring the standard error of the sample and finding the most attractive pairs helps in mean-reverting carry strategies.

5.2.3 Equities in Global Macro

In macro, a trader should understand equity in terms of market behaviour in the country being traded, the sectors that make up those indices, and correlation risk.

Remark 5.2.19. *Top-Down Macro Approach to Equities*
Many macro traders take a top-down approach to trading equities:

i. Start with a global view
ii. Narrow to countries expected to outperform
iii. Select the best sectors within those countries
iv. Optionally, select best companies within sectors

If the investor has a knack for knowing where the next opportunity lies, the most reward typically comes from concentrated exposure rather than diversified index holdings.

Remark 5.2.20. *Major Equity Indices by Region*

i. United States: S&P 500, Nasdaq, DJIA, Russell 2000, Wilshire 5000
ii. Europe: Euro Stoxx 50, DAX, CAC 40, FTSE 100, FTSE MIB, IBEX 35, AEX
iii. Asia Ex-Japan: Shanghai Comp, Hang Seng, KOSPI, ASX 200, NZX 50
iv. Japan: Nikkei, Topix, JASDAQ
v. Emerging Markets: Bovespa, NIFTY, Micex/RTS, Mexico IPC, TOP 40/JALSH, BIST 30/100 etc.

Indices can be traded directionally or as relative value trades (long one index, short another). Different indices have different sector exposures: Bovespa is more commodity-sensitive than NIFTY; S&P 500 has more financial exposure than Nasdaq (technology-weighted).

Remark 5.2.21. *Index Construction Methods*
Index construction methodology affects interpretation:

i. Price-Weighted: considers only stock price, ignores market cap. A single large stock move can dominate.
ii. Market-Cap Weighted: weighted by market capitalisation. Large-cap stocks dominate performance.
iii. Equal-Weighted: each constituent has equal weight regardless of size.

Understanding weighting structure is essential for accurate interpretation of index moves.

Remark 5.2.22. *Sector Rotation Across Economic Cycles*
Different sectors outperform in different economic regimes:

i. Recession/Contraction: Consumer Staples, Utilities, Healthcare (defensive sectors)
ii. Recovery: Financials, Consumer Discretionary, Real Estate
iii. Expansion: Industrials, Materials, Technology
iv. Slowing Growth/Late Cycle: Energy, Materials (inflation hedges)

To best position an equity portfolio, it is critical to know whether the economy is in contraction, recovery, expansion, or slowing growth.

Remark 5.2.23. *Equity Derivatives Overview*
Key equity derivatives for macro trading:

i. ETFs: fast, cheap exposure to indices, sectors, countries, commodities, FX, FI. Trade like stocks.
ii. ADRs (American Depositary Receipts): foreign stocks listed on U.S. exchanges in USD. Subject to FX risk since underlying shares are held locally.
iii. Index Futures: liquid, leveraged exposure to equity indices.
iv. Options: provide asymmetric payoffs for hedging and directional views.

Remark 5.2.24. *The Volatility Index (VIX)*
The VIX measures implied volatility of S&P 500 options one month out. Key properties:

i. Negatively correlated with equity returns (volatility rises when equities fall)
ii. During large selloffs, VIX can move from mid-teens to 30+ in days

iii. VIX options provide asymmetric hedges for equity portfolios

iv. VIX futures typically trade in contango; during sharp selloffs, shift to backwardation

Volatility is measured relative to $\sqrt{252} \approx 16$. A VIX of 16 implies approximately 1% expected daily move; VIX of 24 implies 1.5% daily move ($24/16 = 1.5\%$).
V2X (Euro Stoxx), VXN (Nasdaq), VXD (DJIA), RVX (Russell 2000), GVZ (Gold), OVX (Crude Oil).

Remark 5.2.25. *Variance Swaps*
Variance swaps are OTC products providing pure variance exposure. Advantages over options:

i. Pure volatility exposure without delta hedging

ii. Priced with realised volatility (usually lower than implied)

iii. No interest rate or dividend risk

Payoff structure is convex and nonlinear. Long variance benefits disproportionately from large moves in either direction. Quoted by strike (reference realised volatility), vega notional, variance units, maturity.
Variance units $= \frac{\text{Vega Notional}}{2\times\text{Strike}}$, Payoff $= (\sigma^2_{\text{realised}} - K^2) \times$ Variance Units

Remark 5.2.26. *Dividend Swaps*
An OTC or exchange-traded product that allows one to take a view on dividends of an index to be higher, or lower, than a fixed amount.

Remark 5.2.27. *Equity Valuation Techniques for Macro*

i. Price-to-Book (P/B): P/B $> 2.5\times$ generally overbought; P/B $< 1.5\times$ oversold. Useful for detecting bubbles and value opportunities.

ii. Dividend Yield: dividend/price. Holder gets paid while waiting (long carry). Useful for relative value across markets.

iii. Price-to-Earnings (P/E): price per \$1 of earnings. Best for comparable analysis across countries, sectors, stocks. Forward P/E uses analyst EPS estimates.

iv. Free Cash Flow Yield: FCF/Price. Cash flow from operations minus capital expenditures. Graham and Dodd value investors use this as top screen.

v. Market Cap to GDP: useful for locating potential bubbles. Above 100% warrants caution; rapid rises to 150%+ indicate bubble conditions (Japan 1989).

Remark 5.2.28. *Leading Indicators for Equities*
Useful macro indicators that lead equity performance:

i. PMI (Purchasing Managers' Index): leading indicator scored 0-100. Sharp slope changes and deviations from 50 provide strong signals. ISM below 50 typically indicates U.S. recession.

ii. Baltic Dry Index: shipping prices for raw material dry bulk. Supply is fixed (ships take years to build), demand is inelastic. Leading indicator for raw material demand.

iii. CDX High Yield Index: CDS spread on high-yield corporates. Highly correlated with VIX; inversely related to equity prices. Rising spreads signal risk-off.

iv. Consumer Confidence: in the U.S., consumption accounts for $> 70\%$ of GDP. Higher confidence leads to higher spending and GDP growth.

v. Commodity Prices (YoY): rising commodity prices are inflationary, typically met by central bank tightening, which has bearish equity implications.

Remark 5.2.29. *AUD Volatility as Risk Barometer*
The Australian dollar is a main risk-on currency due to Australia's commodity exports and reliance on Asian economies. AUD moves in tandem with equity prices.
Three-month AUD/USD implied volatility is highly correlated with VIX and inversely correlated with S&P 500. In times of stress, the rate of change in AUD volatility shifts suddenly and aggressively.
Using VIX and AUD volatility combined on an absolute basis smooths out error as a more useful measure.

5.2.4 Fixed Income in Global Macro

Fixed income is the lifeblood of the global financial system. Bank credit and debt capital markets fund governments, corporations, and consumers. Understanding fixed income is essential for macro trading because yields encode growth expectations, inflation expectations, and central bank credibility.

Remark 5.2.30. *Fixed Income Universe*
The fixed income market can be decomposed into:

i. Money Markets: Short-term instruments (maturity < 1 year). Includes T-bills, commercial paper, repos, Fed Funds, SOFR.
ii. Government Bonds: Sovereign debt across the curve (2Y, 5Y, 10Y, 30Y). Risk-free rate benchmark.
iii. Inflation-Linked Bonds: TIPS in US, Linkers in UK, OATi in France. Index to CPI.
iv. Corporate Bonds: Investment grade (IG) and high yield (HY). Credit spread over risk-free rate.
v. Municipal Bonds: State and local government debt, often tax-exempt.
vi. Mortgage-Backed Securities (MBS): Agency and non-agency. Prepayment risk.
vii. Asset-Backed Securities (ABS): Auto loans, credit cards, student loans.

Fixed income is the largest of the four product groups by notional outstanding.

Remark 5.2.31. *Money Markets and Funding*
Money markets allow short-term capital flow between lenders and borrowers:

i. Treasury Bills (T-Bills): Short-term US government obligations (4-week, 13-week, 26-week, 52-week). Risk-free benchmark.
ii. Commercial Paper: Unsecured short-term corporate debt, typically 90–270 days. Lower rates than bank borrowing.
iii. Repurchase Agreements (Repos): Sale of securities with agreement to repurchase. Critical for short-term funding and leverage.
iv. Certificates of Deposit (CDs): Bank time deposits, 30 days to 5 years.
v. Banker's Acceptances: Promissory notes issued by firms to banks, typically for trade finance.

Money market stress (rising repo rates, LIBOR-OIS widening) often precedes broader financial stress.

Remark 5.2.32. *Key Reference Rates*
Reference rates anchor pricing across the fixed income universe:

i. Fed Funds Rate: Unsecured overnight lending rate between depository institutions.
ii. SOFR (Secured Overnight Financing Rate): Overnight Treasury repo rate as USD benchmark.
iii. LIBOR (Legacy): London Interbank Offered Rate. Historically the most important rate in global finance. Phased out post-2021 due to manipulation scandals.
iv. EONIA/EURSTR: Euro overnight rates. EURSTR replaced EONIA in 2022.
v. SONIA: Sterling overnight rate for GBP.

The transition from LIBOR to risk-free rates (SOFR, SONIA, EURSTR) is one of the largest market structure changes in decades.

Remark 5.2.33. *Overnight Index Swaps (OIS)*
OIS is a fixed-for-floating interest rate swap indexed against an overnight rate (Fed Funds effective in US, EONIA/EURSTR in Europe):

i. Excellent tool to hedge or speculate on central bank action
ii. Typically maturity less than two years
iii. Customisable dates for precise central bank meeting targeting
iv. Less volatile than LIBOR, especially in periods of stress

Example: If BOE rate is 0.50% and December MPC OIS trades at 0.625%, the market prices 50% chance of a 25bp hike.

Remark 5.2.34. *LIBOR-OIS Spread*
The LIBOR-OIS spread is a key gauge of funding stress:

i. LIBOR is unsecured interbank lending; OIS is effectively risk-free overnight rate

ii. Spread widens during credit stress as banks demand higher compensation for counterparty risk

iii. 2008 financial crisis: LIBOR-OIS spiked as institutions stopped trusting each other

iv. European crisis: Spread widened as European banks faced USD funding stress

A widening spread does not automatically indicate crisis. Serves as useful barometer for funding market pressure.

Remark 5.2.35. *Eurodollar Futures*
Eurodollar deposits are USD time deposits at banks outside Fed jurisdiction. Eurodollar futures are the most heavily traded futures contracts in the world:

i. Contract on 3-month LIBOR (transitioning to SOFR futures)

ii. Contract size: $\$1,000,000$ notional

iii. Price: $100 -$ 3-Month LIBOR Yield

iv. Tick value: 1 basis point = \$25

v. Months: March (H), June (M), September (U), December (Z)

vi. Colour codes by year: White (Y1), Red (Y2), Green (Y3), Blue (Y4), Gold (Y5)

Eurodollar futures used to hedge LIBOR exposure, manage weighted average maturity of short-term liabilities.

Remark 5.2.36. *Interest Rate Swaps*
Interest rate swaps allow exchanging one set of interest payments for another:

i. Fixed-for-Floating: Most common. Exchange fixed rate for floating rate (SOFR, EURSTR).

ii. Receiver: Receives fixed, pays floating. Profits if floating rates fall.

iii. Payer: Pays fixed, receives floating. Profits if floating rates rise.

iv. Swap curve driven by future rate expectations; increasingly exchange-traded post-Dodd-Frank.

Interest rate swaps are the largest derivatives market by notional. Used for hedging, speculation, and transforming asset/liability profiles.

Remark 5.2.37. *Forward Rate Agreements (FRAs)*
FRAs are OTC contracts to exchange a reference rate (SOFR, formerly LIBOR) for a fixed rate:

i. More customisable than Eurodollar futures (pick specific dates)

ii. Notation: "3×6" means 3-month forward, 3-month rate

iii. Fixing date, settle date, and maturity date are key terms

iv. Used for hedging or speculating on short-term rate movements

FRAs allow precise targeting of rate exposure around central bank meetings or data releases.

Remark 5.2.38. *US Treasury Futures*
Treasury futures are standardised contracts for trading various maturities:

i. 2-Year (TU): Contract size $\$200,000$, tick \$15.625

ii. 5-Year (FV): Contract size $\$100,000$, tick \$7.8125

iii. 10-Year (TY): Contract size $\$100,000$, tick \$15.625

iv. 30-Year (US): Contract size $\$100,000$, tick \$31.25

v. Ultra-Long (WN): 25+ years, contract size $\$100,000$, tick \$31.25

Treasury futures are physically settled (cheapest-to-deliver mechanics). Most traders roll positions before delivery. Futures curve typically in backwardation due to term premium.

Remark 5.2.39. *German Bund Futures*
German government bonds (Bunds) are the European benchmark:

i. Schatz (DU): 1.75–2.25 year maturity

ii. Bobl (OE): 4.5–5.5 year maturity

iii. Bund (RX): 8.5–10.5 year maturity

iv. Buxl (UB): 24–35 year maturity

Bunds trade as "risk-free" for the Eurozone. Spreads to Bunds (e.g., BTP-Bund spread for Italy) are key measures of peripheral sovereign risk.

Remark 5.2.40. *Yield Curve Trades*
Curve trades express views on the shape of the yield curve (done DV01 neutral):

i. Steepener: Long short-end, short long-end. Profits from curve steepening.
ii. Flattener: Short short-end, long long-end. Profits from curve flattening.
iii. Bull Flattener: Rates fall, long end falls more than short end.
iv. Bear Flattener: Rates rise, short end rises more than long end (often Fed hiking).
v. Bull Steepener: Rates fall, short end falls more than long end (often Fed cutting).
vi. Bear Steepener: Rates rise, long end rises more than short end (inflation fears).

The 2s10s spread (10Y yield minus 2Y yield) is a classic recession indicator. Inversion has preceded every US recession since 1970.

Remark 5.2.41. *Carry and Rolldown*
Fixed income positions have carry and rolldown components:

i. Carry: Coupon income − financing cost. + for steepeners, − for flatteners in normal yield curve.
ii. Rolldown: Return from bond "rolling down" a positively sloped yield curve as it approaches maturity.
iii. In DV01-neutral trades, carry differential between legs must be accounted for.
iv. Rolldown is deterministic if curve shape unchanged; carry is known at trade inception.

Total return = Price change + Carry + Rolldown. Carry and rolldown dominate in low-volatility environments.

Remark 5.2.42. *Treasury Inflation-Protected Securities (TIPS)*
TIPS are US government bonds indexed to inflation (CPI-U NSA):

i. Principal adjusts with CPI; coupon is fixed rate applied to adjusted principal
ii. 5Y, 10Y, and 30Y maturities issued quarterly
iii. Floor at par: At maturity, receive greater of par value or inflation-adjusted principal
iv. 3-month lag between CPI and index calculation

TIPS allow investors to take views on real rates or inflation. Short-end TIPS trade like food and energy.

Remark 5.2.43. *Breakeven Inflation*
Breakeven inflation rate is the implied inflation rate from TIPS:

$$\text{Breakevens} = \text{Nominal Treasury Yield} - \text{TIPS Real Yield}$$

For example, if 10Y Treasury yields 4.0% and 10Y TIPS yields 1.5%, 10Y breakeven is 2.5%.

i. Breakevens reflect market-implied inflation expectations
ii. Can be traded directly via breakeven swaps
iii. Rising breakevens = market expects higher inflation
iv. Fed monitors breakevens for inflation expectations anchoring

Negative real rates (TIPS yield < 0) indicate investors accepting guaranteed loss of purchasing power for safety.

Remark 5.2.44. *Sovereign Credit Risk*
Sovereign debt is issued by national governments. Key concepts:

i. Reference Entity: Legal name for CDS purposes (e.g., Hellenic Republic for Greece)
ii. Curve Inversion: When short-term yields exceed long-term yields, market prices default risk
iii. Credit Events: Failure to pay, restructuring, moratorium
iv. Recovery Rate: Typically assumed 40% for sovereigns; historical average 55% with high variance

Peripheral European spreads (Italy, Spain, Portugal, Greece) to German Bunds are key risk indicators.

Remark 5.2.45. *Credit Default Swaps (CDS)*
CDS provides insurance-like protection against credit events:

i. Protection Buyer: Pays spread (premium), receives par if credit event occurs

ii. Protection Seller: Receives spread, pays par minus recovery if credit event occurs

iii. Premium quoted as annual spread, paid quarterly (March, June, September, December)

iv. Can buy protection without owning underlying bond ("naked CDS")

CDS provides asymmetric payoff: pay small premium for potentially large payout. Attractive risk/reward for tail hedging. Counterparty risk is critical.

Remark 5.2.46. *CDS Credit Events*
Credit events trigger CDS settlement:

i. Failure to Pay: Sovereign/corporate misses interest or principal payment

ii. Restructuring: Postponement, reduction, or deferral of obligations

iii. Bankruptcy: Corporate only (sovereigns cannot file bankruptcy)

For sovereigns, restructuring is not automatic trigger. Both buyer and seller have right (not obligation) to trigger CDS. Greece 2012 required ISDA determination committee ruling.

Remark 5.2.47. *Historical Sovereign Defaults*
Recent sovereign defaults and restructurings:

i. Russia 1998: Devaluation and default on local currency obligations, $73B

ii. Argentina 2001: Largest sovereign default at time, $82B, 70% haircut

iii. Uruguay 2003: Contagion from Argentina, 5-year maturity extension

iv. Greece 2012: PSI restructuring, largest sovereign default in history, $200B+

v. Argentina 2020: Ninth default, $65B restructured

Recovery rates vary widely (20%–80%). "Orderly" restructurings tend to have higher recovery.

Remark 5.2.48. *Fixed Income ETFs*
Fixed income ETFs provide liquid, diversified exposure:

i. Treasury: SHY (1–3Y), IEF (7–10Y), TLT (20+Y), TBT (2x inverse 20+Y)

ii. TIPS: TIP (broad TIPS)

iii. Investment Grade: LQD (corporate IG), AGG (aggregate bond)

iv. High Yield: HYG, JNK (high yield corporate)

v. Emerging Markets: EMB, PCY (EM sovereign debt)

vi. Municipals: MUB (muni bonds)

vii. MBS: MBB (mortgage-backed)

ETFs provide quick and cheap exposure to markets otherwise requiring large minimums or dealer relationships.

Remark 5.2.49. *Duration and DV01*
Duration measures sensitivity to interest rate changes:

i. Modified Duration: Approximate percentage price change for 1% yield change

ii. DV01 (Dollar Value of 01): Dollar change for 1bp yield change $= \frac{\text{Duration} \times \text{Price}}{10000}$

iii. Longer maturity = higher duration = more rate sensitivity

iv. Curve trades are structured DV01-neutral to isolate curve shape from level

A 10Y bond with duration 8 loses approximately 8% if yields rise 100bp.

Remark 5.2.50. *Convexity*
Convexity measures how duration changes as yields change:

i. Positive convexity: Bond gains more when yields fall than it loses when yields rise (most bonds)

ii. Negative convexity: MBS, callable bonds. Gains are capped due to prepayment/call risk

iii. Convexity matters more for large yield moves

iv. Long convexity = long volatility; benefits from rate moves in either direction

Convexity adjustment:

$$\Delta P \approx -\text{Duration} \times \Delta y + \frac{1}{2} \times \text{Convexity} \times (\Delta y)^2$$

Remark 5.2.51. *Curve Inversion as Recession Indicator*
Yield curve inversion (short rates exceeding long rates) has predicted every US recession since 1970:

i. 2s10s inversion: 10Y yield below 2Y yield
ii. 3m10s inversion: 10Y yield below 3-month T-bill
iii. Typical lead time: 6–18 months before recession
iv. Mechanism: Markets expect Fed to cut rates in response to coming weakness

Curve inversion is necessary but not sufficient for recession. Duration and depth of inversion also matter.

Remark 5.2.52. *Fixed Income in Different Macro Regimes*

i. Growth Shock (positive): Rates rise, curve steepens, credit spreads tighten
ii. Recession: Rates fall sharply, curve bull steepens then flattens, credit spreads widen
iii. Inflation Shock: Nominal rates rise, real rates may fall initially, curve bear steepens, TIPS outperform nominals
iv. Deflation/Disinflation: Nominal rates fall, breakevens collapse, duration rallies
v. Financial Crisis: Flight to quality (Treasuries rally), credit spreads explode, funding markets seize

Fixed income tells what the market thinks will happen before it happens. The curve summarises the entire macro narrative.

5.2.5 Commodities in Global Macro

Commodities are driven by physical supply and demand, inventories, and geopolitics. They are highly sensitive to growth cycles, inflation shocks, and currency moves (especially USD). Commodities anchor macro views in real economic constraints.

Remark 5.2.53. *Contango and Backwardation*
The shape of the futures curve reveals market structure:

i. Contango: Near-term futures trade cheaper than further-dated contracts. Exists due to storage costs, financing costs, and opportunity cost of capital. Normal state for storable commodities like gold.
ii. Backwardation: Near-term futures trade at premium to further-dated contracts. Occurs during supply shortages or immediate demand spikes.
iii. Agricultural seasonality: Curves can exhibit both. Contango after harvest (abundant supply, storage costs), backwardation before harvest (depleted stocks, tight supply).

Contango creates negative roll yield; backwardation creates positive roll yield.

Remark 5.2.54. *Commodity Indices*
Major commodity indices for benchmarking and trading:

i. CRB (Commodity Research Bureau) Index: 22 commodities, broad representation of prices.
ii. S&P GSCI: Goldman Sachs Commodity Index. Production-weighted, heavily energy-exposed ($> 50\%$).
iii. Bloomberg Commodity Index (BCOM): Diversified, liquidity-weighted.
iv. ETFs: DBC (Deutsche Bank commodities, $> 50\%$ energy), DBA (agriculture only, includes corn, soybeans, sugar, cattle, coffee, cocoa).

Caution for ETFs: contango and roll costs can cause significant performance drag versus spot prices.

Remark 5.2.55. *Commodity Risk and Return Profile*
Commodities have distinct risk characteristics:

i. Higher volatility than FX, equities, and fixed income
ii. Historically attractive Sharpe ratio as a relative comparison
iii. Prior to 2008, exhibited negative correlation with equities (diversification benefit)
iv. Post-2008, correlations with risk assets increased significantly

v. Supply shocks can cause extreme price moves (geopolitics, weather, production disruptions)

Commodities are particularly useful as inflation hedges and for expressing views on global growth.

Remark 5.2.56. *Crude Oil Fundamentals*
Oil is the most important energy commodity, accounting for 33%of global energy demand:

i. Quality measures: API gravity (higher = lighter) and sulphur content (lower = sweeter)
ii. Light sweet crude: Most desirable (WTI, Brent). Heavy sour crude least desirable (Venezuela, Mexico)
iii. Products: Gasoline (cars), kerosene (diesel, jet fuel), heating oil, LPGs
iv. Global consumption approximately 90 million barrels per day

Remark 5.2.57. *Major Oil Benchmarks*
Key oil benchmarks and their characteristics:

i. WTI (West Texas Intermediate): Light sweet crude. Trades on NYMEX (CL). Delivery at Cushing, Oklahoma. Contract: 1,000 barrels. ETF: USO.
ii. Brent Crude: North Sea light sweet crude. Trades on ICE (CO). Prices more than half of world's internationally traded crude. Delivery at Sullom Voe, Scotland.
iii. WTI-Brent Spread: Historically WTI traded at premium (higher grade), but infrastructure constraints at Cushing and Middle East supply sensitivity pushed Brent to premium.

Remark 5.2.58. *OPEC and Oil Supply*
OPEC controls significant portion of global oil supply:

i. 12 member countries controlling $> 35\%$ of global production, 73% of reserves
ii. Saudi Arabia has largest spare capacity. "The Call on OPEC" typically means Saudi increases production
iii. Key NOCs: Saudi Aramco (largest company by value, 1/6 of global reserves), NIOC (Iran), PDVSA (Venezuela), INOC (Iraq), KOC (Kuwait)
iv. OPEC quota system attempts to coordinate production to stabilise prices

Political risk in OPEC members is critical for oil price forecasting.

Remark 5.2.59. *Oil Chokepoints and Infrastructure*
Critical infrastructure for global oil flows:

i. Strait of Hormuz: Most important chokepoint. 17 million bbl/d (20% of global traded oil). Between Iran and Oman. Geopolitical flashpoint.
ii. Strait of Malacca: 14 million bbl/d. Connects Indian and Pacific Oceans. Critical for Asian supply.
iii. Mandab Strait/Suez Canal: 3 million and 2 million bbl/d respectively. Piracy risk.
iv. Pipelines: Druzhba (Russia to Europe, 2,300 miles, 1.4 million bbl/d). Keystone (Canada to US).

Supply disruption at chokepoints causes immediate price spikes, especially in Brent.

Remark 5.2.60. *Crack Spread*
The crack spread measures refining margins:

i. Spread between crude oil input and refined product output (gasoline, heating oil)
ii. $3:2:1$ Crack Spread: 3 barrels crude vs. 2 barrels gasoline + 1 barrel heating oil (most common)
iii. High crack spread: refiners profitable, incentive to increase production
iv. Low/negative crack spread: margins squeezed, refiners slow production
v. Factors: Cushing supply, driving demand, refinery shutdowns, hurricanes

Conversion: gasoline/heating oil in cents/gallon, crude in dollars/barrel. 1 barrel = 42 gallons.

Remark 5.2.61. *Natural Gas*
Natural gas is the cleanest fossil fuel (30–50% less CO2 than gasoline/coal):

i. US and Russia account for 38% of global production
ii. Shale revolution: Horizontal drilling and hydraulic fracturing transformed US from importer to exporter
iii. Flaring: Burning unwanted natural gas, common in US due to regulatory requirements

iv. LNG: Liquefied at $-260°$F for transport, 600x less volume than gas state

v. Uses: Power generation, heating, fertiliser (ammonia), transportation

Natural gas futures (NG) trade on NYMEX. Contract: 10,000 MMBtu, delivery at Henry Hub, Louisiana. Quoted in \$/MMBtu. ETF: UNG (subject to contango drag).

Remark 5.2.62. *Gold*
Gold is both precious metal and monetary asset. Currency symbol: XAU.

i. Jewelry: 40%+ of demand. Investment: second largest demand driver

ii. Central banks hold 31,920 tonnes. US largest holder (8,134 tonnes, 73%of reserves)

iii. China overtook South Africa as largest producer

iv. India largest consumer (50% used for weddings)

v. Real rates: Gold inversely correlated with real interest rates. Negative real rates support gold.

Trading: GLD ETF (1/10 oz per share), gold futures (GC) on COMEX (100 troy oz contract).

Remark 5.2.63. *Silver*
Silver has both monetary and industrial properties. Currency symbol: XAG.

i. Highest electrical conductivity of all metals

ii. Industrial applications: 44% of demand (batteries, electronics, solar panels)

iii. Jewelry/silverware: 22% of demand

iv. Mexico and China account for 47% of global production

v. More volatile than gold; often trades as leveraged gold play

Trading: SLV ETF (1 oz per share), silver futures (SI) on COMEX (5,000 troy oz). "Triple 9s" (99.9% purity) is deliverable grade.

Remark 5.2.64. *Platinum*
Platinum is extremely rare with concentrated supply:

i. South Africa holds 80% of global reserves

ii. Autocatalyst (catalytic converters): 38% of demand

iii. Jewelry: 31% of demand

iv. Almost twice as heavy as gold (density 21 vs. 19)

v. Play on emerging market growth and automotive demand

Platinum futures (PL) trade on COMEX (50 troy oz). Platinum/gold ratio is a key relative value indicator.

Remark 5.2.65. *Copper*
Copper is considered a leading economic indicator:

i. Chile dominates production (27%); China dominates consumption (43%), stockpiles are closely watched

ii. Uses: Building construction (33%), infrastructure (15%), equipment manufacturing (52%)

iii. Major input for urbanising economies, plays into EM growth story

iv. Recycled copper ("secondary copper") represents significant market share

Trading: LME (LMCADS03, \$/ton, 25 tons/contract) or NYMEX (HG, cents/lb, 25,000 lbs/contract).

Remark 5.2.66. *Aluminium*
Aluminium is the most abundant metal in Earth's crust:

i. China accounts for 40%+ of both production and consumption

ii. Lightweight, durable, does not corrode

iii. Uses: Construction, transportation (less energy needed to move lighter vehicles), packaging

iv. One of few commodities where China is self-sufficient

v. Energy-intensive to produce (aluminium smelters locate near cheap power)

Trading: LME (LMAHDS03, \$/ton, 25 tons/contract).

Remark 5.2.67. *Agricultural Commodities Overview*
Key characteristics of agricultural trading:

i. Supply determined by weather, planting decisions, yields
ii. Demand relatively inelastic (food consumption stable)
iii. US is largest player in most agricultural products
iv. Best agriculture traders employ full-time meteorologists
v. Crop calendars define sensitive periods for each commodity
vi. Planting, flowering/silking, and harvest are critical stages

Remark 5.2.68. *Corn*
Corn is the largest cereal crop globally:

i. US accounts for 35%+ of production, 30%+ of consumption, largest exporter
ii. Western Corn Belt (Iowa, Illinois, Nebraska) produces 47% of US corn
iii. Uses: Livestock feed, ethanol (27% of US demand, growing), food
iv. Ethanol mandate: US Energy Policy Act 2005 requires ethanol blending, creating structural demand
v. China demand growth: feedstock demand from 0% (2007) to 5%+ of imports

Trading: CBOT (C), cents/bushel, 5,000 bushels/contract. 1 bushel = 56 lbs.
Delivery months: H, K, N, U, Z. Critical period: June–August (silking).

Remark 5.2.69. *Wheat*
Wheat has been cultivated for $10,000+$ years:

i. EU, China, India are largest producers; US is fourth largest but largest exporter
ii. Classifications: season (winter/spring), gluten content (hard/soft), colour (red/white/amber)
iii. Uses: Bread, flour, noodles, beer, feedstock
iv. Close substitute for corn. Prices move together
v. Russia export bans have caused supply shocks

Trading: CBOT (W), cents/bushel, $5,000$ bushels/contract. 1 bushel = 60 lbs.
Critical period: heading stage.

Remark 5.2.70. *Soybeans*
Soybeans are a critical oilseed with high protein content:

i. US largest producer (31%), Brazil second largest but largest exporter
ii. China largest consumer (crush) and importer (63% of global imports)
iii. Products: Soybean meal, vegetable oil, soy milk, tofu, edamame
iv. 90%+ of soybeans are genetically engineered (lower pesticide costs)
v. Brazil/Argentina weather critical, neighbouring countries, correlated weather risk

Trading: CBOT (S), cents/bushel, 5,000 bushels/contract. 1 bushel = 60 lbs. Most vulnerable during flowering.
Delivery months: F, H, K, N, Q, U, X.

Remark 5.2.71. *Cotton*
Cotton is the most important textile fiber:

i. China, India, US are largest producers (65% of production)
ii. China largest producer and consumer (textile industry)
iii. US third largest producer but largest exporter (30% of exports)
iv. Classifications: character (strength), grade (colour/purity), staple (fibre length)
v. More drought-resistant than other crops

Trading: ICE (CT), cents/lb, $50,000$ lbs/contract (≈ 104 bales). 1 bale = 480 lbs.
Delivery months: H, K, N, V, Z.

Remark 5.2.72. *Coffee*
Coffee has distinct supply characteristics:

i. Two types: Arabica (60% of production, higher grade, higher altitude) and Robusta (stronger taste, more caffeine, lower altitude)

ii. Brazil largest producer/exporter (37%). Vietnam second (mostly robusta)

iii. EU and US are largest importers (46% and 23% respectively)

iv. Biennial cycle: Coffee trees alternate high and low yield years

v. Tree life cycle approximately 20 years

Trading: ICE (KC), cents/lb of arabica, $37,500$ lbs/contract. Industry measures in bags (60 kg = 132 lbs). Critical period: blooming (water is critical).

Remark 5.2.73. *Ending Stocks and Supply/Demand Balance*
Ending stocks (inventory) are the most important driver of commodity prices:

i. Ending Stocks = Beginning Stocks + Production − Consumption

ii. Low stocks relative to consumption (stocks-to-use ratio) = price support

iii. High stocks = price pressure

iv. USDA WASDE reports provide monthly supply/demand estimates

v. Small changes in production or consumption can cause large price moves when stocks are tight

Remark 5.2.74. *Weather Risk in Agriculture*
Weather is the primary supply shock driver in agriculture:

i. Drought during critical growth stages (flowering, silking) is most damaging

ii. Livestock cascade: high grain prices → expensive feed → early livestock slaughter → short-term meat supply up, future supply down

iii. El Niño/La Niña cycles affect global weather patterns

iv. Monitor: NOAA drought monitors, crop condition reports, weather forecasts for key growing regions

Remark 5.2.75. *Commodities in Different Macro Regimes*

i. Growth Shock: Commodity demand rises, prices increase, especially energy and industrial metals

ii. Recession: Demand destruction, prices fall (except gold as safe haven)

iii. Inflation Shock: Commodities outperform as inflation hedge; energy and agriculture lead

iv. USD Weakness: Commodity prices rise (inverse correlation with dollar)

v. Geopolitical Crisis: Energy spikes on supply fears; gold rises on safe-haven demand

Commodities are the inflation and real-economy link in the macro framework.

5.3 Practitioner Frameworks

This section distills operational frameworks from leading global macro practitioners, drawn primarily from Steven Drobny's interviews [s7-drobny]. Focus is on actionable principles rather than historical narratives.

5.3.1 Portfolio Construction and Risk Frameworks

Remark 5.3.1. *Systematic Risk Premia Allocation (Leitner Framework)*

i. Equal 20% risk allocation across: Equities (value screens), Fixed Income (5x leveraged short-duration), FX (carry + vol premia), Commodities (roll yield), Real Estate

ii. Additional 10% for 1–2 high-conviction opportunities annually

iii. Anchor to data first, then narratives; actively seek disconfirming evidence

iv. Not being invested = foregoing risk premia; short positions pay away premia

Remark 5.3.2. *Convexity and Optionality (Siva-Jothy/Anonymous Currency Manager)*

i. Maintain long gamma; never short gamma in macro portfolios

ii. 50–60% options (3–6 months); 75% of price moves occur 10–15% of time around data releases

iii. Systematically long options to capture regime shifts; leverage is function of correct macro analysis

iv. Never forced seller; liquidity-only mandate (G10 only)

v. Timing: buy options when consensus says inevitable and market at extreme

Remark 5.3.3. *Trade Construction Discipline (Drobny/Wadhwani)*

i. Views without actionable trades are useless; compare fundamentals to market perception

ii. Look for positioning extremes (2–5 year timeframes); CA deficit currency rising = longs building

iii. Minimum risk/reward $4:1$; after two stop-outs on same thesis, move on

iv. 3–4 major opportunities annually; overtrading as damaging as running losers

v. Positioning and sentiment dominate fundamentals short-term

vi. Stop-loss most critical; exit immediately when thesis invalidated

Remark 5.3.4. *Fixed Income Specific (Porter)*

i. Time horizon arbitrage: institutional short-termism creates medium-term opportunity

ii. Focus on one-year, one-year forwards (most volatile, greatest overshooting)

iii. Interest rate cycles 16–24 months; maximise income per VaR via front-end

iv. Curve steepener $\approx$ 90% bullish bonds; flattener $\approx$ bearish

5.3.2 Asset-Class Specific Frameworks

Remark 5.3.5. *Commodity Framework (Anderson)*

i. Pure microeconomics: supply/demand, cost curves, competitive positioning

ii. Fundamentals over price: ignore day-to-day moves, focus on economics

iii. Mean reversion: margins revert to mean, not prices

iv. Contrarian timing: front-page hysteria = reversal indicator (sell euphoria, buy panic)

v. Micro-economic inevitability: stress-test thesis against adverse macro scenarios. Long = assume low growth; short = assume high growth

vi. On-the-ground research: observe anecdotal evidence, see fundamental change before markets price it

vii. Only invest low-cost producers; high-cost assets can't be bought cheap enough

viii. Reject VaR for concentrated portfolios (assumes diversification); use volatility-adjusted sizing

Remark 5.3.6. *Equity Macro Framework (Bessent)*

i. Express macro views through equities: 50% of stock move = market, 30% = sector, 20% = stock-specific

ii. Bottom-up research for macro confirmation: micro insights reveal macro picture

iii. Entry points critical: good entry means rarely stopped out

iv. Concentrated portfolio: 8–14 large positions, not equally weighted

v. Paradigm shifts = major profits: identify when game changes, not how to play game better

Remark 5.3.7. *FX Framework (Anonymous Currency Manager)*

i. FX is tail of credit curve: Fed funds → short rates → govts → credit → equities → FX

ii. Currencies reflect relative sentiment/credit spreads between economies

iii. Three currency blocs: USD, EUR, Asia

iv. Directional concentration: 3–4 macro calls maximum (bonds, equities, USD vs blocs)

v. Five-month time horizon for thesis validation

vi. Basic law: bullish = long or flat, bearish = short or flat. Never relative value

Remark 5.3.8. *Emerging Markets Crisis Framework (Dimitrijević)*

i. Country implosion acts like company collapse: all instruments correlate

ii. Strict limits: 10% global macro fund, 15% EM fund per country

iii. Best EM trades after panics/disasters

iv. Clean sheet valuation: ignore yesterday's price. Ask: "would we invest at current price with clean sheet?"

v. Stop-loss mechanics: work in continuous markets, dangerous in distressed/value plays

vi. Internal stops only (counterparties invariably hit external stops)

vii. One-month cool-off after stop-out

viii. Hedge mismatch: short rich asset class as hedge to long attractive asset class within same country

5.3.3 Execution and Flow Analysis

Remark 5.3.9. *Flow and Cross-Market Linkages (Harris)*

i. Flow matrix: mental model tracking global money flows seeking highest risk-adjusted return

ii. Money craves stability; fear greater than greed drives capital flight

iii. Cross-market linkages: geopolitical events ripple through asset classes (e.g., Brazilian real devaluation → sell soybeans as farmers hedge USD revenues)

iv. Slower Fool Theory: exit before discontinuous leap into illiquidity

v. Risk management: 80% trades wrong, 20% profitable must exceed losses

vi. Small probes (reconnaissance) test thesis before scaling

vii. Never pray trades back to breakeven; clear books, start fresh

viii. Best at 3–4 concentrated positions; avoid overtrading

Remark 5.3.10. *Contrarian Indicators*

i. Concept exhaustion: magazine covers = fully discounted

ii. Gestalt shifts: volatility spikes when protection buying signals crowded positioning

iii. "Frozen ice theory": strong trades at edges, crowded in middle on proxies

iv. Central banks often counter-indicators in market timing

Chapter 6

Portfolio construction and risk

6.1 Overview of Systematic Investments

A schematic of a live 'production' trading strategy is shown below, but does not include everything else necessary to create the strategy (i.e., research tools).

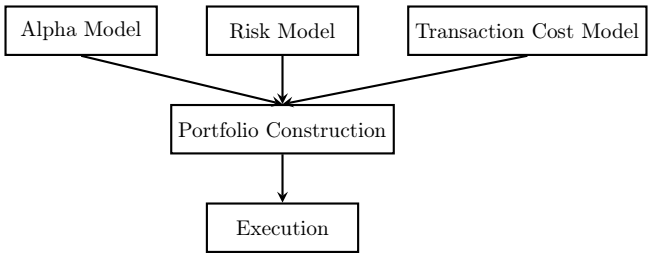


Figure 6.1: Live Production Trading Strategy Overview

At its core, the trading system is organised into three primary modules:
The trading system has three modules:

i. Alpha model: predicts the future of the instruments considered for trading, i.e. directional alpha

ii. Risk model: limits amount of exposure to factors that are unlikely to generate returns but could drive losses, i.e. directional exposure limit on an asset class

iii. Transaction cost model: determine if the cost of the trades needed to migrate from current portfolio to new portfolio is desirable to the portfolio construction model.

These models feed into a portfolio construction model that balances the tradeoffs of profit and risk to determine the best portfolio to hold. The model finds the differences in trades that need to be executed.
The execution model then takes the required trades, and using inputs such as urgency in which the trades need to be executed and dynamics of liquidity in the markets, executes the trades in an efficient and low cost manner.

Method 6.1.1. *Chains of Production for Alpha Signals*

i. Data Curation: for collecting, cleaning, indexing, storing, adjusting, and delivering all data to production chain. Requires experts in market microstructure and data protocols such as FIX.

ii. Feature Analysis: transform raw data into informative signals. Requires experts in information theory, signal extraction and processing, visualisation, labelling, weighting, classifiers, feature importance techniques. Feature analysts collect and catalogue libraries of findings.

iii. Strategists: informative features are transformed into actual investment algorithms. Strategists will parse libraries of features for ideas to develop an investment strategy. Require data scientists with deep knowledge of financial markets and economy. Features may be discovered by black box, but strategy is developed in a white box.

iv. Back-testers: assess profitability of investment strategy under various scenarios. Requires data scientists with deep understanding of empirical and experimental techniques. Good back-tested incorporates in analysis meta-information on how strategy was created.

v. Deployment Team: integrate strategy code into production line. Requires algorithm specialists and mathematical programmers. To ensure deployed solution is logically identical to prototype, and to optimise implementation sufficiently such that production latency is minimised.

vi. Portfolio Oversight: once strategy is deployed, follows lifecycle.

1. Embargo: initially, strategy is run on data observed after end date of backtest. If embargoed performance is consistent with backtest, strategy is promoted to next stage.
2. Paper Trading: strategy run on live, real-time feed. Performance accounts for data parsing latencies, calculation latencies, execution delays, and other time lapses between observation and positioning.
3. Graduation: strategy manages real position, whether in isolation of as part of ensemble. Performance evaluated precisely, including attributed risk, returns, and costs.
4. Re-allocation: based on production performance, allocation is re-assessed frequently and automatically. Strategy allocation follows a concave function, Initial allocation is small. As time passes and strategy performs as expected, allocation is increased. Over time, performance decays and allocations become gradually smaller.
5. Decommission: if strategy perform below expectations for sufficiently extended period of time, strategy is discontinued.

6.1.1 Alpha Models Overview

Theory-driven models tests theories of why markets behave in a manner, and see if they can be used to predict the future. Strategies utilising price-related data are trend and mean reversion; strategies utilising fundamental data are value/yield, growth and quality. Usually more than one model is used in combination.

Definition 6.1.2. *Theory Driven Models*

i. Trend Following: markets move in given direction long enough that the trend can be identified. As more data support the bull/bear thesis in an uncertain market, more market participants will adopt the same thesis and hence move the asset price to a new equilibrium.
Moving average crossover indicator strategy has less than one point of return for every point of downside risk taken, as market behaviour are unstable and episodic.

ii. Mean Reversion: markets move in opposite direction to the prevailing trend. Short-term imbalances between buyers and sellers due to liquidity forces prices to move abruptly in one direction, which increases probability of trend reversion as liquidity issue is resolved.
Statistical arbitrage bets on convergence of prices of similar stocks whose prices have diverged.
Longer-term trends can occur despite smaller oscillations around these trends occurring in the shorter term, hence both strategies may be used in conjunction.

iii. Value/Yield: value strategies uses ratios of fundamental factor against the price of the instrument, inverted to keep the ratio consistent. The higher the yield, the cheaper the instrument.
Buying undervalued security and selling overvalued security is a *carry trade*. The difference between yield received and yield paid is the *carry*.
Quant Long Short (QLS) ranks stocks by attractiveness based on various factors such as value, then buy the higher-ranked stocks while shorting the lower-ranked stocks.

iv. Growth: make predictions based on asset's expected or historically observed level of economic growth. Forward-looking growth expectations are typically used as a metric.
Growth is trending, and strongest growers are becoming more dominant relative to competitors. Macro growth factors may be used on foreign exchange, while micro growth factors may be used on companies.

v. Quality: All else being equal, it is better to long high quality and short low quality. Capital safety is important. Factors include earnings quality, equity-to-debt ratios etc.

Data-driven models are more difficult to understand, with more complicated mathematics. Relies on data mining, more technically challenging and far less widely practiced. Typically more used in high-frequency space, as they can discern how market behaves without caring about the economic theory or rational.

Method 6.1.3. *Strategy Parameters*
An implementation approach requires a forecast target, time horizon, bet structure, investment universe, model specification, and run frequency.

i. Forecast Target: models may forecast direction, magnitude, duration of move, and may include probability into the forecast. Signal strength is of importance, defined by a larger expected return and/or higher likelihood of return. A higher level of signal strength results in a bigger bet taken on the position.

ii. Time Horizon: models may have forecast horizons ranging from microseconds to years. There are more variability between short-term and long-term strategies, as short-term strategies are making very large number of trades compared to long-term strategies.

iii. Bet Structure: models can be made to forecast an instrument relative in itself or to others. For relative forecasts, smaller clusters (pairs) or larger clusters (sectors) may be used. For pairs, few assets can be compared precisely and directly. Large cluster grouping may eliminate impact of general movement of the sector and hence focus on the relative movement of stocks within the sector, allowing for clearer distinction between group behaviour and idiosyncratic behaviour. Clusters may be created either via statistical methods or using heuristics (i.e., fundamentally defined industry groups).
Statistical methods may be fooled by data, leading to bad grouping. Heuristic grouping may be imprecise for conglomerates, and may be too rigid. Relative alpha strategies tend to exhibit smoother returns during normal times than intrinsic alpha strategies, but may face incorrect groupings during stressful periods. This may be mitigated by utilising several grouping techniques in concert.

iv. Investment Universe: choices made on geography, asset class, instrument class, and exclusions. Liquidity is preferred so estimations of transaction costs are reliable. Large quantities of high quality data is required, which is found in highly liquid and developed markets. Instruments with consistent behaviour is preferred, hence biotech stocks are excluded due to sudden, violent price changes. Hence, the most common asset classes and instruments modelled are common stocks, futures (on bonds and equity indices) and forex.

v. Model Specification: focuses on definition of the strategy mathematically, and may be the source of alpha. Specification details in terms of machine learning or data mining techniques are also defined, to assist in fitting models to the data and setting parameter values. Refitting frequency is also defined to refresh the model and make it adapt to current market conditions; may lead to greater risk of overfitting.

vi. Run Frequency: defined from monthly to real time frequency. Increasing frequency of runs lead to greater number of transactions and hence higher transaction costs, and risk of moving portfolio based on noisy data. Less frequency of runs lead to smaller number of larger-sized trades, hence may move the market with block trades; may also miss opportunities to trade at more favourable prices.

Method 6.1.4. *Blending of Models*
Most common approaches are linear models, nonlinear models, and machine learning models. If models are not combined, then several portfolios are constructed based on output from each model, then combined using portfolio construction techniques. The best method depends on the model.

i. Linear Models: require independence of factors, and each factor to be additive. To determine the weight of each alpha factor, multiple regression techniques may be used.

ii. Nonlinear Models: used when factors are not independent, or the relationship changes over time. Conditional models base the weight of one factor on the reading of another factor. Rotational models assign weights of factors that fluctuate over time based on updated calculations of the various signal's weights, giving higher weights to factors with better performance recently.

iii. Machine Learning Models: developing machine learning strategies takes as much effort to produce one true investment strategy as to produce a hundred. The complexities include data curation and processing, HPC infrastructure, software development, feature analysis, execution simulations, backtesting etc.
Decades ago, macroscopic alpha based on simple tools like econometrics are common, but this is quickly diminishing. Microscopic alpha however, becomes more abundant, but requires heavy ML tools.

6.1.2 Risk Models

Risk model concerns the intentional selection and sizing of exposures to improve the quality and consistency of returns. By pursing an alpha, we want to be invested in the movement of the exposure to profit in the long run.

Method 6.1.5. *Factor-Based Models*
Factor-based models decompose asset returns into contributions from systematic factors and idiosyncratic components. The most common factors include:

i. Market Factor: Captures the overall movement of the market.

ii. Size Factor: Reflects the differential risk associated with companies of varying market capitalizations.

iii. Value Factor: Accounts for risk due to discrepancies between market prices and fundamental valuations.

iv. Momentum Factor: Measures the tendency of asset prices to continue in their current trajectory.

This allows traders to understand which elements drive portfolio risk and adjust exposures accordingly.

Method 6.1.6. *Statistical Models*
Statistical risk models leverage historical data and probabilistic techniques to quantify risk parameters.

i. Historical Simulation: Directly computing risk metrics from past return distributions.

ii. Monte Carlo Simulation: Generating a large number of potential future return scenarios to estimate risk under diverse conditions.

iii. Parametric Methods: Employing analytical formulas based on assumed return distributions to calculate key risk measures.

These are useful for dynamically updating risk assessments as new market data become available.

Method 6.1.7. *Limiting Size of Risk*
The quantitative risk models that limit the size of risk varies by the manner in which size is limited, how risk is measured, and what is having its size limited.
Size limits can be limited by hard constraints and penalties. A hard limit may be arbitrary, hence penalty functions may be built to allow a position to increase beyond the limit level, only if the alpha model expects a significantly larger return. The levels of limits and penalties may be determined from either theory or data.
To measure risk, there are two methodologies. The first is longitudinal, and measures risk through the volatility of an instrument. The second is to measure the correlation or covariance between assets (dispersion).
Size limiting may be applied to single positions and groups of positions (sectors, asset classes). It may also be applied to various types of risks and the amount of portfolio leverage.

Method 6.1.8. *Limiting the Types of Risk*
To eliminate unintentional exposure as there is no expectation of being compensated sufficiently for accepting them. This can be achieved through theoretical or empirical risk models.

i. Theory-Driven Risk Models: focuses on systematic risk factors, derived from economic theory. Systematic risks cannot be diversified away. Equity may have market risk, sector risk, market capitalisation risk etc. Fixed income may have interest rate risk.

ii. Empirical Risk Models: uses historical data to determine the unnamed systematic risks that should be measured and mitigated. Uses principal component analysis (PCA) to discern unnamed systematic risks that may correspond to named risk factors. Used by statistical arbitrage traders who are betting on exactly the component of an asset's return not explained by systematic risks.

6.1.3 Transaction Cost Models

Trade is made only if it increases the odds or magnitude of return (from alpha model), or if it decreases the odds or magnitudes of loss (from risk model). However, this improvement should be higher than cost of trading. The transaction cost model is not designed to minimise cost of trading, only to inform portfolio construction engine the cost of making any given trade.

Remark 6.1.9. *Transaction Cost Components*

i. Commissions and Fees: paid to brokerages (access to other market participants), exchanges (improved transaction security) and regulators (operational infrastructure) for the services provided. The bank's infrastructure is used by quants, where the brokerage commissions are rather small on a per-trade basis. Brokers also collect clearing and settlement fees. Clearing is the activity involving regulatory reporting and monitoring, tax handling, and handling failure, taken place in advance of settlement. Settlement is the delivery of securities in exchange for payment in full.

ii. Slippage: the change in price between the time the quant system decides to transact and the time when the order is at the exchange for execution. Trend-following strategies suffer most from slippage as assets are already moving in desired direction; mean-reverting strategies suffer the least from slippage. The lower the latency to market, the smaller the slippage. The more volatile an asset, the bigger the slippage.

iii. Market Impact: measures how much an order moves the market by its demand for liquidity. The impact of the trade on the market is unknown until the trade has already been completed. There may also be interaction between slippage and market impact (i.e., selling when a stock is trending upwards).

Definition 6.1.10. *Types of Transaction Cost Models*

i. Flat Model: cost of trading is the same, regardless of size of order. Model is reasonable if size traded is nearly always about the same, and liquidity remains sufficiently constant.

ii. Linear Model: cost of trading increases at a constant rate relative to size of order. Better estimate than flat transaction cost model.

iii. Piece-Wise Linear Model: using piece-wise linear functions to model costs. Balance between simplicity and accuracy; better accuracy than flat or linear models.

iv. Quadratic Model: most computationally intensive, but also most accurate.

6.1.4 Portfolio Construction Models

Comes in two major forms: rule-based, optimisers. Rule-based models are based on heuristics, can be exceedingly simple or rather complex, and derived from human experience (trial and error). Optimisers comprises of an objective function and uses algorithms to reach the end goal.

Definition 6.1.11. *Rule-Based Models*

i. Equal Position Weighting: used if portfolio manager believes that if a position is good enough to own, no other information is needed in determining its size. Strength of signal is not used as input in weighting. Model assumes that there is sufficient statistical strength and power to predict not only direction but also magnitude relative to other forecasts in the portfolio. Portfolio takes few large bets on 'best' forecast, many smaller bets on less dramatic forecasts; may take excess risk in an idiosyncratic event on a seemingly attractive position, resulting in adverse selection bias.

ii. Equal Risk Weighting: adjust position sizes inversely to volatilities or a measure of risk. More volatile positions given smaller allocations, less volatile positions given larger allocations.
When unit of risk is equalised, it is almost always a backward-looking measurement such as volatility. If volatility changes with time, then model will be misled.

iii. Alpha-Driven Weighting: position size based primarily on alpha model. Alpha signal determines size of position, but usually with size limits. Constraints used also includes limits on size of total bet on a group. May also have a function that relates the magnitude of forecast to size of position.
If model used in futures trend following, might suffer sharp drawdowns. Reliance on accuracy of alpha.

iv. Decision-Tree Weighting: decision path to arrive at the allocation for given instrument, depending on type of alpha model and type of instrument. Constraints may include percentage limits for allocation.
Model size grows dramatically if more alpha models or mode types of positions are included.

Remark 6.1.12. *Optimisers Models Parameters*
Harry Markowitz's mean variance optimisation (MVO) as the pioneer model. Models are based on principles of modern portfolio theory (MPT). Inputs include asset expected return (mean), asset variance, expected correlation matrix. Other inputs include size of portfolio in currency terms, desired risk level (volatility or expected drawdown), and other constraints such as liquidity, universe limits.
Model uses an objective function and an algorithm to seek the goal, usually maximising return of portfolio relative to volatility of portfolio returns.

i. Expected Return: alpha models as basis of expected return, which also includes expected direction.

ii. Expected Volatility: stochastic volatility forecasting methods is commonly used, as volatility may have high and low periods, with occasional jumps. GARCH model is most used.

iii. Expected Correlation: as instrument correlations are not stable over time, it is more appropriate to group assets together before computing correlation within the group.

Method 6.1.13. *Optimisation Techniques*

i. Unconstrained Optimisation: most basic form with no constraints. Might provide a single-instrument portfolio, where all money will be invested in instrument with highest risk-adjusted return.

ii. Constrained optimisation: constraints include position limits, limits on various groupings of instruments. Might result in constraints driving the portfolio construction more than the optimiser.

iii. Black-Litterman Optimisation: blends investor expectations with a degree of confidence about those expectations, and these with historical precedent evident in the data. Adjusts historically observed correlation levels by utilising investor's forecast of return for the various instruments.

iv. Grinold and Kahn's Approach: builds a portfolio of signals, instead of sizing positions. To build factor portfolios, each of which are usually rule-based portfolios based on a single type of alpha forecast. Each portfolio backtested, then series of returns are then treated as instruments of a portfolio by the optimiser. Number of factor portfolios is more manageable, usually not more than 20. What is optimised is then a handful of factor portfolios. The model allows for inclusion of risk model, transaction cost model, portfolio size, and risk targets as inputs.

v. Resampled Efficiency: to improve the inputs to optimisation by addressing oversensitivity to estimation error. To resample data using Monte Carlo simulation to reduce estimation error in inputs to the optimiser.

vi. Data-Mining Approaches: machine learning techniques such as supervised learning or genetic algorithms used, as MVO involves searching many possible portfolios to find the best.

6.1.5 Execution Model

Two basic ways to execute trade: through electronic, or through human intermediary. For electronic execution, achieved through direct market access (DMA), which allows traders to utilise the infrastructure and exchange connectivity of brokerage firms to trade directly on electronic markets.
Execution algorithms can be acquired through building, using broker's, or a third-party software vendors.
Brokerages offer portfolio bidding, where the 'blind' portfolio for transaction is described by characteristics such as valuation ratios of longs and shorts, sector breakdown, market capitalisation etc. Broker then quote a fee in basis points in terms of the gross market value of portfolio traded. Hence, certainty is provided by the broker to the trader. Once agreement reached, broker receives fee and assumes risk of trading out the portfolio at future market prices, which may be better or worse than prices guaranteed.

Remark 6.1.14. *Order Execution Algorithm Parameters*

i. Aggressive vs Passive: algorithm make decision of passive vs aggressive order, depending on how immediately the trader wants to do the trade. Market orders are considered aggressive. Limit order at current best order is fairly aggressive, while limit order below current bid is passive.
Many exchanges pay providers of liquidity for placing passive orders, charging traders for using liquidity provided. Orders that cross the spread are using liquidity by using a passive order placed by another trader, reducing liquidity available. Paying for liquidity sweetens deal for passive order, only if order is actually executed; passive trader gets better transaction price and a commission rebate from the exchange. Momentum strategies uses more aggressive orders; mean reversion uses more passive orders. A stronger, more certain signal will be executed with greater aggressiveness than a weaker or less certain signal. A middle ground will be to put limit orders between best current bid and offer.

ii. Large vs Small Order: a large order may be broken into many smaller orders over a window of time, but risk price moving in adverse direction. Size of chunk depends on transaction cost model estimate, and analysis of correct level of aggressiveness.

iii. Hidden vs Visible Order: a queue as a visible order gives away a bit of information. Hidden order will provide no information to the market, staving off imbalances, but reduces priority of trade in the queue. Algorithmic trading utilising hidden order is 'iceberging', which is taking a single larger order and chopping it into many smaller chunks, most posted to order book as hidden orders.

iv. Order Routing: if there are several pools of liquidity for the same instrument, smart order routing will be used, which determines which pool of liquidity is most suitable for sending a given order. Depth of liquidity on various ECNs and connectivity speeds are also considered in smart order routing.

v. Cancelling and Replacing Orders: traders may place larger number of orders with no intention of execution, then rapidly cancelling them and replacing them with other orders. This allows gaining of information on how market responds to the changing depth of the book, providing information on how to profit from the pattern of reaction. If trader wants to buy a large number of shares, he may enter a large number of small orders to sell the shares further away from market and cancel, improving market perception.

Definition 6.1.15. *High Frequency Trading*
Alpha driving strategies on extremely near-term bets (seconds or less) are *microstructure alphas*, focusing on liquidity patterns in order book. Larger quants may also use this to guide execution models, improving costs of entering trades. Small differences over a single trade add up significantly in the long run.To trade microstructure alpha as independent high frequency strategies, large investments in infrastructure and research must be done. Machine learning techniques may also be used to discern patterns in execution of other player orders. The more inferior the execution models, the easier it is to discern the pattern, allowing the ML strategy to profit from these patterns in the future. Patterns in the shorter timescale are somewhat stable.

Definition 6.1.16. *HFT Shark Strategy*
Designed to detect large orders that are iceberged, by sending series of very small trades; if each of these small orders get filled quickly, this may be a sign of a large and iceberged order. The shark simply front-run this large, hidden order by placing visible trades in front of the iceberged order. The iceberg strategy must then push prices up to execute trades. When the iceberged order is complete, prices will be pushed up favourably for the shark, which can then exit the position with a quick and relatively riskless profit.

Remark 6.1.17. *HFT Trading Infrastructure*
Using a broker that act as trading agent allows the infrastructure requirements to be handled by the broker, instead of dealing with the regulatory and other constraints.
High frequency strategies may use colocation or sponsored access. Colocation setup is where trader attempts to place trading servers as physically close to the exchange as possible.
Financial Information eXchange (FIX) protocol is the choice of real-time electronic communication among users. The software that implements the FIX protocol is free and open source (FIX engine). High frequency traders will likely build their own FIX engines to ensure optimal speeds.

6.1.6 Research

Definition 6.1.18. *Scientific Method*

1. Researcher observe a phenomenon in the market and construct a theory.
2. Researcher seeks out information to test the theory.
3. Researcher tests the theory, and with enough confidence, risk some capital on the validity of the theory.

Remark 6.1.19. *Sources of Alpha Idea Generation*

1. Observing the market, using the scientific method to test the theory
2. Academic literature, requiring significant time to read academic journals, working papers, and conference presentations for ideas. Literature from other fields such as astronomy, physics, or psychology, may provide ideas relevant to quant finance problems.
3. Migration of a researcher or portfolio manager from one quant shop to another.
4. Lessons from activities of discretionary traders

Remark 6.1.20. *Model Quality Assessment*

i. Cumulative profit graph: if profit profile is not smooth, with long periods of inactivity, sharp losses and gains, then the model may have issues
ii. Average annual rate of return: indicates how well the strategy made on historical data
iii. Variability of returns: the less variable the level of returns, the better the strategy. May look at lumpiness of returns, which is the portion of strategy's total returns that comes from periods that are significantly above average (measures consistency of returns).
iv. Worse Peak-to-Valley Drawdowns: measures maximum decline from any cumulative peak in profit curve. The lower the drawdown the better the strategy. Also, to measure recovery period after drawdowns; the shorter the recovery period the better the strategy.
v. Predictive Power: R-squared statistic may be used, which shows how much of the variability of the predicted asset have been accounted for. A exceedingly high R^2 in would be 0.05 out of sample. Instrument returns may be bucketed by deciles; a model with reliable predictive power is one that appropriately buckets the instruments correctly.
vi. Percentage Winning Trades, Winning Time Periods: whether the strategy tends to make profits from a small portion of trades that do very well, or from a large number of trades.
vii. Ratios of Returns vs Risk: Statistics such as risk-adjusted return, Sharpe ratio, information ratio, Sterling ratio, Calmer ratio, Omega ratio.
viii. Relationship with Other Strategies: value-add of new strategy compared with results of existing strategy with and without the new idea.
ix. Time decay: understand strategy returns if trades are initiated on lagged basis after receiving a trading signal. Determine strategy sensitivity to timeliness with information received, and crowdedness of strategy.
x. Sensitivity to specific parameters: high quality strategy has small changes in outcomes from slight changes in parameters. Or else this may be a sign that model may be overfitted.
xi. Overfitting: plot a graph of parameter value vs function outcome; a good model has a flatter curve with no jumps. Models that are parsimonious (less parameters) uses less assumptions, hence less overfitting.

Remark 6.1.21. *Other Considerations in Model Testing*
Overestimation of trading costs may cause portfolio to hold positions for longer than optimal, and underestimation may result in high portfolio turnover and bleed from trading costs. Assumptions on availability of short positions must also be made; hard-to-borrow lists must be taken into consideration.

6.1.7 Risk Assessment

Definition 6.1.22. *Model Risks*
Quant models has model risk, the risk that the model does not accurately describe, match, or predict the real-world phenomenon. Each component of the quant model may all have model risk.

i. Inapplicability of Modelling: occurs when quant model is mistakenly applied to a problem. May also occur with misapplication of a technique to a given problem.

ii. Model Misspecification: occurs when the model doesn't fit the real world. Model may work fine most of the time, but fail when an extreme event occurs.

iii. Implementation Errors: erors in programming or architecting systems. Architectural error may also occur when models are loaded in a wrong sequence.

Definition 6.1.23. *Regime Change Risk*
Quant models are based on relationships prevalent in historical data. If there is a regime change, the historical relationships and behaviour may be altered, hence the model may lose effectiveness.

Definition 6.1.24. *Exogenous Shock Risk*
Risks driven by information that is not internal to the market, i.e., terrorist attacks, start of wars, bank bailouts, change in regulation such as in shorting rules. May require discretionary overrides.

Definition 6.1.25. *Contagion Risk*
Happens when other investors hold the same strategies. First part of risk factor relates to how crowded the quant strategy is. Second part relates to what else is held by other investors that could force them to exit the quant strategy in a panic (ATM effect).
Quant liquidation criss may be driven by size and popularity of quantitative strategies, subpar returns from operators leading up to the crisis, the practice of funds cross-collateralising many strategies against each other, and risk targeting (risk managers target a specific level of volatility for their funds or strategies).

Method 6.1.26. *Risk Monitoring Methods*

i. Exposure Monitoring Tools: with current positions held, the positions are grouped for the various exposures (i.e., valuation, momentum level, volatility) to monitor gross and net exposure to various sectors and industries, buckets of market capitalisation, various style factors.

ii. Profit and Loss Monitors: with current portfolio, compare that with previous day closing price. Intraday performance charts are used. May also look at source of profit, hit rate (percentage of time strategy makes money on a given position).

iii. Execution Monitors: shows progress of executions, i.e., which orders are currently being worked on, which ones are completed, with transaction size and prices. Fill rates for limit orders are used for more passive execution strategies. Slippage and market impact are also monitored.

iv. System Performance Monitors: checks for software and infrastructure errors. Checks performance of CPUs, speeds of various stages of automated processes, latency in communication of messages.

Chapter 7

Market microstructure and market making

Notes awaiting migration. Entries already migrated from this chapter are in Chapter 1; the record of each block's destination is in `notes/migration.md`.

Market microstructure studies how trading rules, intermediaries, and order flow transform latent trading demands into transactions, prices, and allocations. The central departure from frictionless asset pricing is that the trading mechanism is not neutral: priority rules, transparency, inventory, information, and immediacy affect the prices at which trades occur.

This chapter treats a market as an engineered trading system rather than a single supply-demand crossing. The practical questions are: who wants immediacy, who absorbs the other side, what information is revealed by the order, how the venue ranks competing messages, and how much of the final execution cost comes from spread, market impact, delay, adverse selection, and operational frictions.

The source base is front-loaded here to avoid repeated citations throughout the section. Demsetz provides the canonical transaction-cost view of why immediacy has a price [demsetz-1968]. O'Hara and Madhavan frame the theoretical links among information, strategic trading, dealer behavior, and price discovery [catalogue-0496]; [madhavan-2000]. Harris supplies the practitioner taxonomy of traders, orders, markets, and trading costs [alpha-harris]. Hasbrouck anchors the empirical treatment of trades, quotes, order flow, and high-frequency data [s6-hasbrouck]. Lehalle and Laruelle connect these ideas to fragmented electronic markets, smart routing, latency, and modern execution practice [alpha-laruelle].

The section therefore moves from market organization to order mechanics, then to information and spreads, empirical order-flow analysis, and market making; algorithmic execution and asset-class-specific microstructure follow in Chapter 8. The common thread is implementation: every model is useful only insofar as it explains how real orders become real fills under a particular market design.

7.1 Foundations of Trading and Market Organization

7.1.1 Roles of Financial Markets

Principle 7.1.1. *Core Market Functions*
A financial market is useful when it performs four linked functions:

i. *liquidity provision*: converts desired portfolio changes into executable trades;

ii. *price discovery*: aggregates dispersed beliefs and information into observable prices;

iii. *risk transfer*: reallocates exposures to agents more willing or able to bear them;

iv. *capital allocation*: directs savings toward issuers and projects with attractive risk-adjusted uses of capital.

Microstructure studies the frictions inside these functions rather than assuming they occur automatically.

Principle 7.1.2. *Immediacy Has a Price*
Asynchronous order arrival makes immediacy costly. The bid-ask spread compensates traders who absorb inventory, funding, processing, and adverse-selection risks.

Proposition 7.1.3. *Zero-Sum Trading Accounting*
For a trade at P with common terminal value V, buyer and seller gains sum to zero: $(V - P) + (P - V) = 0$. Frictions make trading negative-sum unless it creates information, liquidity, risk-transfer, or allocation value.

Proof. Buyer gain plus seller opportunity gain is zero before costs; costs subtract. □

Taxonomy 7.1.4. *Trading Benefit Segmentation*

i. *private benefits*: accrue to traders through exchange, hedging, funding, immediacy, or speculation;

ii. *public benefits*: accrue broadly when liquid markets and informative prices improve allocation decisions.

Definition 7.1.5. *Liquidity Provision*
Liquidity is the ability to trade size quickly, cheaply, and with limited price impact.
Supplied by limit orders, dealers, market makers, internalisers, and crossing systems, and consumed by marketable orders and urgent traders.

Mechanism 7.1.6. *Price Discovery*
Trades, quotes, cancellations, and depth transmit information about value, urgency, and risk-bearing capacity. Price discovery is strong when prices update quickly without being dominated by transient imbalances.

Mechanism 7.1.7. *Risk Transfer*
Markets separate cash-flow ownership from risk-bearing. Hedgers shed unwanted exposure; speculators, dealers, and market makers accept it for compensation.

Principle 7.1.8. *Capital Allocation*
Primary markets move capital from savers to issuers. Liquid and informative secondary markets provide exit prices, lower financing costs, and discipline poor uses of capital.

Mechanism 7.1.9. *Informative Prices and Allocation*
Primary prices screen new projects; secondary prices help investors judge managers, compensate performance, and reallocate control. Noisy prices weaken all three channels.

Principle 7.1.10. *Mechanism Relevance*
In Walrasian abstraction, the mechanism disappears after supply and demand clear. In real markets, sequencing, asymmetric information, costly immediacy, and priority rules make the mechanism part of price formation.

Caveat 7.1.11. *Microstructure Trade-Off*
Market design rarely improves every function at once. Transparency, spread size, venue opacity, and priority rules trade off liquidity provision, execution cost, information leakage, and public price discovery.

7.1.2 Market Participants and Trading Motives

Model 7.1.12. *Trading Game View*
A trading mechanism is a game in which participants submit orders under venue rules. The players include customers, brokers, dealers, market makers, and traders acting for their own account. Some demand liquidity, some supply liquidity, some carry information, and some process orders.
Intuition: a trader's best order depends on what others may know, how fast they can react, and how the venue ranks competing messages.

Notation 7.1.13. *Long and Short Positions*
A trader with a *long position* owns the asset and benefits when its price rises. A trader with a *short position* has sold an asset not owned, or has equivalent negative exposure, and benefits when its price falls. Closing a short position is called *covering*.

Taxonomy 7.1.14. *Motive Segmentation*

i. *utilitarian*: solve an outside problem;

ii. *profit-motivated*: expect trading profits above costs;

iii. *futile*: expect profit without enough information, skill, discipline, or agency control.

Motive is inferred from context, horizon, and exposure, not order side alone.

Caveat 7.1.15. *Multiple Trader Identities*
Traders often combine motives. A pension fund may invest, hedge FX, and time factors; an HFT may make markets in one book and arbitrage another.

Principle 7.1.16. *Patience Defines the Liquidity Side*
Impatient traders demand liquidity by initiating trades. Patient traders supply liquidity by waiting, posting, quoting, or making themselves discoverable.
The same institution can demand liquidity for one order and supply it for another.

Taxonomy 7.1.17. *Buy-Side Motives*
Buy-side traders use markets to solve problems outside the market itself:

i. investors move wealth across time;

ii. borrowers move wealth from the future to the present;

iii. hedgers reduce business or portfolio risk;

iv. asset exchangers acquire assets they value more than what they tender;

v. gamblers trade for entertainment or payoff convexity rather than investment efficiency.

These traders buy liquidity services rather than sell them.

Principle 7.1.18. *Instrument Choice Follows the Problem*
Investors and borrowers care about after-cost return and funding access. Asset exchangers need the right asset. Hedgers need correlation, maturity, and liquidity. Gamblers care about payoff shape.

Taxonomy 7.1.19. *Hedge-Vehicle Fit*

i. *forwards*: customise asset, date, size, and delivery, but create search, credit, and exit frictions;

ii. *futures*: standardise terms and clearing, improving liquidity while introducing basis and delivery mismatch;

iii. *options*: cap downside for a premium, preserving convex upside or nonlinear exposure;

iv. *swaps*: exchange cash-flow risks over time, especially interest-rate, currency, credit, or inflation exposures.

The best hedge removes the intended risk without adding larger liquidity, basis, or counterparty risks.

Taxonomy 7.1.20. *Information-Motivated Traders*
Informed traders trade when prices lag information. Value traders use fundamentals; news traders use recent information; arbitrageurs use relative mispricing.
Their orders improve prices only as the market absorbs the information.

Taxonomy 7.1.21. *Informed Trader Liquidity Role Segmentation*

i. *value traders*: supply liquidity when uninformed demand pushes price away from fundamental value;

ii. *news traders*: demand liquidity when new information has not yet been incorporated into price;

iii. *arbitrageurs*: supply or demand liquidity across related instruments to close relative mispricing.

The same trader can switch roles. The useful distinction is whether the order corrects stale prices, absorbs non-value flow, or links related prices.

Mechanism 7.1.22. *Arbitrageur Economic Function*
Arbitrageurs trade relative value. They connect buyers and sellers across instruments, venues, maturities, currencies, or risk forms. Their activity enforces the law of one price only after costs, funding, hedge error, and execution risk are small enough for the trade to be worth doing.

Taxonomy 7.1.23. *Liquidity Supplier Segmentation*

i. *market makers*: supply immediacy for small to moderate size at narrow spreads, but lose to informed flow and adverse inventory moves;

ii. *block dealers*: supply depth to known clients by warehousing risk, but lose when client information or unwind cost is misread;

iii. *value traders*: supply depth and resiliency when price deviates from fundamental value, but lose to better-informed value traders or financing pressure;

iv. *precommitted traders*: post aggressive limit orders to lower the cost of trades they already want to do, but risk non-execution or adverse selection;

v. *arbitrageurs*: move liquidity across related markets, but lose to faster arbitrageurs, legging risk, funding shocks, and stale hedges.

Treating all passive flow as "market making" hides the source of depth.

Taxonomy 7.1.24. *Profit-Motivated Trader Segmentation*

i. *informed speculators*: profit when prices move toward information they already understand;

ii. *order anticipators*: profit by predicting trades other participants must make;

iii. *bluffers and manipulators*: profit only if others misread false signals;

iv. *dealers*: sell immediacy and inventory capacity rather than directional views.

Informed speculators aid price discovery; dealers supply liquidity. Middle categories mainly redistribute surplus.

Principle 7.1.25. *Speculation Requires Comparative Advantage*
Profitable speculation requires more than being intelligent or informed in isolation. The trader must have an advantage over the marginal counterparty after spread, fees, financing, market impact, and operational error.

Taxonomy 7.1.26. *Trading Edge Segmentation*

i. *information*: faster, cleaner, or more exclusive interpretation of value-relevant facts;

ii. *model*: better mapping from data to price, hedge, or risk;

iii. *execution*: lower leakage, better routing, queue skill, or lower impact;

iv. *risk bearing*: ability to hold inventory or convergence risk when others cannot;

v. *discipline*: sizing, stopping, and avoiding trades outside the edge.

An edge that cannot survive trading costs is not a trading edge.

Caveat 7.1.27. *Skill and Luck Look Similar Ex Post*
Large realised profits may come from skill, favourable exposure, or simple luck.
A useful skill claim must explain why the decision was right before the outcome was known, why the edge should persist, and why it is not already competed away.

Proposition 7.1.28. *Past Profit Alone Does Not Identify Skill*
Let future excess profit be $\Pi_{t+1} = S_{t+1} + L_{t+1}$, where S is persistent skill and L is zero-mean luck.
If S_{t+1} is independent of past realised profits conditional on current information, then past profit has no incremental predictive value for Π_{t+1}.

Proof. Conditional on current information $\mathcal{F}_t$, $\mathbb{E}[\Pi_{t+1} \mid \mathcal{F}_t, \Pi_t] = \mathbb{E}[S_{t+1} \mid \mathcal{F}_t] + \mathbb{E}[L_{t+1} \mid \mathcal{F}_t] = \mathbb{E}[\Pi_{t+1} \mid \mathcal{F}_t]$. □

Corollary 7.1.29. *Profit-Motivated Flow Needs Non-Profit Motives*
Speculators and dealers cannot all profit by trading only with each other. Their gains require traders who pay for immediacy, risk transfer, information, or entertainment.

Proof. Let G_i be trader i's gross trading gain in a closed set of profit-motivated traders, and let $C_i \geq 0$ be trading costs. By zero-sum accounting, $\sum_i G_i = 0$, so aggregate net profit is $\sum_i (G_i - C_i) = -\sum_i C_i \leq 0$.
Thus the group cannot earn positive aggregate net trading profit from itself; positive profits require counterparties with offsetting trading losses or non-trading service value. □

Corollary 7.1.30. *Turnover Drag for Active Management*
Let R_M be the gross market return, a_i active manager i's gross active excess return, $c_i \geq 0$ the manager's trading and fee drag, and w_i value weights. If active managers aggregate to the market portfolio, $\sum_i w_i a_i = 0$, so aggregate net active excess return is $\sum_i w_i(a_i - c_i) = -\sum_i w_i c_i \leq 0$.
A single active manager beats a passive fund with drag c_P only if $a_i > c_i - c_P$.

Proof. Active deviations clear against each other in aggregate, giving $\sum_i w_i a_i = 0$. Subtracting costs gives the aggregate inequality. Relative to passive exposure, manager i must cover the incremental drag $c_i - c_P$. □

Definition 7.1.31. *Noise and Liquidity Traders*
Noise traders trade for non-value reasons: rebalancing, cash needs, index changes, behavioural demand, or errors. They mask informed flow and give liquidity suppliers less toxic orders.

Taxonomy 7.1.32. *Futile Trader Segmentation*

i. *pseudo-informed*: trade stale or already-public information as if it were private;

ii. *inefficient*: have a plausible strategy but execute, size, or control risk poorly;

iii. *victimised*: rely on advisers, brokers, or agents whose incentives are misaligned;

iv. *rogue*: hide losses or risks because internal controls fail.

Futile traders matter because their losses can fund informed traders and dealers, while their distress can still damage clients, firms, or market confidence.

Mechanism 7.1.33. *Dealers and Market Makers*
Dealers trade for their own account. Market makers quote prices or otherwise stand ready to trade, often holding inventory only until an offsetting order arrives. Their compensation comes from spreads, rebates, information, or client relationships; their risks are inventory, adverse selection, funding, and operational failure. They are paid for immediacy: the client trades now, while the market maker manages the uncertainty of unwinding later.

Taxonomy 7.1.34. *Sell-Side Intermediaries*
Brokers arrange trades for clients without intending to own the position. Dealers trade as principal. Broker-dealers combine both functions, sometimes arranging agency trades and sometimes committing their own capital. The sell side exists because buy-side traders pay for search, immediacy, execution expertise, clearing access, and balance-sheet intermediation.

Taxonomy 7.1.35. *Broker Service Segmentation*

i. *search*: find counterparties, dealers, venues, or natural block liquidity;
ii. *representation*: submit, monitor, and negotiate client orders;
iii. *access*: connect clients to exchanges, dealers, ECNs, and clearing members;
iv. *post-trade*: manage confirmation, clearing, settlement, custody, margin, and records;
v. *information control*: expose enough order interest to trade without revealing the full parent order.

Taxonomy 7.1.36. *Block-Trading Role Segmentation*

i. *block initiator*: trader with a parent order too large for normal displayed liquidity;
ii. *block liquidity supplier*: trader, dealer, or institution willing to fill the block;
iii. *block dealer*: principal that facilitates the block and then unwinds inventory;
iv. *block broker*: agent that assembles counterparties without intending to own the risk;
v. *upstairs market*: relationship and telephone market used to find size without public exposure.

Block trading is mostly a search and information-control problem, not a different asset.

Principle 7.1.37. *Block Dealers vs Block Brokers*
Dealers are useful when immediacy matters and balance sheet can absorb the position.
Brokers are useful when the block is too large or information-sensitive for one principal, so assembling natural counterparties is cheaper than warehousing risk.
Broker-dealers can be credible when they are willing to participate in the trade they recommend.

Taxonomy 7.1.38. *Block Trader Skill Segmentation*

i. *latent demand discovery*: know who might trade if contacted;
ii. *motive audit*: infer whether the initiator is uninformed, informed, honest, or strategic;
iii. *exposure control*: reveal size only to counterparties who will not trade ahead;
iv. *transaction-cost analysis*: price facilitation, unwind risk, and likely market impact;
v. *reputation management*: make future counterparties willing to participate.

Caveat 7.1.39. *Broker Agency Conflict*
Brokers are agents, but commissions, soft dollars, payments for order flow, inventory, and affiliated dealers can pull routing decisions away from pure client execution quality.

Practitioner Note 7.1.40. *Institutional Investors and HFTs*
Institutional investors often trade large parent orders that must be hidden, sliced, or negotiated to reduce market impact. High-frequency traders use speed, automation, and short holding periods; they may act as market makers, arbitrageurs, statistical traders, or order anticipators. Their role depends on strategy.

Taxonomy 7.1.41. *HFT Identification Markers*

i. automated order generation, routing, execution, and cancellation;
ii. low-latency infrastructure, co-location, and direct market-data feeds;
iii. short holding periods and low overnight inventory;
iv. high message counts relative to executed trades;
v. market making, cross-venue arbitrage, or very short-horizon statistical trading.

These markers describe technology and horizon, not whether the strategy is beneficial.

Mechanism 7.1.42. *HFT as Fragmentation Link*
Fast market makers connect fragmented venues by quoting related prices, arbitraging stale quotes, and hedging inventory across books. Their compensation is the spread, rebates, and short-lived information advantages; their cost is technology, inventory risk, and adverse selection. They act like moving bridges between books: useful when they keep prices aligned, fragile when those bridges are cancelled under stress.

Caveat 7.1.43. *Tighter Spread, Different Cost*
An HFT can improve the displayed spread by stepping inside an older quote, but investor execution cost may not fall by the same amount. Smaller trades, more prints, and higher routing complexity can shift fixed and reporting costs toward slower investors.

Caveat 7.1.44. *Investor Size Asymmetry*
The uncertainty created by meeting a fast counterparty is not borne equally. A small investor sees high variance in a small number of trades; a large institution diversifies this uncertainty across many child orders but still pays aggregate spread, impact, and signalling costs.

Caveat 7.1.45. *Blurring of Roles*
Fragmentation blurs the distinction between liquidity takers and providers. The same agent may consume liquidity in one venue, supply it in another, and intermediate price formation through routing speed.

7.1.3 Trading Venues and Market Architecture

Principle 7.1.46. *Venue as Rule System*
A market is a rule system that governs access, order submission, visibility, execution, clearing, and settlement.

Taxonomy 7.1.47. *Market-Structure Segmentation*

i. *session*: when orders may enter and trades may occur;
ii. *execution*: who arranges the match and which rule prices the trade;
iii. *information*: who observes orders, quotes, trades, and when;
iv. *access*: how traders send orders, cancels, and reports;
v. *price grid*: the tick sizes and admissible price points.

Market structure matters because it defines both tradability and information access.

Taxonomy 7.1.48. *Access Form Segmentation*

i. *physically convened*: traders meet on a floor or pit;
ii. *distributed*: traders negotiate by phone, message, or bilateral networks;
iii. *screen-based*: orders, quotes, and trades are submitted through electronic interfaces.

Taxonomy 7.1.49. *Market-Centre Segmentation*

i. *market centre*: any venue, dealer, broker system, or crossing mechanism that can arrange trades;
ii. *consolidated market*: traders for same instrument meet on one dominant centre or tightly linked book;
iii. *fragmented market*: the same instrument trades across several centres with shared information or routing;
iv. *segmented market*: traders select centres because order size, information, patience, access, or credit differs.

Fragmentation is not automatically bad; it is costly when the links among centres fail.

Definition 7.1.50. *Exchanges*
Exchanges centralise order interaction under membership, listing, trading, reporting, and clearing rules. They standardise access and make execution more reliable.

Mechanism 7.1.51. *Exchanges as Search Engines*
An exchange lowers bilateral search cost by collecting trading interest, ranking it under common rules, and publishing enough information for traders to find the best available counterparty.
Its economic service is not only matching; it is making liquidity searchable.

Definition 7.1.52. *OTC Markets*
OTC markets use bilateral or dealer-mediated negotiation rather than a central book. They fit bonds, swaps, FX, bespoke derivatives, and heterogeneous contracts.

Taxonomy 7.1.53. *Electronic Communication Networks (ECNs) and Alternative Trading Systems (ATSs)*
ECNs organise electronic matching outside traditional exchange structures. ATSs include dark pools, crossing systems, and specialised venues competing on anonymity, fees, block handling, or lower information leakage.

Taxonomy 7.1.54. *Dark Pools, Internalizers, and Crossing Networks*
Dark pools hide pre-trade interest; internalizers match against broker-dealer inventory or internal flow; crossing networks match natural buyers and sellers, often at lit-market reference prices. These venues reduce signalling but can weaken public price discovery.

Taxonomy 7.1.55. *Order-Flow Arrangement Segmentation*

i. *internalisation*: broker-dealer fills client orders as principal or against affiliated liquidity;
ii. *preferencing*: broker routes orders to a selected dealer or venue by arrangement;
iii. *payment for order flow*: liquidity supplier pays broker for routed orders;
iv. *internal crossing*: broker matches client orders against each other away from public books.

These arrangements lower search and commission costs when flow is uninformed, but fragment order interaction and create best-execution conflicts.

Taxonomy 7.1.56. *Brokered Transaction Segmentation*

i. *order flow*: broker receives client orders and matches, routes, or exposes them;
ii. *block*: broker searches quietly for the other side of a large trade;
iii. *offering*: broker distributes new or seasoned securities to investors;
iv. *corporate-control*: broker advises or introduces parties in merger and acquisition trades.

Taxonomy 7.1.57. *Request for Quotation (RFQ) vs Central Limit Order Book (CLOB)*
RFQ markets request quotes from dealers, often bilaterally. A CLOB exposes standing interest multilaterally; RFQ protects relationships and discretion, while CLOBs improve comparability and public interaction.

Taxonomy 7.1.58. *Designated Market Maker Segmentation*

i. *single specialist*: one member carries quoting, auction, and order-book duties for an instrument;
ii. *multiple DMM*: several assigned dealers share liquidity obligations and compete for flow;
iii. *third-market dealer*: quotes away from the listing exchange, usually with weaker formal duties;
iv. *pure CLOB*: no assigned dealer; public orders and voluntary market makers supply liquidity.

Design choice is whether mandatory liquidity support is worth granting privileged access or order-flow rights.

Taxonomy 7.1.59. *Market Information Systems*

i. *collection*: capture orders, quotes, trades, and participant reports;
ii. *distribution*: publish or sell feeds, tapes, and reference data;
iii. *routing*: transmit orders, cancels, acknowledgements, and fills;
iv. *presentation*: display books, quotes, trades, indications, and alerts;
v. *storage*: preserve audit trails and historical market data.

Mechanism 7.1.60. *Trading Systems and Technology*
Communications technology lowers the cost of knowing where liquidity is; computing lowers the cost of acting on it. Modern trading systems therefore compete not only on matching rules, but on market-data quality, order-routing speed, conditional order handling, and auditability.

Mechanism 7.1.61. *Automated Reporting Advantage*
When orders and trades are born digital, quotes, executions, order status, and timestamps can be reported faster and with fewer transcription errors. Floor systems need manual entry or support systems to reach comparable audit and market-data quality.

Caveat 7.1.62. *Stale Quotes and Vanishing Liquidity*
RFQ quotes can become stale before acceptance; displayed CLOB liquidity can vanish before an order arrives. In fragmented electronic markets, observed liquidity is an estimate, not a guarantee.

Process 7.1.63. *Post-Trade Infrastructure*
Trading does not end at execution:

i. *clearing* matches buyer and seller records and validates trade terms;

ii. *settlement* exchanges cash and securities;

iii. *custody* holds assets on behalf of clients;

iv. *depositories* immobilise or dematerialise securities so ownership can transfer by book entry.

Clearinghouses may become buyer to every seller and seller to every buyer, reducing counterparty risk through margin and default management.

Proposition 7.1.64. *Brokered Credit Compression*
With n traders, direct bilateral trading can require $n(n-1)/2$ credit relationships.
If clients trade through B clearing brokers, the relationship count falls to $n + B(B-1)/2$ when each client relies on one broker and brokers check each other.

Proof. The direct market must check every unordered trader pair. In the brokered market, each client checks one broker, and the remaining checks are among broker pairs. □

Practitioner Note 7.1.65. *Architecture Choice*
Venue choice depends on order size, urgency, information sensitivity, instrument standardisation, and counterparty risk. Small urgent orders favour visible and immediate venues; large or information-sensitive orders often favour dark, negotiated, or sliced execution.

Mechanism 7.1.66. *Liquidity Bridges*
Fragmented venues require routing, arbitrage, and market-making links between books. These links reduce price dispersion, but their compensation is paid through spreads, fees, latency races, or information leakage.

Practitioner Note 7.1.67. *Straight-Through Processing*
Straight-through processing sends trade details from execution to clearing, settlement, custody, and reporting with minimal manual intervention.

7.1.4 Auction and Dealer Market Structures

Taxonomy 7.1.68. *Execution-System Segmentation*

i. *quote-driven*: dealers supply liquidity through tradable quotes and balance sheets;

ii. *order-driven*: rules match public orders under priority and pricing rules;

iii. *brokered*: brokers search for counterparties and reveal interest selectively;

iv. *hybrid*: venue rules combine books, dealers, auctions, and negotiation.

Taxonomy 7.1.69. *Trading Session Segmentation*

i. *continuous*: trades can occur whenever compatible orders arrive;

ii. *call*: orders batch until a scheduled or triggered uncrossing;

iii. *rotation*: instruments are called sequentially;

iv. *extended*: trading occurs outside the regular session, usually with thinner liquidity.

Mechanism 7.1.70. *Open-Outcry Oral Auctions*
Open-outcry markets arrange trades through public bids, offers, and acceptances on a floor or pit. The economic idea is still useful electronically: trading interest should be exposed enough for competing traders to react, while the rule book decides whose interest has precedence.

Rule 7.1.71. *Public Exposure Rule*
An auction is open when bids, offers, and acceptances are announced to the crowd before a trade is final. Public exposure supports fairness and price competition, but it also reveals trading interest.

Taxonomy 7.1.72. *Floor vs Automated Auction Segmentation*

i. *operational fairness*: automated rules are easier to apply uniformly and audit;

ii. *fair access*: floors privilege physical presence; screens distribute access through networks;

iii. *capacity*: oral auctions are bounded by human attention; matching engines scale with system capacity;

iv. *speed*: electronic systems record and match standard orders faster;

v. *information exchange*: floor brokers can exchange soft information useful for blocks;

vi. *cost*: floor systems carry labour and real-estate costs; automated systems carry fixed technology and reliability costs.

Neither form dominates every use case; the valuable question is which trading problem the structure solves.

Proposition 7.1.73. *Fast-Market Capacity Condition*
If N traders submit messages at average rate λ and the auction mechanism processes messages at rate μ, orderly real-time handling requires $N\lambda < \mu$.

Proof. The aggregate arrival rate is $N\lambda$. When arrivals exceed processing capacity, the backlog grows on average, so quotes, order reports, and time precedence become unreliable. □

Taxonomy 7.1.74. *Order-Driven Rule Stack*

i. *precedence rules*: rank orders when more than one trader wants the same trade;
ii. *matching rules*: decide which buy and sell orders can trade;
iii. *pricing rules*: determine the transaction price once a match is feasible;
iv. *access rules*: decide who may submit, observe, cancel, or route orders.

Mechanism 7.1.75. *Continuous Auctions*
Orders execute whenever they arrive and match resting interest. Continuous trading provides immediacy but exposes prices to asynchronous order-flow imbalances.

Mechanism 7.1.76. *Call Auctions*
Orders accumulate, then execute at a single clearing price. Opening and closing calls concentrate liquidity when uncertainty and imbalance are high.

Mechanism 7.1.77. *Fixing Auctions*
Fixing auctions are call auctions used to create reference prices.
They are valuable when investors need one public price for index valuation, NAV computation, derivative settlement, opening risk transfer, or end-of-day inventory adjustment.

Principle 7.1.78. *Reference Price Monopoly*
Even when continuous trading fragments, primary exchanges often retain the auction flow that defines official reference prices. This makes auction volume structurally different from continuous-session volume and gives the primary market a separate source of liquidity concentration.

Taxonomy 7.1.79. *Auction Session Segmentation*

i. *opening call*: rebuilds the book after overnight information and sets the first tradable price;
ii. *intraday call*: concentrates liquidity around scheduled events or low-liquidity periods;
iii. *closing call*: sets the end-of-day reference price and absorbs benchmark-driven flow;
iv. *volatility call*: pauses continuous trading when price moves or imbalances breach venue rules.

Mechanism 7.1.80. *Dealer Markets*
Prices form through dealer quotes rather than a single public book. Dealers absorb timing gaps, manage inventory, and revise quotes as information and inventory change.

Mechanism 7.1.81. *Brokered Markets*
Brokers create liquidity by search rather than inventory commitment. This fits blocks, unique assets, and illiquid instruments where public standing orders would reveal too much or attract too little interest.

Caveat 7.1.82. *Floor Information Exchange*
Floor and brokered systems can reveal who might trade, whether more size remains, and whether the contra side appears informed. This helps blocks and relationship trading, but weakens equal access and creates audit, exposure, and allocation risk.

Taxonomy 7.1.83. *Hybrid Markets*
Hybrid markets combine order books, dealers, auctions, hidden orders, and off-book negotiation. They mix public order interaction with intermediary support.

Taxonomy 7.1.84. *Specialist Role Segmentation*

i. *dealer*: trades as principal when public liquidity is absent or imbalanced;
ii. *broker*: represents routed orders and searches for contra-side interest;

iii. *auctioneer*: conducts openings, rotations, and imbalance-resolution calls;

iv. *exchange official*: helps maintain orderly markets and disseminated quotes.

The same agent may protect public orders, trade against them, and observe order flow first.

Rule 7.1.85. *Priority and Access Rules*
Priority, tick size, order type, display, halt, and access rules determine how trades are resolved. The same latent supply and demand can produce different prices, quantities, and costs under different rules.

Case Study 7.1.86. *Gold Fixing as Sequential Auction*
The London gold fixing resembles a call process: participants submit net interest, revise after tentative prices, and stop when net demand and supply balance. It produces a clearing price, not continuous immediacy.

Practitioner Note 7.1.87. *Why Large Orders Like Calls*
Calls reduce the need to reveal urgency sequentially. A large order can interact with accumulated opposite interest at one price, often with less signalling than chasing liquidity through the continuous book.

Caveat 7.1.88. *Auction Transparency Trade-Off*
Pre-auction imbalance messages help traders converge to a clearing price, but they can also invite probing, cancellation, and last-second repricing. More transparency improves price discovery only when participants cannot cheaply game the signal.

7.1.5 Market Quality and Efficiency

Definition 7.1.89. *Market Quality*
Market quality measures how well trading interest becomes low-cost execution and informative prices. Include liquidity, tightness, depth, resiliency, transparency, volatility control, and informational efficiency.

Definition 7.1.90. *Liquidity*
Liquidity is the ability to trade quickly, in size, at low cost, and with limited price impact. Spreads, depth, volume, turnover, fill rates, and price impact each capture only one face of it.

Mechanism 7.1.91. *Liquidity as Bilateral Search*
Liquidity is the object of a bilateral search: buyers search for sellers, sellers search for buyers, and both sides decide how much interest to reveal. Markets, brokers, dealers, and routers are useful when they reduce this search cost without leaking too much information.

Proposition 7.1.92. *Search Stopping Rule*
Let B be the expected improvement from one more search attempt, c the direct cost of searching, and L the expected loss from the current best match disappearing while the trader searches. Continue searching iff $B > c+L$.

Proof. The incremental expected gain from another search is the improvement B less the search cost c and lost-option cost L. Searching is optimal exactly when this net gain is positive. □

Taxonomy 7.1.93. *Liquidity Search Strategy Segmentation*

i. *active search*: cross the spread, call dealers, expose interest, or route aggressively;

ii. *passive search*: post limits, show indications, wait for contra-side interest, or reveal only to brokers;

iii. *selective search*: ask trusted counterparties first, then broaden exposure if cost remains acceptable.

Active search buys speed with leakage. Passive search buys discretion with non-execution risk.

Mechanism 7.1.94. *Public Benefits of Liquid Markets*
Liquid markets lower the cost of exchange, hedging, risk sharing, and informed trading. They can benefit even non-traders through cheaper goods, lower capital costs, and better prices.

Taxonomy 7.1.95. *Four Liquidity Variables*

i. *traded volume*: realised liquidity, where traders actually agreed to trade;

ii. *quoted quantity*: offered liquidity at the best limits or across the visible book;

iii. *bid-ask spread*: price of immediacy for small trades;

iv. *volatility*: uncertainty faced by liquidity providers and liquidity takers.

Past volume is not enough: a market can trade actively and still have poor offered liquidity for the next order.

Definition 7.1.96. *Liquidity Risk*
Liquidity risk is uncertainty about the cost of turning a desired position into an executed trade. It includes spread risk, depth risk, market-impact risk, timing risk, and the risk that displayed liquidity disappears.

Taxonomy 7.1.97. *Tightness, Depth, and Resiliency*
Tightness is cost of an immediate small round trip, proxied by bid-ask spread. Depth is the quantity available at or near current prices. Resiliency is the speed with which liquidity returns after a shock or large trade.

Definition 7.1.98. *Price Continuity*
Price continuity is the absence of unnecessary jumps and reversals for ordinary order flow.
It is useful when jumps reflect missing liquidity rather than new information; forcing continuity against informed flow only transfers losses to liquidity suppliers.

Taxonomy 7.1.99. *Search-Output Liquidity Dimensions*

i. *immediacy*: time required to arrange the trade at acceptable terms;
ii. *width*: cost of trading a given size, including spread, fees, and price concession;
iii. *depth*: size available at a given cost or price concession;
iv. *resiliency*: speed with which prices and depth recover after order-flow shocks.

For small trades, width is often the spread. For large trades, depth and information leakage dominate.

Proposition 7.1.100. *Depth-Width Duality*
For a fixed horizon, let $C(q)$ be the nondecreasing all-in cost of trading size q, and let $D(c) = \max\{q : C(q) \leq c\}$ be executable depth at cost budget c. Then $q \leq D(c)$ iff $C(q) \leq c$.

Proof. If $C(q) \leq c$, then q is feasible under the cost budget, so $q \leq D(c)$.
If $q \leq D(c)$, monotonicity gives $C(q) \leq C(D(c)) \leq c$. □

Taxonomy 7.1.101. *Displayed, Undisclosed, and Latent Liquidity*

i. *displayed*: firm quotes and exposed limit orders that can be taken immediately;
ii. *reserve*: hidden size behind a displayed slice or venue-level hidden interest;
iii. *indicated*: IOIs, RFQs, dealer axes, or broker-distributed interest that must be asked for;
iv. *latent*: buy-side, value-trader, or dealer interest that appears only after contact, price movement, or relationship search.

Displayed liquidity is easiest to measure; latent liquidity is often decisive for blocks.

Definition 7.1.102. *Book Size*
Book size is the quantity available at the best bid and ask, or within a specified number of price levels.
For small-tick instruments, several limits must often be summed; for large-tick instruments, the first limits may capture most executable size.

Caveat 7.1.103. *Book Size Is a Lower Bound*
Displayed book size understates tradable quantity when hidden, reserve, midpoint, or fast-replenishing liquidity exists. It also overstates certainty when posted quantities are short-lived.

Mechanism 7.1.104. *Market Impact and Decay*
Market impact is the price movement caused by trading pressure. Temporary impact decays when other traders disagree with the move; permanent impact remains when the order flow reveals information.

Principle 7.1.105. *Resiliency as Synchronisation*
Liquidity is resilient when opposing trading interests arrive before a price move becomes self-reinforcing. Poor synchronisation turns temporary imbalances into persistent price pressure.

Mechanism 7.1.106. *Value-Trader Resiliency*
Value traders make markets more resilient when they buy after uninformed selling pushes price below value, or sell after uninformed buying pushes price above value. Their balance sheet is a delayed liquidity buffer: useful only when valuation confidence, funding, and risk limits survive the shock.

Taxonomy 7.1.107. *Liquidity State Drivers*

i. *trader identity*: uninformed flow is easier to absorb than informed flow;
ii. *instrument attention*: widely followed instruments attract more latent counterparties;

iii. *valuation uncertainty*: unclear fundamentals make suppliers demand larger concessions;

iv. *timing*: closed markets, holidays, and event windows raise search cost;

v. *same-side imbalance*: one-sided displayed interest makes the other side expensive;

vi. *counterparty quality*: reputable traders find liquidity because others expect fairer interaction.

These drivers explain why a market can look liquid in normal prints but become expensive for a specific order.

Principle 7.1.108. *Informational Efficiency*
Informational efficiency means prices incorporate available information quickly and with limited noise. Visibility, update speed, and uninformed flow determine how much information enters prices.

Mechanism 7.1.109. *Bid-Ask Bounce*
Transaction prices alternate between bid and ask even when the efficient price is stable.
A simple bounce statistic compares alternations with continuations: if N^a is the number of price alternations and N^c the number of same-direction continuations, then $\eta = \frac{N^c}{2N^a}$.
Lower η means alternations dominate continuations, so more observed price movement is microstructure noise rather than information.

Caveat 7.1.110. *Transparency Paradox*
More shared information can improve price formation, but each trader prefers to reveal as little as possible before execution. Market design balances public price discovery against private information leakage.

Taxonomy 7.1.111. *Transparency Segmentation*

i. *pre-trade*: orders, quotes, and depth visible before execution;

ii. *post-trade*: prices, sizes, venues, and trade reports after execution;

iii. *top-of-book*: best bid and ask only;

iv. *depth-of-book*: multiple price levels or order-level detail;

v. *timeliness*: real-time, delayed, batched, or end-of-day publication.

Caveat 7.1.112. *Latency and Book State*
When order books update faster than an observer can receive data, decide, and route an order, the displayed book is stale. Available liquidity becomes probabilistic rather than deterministic.

Definition 7.1.113. *Nyquist-Shannon Latency Bound*
For an order-book update frequency f, the latency bound is $\tau^\star = \frac{1}{2f}$. If round-trip observation and routing time exceeds this scale, deterministic use of the displayed book is no longer reliable.

Caveat 7.1.114. *Quality Trade-Offs*
Market quality criteria conflict. Displayed depth, tick size, dark trading, and transparency shift costs across liquidity provision, queue priority, information leakage, and price discovery.

Empirical Finding 7.1.115. *Intraday Liquidity Seasonality*
Volume, volatility, spreads, and book size are time-of-day conditional. Opens tend to have high volume, high volatility, and wide spreads; the middle of the day is closer to steady state; the close attracts benchmark flow, higher volume, and often larger best-limit size.

Principle 7.1.116. *Spread and Depth Substitute Imperfectly*
When the spread cannot narrow further because the tick binds, liquidity providers compete by increasing size at the best bid and ask. A larger book can then play a role similar to a smaller spread, but only for orders small enough to fit inside displayed depth.

Mechanism 7.1.117. *Spread-Volatility Link*
Market makers ask for wider spreads when volatility rises because adverse selection and inventory risk increase. The relation is cleaner after controlling for tick size, trading frequency, and volatility per trade.

Definition 7.1.118. *Flow Reactivity*
Flow reactivity measures how strongly trade counts rise with high-frequency volatility. A resilient venue keeps market makers active when uncertainty rises; a fragile venue loses displayed liquidity when it is most valuable.

Empirical Finding 7.1.119. *Volume Is Not Directly Comparable*
Reported volume depends on architecture. A direct match may count once, while dealer intermediation may count multiple legs, so cross-venue comparisons need the counting convention.

Caveat 7.1.120. *Visible Activity vs Economic Importance*
Visible markets are not always the largest exposures. Stocks attract attention, while real estate, bonds, currencies, swaps, and futures may dominate capital value, risk transfer, or notional activity.

7.1.6 Regulatory and Structural Evolution

Principle 7.1.121. *Regulation as Market Design*
Regulation is market design by law. Good rules support communication, performance, investor protection, competition, and access; bad rules protect incumbents or raise trading costs.

Taxonomy 7.1.122. *Market-Design Question Segmentation*

i. *consolidation*: central book versus fragmented venues;
ii. *intermediation*: order-driven, quote-driven, brokered, or hybrid;
iii. *access*: internalisation, preferencing, linkages, and who may trade;
iv. *transparency*: order-book visibility, trade reporting, and market-data ownership;
v. *controls*: tick sizes, price limits, halts, and listing permissions.

Process 7.1.123. *Welfare Evaluation*

i. identify feasible market structures;
ii. state the objective function and trade-offs;
iii. estimate effects on liquidity, information, volatility, and distribution;
iv. choose the rule with the best expected social welfare.

Taxonomy 7.1.124. *Policy Objective Segmentation*

i. *utilitarian traders*: protect the users whose needs make markets exist;
ii. *public benefits*: promote informative prices and low-cost liquidity;
iii. *profit-motivated traders*: support them only when they supply liquidity or information;
iv. *predatory activity*: discourage manipulation, bluffing, and front-running.

Rule 7.1.125. *Insider-Trading Restrictions*
Rules restrict trading or tipping on material nonpublic information obtained through duty, confidence, or misappropriation. Insiders may still trade under disclosure, prearranged plans, trading windows, and approvals.

Taxonomy 7.1.126. *Insider-Trading Control Segmentation*

i. *disclosure controls*: broad public release and selective-disclosure limits;
ii. *ownership reports*: public filings of insider holdings and changes in ownership;
iii. *prearranged plans*: good-faith trading plans adopted before the trader has material nonpublic information;
iv. *blackout windows*: firm-level restrictions around earnings, deals, and other sensitive events;
v. *short-swing rules*: disgorge short-horizon gains by statutory insiders;
vi. *surveillance and audit trails*: link abnormal trading to information access and communications.

The controls separate legitimate insider liquidity needs from trades that monetise privileged information.

Process 7.1.127. *Insider-Trading Enforcement Logic*

i. detect abnormal price, volume, option activity, or order flow before a material event;
ii. identify the public event and the set of people who knew the information beforehand;
iii. match suspicious traders to access, relationships, communications, and beneficial owners;
iv. test alternative explanations such as public analysis, hedging, or unrelated liquidity demand;
v. pursue disgorgement, penalties, injunctions, or criminal referral when evidence supports intent.

Surveillance starts from market data, but proof usually requires the information path.

Taxonomy 7.1.128. *Insider-Trading Debate Segmentation*

i. *fairness and confidence*: restrictions can make outsiders more willing to supply capital;
ii. *liquidity*: restrictions remove a class of toxic informed flow and can narrow spreads;
iii. *corporate control*: restrictions reduce incentives for managers to hide information or front-run their firms;
iv. *price efficiency*: permissive trading can move prices toward value faster;
v. *enforcement cost*: detecting remote tippees and slow trading is expensive and imperfect;

vi. *incentives*: allowing some insider gains can resemble compensation for entrepreneurial information.

The welfare question is not whether insider trading is informative, but whether the information benefit exceeds liquidity, control, confidence, and enforcement costs.

Proposition 7.1.129. *Insider-Trading Policy Trade-Off*
Let L be the liquidity and confidence gain from restricting insider trading, C the corporate-control gain, E the enforcement cost, and D the cost of delayed price discovery.
A restriction improves market welfare when $L + C > E + D$.

Proof. The policy's net welfare change is the benefit from lower adverse selection and better control, $L + C$, less enforcement and information-delay costs, $E + D$.
The change is positive exactly when $L + C > E + D$. □

Taxonomy 7.1.130. *Exposure-Protection Rule Segmentation*

i. *time precedence*: protects displayed orders from later traders at the same price;
ii. *minimum price increment*: makes stepping ahead costly enough to preserve queue value;
iii. *undisplayed orders*: let traders make firm commitments without full pre-trade exposure;
iv. *audit trail*: deters broker front-running, dual-capacity abuse, and allocation manipulation;
v. *delayed block reporting*: can protect large trades while they are being completed.

These reduce exposure costs, but each also trades off public transparency, price discovery, or queue competition.

Rule 7.1.131. *Decimalization*
Decimalization reduced U.S. equity tick sizes from fractions to cents. Smaller ticks narrowed quoted spreads but reduced displayed depth and queue stability.

Rule 7.1.132. *Reg NMS*
Reg NMS links U.S. equity venues through protected quotes and routing obligations. It protects displayed top-of-book prices, while depth, fees, latency, and hidden liquidity still differ across venues.

Rule 7.1.133. *MiFID and MiFID II*
MiFID broke exchange monopolies by allowing trading outside historical exchanges; MiFID II tightened best-execution, transparency, and reporting duties. The combined effect was more competition, more venue types, and greater reliance on routing, reporting, and post-trade data quality.

Principle 7.1.134. *Best Execution After Fragmentation*
Best execution is not only the best displayed price. In fragmented markets it also depends on fees, likelihood of execution, latency, hidden liquidity, venue access, order size, and the investor's horizon.

Mechanism 7.1.135. *Fragmentation*
Fragmentation means the same instrument trades across multiple venues. It can lower fees and specialise execution, but splits liquidity and complicates best execution.

Proposition 7.1.136. *Order-Flow Externality*
Let venue i give trader utility $U_i = s_i - c_i + \ell(F_i)$, where s_i is service value, c_i is access and trading cost, F_i is order flow, and ℓ is increasing. A technically better entrant with $s_e - c_e > s_m - c_m$ will still fail to attract a trader from incumbent m when $\ell(F_m) - \ell(F_e) > (s_e - c_e) - (s_m - c_m)$.

Proof. The trader chooses the entrant only if $U_e > U_m$.
Rearranging $s_e - c_e + \ell(F_e) > s_m - c_m + \ell(F_m)$ gives the stated threshold. □

Taxonomy 7.1.137. *Levels of Fragmentation*

i. *operators*: new market operators compete with incumbents, then often merge as fixed costs bite;
ii. *venues*: one operator may run lit books, dark books, periodic auctions, and internalisation channels;
iii. *order books*: one venue may maintain several books with different access, transparency, or priority rules;
iv. *orders*: large parent orders are split through time and across compatible venues;
v. *technology*: protocols, market data feeds, and native gateways become part of venue competition.

Mechanism 7.1.138. *Structural Evolution of Fragmentation*
Post-MiFID fragmentation first reduced the monopoly share of historical exchanges, then consolidated around large pan-European operators. Complexity did not disappear: liquidity was segmented into lit books, dark pools, broker crossing systems, systematic internalisers, and specialised order books.

Taxonomy 7.1.139. *Post-MiFID Phases*

i. *entry*: new MTFs competed for order flow, often with bank sponsorship or imported technology;

ii. *convergence*: successful venues moved toward pan-European offers and common functionality;

iii. *dark segmentation*: broker crossing networks, dark MTFs grew around first-hand liquidity, lower signalling;

iv. *consolidation*: fixed technology, data, and connectivity costs pushed operators toward mergers.

Caveat 7.1.140. *Early MiFID Failures*
Turquoise and NASDAQ OMX Europe show that technology and sponsorship were not sufficient. Turquoise gained relevance only after its liquidity agreement ended and it became part of the LSE group; NASDAQ OMX's European MTF and router never reached durable scale.

Caveat 7.1.141. *Liquidity Segmentation*
Venue competition separates natural investor liquidity from opportunistic market-making liquidity. This can improve matching for specific order types, but the trader now pays for aggregation through routing logic, data, connectivity, and information leakage.

Taxonomy 7.1.142. *Market-Segmentation Motives*

i. *size*: small orders want broad exposure; blocks want controlled exposure;

ii. *information*: informed flow wants anonymity; uninformed flow wants identification and price improvement;

iii. *patience*: impatient traders buy immediacy; patient traders prefer orders with time precedence;

iv. *access*: members, floor traders, prime-brokered clients, and retail accounts do not see the same facilities;

v. *credit and trust*: bilateral markets exclude weak counterparties or make them trade through guarantors.

These differences explain why several market structures can survive for the same instrument.

Taxonomy 7.1.143. *Technology Fragmentation*

i. *application protocols*: venue APIs and market-data formats such as ITCH, XDP, EMDI, or RWF;

ii. *network protocols*: multicast UDP, TCP, replay channels, and venue-specific recovery logic;

iii. *compression protocols*: FAST, ZLIB, or uncompressed feeds.

Even when trading rules converge, access remains fragmented through protocols, feeds, and operational detail.

Rule 7.1.144. *Transparency and Dark Trading*
Pre-trade transparency reveals resting interest; post-trade transparency reveals completed transactions. Dark trading reduces information leakage, so regulators often pair it with caps, waivers, or stronger reporting.

Mechanism 7.1.145. *Broker Crossing Networks*
Broker crossing networks internalise client flow before exposing it to lit markets.
They can protect large investors from signalling, but they also raise conflict questions when broker prop flow, market-making flow, and client flow meet in the same pool.

Rule 7.1.146. *MiFID II Dark Caps*
MiFID II constrained dark trading through double volume caps: a venue-level cap and an aggregate market-level cap. The intent is to preserve price formation on lit venues while still allowing limited dark execution.

Taxonomy 7.1.147. *MiFID Venue Statuses*
European transactions may occur on regulated markets, multilateral trading facilities, systematic internalisers, or OTC channels. Regulated markets and MTFs use multilateral rule books; systematic internalisers trade bilaterally against client flow; OTC trading keeps negotiation bilateral but still faces reporting obligations.

Rule 7.1.148. *Tick Size Reforms*
Tick size rules balance price competition against queue discipline. Large ticks widen spreads but support displayed queues; small ticks narrow spreads but encourage queue jumping, cancellation, smaller displayed size.

Caveat 7.1.149. *Tick Size as Market Share Tool*
In fragmented markets, venues can use smaller ticks to undercut incumbent quotes and attract flow. This may improve displayed prices in the short run, but can also start a race to economically meaningless price increments.

Rule 7.1.150. *Sub-Penny and Tick-Size Wars*
Sub-penny rules restrict displayed price improvement, but off-exchange venues can still internalise flow at tiny increments. In Europe, early post-MiFID venues used different tick tables to compete until harmonisation reduced tick-size competition.

Taxonomy 7.1.151. *Who Gains From Tick Changes*

i. *liquidity takers*: may gain from narrower spreads on small trades;

ii. *liquidity providers*: lose queue rents when stepping ahead becomes cheap;

iii. *large investors*: may face less displayed depth and higher search or slicing costs;

iv. *venues*: can gain share by offering finer price grids or specialised order types.

Principle 7.1.152. *Tick Policy Design*
Tick tables should vary with price and liquidity, stay stable enough for adaptation, and be revised using spreads, depth, order-to-trade ratios, and standard-size execution costs.

Caveat 7.1.153. *Cross-Market Precedence Failure*
Time precedence protects a displayed order only inside the venue that enforces it.
In fragmented markets, quote matchers can bypass an older queue by trading elsewhere unless tick, routing, and trade-through rules restore economic precedence across venues.

Rule 7.1.154. *Circuit Breakers and Structural Controls*
Halts, limit moves, short-sale constraints, margin rules, and position limits do not remove volatility. They change when and how traders express demand, supply, and leverage.

Taxonomy 7.1.155. *Circuit Breaker Scope*

i. *local*: halt one instrument or one venue;

ii. *related-instrument*: halt derivatives, ETFs, or baskets tied to the halted instrument;

iii. *market-wide*: halt a broad market after index-level stress;

iv. *dynamic bands*: reject or pause trades outside a price corridor;

v. *static thresholds*: trigger from a fixed reference price or previous close.

Taxonomy 7.1.156. *Extreme-Volatility Control Segmentation*

i. *trading halts*: stop trading until information, order imbalance, or system stress is resolved;

ii. *price limits*: permit trading only inside a daily or intraday price band;

iii. *transaction taxes*: raise the cost of trading, especially high-turnover strategies;

iv. *margin and position limits*: cap leverage and forced-loss exposure;

v. *collars*: restrict specific order types or program trades under defined market moves.

The same rule can damp transitory volatility while delaying fundamental repricing.

Principle 7.1.157. *Circuit Breaker Design*
Circuit breakers should stop disorderly liquidity demand, not ordinary price discovery. Useful triggers combine price movement with liquidity, volatility, correlation, or related-instrument stress.

Proposition 7.1.158. *Halt Value Depends on Volatility Source*
A halt improves price quality only if the expected reduction in transitory noise and operational stress exceeds the cost of delayed fundamental price discovery and pent-up order flow.

Proof. Let benefit B be the avoided transitory trading error and system overload. Let cost C be stale prices, delayed hedging, and accumulated imbalance.
The halt improves expected price quality exactly when $B > C$. □

Caveat 7.1.159. *Gravitational Effect*
If traders expect a halt or limit to prevent later execution, they may submit orders earlier. The rule can therefore pull liquidity demand forward and increase volatility near the trigger.

Caveat 7.1.160. *Propagation Problem*
Halting one instrument can push pressure into substitutes. If ETFs, futures, options, or highly correlated names remain open, the halt may move price discovery elsewhere instead of calming it.

Mechanism 7.1.161. *Price Limits and Margin Waterfall*
Price limits can slow default losses by forcing losing traders to post margin over several sessions instead of one jump. They also delay liquidation, so unresolved losses and pent-up orders may reappear when trading resumes.

Taxonomy 7.1.162. *Trading-Restriction Incidence*

i. *informed traders*: restrictions slow incorporation of value-relevant information;

ii. *dealers*: higher costs reduce supplied immediacy and widen spreads;

iii. *order anticipators*: restrictions may reduce front-running and forced-flow extraction;

iv. *bluffers*: higher manipulation cost can reduce transitory volatility;

v. *utilitarian traders*: hedgers, investors, and borrowers may lose useful risk-transfer access;

vi. *gamblers*: discouraging stale-information trading can reduce noise.

Good policy discriminates by harmful behaviour, not by trading frequency alone.

Rule 7.1.163. *Post-Flash-Crash Controls*
After the 2010 Flash Crash, U.S. market structure moved toward single-stock pauses, limit-up/limit-down bands, consolidated audit trail reporting, stub-quote restrictions, and tighter controls on naked market access.

Caveat 7.1.164. *Stub Quotes*
Stub quotes satisfy a formal two-sided quoting duty at prices far from the market. They are not real liquidity; executions against them reveal missing protection in the trading mechanism.

Principle 7.1.165. *Consolidated Tape as Resilience*
Helps fragmented markets retain a shared post-trade reference. Without it, stress or outages can leave participants unsure whether price discovery is happening in one market or several disconnected ones.

Taxonomy 7.1.166. *Regulatory Institutions*
Market oversight is shared across:

i. governmental regulators, such as securities and futures commissions;

ii. self-regulatory organisations, such as exchanges, clearinghouses, and trader associations;

iii. private standard setters, such as accounting, performance-reporting, and professional bodies;

iv. international organisations, such as IOSCO and the World Federation of Exchanges.

This division creates expertise, jurisdictional overlap, and regulatory competition.

Taxonomy 7.1.167. *Regulatory Political-Economy Risks*

i. *regulatory risk*: officials adopt visible rules after crises because inaction is blamed more than costly action;

ii. *regulatory capture*: rules favour incumbents over diffuse market users;

iii. *measurement risk*: policy is adopted before transitory and fundamental volatility can be cleanly separated.

Post-crash rules should therefore be judged by mechanism, not only by public reassurance.

Caveat 7.1.168. *Unintended Consequences*
Rules change incentives. Taxes may move liquidity abroad, bans may move risk into substitutes, protected quotes may increase routing complexity, and exchange rules may protect incumbents.

Taxonomy 7.1.169. *Regulatory-Service Segmentation*

i. *private services*: connectivity, routing, order management, clearing access, account tools paid by users;

ii. *public services*: surveillance, audit trails, price continuity, disclosure, and manipulation controls.

Competition supplies private services readily.
Public services usually need exchange rules, regulation, or industry coordination.

Mechanism 7.1.170. *Competition to Merger*
Opening trading to competition first produces innovative venues and lower explicit costs. Once practices converge, scale becomes decisive: mergers reduce fixed technology costs, expand connectivity, and preserve distinct order books under larger operating groups.

Practitioner Note 7.1.171. *Electronification Beyond Equities*
Electronification does not imply one architecture. Equities move to lit books and dark pools; fixed income and derivatives retain RFQ, dealer networks, and central clearing because contracts and counterparty risks differ.

7.2 Orders, Matching, and Limit Order Books

7.2.1 Order Types and Instructions

Definition 7.2.1. *Order*
An order is a trade instruction. At minimum it identifies the instrument, side, and quantity; practical orders also specify price limits, activation rules, validity, display, routing, settlement, or broker discretion.

Taxonomy 7.2.2. *Order-Instruction Segmentation*

i. *economic terms*: instrument, side, quantity, and price constraint;
ii. *timing*: submission time, activation trigger, validity, and expiry;
iii. *execution constraint*: immediate execution, full execution, minimum size, or partial-fill permission;
iv. *display*: fully displayed, hidden, reserve, or iceberg quantity;
v. *handling*: held by strict rules, or not-held with broker discretion;
vi. *post-trade terms*: settlement, clearing, and account capacity.

The label of an order is therefore only a shorthand for a bundle of instructions.

Taxonomy 7.2.3. *Core Order-Type Segmentation*

i. *market*: pay immediacy and consume the best available liquidity;
ii. *limit*: specify a worst acceptable price and wait in the book if not marketable;
iii. *stop*: become active only after a trigger price is reached;
iv. *market-if-touched*: become active when price moves favourably to the touch price;
v. *IOC/FOK*: cancel any unfilled quantity immediately, or require full immediate execution;
vi. *pegged*: reference the midpoint, best bid, best offer, or another venue-derived price;
vii. *hidden/iceberg*: reduce displayed size while keeping some executable interest undisplayed.

Principle 7.2.4. *Order Choice Trades Certainty for Leakage*
Aggressive orders buy execution certainty but reveal urgency. Passive and hidden orders reduce signalling, but they introduce queue risk, missed fills, and adverse selection.

Mechanism 7.2.5. *Market Order Cost Segmentation*

i. *spread cost*: a small market buy normally pays the ask and a small market sell receives the bid;
ii. *price improvement*: negotiation, midpoint interaction, or discretionary handling may beat displayed quote;
iii. *market impact*: large orders walk the book and move prices to attract the other side;
iv. *execution-price uncertainty*: the quote can change between submission and execution.

The market order solves execution uncertainty by accepting price uncertainty.

Definition 7.2.6. *Marketable Limit Order*
A marketable limit order is priced aggressively enough to trade immediately if available liquidity is still present. It is like a market order with a price cap for buys or a price floor for sells; unfilled residual quantity either rests or cancels according to its other instructions.

Principle 7.2.7. *Market vs Limit Order Choice*
Market orders exchange price control for execution certainty. Limit orders exchange execution certainty for price control. Impatient traders, informed traders with decaying signals, and traders facing narrow spreads tend to cross the spread; patient traders, liquidity suppliers, and traders facing wide spreads tend to post.

Proposition 7.2.8. *Limit-Order Fill Threshold*
For a buy order, let p be the probability that a passive limit order fills, S the spread saved if it fills, M the expected cost of chasing the trade if it does not fill, and A the adverse-selection cost conditional on being filled. Posting is preferred to crossing when $p(S - A) > (1 - p)M$.

Proof. Relative to crossing immediately, posting gains S and loses A when it fills.
If it misses, the trader pays expected chase cost M. The expected relative gain is $p(S - A) - (1 - p)M$, so posting is preferred exactly when this is positive. □

Caveat 7.2.9. *Spread-Only Rules Fail*
Rules such as "cross when spreads are narrow, post when spreads are wide" work only when the trader has no valuation information. A trader who knows the current quote is stale may rationally demand liquidity even when the spread looks wide, or supply liquidity even when the spread looks narrow.

Taxonomy 7.2.10. *Conditional Activation Segmentation*

i. *stop order*: activates after adverse price move; buy stops trigger after price rises, sell stops after price falls;
ii. *stop-limit order*: activates after the stop, but can execute only within the specified limit price;
iii. *market-if-touched order*: activates after a favourable price move, then behaves like a market order.

Stops tend to demand liquidity in the direction of the move; market-if-touched orders are more contrarian.

Caveat 7.2.11. *Stops Are Not Limits*
A stop price controls activation, not execution price. A stop-limit adds a limit price after activation, but then loses execution certainty when the market gaps through the limit.

Rule 7.2.12. *Tick-Sensitive Orders*
Tick-sensitive orders condition execution on the direction of the last price change. A buy downtick order waits for a downtick or zero downtick; a sell uptick order waits for an uptick or zero uptick. These instructions are mainly useful when tick sizes are large enough that one tick is economically meaningful.

Taxonomy 7.2.13. *Time-in-Force Segmentation*

i. *day*: valid until the session close;
ii. *good-till-cancel*: valid until the trader or broker cancels it;
iii. *good-till-date*: valid until a specified date or time;
iv. *IOC*: execute immediately and cancel the unfilled balance;
v. *FOK*: execute the full quantity immediately or cancel;
vi. *market-on-open/close*: execute in the opening or closing auction.

Validity instructions decide when an order can become stale enough to disappear.

Taxonomy 7.2.14. *Quantity and Handling Segmentation*

i. *all-or-none*: the whole quantity must execute;
ii. *minimum acceptable quantity*: each fill must meet a specified minimum size;
iii. *display size*: only part of the order is shown;
iv. *not-held*: the broker may use discretion over timing, price, and exposure.

These instructions trade fill probability against information leakage and accountability.

Taxonomy 7.2.15. *Special Instruction Segmentation*

i. *spread orders*: trade two related instruments as one package, usually with a limit on the price difference;
ii. *substitution orders*: let the broker choose among acceptable substitutes;
iii. *special settlement*: request nonstandard settlement, often around record dates or corporate actions.

Caveat 7.2.16. *Instruction Semantics Are Venue-Specific*
The same label can behave differently across venues. Minimum quantity, display refresh, priority loss after modification, pegging reference, and cancellation rules must be read as part of the market design.

7.2.2 Order Lifecycle

Process 7.2.17. *Order State Segmentation*

i. *pending*: submitted but not yet accepted by the broker or venue;
ii. *accepted*: valid for handling, but not necessarily executable yet;
iii. *working*: live in the market or with a broker seeking execution;
iv. *part-filled*: some quantity executed, with residual quantity still governed by the order instructions;
v. *terminal*: filled, cancelled, expired, rejected, or killed by a contingency.

The economic order exists before the public book sees it, but the market only reacts once it is working.

Process 7.2.18. *Limit-Order Lifecycle*

i. *insert*: add displayed or hidden liquidity at a chosen price and size;

ii. *modify*: change price, quantity, instructions, or expiry. Loses priority when price worsens or size increases;

iii. *cancel*: remove resting quantity before execution;

iv. *execute*: trade when an incoming marketable order crosses the resting quote.

Taxonomy 7.2.19. *Message-Channel Segmentation*

i. *private channel*: order entry, acknowledgements, rejects, cancels, and fills between member and venue;

ii. *public channel*: trades, best quotes, depth updates, status messages shared with market-data subscribers.

Private messages control the order; public messages update the common view of the book.

Principle 7.2.20. *Absent Trader Representation*
Limit, stop, validity, and quantity instructions encode future trading contingencies. They let an absent trader participate, but stale contingencies can trigger at bad times.

Caveat 7.2.21. *Lifecycle Tags Change Priority*
Time-in-force, display quantity, hidden flags, minimum quantity, and cancel-on-disconnect rules can change whether an order rests, executes, loses priority, or vanishes under stress.

7.2.3 Limit Order Book Mechanics

Definition 7.2.22. *Best Quotes and Spread*
The *best bid* is the highest displayed buy price and the *best ask* is the lowest displayed sell price. The *bid-ask spread* is $a_t - b_t$, and the quoted midpoint is $m_t = (a_t + b_t)/2$.

Principle 7.2.23. *Liquidity Supply and Demand*
Resting limit orders supply immediacy to other traders; marketable orders demand it.
A standing quote is therefore both a price and a service: it lets someone else trade now.

Taxonomy 7.2.24. *Limit Price Placement Segmentation*

i. *marketable*: crosses the opposite best quote and can execute immediately;

ii. *at the market*: joins the current best bid or ask;

iii. *inside the market*: improves the best bid or ask and narrows the spread;

iv. *behind the market*: rests away from the best quote;

v. *far from the market*: carries low execution probability but low immediacy cost.

Aggressiveness raises fill probability and lowers waiting time, but it gives up more price concession.

Proposition 7.2.25. *Standing Limit Orders Are Trading Options*
Let q be order size, L the limit price, τ the cancellation or expiry time, and V_t the contra-side trader's value at time t. While live, a sell limit order gives the market a call-like payoff $q(V_t - L)^+$; a buy limit order gives the market a put-like payoff $q(L - V_t)^+$, where $x^+ = \max\{x, 0\}$.

Proof. For $t \leq \tau$, an incoming buyer exercises a sell limit only when buying at L gives non-negative surplus, so the exercise value is $\max\{q(V_t - L), 0\}$. The buy-limit case is symmetric, with exercise value $\max\{q(L - V_t), 0\}$. □

Caveat 7.2.26. *Limit-Order Regret*
Limit orders reduce price concession but introduce two risks. The order may not execute and the trader must chase the price later; or it may execute exactly because better-informed flow is trading through it.

Principle 7.2.27. *Equilibrium Spread as Order-Choice Indifference*
In a continuous order-driven market, the spread adjusts until the marginal trader is indifferent between demanding liquidity with a marketable order and supplying liquidity with a limit order.
If spreads are too wide, more traders post limits and compete them down; if spreads are too narrow, passive traders withdraw and marketable flow widens them.

Mechanism 7.2.28. *Limit Orders Give Timing Options*
A standing limit order gives other traders the option to trade later at a stale price. The option is more valuable when volatility is high, cancellation is slow, informed traders are active, or the contra-side trader can monitor the book faster than the resting order owner.

Definition 7.2.29. *Tick Size*
Tick size is the minimum price increment for limit orders.
It discretises prices and can be uniform, price-banded, or liquidity-banded.

Empirical Finding 7.2.30. *Price Clustering*
Even when the grid permits many prices, quotes often cluster at round numbers. Clustering reflects communication, search, and queueing convenience, especially when the tick is small relative to price.

Mechanism 7.2.31. *Binding Tick Constraint*
When the tick is large relative to the latent spread, the quoted spread is floored at one tick. Once the tick no longer binds, further reductions mostly change queue behaviour.

Model 7.2.32. *Harris Tick Discretisation Model*
Let $Ask_{j,t}$ and $Bid_{j,t}$ be the best ask and bid for stock j on trade t of day d, and let $\Psi_{j,t} = (Ask_{j,t} - Bid_{j,t})/tick_{j,d}$ be the observed spread in ticks. The model assumes an unobserved continuous spread $\widetilde{\Psi}_{j,t}$, then rounds it through a discretisation map $\Psi_{j,t} = \Phi(\widetilde{\Psi}_{j,t})$.
The tick is therefore not just a unit of account; it censors small spreads into one-tick quotes.

Definition 7.2.33. *Continuous Spread Expectation*
With turnover proxy T_j, volatility $\sigma_{j,d}$, price $P_{j,d}$, and tick size $tick_{j,d}$,
a compact specification is $\mathbb{E}[\widetilde{\Psi}_{j,t} \mid T_j, \sigma_{j,d}] = g(T_j, \sigma_{j,d}) \frac{P_{j,d}}{tick_{j,d}}$, $g(T_j, \sigma_{j,d}) = \alpha T_j^{\beta} \sigma_{j,d}^{\gamma}$.
Here g is the latent relative spread, and α, β, and γ are fitted spread-response parameters.
Multiplying by $P/tick$ converts it into tick units.

Mechanism 7.2.34. *Tick-Size Counterfactual*
To estimate the effect of a smaller tick, keep the latent spread distribution fixed and change only the rounding grid. If δ is the tick-reduction factor and the new tick is $tick/\delta$, an observed k-tick spread becomes roughly k/δ old-tick units before re-rounding.
The largest expected spread decrease occurs when current spreads are near one tick.

Taxonomy 7.2.35. *Harris Model Inputs*

i. *liquidity*: turnover T_j, usually lowering latent spread;
ii. *risk*: volatility $\sigma_{j,d}$, usually raising latent spread;
iii. *price grid*: $P_{j,d}/tick_{j,d}$, controlling how strongly ticks discretise spreads;
iv. *distribution*: a chosen positive density for latent continuous spreads;
v. *fit check*: compare observed relative spreads with fitted discrete-spread expectations.

Principle 7.2.36. *Tick Size and Queue Discipline*
The tick defines how expensive it is to step ahead of an existing limit order.
Large ticks protect time priority by making queue jumping costly; small ticks make one-tick improvement cheap and weaken the value of being first in queue.

Mechanism 7.2.37. *Quoted Depth Under Small Ticks*
When the price grid becomes finer, liquidity providers can spread orders across more price levels. Best-limit depth often falls, order sizes shrink, and displayed liquidity must be refreshed more frequently.

Caveat 7.2.38. *Order-to-Trade Ratio*
A smaller tick can increase cancellations and modifications relative to trades. More messages do not necessarily mean more economic liquidity; they may simply reflect shorter quote life and more frequent repricing.

Mechanism 7.2.39. *Cumulative Depth*
To compare book quality across tick regimes, measure the cost of executing a fixed monetary size by rebuilding cumulative depth around the best bid and offer. Top-of-book spread alone ignores hidden liquidity, resiliency, and depth beyond the first level.

Principle 7.2.40. *CLOB as Price-Formation Device*
A central limit order book stores unmatched buying and selling interest until compatible prices meet. Each transaction therefore both transfers inventory and updates the public state of price formation.

Taxonomy 7.2.41. *Order-Book Visibility*

i. *open book*: broad participants can observe standing interest;
ii. *closed book*: only the broker, dealer, or venue sees resting orders;

iii. *top-of-book*: only the best limits are visible;

iv. *depth book*: multiple levels, aggregate size, or order-level detail are visible.

Mechanism 7.2.42. *LOB Event Types*
Order-book dynamics are driven by three event families: insertions add liquidity, cancellations remove liquidity without trading, and marketable orders remove liquidity through transactions. Price changes occur when the best queue is depleted or a better price is inserted.

Taxonomy 7.2.43. *Tick-Size Quality Proxies*

i. *binding probability*: share of time or trades with spread equal to one tick;

ii. *spread leeway*: spread in ticks minus one;

iii. *depth*: displayed and cumulative size near the midpoint;

iv. *message intensity*: order-to-trade ratio, cancellations, and modifications.

Caveat 7.2.44. *Value of a Quote*
As ticks and displayed sizes shrink, quotes can become shorter-lived and less informative. A consolidated tape of trades cannot fully replace quote data if only the best limit is visible and deeper book information is lost.

Caveat 7.2.45. *RFQ Can Mimic a Book*
In RFQ markets, dealers answer bilateral requests rather than post firm multilateral quotes.
Aggregating several responses can look like a book, but the liquidity is conditional on the request, relationship, size, and information embedded in the enquiry.

7.2.4 Matching Algorithms

Principle 7.2.46. *Matching Is Market Design*
Matching rules decide which orders trade, at what price, and in what priority. Small rule differences can change displayed liquidity, queue value, auction participation, and the usefulness of a venue for large orders.

Process 7.2.47. *Rule-Based Matching Procedure*

i. rank buy and sell orders by the venue's precedence rules;

ii. pair the highest-ranked buy with the highest-ranked sell when their prices cross;

iii. price the trade using the venue's trade-pricing rule;

iv. leave any residual quantity in the book unless the order instruction cancels it.

The matching loop stops when the best bid is below the best offer, or when no eligible order remains.

Mechanism 7.2.48. *Continuous Matching*
In continuous trading, an incoming marketable order executes immediately against resting interest. The trade-off is speed versus noise: prices update continuously, but liquidity can be thin when order flow is asynchronous.

Mechanism 7.2.49. *Call Matching*
In a call, orders are batched until an uncrossing time and matched at one auction price. The batch creates a thicker crossing point, making calls useful at openings, closings, expiries, and volatility interruptions.

Mechanism 7.2.50. *Frequent Batch Auctions*
Frequent batch auctions group arrivals into short calls. Price priority remains, but within-batch time priority is muted, reducing pure latency value.

Taxonomy 7.2.51. *Trade-Pricing Rule Segmentation*

i. *uniform*: all trades from the same call execute at one clearing price;

ii. *discriminatory*: each trade prices from the standing order, accepted quote, or local match;

iii. *derivative*: trades use an external reference price, such as midpoint, primary close, or VWAP.

The same matched quantity can therefore distribute surplus differently across traders.

Principle 7.2.52. *Pricing-Rule Incidence*
Discriminatory pricing favours aggressive size; uniform pricing favours resting orders by giving all auction fills the same clearing price.

Process 7.2.53. *Call Auction Price Rules*

i. *maximum executable volume*: choose prices that match the largest quantity;

ii. *minimum surplus*: among those prices, minimise unmatched quantity;

iii. *market pressure*: if surplus is only on the buy side choose the higher candidate, and if only on the sell side choose the lower candidate;

iv. *reference price*: if still tied, choose the candidate closest to the venue reference price.

If buy and sell interest do not cross, no auction price can be formed.

Definition 7.2.54. *Auction Surplus*
Auction surplus is the quantity left unmatched at a candidate clearing price. It is not excess volume in the day; it is the residual buy or sell pressure at the auction price.

Model 7.2.55. *Single-Price Supply and Demand Schedules*
For candidate price p, let $S(p) = \sum_i q_i^s \mathbf{1}_{\{s_i \le p\}}$ and $D(p) = \sum_j q_j^b \mathbf{1}_{\{b_j \ge p\}}$.
Matched volume is $M(p) = \min\{S(p), D(p)\}$ and excess demand is $E(p) = D(p) - S(p)$.
A single-price auction chooses $p^\star \in \arg\max_p M(p)$, then applies imbalance, pressure, or reference-price tie-breaks.

Lemma 7.2.56. *Trade Price Redistributes Surplus*
If a buyer with value v_b trades with a seller with value v_s at any price $p \in [v_s, v_b]$, total trader surplus is $v_b - v_s$ and does not depend on p.

Proof. Buyer surplus is $v_b - p$ and seller surplus is $p - v_s$. Their sum is $v_b - v_s$. □

Proposition 7.2.57. *Price Priority Maximizes Auction Surplus*
Unit-expand orders. Sort buyers $b_1 \ge b_2 \ge \cdots$ and sellers $s_1 \le s_2 \le \cdots$. The surplus-maximizing auction trades the largest Q with $b_Q \ge s_Q$, matching the Q highest buyers with the Q lowest sellers.

Proof. Exchange any included lower buyer for an excluded higher buyer, and vice versa. Each exchange weakly raises surplus. Additional units are beneficial exactly while $b_Q \ge s_Q$. □

Rule 7.2.58. *Randomised Uncrossing*
Some venues randomise the exact end of the call after the theoretical auction time. This weakens last-second order placement and cancellation, but it also makes the final book state less deterministic for participants.

Taxonomy 7.2.59. *Priority Segmentation*

i. *price priority*: better-priced orders execute first;

ii. *FIFO time priority*: earlier orders at the same price execute first;

iii. *pro-rata priority*: fills at one price are allocated partly by displayed size;

iv. *size priority*: larger orders receive explicit rank or allocation advantages;

v. *venue-specific priority*: hidden, displayed, broker, or auction-only flags may change rank.

Rule 7.2.60. *Public Order Precedence*
At equal prices, some venues rank public customer orders ahead of member or dealer proprietary interest. This limits access-based queue jumping.

Model 7.2.61. *Pro-Rata Allocation*
At one price level, suppose same-priority resting orders have displayed sizes $q_1, \ldots, q_n$ and compete for incoming quantity $Q < \sum_j q_j$. Under pure pro-rata allocation, order i receives $f_i = Qq_i / \sum_j q_j$ before venue-specific rounding, minimum-fill, and residual rules.

Definition 7.2.62. *Theoretical Matching Curve*
During a call, the theoretical matching curve tracks how close the current indicative matched volume is to the final auction matched volume. It reveals whether the book builds smoothly, jumps at information times, or overshoots before orders are cancelled.

Caveat 7.2.63. *Indicative Volume Is Strategic*
Pre-auction indicative prices, volumes, and imbalances are signals, not commitments. A book can look deeper than the final auction if traders test information by submitting orders they later cancel or modify.

7.2.5 Queue Dynamics and Latency

Principle 7.2.64. *Condition on the Queue*
Queue dynamics are less random once the state of the book is specified. A cancellation at the best bid is not comparable when that queue has one lot, ten lots, or a much larger opposite queue.

Definition 7.2.65. *Queue Imbalance*
Queue imbalance compares available quantity on the best bid and ask. Predictive because a small best queue is easier to deplete, while a large opposite queue can make the next price move less likely to reverse immediately.

Mechanism 7.2.66. *Arrival-Departure Ratio*
For a queue, the arrival-departure ratio compares insertion intensity with cancellation plus transaction intensity. Ratios above one imply queue growth; ratios below one imply depletion.

Caveat 7.2.67. *Adverse Queue Selection*
A limit order is most likely to fill when marketable flow reaches it. That event is not neutral: fills are often more valuable to the taker precisely when the resting quote is stale, underpriced, or exposed to informed flow.

Process 7.2.68. *Queue-Reactive Estimation*

i. choose a starting time and record the queue state;
ii. wait for the next insertion, cancellation, or transaction;
iii. store the waiting time by event type and queue-size bucket;
iv. estimate state-dependent intensities from the average waiting times.

Caveat 7.2.69. *Mechanical Mean Reversion*
Queue-reactive simulations can generate too much price mean reversion: after a best queue disappears, the newly promoted queue may be large enough to pull prices back. Models often need memory, Hawkes-style excitation, or state resets after price moves.

Principle 7.2.70. *Latency Race Boundary*
Latency matters most when a trader can change queue position before others react. It matters less when the tick is wide, the queue is deep, or the order is deliberately not competing for first priority.

7.2.6 Hidden Liquidity and Dark Trading

Definition 7.2.71. *Dark Pool*
A dark pool is a trading venue where available liquidity is not displayed before execution. It is mainly used to reduce signalling, market impact, and exposure to opportunistic short-horizon traders.

Principle 7.2.72. *Reference Price Trading*
Dark pools usually do not form prices independently. Equity dark trades often execute at the midpoint or another reference price from a reliable lit market, so dark liquidity depends on lit price formation.

Taxonomy 7.2.73. *Crossing Network Rule Segmentation*

i. *crossing price*: midpoint, primary close, auction price, VWAP, or another derived reference;
ii. *crossing schedule*: continuous, periodic, after-hours, or event-triggered;
iii. *allocation*: time priority, pro-rata, minimum quantity, or size preference;
iv. *confidentiality*: displayed, indicated, anonymous, or fully dark before execution.

Principle 7.2.74. *Price Ownership Problem*
Reference-price crosses use prices made elsewhere. The dispute is whether dark venues free-ride on lit price discovery or simply cross at a public benchmark.

Taxonomy 7.2.75. *Dark Liquidity Pools*

i. *crossing networks*: match buy and sell orders, often at the primary midpoint;
ii. *internalisation pools*: brokers cross client flow or trade against broker inventory;
iii. *electronic market-makers*: operators or selected liquidity providers accept or reject incoming flow;
iv. *exchange dark books*: hidden order types inside regulated venues;
v. *dark aggregators*: route one order across several dark pools;
vi. *block pools*: focus on large institutional orders and natural liquidity.

Mechanism 7.2.76. *Integrated vs Midpoint Books*
Large hidden orders may interact with lit and hidden liquidity in integrated books. Smaller dark orders are typically crossed at the midpoint with basic limit, MAQ, and IOC controls.

Mechanism 7.2.77. *Reserve and Iceberg Display*
A reserve order displays $d < q$ and refreshes after the displayed slice trades. It hides total size, but repeated refreshes can reveal the parent and may reset priority.

Caveat 7.2.78. *Discovering Hidden Size*
Hidden size is learned by trading into it, not by observing the public queue. Large IOC or minimum-quantity probes can reveal reserve liquidity, so display instructions reduce but do not remove information leakage.

Caveat 7.2.79. *Dark Pools and Price Discovery*
Dark pools can reduce market impact, but they rely on lit quotes for reference prices. If too much flow moves dark, public price discovery may weaken; if dark flow is selective and low-impact, lit market quality can improve.

Caveat 7.2.80. *Derivative Pricing Risks*
Derived prices can be stale or manipulable: stale prices select informed flow; manipulable benchmarks invite small lit trades before a larger cross.

Empirical Finding 7.2.81. *Dark Activity Drivers*
Dark activity is usually concentrated in liquid stocks, high-volume days, volatile periods, and high-spread names. It tends to be lower when order imbalances are large relative to volume.

Mechanism 7.2.82. *Why Investors Use Dark Pools*
The main incentives are secrecy and price improvement. A midpoint execution lets a liquidity taker avoid paying the full spread while hiding the order from traders who might infer and trade against the parent order.

Taxonomy 7.2.83. *Block Exposure-Control Segmentation*

i. *sunshine trading*: announce identity, intention, and full size to invite natural liquidity;
ii. *upstairs negotiation*: expose the block selectively through trusted brokers or dealers;
iii. *crossing and dark pools*: seek natural contra flow without displayed pre-trade interest;
iv. *slicing and reserve display*: trade through normal venues while hiding the parent order.

Taxonomy 7.2.84. *Order Exposure Decision Segmentation*

i. *search*: actively look for the other side or wait for it to arrive;
ii. *display*: show full interest, partial interest, indications, or nothing;
iii. *slicing*: expose the parent all at once or break it across time;
iv. *broker concentration*: use one broker for control or many brokers for concealment;
v. *venue choice*: trade in one pool, many lit venues, dark venues, or upstairs.

Large-order execution is mostly the art of exposing enough information to find liquidity without giving away a free trading option.

Taxonomy 7.2.85. *Exposure Benefit-Cost Segmentation*

i. *benefit*: exposed interest attracts latent contra-side traders and lowers search cost;
ii. *motive leakage*: others infer why the trader wants to trade;
iii. *impact leakage*: others infer future price pressure from the size and urgency;
iv. *option leakage*: quote matchers and anticipators trade before the parent can complete.

Definition 7.2.86. *Parasitic and Defensive Liquidity Suppliers*
Parasitic traders use exposed order information to trade ahead, quote match, withdraw liquidity, or otherwise profit at the exposed trader's expense. Defensive liquidity suppliers use the same information to avoid being price-discriminated against by informed or urgent traders.

Caveat 7.2.87. *Sunshine Trading Credibility*
Sunshine trading works only when the initiator is credibly uninformed and honest about size. Otherwise the announcement gives order anticipators a free option to trade ahead, quote-match, or withdraw liquidity.

Mechanism 7.2.88. *Block Hanging Over the Market*
Once a large parent order is known, three reactions matter: front runners trade on the same side before the block, same-side traders accelerate their own trades, and opposite-side traders delay to capture price impact. All three make execution more expensive for the initiator.

Taxonomy 7.2.89. *Exposure Defence Segmentation*

i. *evasive*: use anonymity, selective disclosure, dark pools, reserve orders, and broker discretion;

ii. *deceptive*: create noisy signals about side, size, completion, or venue interest;

iii. *offensive*: use stings or controlled exposure to identify and punish predatory traders.

Evasive methods are ordinary execution tools.
Deceptive and offensive methods carry legal, compliance, and relationship risk.

Caveat 7.2.90. *Dark Liquidity Is Conditional*
Dark liquidity is hard to estimate: misses are often invisible, and fills can still be adversely selected.

Caveat 7.2.91. *Toxic Liquidity and Gaming*
Dark pools can be gamed through probing, leakage, and adverse selection. Protections include minimum sizes, access controls, and filtering toxic flow.

Taxonomy 7.2.92. *Bilateral vs Multilateral Liquidity*
Multilateral books let many participants provide liquidity under common rules. Bilateral trading preserves selection and relationship control, which can help avoid toxic counterparties but may reduce open price discovery.

Caveat 7.2.93. *Internal Crossing Agency Problem*
Internal crossing can give clients low-impact fills and save commissions, but the broker controls who sees the order and who receives the match. The main risks are selective exposure, conflicted allocation, and missed price improvement from external liquidity.

7.2.7 Cross-Venue Routing and Fragmentation

Principle 7.2.94. *Routing After Fragmentation*
In a fragmented market, liquidity is no longer represented by one order book. The execution problem becomes a joint choice of price, size, time, venue, fees, and information exposure.

Definition 7.2.95. *Smart Order Routing*
A smart order router (SOR) maps a trading instruction to venues according to displayed quotes, depth, historical fill rates, queue position, fees, latency, and order-type constraints. The execution algorithm usually chooses the parent schedule; the router chooses the spatial allocation of each child order.

Principle 7.2.96. *Temporal Then Spatial Split*
Optimal execution first chooses how fast to trade, balancing market impact against timing risk. In a fragmented market, each child order then needs a spatial split across venues, where the SOR tries to obtain at least the limit price and finish the child order quickly.

Process 7.2.97. *SOR Feedback Loop*

i. observe venue quotes, depths, fees, and recent fills;

ii. decide the venue split under the execution policy;

iii. send orders through venue gateways;

iv. receive execution reports and market-data updates;

v. update the next routing decision.

The decision loop is bounded by CPU time, network latency, matching-engine latency, and feed latency.

Model 7.2.98. *Toy SOR Allocation Model*
For K venues, let r_k be the fraction routed to venue k, $V(\omega)$ the child size, and $D_k(\omega)$ the executable liquidity under liquidity state ω. Captured liquidity is $C(r,\omega) = \sum_{k=1}^{K} \min\{r_k V(\omega), D_k(\omega)\}$, $\sum_{k=1}^{K} r_k = 1$.
An SOR learns $r = (r_1, \ldots, r_K)$ by increasing allocation to venues that fill reliably and reducing allocation to venues that disappoint. The min term is the key intuition: a router cannot capture more than it sends to a venue or more than the venue actually has available.

Mechanism 7.2.99. *SOR Share Feedback*
If $p_k(t)$ is venue k's market share and a is the fraction of participants using SORs, a simple share update is $p_k(t+1) = (1-a)p_k(t) + ar_k(t)$. SORs therefore turn routing weights into market share; venues with reliable liquidity gain flow, while noisy venues lose allocation slowly. When a is small, historical share anchors the system; when a is large, routing quality changes share quickly.

Taxonomy 7.2.100. *SOR Fragmentation Drivers*

i. *initial share*: incumbent venues start with the observed flow prior $p(0)$;
ii. *liquidity quality*: stable depth is easier for routers to learn and trust;
iii. *liquidity variance*: noisy fills push routers toward diversification;
iv. *SOR adoption*: higher a makes market share follow routing weights faster;
v. *reassessment speed*: faster learning accelerates convergence or instability.

Taxonomy 7.2.101. *Venue Selection Segmentation*

i. *price and depth*: displayed best quotes, hidden liquidity, and available queue size;
ii. *execution probability*: fill rate, cancellation rate, queue priority, and adverse selection;
iii. *economics*: commissions, access fees, maker-taker rebates, and clearing costs;
iv. *access*: latency, connectivity, order types, and venue-specific restrictions;
v. *information*: pre-trade visibility, post-trade reporting, and leakage from multi-venue signalling.

Mechanism 7.2.102. *Fragmented-Market Consolidation Loop*

i. *local adjustment*: quotes and orders inside each venue react to information from other venues;
ii. *routing*: liquidity demand moves toward the segment with the best expected net execution;
iii. *arbitrage*: traders move inventory and liquidity across segments when prices diverge enough.

When an order cannot move to the best market, the best market may still move to the order through arbitrage.

Proposition 7.2.103. *Fragmented Consolidation Bound*
Suppose traders can observe segment prices and route or arbitrage between two venues at all-in cost k. If the same asset has executable ask a_1 on venue 1 and executable bid b_2 on venue 2, a persistent gap $b_2 - a_1 > k$ cannot survive competitive arbitrage.

Proof. Buying on venue 1 and selling on venue 2 earns $b_2 - a_1 - k > 0$ before inventory limits. Competitive arbitrage increases demand at venue 1 and supply at venue 2, moving prices until the gap is at most k or capacity binds. □

Definition 7.2.104. *Market Share as Routing Prior*
Market share is a rough prior for where trading occurs. For stock k and venue n, value share over $[t_1, t_2]$ is

$$M_k(n) = \left(\sum_{t_1 \le \tau_\ell^k \le t_2} P_\ell^k V_\ell^k \mathbf{1}_n(\delta_\ell^k) \right) / \left(\sum_{t_1 \le \tau_\ell^k \le t_2} P_\ell^k V_\ell^k \right)$$

Here $M_k(n)$ is stock k's value share on venue n, ℓ indexes trades, τ_ℓ^k is trade time, P_ℓ^k is price, V_ℓ^k is size, δ_ℓ^k is venue, and $\mathbf{1}_n(\delta_\ell^k)$ equals 1 if trade ℓ is on venue n and 0 otherwise.
Trade-count share replaces traded value by $\mathbf{1}_n(\delta_\ell^k)$ and divides by the number of trades. Both measures should separate continuous trading from fixing auctions.

Model 7.2.105. *Venue-Share Estimator*
For asset k and venue i, let T_k^i be turnover on venue i inside a fixed segment.
The estimated venue share is $\hat{p}_k^i = \frac{T_k^i}{\sum_j T_k^j}$. The same ratio can be built from trade count, displayed depth, or executable volume, but each choice answers a different routing question.

Model 7.2.106. *Index-Level Venue Share*
For an index universe $\mathcal{K}$ with weights w_k, aggregate asset-level shares by $\hat{p}_{\mathcal{K}}^i = \sum_{k \in \mathcal{K}} w_k \hat{p}_k^i$.

Here $\mathcal{K}$ is the index constituent set, w_k is asset k's index weight, $\hat{p}_k^i$ is asset k's estimated share on venue i, and $\hat{p}_{\mathcal{K}}^i$ is the index-level share on venue i. Index weights measure benchmark exposure; turnover weights measure where actual trading activity is concentrated.

Practitioner Note 7.2.107. *Estimator Comparison*
To compare an asset with its universe, use $d_k^i = \hat{p}_k^i - \hat{p}_{\mathcal{K}}^i$. Here d_k^i is the excess venue share of asset k on venue i relative to the chosen universe. If asset k is included in the benchmark, the two estimators are mechanically correlated; a leave-one-out universe gives a cleaner comparison.

Caveat 7.2.108. *Metric Choice*
The correct fragmentation metric depends on the question. Use all-in market share for venue revenue; use continuous-session trade share for the probability that an order executes on a venue; use value share when the unit of interest is currency traded.

Taxonomy 7.2.109. *Market Share Segmentation*

i. *continuous ex-fixing*: best for venue choice and execution probability;
ii. *opening, closing, and intraday calls*: best for reference-price and auction-liquidity questions;
iii. *traded value*: best for economic exposure and venue revenue;
iv. *trade count*: best for fill-frequency and small-order routing questions;
v. *stock or sector buckets*: needed because liquidity and capitalisation affect fragmentation.

Combining these segments can make a primary venue look more competitive because it owns the fixing auctions.

Empirical Finding 7.2.110. *Intraday Venue-Share Shape*
Primary markets often regain share near the open and close because auctions, reference-price demand, and early order-book formation concentrate flow. MTF shares can be flatter, rising, or inverted depending on SOR adoption, fee agreements, and the presence of professional liquidity providers.

Caveat 7.2.111. *Market Share Is Conditional*
Intraday market share is not a stable venue attribute. It changes with time of day, expiry windows, benchmark prices, news, and whether investors must trade on the venue that defines the relevant reference price.

Definition 7.2.112. *Fragmentation Efficiency Index*
Let q_n be the share of trading or liquidity observed in venue n, with $\sum_{n=1}^{N} q_n = 1$.
The Fragmentation Efficiency Index (FEI) is normalised entropy: $\mathfrak{F}(q_1, \ldots, q_N) = \frac{-\sum_{n=1}^{N} q_n \log q_n}{\log N}$.
Here N is the number of venues and $\mathfrak{F}$ is the dispersion score for the share vector q.
$\mathfrak{F} = 0$ when all flow is in one venue and $\mathfrak{F} = 1$ when flow is evenly split across venues.

Taxonomy 7.2.113. *Fragmentation Measurement Segmentation*

i. *trade-count FEI*: probability that the next print occurs away from the primary venue;
ii. *value-share FEI*: economic dispersion of traded notional;
iii. *displayed-liquidity FEI*: dispersion of visible book quantity, usually by price level;
iv. *session FEI*: separate continuous trading from auctions and benchmark windows;
v. *universe FEI*: compare large caps, small caps, sectors, or index constituents separately.

The same formula answers different questions only after the underlying share definition is fixed.

Caveat 7.2.114. *FEI Venue Count*
FEI depends on the considered venue set. If t is the measurement date and N is the venue count, adding an empty venue mechanically lowers the index from $\mathfrak{F}(t)$ to $\frac{\log N}{\log(N+1)}\mathfrak{F}(t)$. Compare FEI through time only after controlling for venue openings, closures, and data coverage.

Caveat 7.2.115. *Fragmentation of What*
Market-share and entropy metrics describe how traded value or trades are dispersed. They do not by themselves measure protocol costs, connectivity costs, hidden liquidity, or the difficulty of accessing each pool.

Mechanism 7.2.116. *Aggregated Liquidity*
Consolidating visible depth across venues can reduce impact. A market order that would walk the book on one venue may execute at the same or better price when routed across several books. The benefit comes from replacing one steep book with several shallower local walks.

Caveat 7.2.117. *Intraday Non-Stationarity*
Venue share and traded volume are time-of-day conditional.
Opens, closes, lunch periods, linked-market openings, scheduled news, and expiry dates can create repeatable volume spikes, so a single daily average is a weak routing prior.

Taxonomy 7.2.118. *Volume-Curve Segmentation*

i. *session*: open, continuous core, lunch break, close, and after-hours;

ii. *event*: macro releases, earnings, index rebalances, and derivative expiries;

iii. *reference price*: auction close, VWAP window, snapshot median, or settlement formula;

iv. *calendar*: weekday, month-end, daylight-saving shifts, holidays, and local trading hours;

v. *venue*: primary auctions, lit continuous books, dark pools, and internalised flow.

Caveat 7.2.119. *Average Trade Size*
Falling average trade size does not mechanically mean smaller investor orders.
Fragmentation can split one aggressive order across several passive orders and venues, so trade prints reveal book mechanics as well as investor size.

Empirical Finding 7.2.120. *Liquidity and Capitalisation Segmentation*
Across stocks, fragmentation tends to increase with liquidity and market capitalisation.
For a given stock, high-volume or stressed days can reduce fragmentation as flow returns to the historical venue for depth, certainty, or familiar trading processes.

Mechanism 7.2.121. *Why Large Caps Fragment First*
HFTs and liquidity arbitrageurs warehouse large-cap risk more easily: stock borrow is deeper, index or futures hedges are cleaner, inventory can be offset in correlated instruments. Small caps therefore catch up more slowly.

Rule 7.2.122. *Maker-Taker Pricing*
Maker-taker venues charge liquidity takers and pay rebates to liquidity makers. The displayed price can therefore differ from the economic price after fees and rebates.

Caveat 7.2.123. *Smart Routers vs Smart Fee Savers*
Routing cannot ignore fee schedules. Clearing and settlement may be netted by member inventory, while venue fees can depend on monthly turnover, cross-venue volume, or maker-taker status. The marginal cost of one child order is therefore path-dependent.

Mechanism 7.2.124. *Rebate Effects*
Rebates attract passive flow and can deepen displayed queues, but they also change routing incentives. A broker or router must compare net execution quality, not only the visible quote or the nominal likelihood of a fill. The economic best venue is the one with the best expected all-in fill, not necessarily the best displayed price.

Caveat 7.2.125. *Subsidised Share*
Maker agreements and fee holidays can create market share before a venue has durable natural flow. When the subsidy ends, share can fall quickly unless the venue has become part of traders' routing logic.

Caveat 7.2.126. *Post-Trade Barriers*
Fragmentation also depends on clearing, settlement, membership, and registration rules. A venue may fail to gain market share if post-trade access makes trading operationally expensive.

Case Study 7.2.127. *Spain as Fragmentation Control Case*
Spain's delayed MTF entry after MiFID makes it useful for separating fragmentation effects from broader European market stress. When alternative venues could not clear and settle efficiently, primary-market dominance persisted even while fragmentation advanced elsewhere.

Mechanism 7.2.128. *Event-Driven Volume Clustering*
Volume clusters when many traders need to act at the same reference time.
Linked market openings import information; scheduled macro releases create common reassessment points; derivative expiries force cash-future convergence and settlement trades.

Caveat 7.2.129. *Reference Rules Create Volume*
If a benchmark is computed from a close, auction price, or short snapshot window, traders with benchmark risk are pulled toward that window. Volume then reflects the rule as much as investor preference.

Caveat 7.2.130. *Duplicate Liquidity*
The same HFT market maker may quote on several venues and cancel remaining quotes once one venue trades. Aggregated displayed depth may overstate executable liquidity unless router discounts correlated cancellations.

Mechanism 7.2.131. *Spread-Share Feedback*
Best-execution and SOR logic move flow toward venues with better net spreads and reliable fills.
More flow attracts market makers, which can tighten spreads further. The same loop can run backward when spreads deteriorate or fills become volatile. This is why venue quality can become self-reinforcing after a venue crosses a minimum flow threshold.

Taxonomy 7.2.132. *Venue Quality Segmentation*

i. *tightness*: quoted and effective bid-ask spread;

ii. *depth*: displayed size and replenishment near the best limits;

iii. *resilience*: continuity of liquidity during volatile intervals;

iv. *coverage*: how evenly liquidity is supplied across index constituents;

v. *access*: members, connectivity, latency, order types, and data quality.

Definition 7.2.133. *Coverage Liquidity Metric*
Coverage asks how many stocks are needed to explain a target share of index turnover, commonly 80%.
Low coverage means liquidity is concentrated in a few names; high coverage means liquidity is more balanced across constituents.

Definition 7.2.134. *Gini Liquidity Imbalance*
A turnover Gini index summarises inequality in trading activity across stocks. A value near zero means turnover is evenly distributed; a value near one means a small subset of stocks dominates index liquidity.

Model 7.2.135. *Lorenz-Curve Gini*
Let $L(u)$ be the Lorenz curve for venue share, turnover, or actor activity after observations are ordered from smallest to largest contribution. The Gini coefficient is $G = 1 - 2\int_0^1 L(u)\,du$.
Here G is the Gini coefficient, u is the cumulative fraction of observations, and $L(u)$ is the cumulative fraction of the measured quantity. Equal contribution gives $G = 0$; complete concentration gives G close to 1.

Taxonomy 7.2.136. *Gini Measurement Segmentation*

i. *venue share*: concentration of trading across venues for one asset or universe;

ii. *asset turnover*: concentration of liquidity across index constituents;

iii. *participant activity*: concentration of messages, trades, or liquidity supply across actors;

iv. *time segment*: separate open, continuous session, close, and event windows.

The same Gini value is meaningful only after fixing the unit being ranked.

Model 7.2.137. *Kolmogorov-Smirnov Distribution Test*
For observations $X_1, \ldots, X_n$, the empirical distribution is $F_n(x) = \frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_{\{X_k \le x\}}$.
For two samples with empirical distributions F_n^X and F_m^Y, two-sample statistic is $D_{n,m} = \sup_x \left|F_n^X(x) - F_m^Y(x)\right|$.
Here X_k is observation k, n and m are sample sizes, F_n^X and F_m^Y are empirical distribution functions, and $D_{n,m}$ is largest vertical gap. Reject equal distributions at α when $\sqrt{\frac{nm}{n+m}}D_{n,m} >$ Kolmogorov critical value.

Practitioner Note 7.2.138. *Venue-Share Distribution Comparison*
Use the KS test to compare cross-sectional venue-share distributions across venues or dates. A high Gini says trading is concentrated; a rejected KS test says the whole distribution differs.

Mechanism 7.2.139. *HFT Perimeter Extension*
Fast market makers usually enter liquid large caps first because hedging, borrowing, and inventory recycling are easier. As competition compresses returns, they extend toward less liquid names, improving coverage until hedging cost and inventory risk become uneconomic.

Caveat 7.2.140. *Primary Outage Dependency*
Many alternative venues peg orders, route decisions, or price controls to primary-market quotes. In a primary outage, those may pause alternative trading or create separate prices on venues that continue to trade.

Principle 7.2.141. *Fragmentation Needs Shared References*
Fragmented markets work best when participants share timely quotes, trades, and reference prices. Competition adds liquidity only when the information layer keeps venues synchronised.

Caveat 7.2.142. *Transparency Is Not Certainty*
Pre-trade transparency reveals quotes, not guaranteed liquidity.
Post-trade transparency gives more reliable prints after the fact, so small dark pools with weak pre-trade disclosure may still be informative through execution records.

Practitioner Note 7.2.143. *Fragmented Routing Objective*
A practical router ranks venues by expected net cost:
expected cost = spread/impact + fees − rebates + latency risk + information cost.
Patient orders can seek rebates and hidden liquidity; urgent orders pay for immediacy and certainty.

Empirical Finding 7.2.144. *Primary-Market Variance Feedback*
If an alternative venue delivers volatile fills, rational routers reduce future allocation to it.
Lower variance makes routing fractions easier to estimate; higher variance pushes flow back to the primary market and can lower the alternative venue's share.

7.3 Information, Price Formation, and Spreads

7.3.1 Price Discovery

Mechanism 7.3.1. *Fragmentation and Price Formation*
Fragmentation can improve prices when venues remain synchronised by routing, arbitrage, and consolidated information. It can damage price formation when venues lose a common reference, when primary quotes become stale, or when market participants pause routing during stress.

Mechanism 7.3.2. *Fast Liquidity and Price Discovery*
Fast market makers help price discovery when they update quotes and arbitrage stale venues. They hurt it when speed investment mainly earns queue priority or fee rebates without adding durable liquidity. Fee schedules therefore affect the price formation process, not only venue revenue.

Caveat 7.3.3. *Outage Price Discovery*
If alternative venues keep trading while primary data or primary matching is unavailable, prices may temporarily split into parallel formation processes rather than one consolidated market price.

7.3.2 Strategic Trading Models

Taxonomy 7.3.4. *Informed Trader Style Segmentation*

i. *value*: estimate total fundamental value from broad information;
ii. *news*: estimate value changes from recent information;
iii. *information-oriented technical*: exploit predictable errors by other traders;
iv. *arbitrage*: trade relative mispricing across instruments with common valuation factors.

Taxonomy 7.3.5. *Informed Trading Speed Segmentation*

i. *aggressive*: trade quickly when information is short-lived or competitors know it;
ii. *stealth*: trade slowly when impact and information leakage dominate urgency;
iii. *inactive*: avoid trading when the price already reflects the information.

Taxonomy 7.3.6. *Insider-Trading Profitability Segmentation*

i. *materiality*: larger value revisions create larger gross opportunity;
ii. *exclusivity*: fewer informed competitors leave more liquidity before price adjustment;
iii. *time to public release*: near-term disclosure makes profits easier to realise but easier to detect;
iv. *liquidity*: deeper markets allow larger positions before impact erodes the signal;
v. *traceability*: communication trails, relationship links, and unusual timing raise enforcement risk.

The cases with the largest private payoff are often the worst compliance fact pattern: large, exclusive, traceable, and close to a public event.

Proposition 7.3.7. *Deterrence Condition for Illegal Information Trading*
Let G be expected gross trading gain from material nonpublic information, K execution cost and leakage, R residual price risk, q detection probability, and F penalty or disgorgement.
A purely self-interested trader is deterred when $G \leq K + R + qF$.

Proof. Expected private surplus is $G - K - R - qF$. Trading has positive expected surplus only when this quantity is positive; deterrence requires the opposite inequality. □

Definition 7.3.8. *Arbitrage Opportunity*
An arbitrage opportunity exists when related instruments trade outside their normal relation after adjusting for carry, conversion, delivery, funding, borrow, and execution costs. The trade is to buy the cheap leg and sell the expensive leg in a hedge portfolio.

Notation 7.3.9. *Basis and Arbitrage Spread*
Let B_t be the observed basis between a hedge portfolio and its reference leg, and let $B_t^\star$ be the fair basis implied by carry, delivery, conversion, dividends, funding, borrow, and other contract terms.
The *arbitrage spread* is $A_t = B_t - B_t^\star$.

Proposition 7.3.10. *No-Trade Band for Basis Arbitrage*
Let $K_t \geq 0$ be all-in round-trip cost and let $R_t \geq 0$ be compensation for residual basis risk. A symmetric basis trade is attractive only when $|A_t| > K_t + R_t$.

Proof. Trading the convergence of one unit of basis gives expected gross surplus $|A_t|$.
Net expected surplus is $|A_t| - K_t - R_t$, which is positive exactly under the stated condition. □

Taxonomy 7.3.11. *Arbitrage Characterisation Segmentation*

i. *price characterisation*: arbitrageurs enforce relative value and the law of one price;
ii. *quantity characterisation*: arbitrageurs supply liquidity, transform risk, and repackage exposures;
iii. *implementation characterisation*: arbitrageurs earn spread only if execution, funding, hedging is feasible.

The price view explains why mispricing closes.
The quantity view explains why the trade is socially useful and why costs bound the spread.

Taxonomy 7.3.12. *Arbitrage Type Segmentation*

i. *pure arbitrage*: basis strongly mean reverting as delivery, conversion, or cash-settlement links instruments;
ii. *speculative arbitrage*: basis expected to mean revert, but hedge portfolio nonstationary or model-dependent;
iii. *risk arbitrage*: payoff depends on a corporate event, legal condition, or deal completion;
iv. *statistical arbitrage*: factor models identify relative mispricing across many instruments.

Pure arbitrage can lose money if convergence is slow, financing fails, or implementation costs exceed spread.

Taxonomy 7.3.13. *Pure Arbitrage Mechanism Segmentation*

i. *shipping*: same asset trades in different locations, bounded by transport and storage costs;
ii. *delivery*: futures and cash prices converge toward delivery or settlement;
iii. *conversion*: one risk form is manufactured from another, such as options, ETFs, or commodity crushes;
iv. *cross-rate*: triangular or multi-leg exchange rates imply an inconsistent synthetic price.

Closest to mechanical arbitrage, but binding constraint is usually speed, balance sheet, or operational capacity.

Taxonomy 7.3.14. *Arbitrage Risk Segmentation*

i. *implementation*: legs fill at worse prices, or one leg fills while another does not;
ii. *basis*: the spread widens before convergence, forcing mark-to-market losses;
iii. *model*: the assumed fair relation is wrong or misses a contract feature;
iv. *carry*: funding, borrow, dividends, storage, insurance, or spoilage costs differ from expectation;
v. *staying power*: margin, financing, or risk limits force liquidation before convergence;
vi. *specific-factor*: an idiosyncratic shock breaks the relative-value relation.

Question is not only whether the basis is cheap, but whether the portfolio can survive the path to convergence.

Principle 7.3.15. *Common vs Specific Factor Test*
Relative-value trades work best when the spread move is caused by uninformed liquidity demand or slow common-factor adjustment. They fail when the move reflects a permanent instrument-specific shock.
Residualisation is therefore not cosmetic; it decides whether the spread is an opportunity or new information.

Principle 7.3.16. *Orthogonal Information Advantage*
The best informed traders are not only accurate; their errors are different from other traders' errors. Orthogonal information is more valuable than crowded information with the same average accuracy.

Principle 7.3.17. *Value Traders vs News Traders*
Value traders profit when price differs from value because liquidity demand or dealer inventory pressure moved price away from fundamentals.
News traders profit when value itself has changed and price has not caught up. They trade against each other when one mistakes stale price pressure for mispricing, or stale news for new information.

Caveat 7.3.18. *Informed Trading Paradox*
Perfectly informative prices eliminate informed profits; zero informed profits stop prices from becoming informative. Markets persist between these limits because information is costly, temporary, and unevenly distributed.

Caveat 7.3.19. *Skilled Speculation vs Inside Information*
Correct forecasts from public filings, alternative data with clean rights, channel checks, or independent modelling can be legitimate informed trading. The same trade becomes a compliance problem when the edge comes from material nonpublic information or breach of confidence. Timing alone cannot distinguish the cases.

Mechanism 7.3.20. *Insider-Trading Market Footprint*
Leaked or illegal information may appear as abnormal signed volume, option activity, price drift, widened spreads, quote fading, or dark-pool activity before news. These are surveillance leads, not proof: rumours, public-information analysis, hedging, and unrelated liquidity demand can look similar.

Definition 7.3.21. *Winner's Curse*
The winner's curse is the adverse information revealed by winning a common-value trade or auction.
A buyer who wins may be the most optimistic bidder; a seller who wins may be the most pessimistic offeror.

Proposition 7.3.22. *Winner's-Curse Bid Adjustment*
Let a buyer's estimate be $\widehat{V}$ and let $\omega_N = \mathbb{E}[\widehat{V} - V \mid \text{buyer wins among } N \text{ buyers}]$ be the expected overvaluation conditional on winning. With all-in cost c, a zero-profit bid is $b^\star = \widehat{V} - \omega_N - c$.

Proof. Conditional expected profit from bid b is $\mathbb{E}[V - b - c \mid \text{win}] = \widehat{V} - \omega_N - b - c$.
Setting this to zero gives $b^\star = \widehat{V} - \omega_N - c$. □

Taxonomy 7.3.23. *Winner's-Curse Severity Segmentation*

i. *competitor count*: more rival bidders increase the chance that winning selects an extreme estimate;
ii. *valuation dispersion*: noisier estimates require larger bid shading;
iii. *information overlap*: common data reduce dispersion; independent but noisy private estimates increase selection risk;
iv. *market power*: a trader facing few competitors may bid less aggressively and still win.

Taxonomy 7.3.24. *Bluffing and Manipulation Segmentation*

i. *rumourmonger*: spreads false or selectively framed information to make others trade;
ii. *price manipulator*: trades to make prices, volume, or timing look informative;
iii. *wash trader*: arranges offsetting trades with confederates to manufacture activity.

The bluff works only if other traders mistake the signal for information or urgent liquidity demand.

Principle 7.3.25. *Bluffers vs Informed Speculators*
Both may trade aggressively and advertise conviction. The difference is informational content: informed speculators trade on value-relevant beliefs, while bluffers create signals they hope others will misread.

Mechanism 7.3.26. *Value Traders Discipline Bluffs*
Value traders break bluffs by trading against clear mispricing. Where valuation, borrow, or funding is constrained, bluffs can persist and attract momentum flow.

Caveat 7.3.27. *Manipulation Evidence Problem*
Price reversals are not enough to prove manipulation. Prosecutors must usually separate a false signal or wash trade from a bad but legitimate valuation view, which is why small or careful bluffs are hard to punish.

Taxonomy 7.3.28. *Order-Anticipation Segmentation*

i. *front running*: trade before a known or inferred order;
ii. *sentiment-oriented technical*: predict trades uninformed traders will make;
iii. *quote matching*: step ahead of passive quotes to extract their option value;
iv. *squeezing*: corner scarce supply and force constrained traders to liquidate;
v. *stop gunning*: push prices to trigger clustered stop orders.

These strategies use information about future order flow rather than fundamental value.

Taxonomy 7.3.29. *Specialist Order-Flow Strategy Segmentation*

i. *flow speculation*: trade from early knowledge of routed orders or imbalance;
ii. *quote matching*: improve by the minimum tick to step ahead of public interest;

iii. *cream skimming*: fill less toxic marketable flow and leave harder flow to the book;

iv. *stopping option*: guarantee a price, then choose execution after observing later order flow.

These strategies monetise privileged sequencing, so audit trails and public-order precedence are central controls.

Mechanism 7.3.30. *Front-Running Effect*
Front runners accelerate anticipated order impact. They can speed price discovery for informed flow, but usually raise execution costs and discourage large-order exposure.

Mechanism 7.3.31. *Sentiment-Oriented Technical Trading*
These traders forecast predictable non-value flow: tax timing, index changes, hedging, funding needs, or behavioural demand. Profits decay when value traders correct the move or the pattern crowds.

Caveat 7.3.32. *Quote Matching and Tick Protection*
Quote matching extracts value from standing limit orders without supplying durable liquidity. Time precedence and meaningful tick sizes protect passive liquidity by making queue jumping costly.

Caveat 7.3.33. *Squeezes and Stop Gunning*
Squeezes and stop gunning profit from forced trades. They usually make prices less informative, raise transaction costs, and reduce liquidity for traders who must close positions.

7.3.3 Liquidity and Market Impact

Taxonomy 7.3.34. *Market Impact Lens Segmentation*

i. *theoretical*: strategic splitting and information inference;

ii. *reduced-form*: price response to signed volume, participation, volatility, and time;

iii. *practitioner*: cost of urgency, benchmark choice, and venue access.

Definition 7.3.35. *Temporary and Permanent Impact*
Temporary impact is the price concession needed to obtain immediacy and is expected to decay as liquidity replenishes. Permanent impact is the residual price shift after decay, but it may reflect information in the order rather than mechanical pressure.

Mechanism 7.3.36. *Propagator Intuition*
A trade has an impulse effect on price, and that effect decays through time. The observed price path is the sum of many signed trade impulses with different decay profiles. This separates immediate pressure from later relaxation: a print can move price now without requiring the full move to be permanent.

Proposition 7.3.37. *Round-Trip Manipulation Profit Test*
Suppose a trader completes a zero-inventory round trip, buying and selling total quantity $Q > 0$. Let $\bar{P}_B$ be the average buy price, $\bar{P}_S$ the average sale price, and c the all-in cost per share.
The round trip is profitable if and only if $\bar{P}_S - \bar{P}_B > c$.

Proof. The trader pays $Q\bar{P}_B$ on buys and receives $Q\bar{P}_S$ on sales. Net profit after costs is $Q(\bar{P}_S - \bar{P}_B - c)$, which is positive exactly under the stated condition. □

Principle 7.3.38. *Impact Symmetry Disciplines Bluffing*
Liquidity suppliers are vulnerable when buys and sells, large and small orders, or fast and slow trades receive inconsistent price impact. A manipulation-resistant rule makes impact depend on net pressure and risk, so a zero-net sequence cannot mechanically create a higher average sale price than buy price after costs.

Caveat 7.3.39. *Permanent Impact Identification*
Calling a long-run price move permanent impact is risky unless market moves, sector moves, autocorrelated flow, and the signal that caused the trade have been separated.

7.3.4 Volatility and Liquidity Interaction

Mechanism 7.3.40. *Liquidity Spiral*
A large order can consume depth, widen spreads, and raise volatility. Higher volatility makes liquidity providers reduce size or cancel quotes, forcing the next order to trade through less liquidity and amplifying the move. The state variable is confidence in replenishment: once traders doubt the book will refill, everyone demands more price concession at the same time.

Case Study 7.3.41. *Flash Crash Mechanism*
The 2010 Flash Crash illustrates a liquidity spiral: a large automated futures sell program met thin liquidity, high-frequency traders recycled inventory quickly, aggressive flow migrated into cash equities, spreads widened, and some routers or venues paused.
The trigger was an order; the crash mechanism was fragile liquidity under linked-market stress.

Model 7.3.42. *Depth-Driven Flash Crash Toy Model*
On one side of the book, let F_t be the market-order arrival rate, $\bar{F}$ the normal refill rate, $r_t = F_t - \bar{F}$ the overflow, and $I_t = 1/D_t$ inverse depth. A compact depth-impact dynamic is $\frac{dI_t}{I_t} = \sigma_I dW_t + r_t dt$.
Here D_t is market depth, I_t is impact, σ_I is liquidity-shock volatility, and W_t captures random liquidity shocks. Positive overflow r_t makes inverse depth compound upward, so each additional trade has larger price impact than the last.

Mechanism 7.3.43. *Hot-Potato Amplification*
If a fraction ρ of traders follows volume mechanically, initial trading rate r_0 can amplify roughly as $r_t \simeq \frac{1-\rho^t}{1-\rho} r_0$.
When ρ is high, overflow consumes depth faster than passive liquidity can refill it. The loop is mechanical: trading creates volume, volume targets force more trading, and depth absorbs the loop until it fails.

Taxonomy 7.3.44. *Flash-Crash Control Segmentation*

i. *rate controls*: cap participation rates or stop volume-following feedback;
ii. *depth controls*: require or incentivise fast refill near the touch;
iii. *price controls*: pause trading after large price moves;
iv. *cross-market controls*: coordinate futures, ETFs, cash equities, and options;
v. *data controls*: detect overflow, quote withdrawal, and broken references in real time.

Caveat 7.3.45. *Systemic Risk Link*
Fragmentation is not systemic risk by itself. It becomes dangerous when market participants react to stress in the same direction, when related venues desynchronise, or when liquidity withdrawal in one instrument propagates to hedges, ETFs, futures, and correlated stocks.

Taxonomy 7.3.46. *Shock Propagation Channels*

i. *cross-asset hedging*: futures, ETFs, options, and cash baskets transmit inventory pressure;
ii. *routing feedback*: routers chase fading liquidity or pause when fills become unreliable;
iii. *quote withdrawal*: market makers widen or cancel when volatility exceeds their risk budget;
iv. *reference-price dependency*: pegged and benchmark orders inherit stale or broken references;
v. *correlated constraints*: margin, risk limits, and kill switches cause simultaneous de-risking.

Empirical Finding 7.3.47. *Trade Count as HFT Proxy*
Rising trades per day can proxy for higher high-frequency participation when trade sizes shrink and message intensity rises. It is not sufficient alone: the same count can reflect genuine liquidity, excessive slicing, or stress-induced churn.

7.4 Order Flow and Empirical Microstructure

7.4.1 Trade Classification and Signed Flow

Taxonomy 7.4.1. *Trade-Sign Classification Segmentation*

i. *quote rule*: compare trade price with the contemporaneous midpoint;

ii. *tick rule*: classify upticks as buyer-initiated and downticks as seller-initiated;

iii. *block-price rule*: for negotiated blocks, compare the block price with the prevailing quote.

Classification is an inference about who demanded immediacy, not a fact printed on the trade.

Method 7.4.2. *Lee-Ready Classification*
Use the quote rule first: trades above the midpoint are buyer initiated, trades below it are seller initiated. If the trade is exactly at the midpoint, use the tick rule: classify an uptick as buyer initiated and a downtick as seller initiated; if price is unchanged, look back to the last non-zero tick.

Definition 7.4.3. *Signed Volume*
For trade sign $\epsilon_t \in \{+1, -1\}$ and size q_t, signed volume is $\epsilon_t q_t$. Aggregating signed volume gives a simple order-flow pressure measure, but classification errors and trade splitting can create false persistence.

Caveat 7.4.4. *Passive-Aggressive Ambiguity*
A limit order can be economically aggressive even when it waits passively in the book. Trade-sign algorithms classify the side that demanded immediate execution, not the trader with the stronger valuation motive.

Empirical Finding 7.4.5. *Block Initiator Asymmetry*
If a block print is closer to bid than ask, the seller likely initiated; if it is closer to ask, the buyer likely initiated. Historical block evidence found many more seller-initiated than buyer-initiated blocks, consistent with latent demand, price-discrimination, and asymmetric-information frictions being easier to solve for sellers than buyers.

7.4.2 Order Flow Imbalance

Definition 7.4.6. *Order Flow Imbalance*
Order flow imbalance measures the net pressure from buyer-initiated and seller-initiated activity over a chosen horizon. It is useful because short-horizon prices often respond more to signed pressure than to unsigned volume.

Model 7.4.7. *Signed-Trade Imbalance*
For trade sign $\epsilon_i \in \{-1, +1\}$ and size q_i, trade imbalance over window W is $OFI_W = \sum_{i \in W} \epsilon_i q_i$.
Normalise by total volume, ADV, or displayed depth when comparing symbols or regimes.

Model 7.4.8. *Queue Imbalance*
At the best quotes, queue imbalance is $QI_t = \frac{B_t - A_t}{B_t + A_t}$, where B_t is displayed bid size and A_t is displayed ask size. Positive imbalance means more displayed demand at the bid than supply at the ask.

Taxonomy 7.4.9. *Imbalance Feature Segmentation*

i. *trade imbalance*: signed executed volume;

ii. *quote imbalance*: size difference at the best bid and ask;

iii. *multi-level imbalance*: weighted depth across several price levels;

iv. *event imbalance*: net add, cancel, and execute messages at each queue;

v. *cross-venue imbalance*: pressure aggregated across fragmented books.

Caveat 7.4.10. *Imbalance Is Horizon-Dependent*
Queue imbalance can predict the next quote move but fail over longer horizons once hidden liquidity, replenishment, and informed trading enter. Always bind the feature to a forecast horizon.

7.4.3 Stylized Facts of High-Frequency Data

Taxonomy 7.4.11. *High-Frequency Stylized Facts*

i. *heavy tails*: short-horizon returns have more extremes than Gaussian returns;

ii. *volatility clustering*: large moves tend to arrive near other large moves;

iii. *long memory*: signs, durations, and volatility can remain persistent;

iv. *intraday seasonality*: volume, spread, volatility, and depth depend strongly on clock time.

These facts are data conditions, not trading signals by themselves.

Model 7.4.12. *Signature Plot*
Let S_t be price, $X_t = \log S_t$ log-price, and Δ the sampling step.
The realised variance estimator is $\hat{V}_R(\Delta) = \sum_{j=1}^{T/\Delta} r_\Delta(j)^2$, where $r_\Delta(j) = X_{j\Delta} - X_{(j-1)\Delta}$. Here T is the horizon and $r_\Delta(j)$ is the jth log return. In frictionless continuous time, $\hat{V}_R(\Delta)$ converges to integrated variance as $\Delta \downarrow 0$; in real data, bid-ask bounce and quote noise make the estimator rise at very fine sampling.

Mechanism 7.4.13. *Microstructure Noise in Volatility*
Observed log-price can be written $X_t = M_t + \epsilon_t$, where M_t is the efficient semimartingale price and ϵ_t is observation noise. At high frequency, returns include both efficient-price changes and noise differences, so realised variance can diverge as sampling becomes too fine.

Definition 7.4.14. *Epps Effect*
The Epps effect is the fall in estimated cross-asset correlation as the sampling interval becomes very short. It is driven by asynchronous trades, lead-lag relations, tick effects, and other microstructure frictions.

Model 7.4.15. *Realised Covariance*
For two log-price processes X^1 and X^2, realised covariance at step Δ is $\hat{C}_R(\Delta) = \sum_{j=1}^{T/\Delta} r^1_\Delta(j) r^2_\Delta(j)$.
Here $r^a_\Delta(j)$ is the jth return of asset a. The estimator needs synchronised observations; refresh-time or previous-tick sampling is usually required before interpreting high-frequency correlations.

Model 7.4.16. *Time-Series Decomposition*
Microstructure time series can be decomposed into trend, seasonality, and residual variation: $X_t = C_t + S_t + \epsilon_t$ or $X_t = C_t S_t + \epsilon_t$. Here X_t is the observed series, C_t is the trend, S_t is the seasonal component, ϵ_t is unexplained noise, and t indexes equally spaced observations.

Definition 7.4.17. *Centred Moving Average*
For odd window length $2k+1$, the centred moving average is $MA_t = \frac{X_{t-k}+\cdots+X_t+\cdots+X_{t+k}}{2k+1}$. For even window length $2k$, use half weight at the endpoints: $A_t = \frac{0.5X_{t-k}+X_{t-k+1}+\cdots+X_{t+k-1}+0.5X_{t+k}}{2k}$.
Here MA_t is the local trend estimate and k controls the smoothing width. The window should match the seasonal period, or an integer multiple of it, when estimating seasonality.

Model 7.4.18. *Additive Deseasonalisation*
For period index i and within-period slot j, write $X_{ij} = C_{ij} + S_j + \epsilon_{ij}$.
If MA_{ij} estimates C_{ij}, then $\hat{S}_j = \frac{1}{n}\sum_{i=1}^{n}(X_{ij} - MA_{ij})$, $S_j = \hat{S}_j - \frac{1}{p}\sum_{k=1}^{p}\hat{S}_k$. Here p is the number of slots per period, n is the number of periods, and S_j is centred so the seasonal effects sum to zero.

Model 7.4.19. *Multiplicative Deseasonalisation*
For proportional seasonality, write $X_{ij} = C_{ij}S_j + \epsilon_{ij}$ and estimate $\hat{S}_j = \frac{1}{n}\sum_{i=1}^{n}\frac{X_{ij}}{MA_{ij}}$, $S_j = \hat{S}_j / \left(\frac{1}{p}\sum_{k=1}^{p}\hat{S}_k\right)$.
Here S_j is normalised to average one. This is natural for volume curves because activity is often a relative multiplier of the day's baseline.

Model 7.4.20. *Volume Seasonality Factorisation*
A compact volume model is $V(y,m,d,n,\omega) = \{f(n)+g(d,n)\}F(m)G(y)v(n,\omega)$, where $f(n)$ is the usual intraday profile, year y, month m, day d, time slot n, and event state ω, $g(d,n)$ captures special-day effects, $F(m)$ is monthly seasonality, $G(y)$ is yearly scale, and $v(n,\omega)$ is residual slot-level variation.

Practitioner Note 7.4.21. *Seasonality Before Signal*
Remove predictable clock-time effects before comparing venues, estimating impact, or training intraday features. Otherwise a noon lull, close auction, expiry day, or macro window can look like alpha.

7.4.4 Price Impact Estimation

Definition 7.4.22. *Execution-Time Impact*
Execution-time impact is the price move observed while the order is being traded, measured from the perspective of the order side and often normalised by spread or volatility.

Empirical Finding 7.4.23. *Impact Has a Hump Shape*
Intraday impact usually rises during execution and partly decays afterwards. The remaining shift is the candidate permanent component; the decayed part is temporary impact and liquidity replenishment.

Taxonomy 7.4.24. *Impact Driver Segmentation*

i. *stock characteristics*: liquidity, volatility, spread, tick size, and typical depth;

ii. *order characteristics*: size, duration, participation rate, side, and aggressiveness;

iii. *timing*: trend, rebalancing pressure, signal arrival, auctions, and market-wide events;

iv. *venue access*: lit depth, dark liquidity, queue priority, and routing constraints.

Model 7.4.25. *Participation-Duration Shape*
A compact empirical specification writes market impact as increasing in duration and participation:

$$\text{Market Impact} \propto \text{Duration}^{\alpha} \times \text{Participation}^{\gamma}.$$

Here α is the duration exponent, γ is the participation exponent. The exponents are calibrated from executions; in many datasets the participation exponent is concave, so extra participation hurts, but less than linearly. The model is not structural; it is a compact way to express the practitioner trade-off between urgency and footprint.

Model 7.4.26. *Order-Flow Impact Regression*
A Glosten-Harris style reduced form writes $\Delta P_t = \lambda\, \epsilon_t q_t + \psi\, \Delta(\epsilon_t q_t) + \eta\, \Delta\epsilon_t + u_t$ where ϵ_t is trade sign and q_t is trade size. The level signed-size term estimates permanent information impact; changes in sign or signed size capture transitory spread and liquidity-supply effects.

Empirical Finding 7.4.27. *Index Reconstitution Impact*
Index additions and deletions create predictable one-sided flow. Price impact is largest when forced index demand is large relative to stock liquidity; partial reversal after the event suggests temporary demand pressure, while the residual move reflects changed ownership, attention, or information.

Caveat 7.4.28. *Order-Flow Regression Identification*
Signed order flow is endogenous: informed traders choose when and how much to trade. Use controls for market moves, time of day, volatility, spread, and order splitting before interpreting coefficients as causal impact.

Caveat 7.4.29. *Timing Is Not Noise*
Two orders with identical size and participation can have different impact because one trades with the market and the other trades against it. Impact estimation therefore has to separate execution mechanics from the signal that caused the trade.

7.4.5 Market Impact Models

Taxonomy 7.4.30. *Model Family Segmentation*

i. *Almgren-Chriss*: optimise a schedule by balancing expected impact cost against execution risk;

ii. *linear impact*: use tractable instantaneous or temporary cost proportional to trading rate or signed size;

iii. *propagator*: model each signed trade as a decaying impulse in future prices;

iv. *empirical square-root*: summarise metaorder cost as concave in order size relative to market volume.

Principle 7.4.31. *Permanent Impact or Price Anticipation*
Long-horizon impact mixes metaorder pressure and information revealed by trading. Remove the information effect before calling the residual permanent impact.

Taxonomy 7.4.32. *Permanent-Impact View Segmentation*

i. *mechanical view*: prices move because buying and selling pressure are imbalanced;

ii. *informational view*: prices move because the trade reveals new expectations about value;

iii. *hybrid view*: executions contain both pressure and information, so identification matters more than labels.

Model 7.4.33. *Propagator Midprice Dynamics*
In trade time, a standard propagator writes the midprice as $m_k = m_{-\infty} + \sum_{k'<k} G(k-k')\epsilon_{k'} + \eta_k$.

Here m_k is the midprice after trade k, $\epsilon_k \in \{-1, +1\}$ is trade sign, $G(\cdot)$ is the decaying impact kernel, η_k is fair-price noise, and $m_{-\infty}$ is an initial price level.

Mechanism 7.4.34. *Trade-Sign Memory and Kernel Decay*
Trade signs are positively autocorrelated because large orders are split and similar agents trade together.
A decaying kernel offsets this persistence: impact must decay enough that predictable order flow does not imply predictable price drift.

Model 7.4.35. *Price Response Curve*
For horizon h, define the response curve by $\mathcal{R}(h) = \mathbb{E}[(m_{k+h} - m_k)\epsilon_k]$.
Here $\mathcal{R}(h)$ measures the expected signed price move after a trade. It is estimated with the trade-sign autocorrelation $C(\ell) = \mathbb{E}[\epsilon_{k+\ell}\epsilon_k]$ and the kernel increments $\Delta G(\ell) = G(\ell+1) - G(\ell)$.

Taxonomy 7.4.36. *Propagator Kernel Segmentation*

i. *non-parametric*: estimate discrete kernel increments directly from response and sign autocorrelation;
ii. *exponential*: impose fast decay with a small number of parameters;
iii. *power law*: allow slow decay and long-memory order flow;
iv. *agent-specific*: estimate separate kernels or intensities by participant class.

Process 7.4.37. *Debiasing Long-Horizon Impact*

i. classify trades by whether they are likely information-driven or liquidity-driven;
ii. remove market and sector moves with a benchmark or factor model;
iii. adjust for autocorrelated metaorder flow after the execution day;
iv. measure the remaining post-execution drift over several days.

Model 7.4.38. *CAPM-Like Post-Execution Decomposition*
For event day D, a stock's post-execution move can be split into systematic and idiosyncratic parts:

$$\log(P_d) - \log(P_{d-1}) = \beta\,[\log(I_d) - \log(I_{d-1})] + \Delta W_d.$$

Here P_d is the stock close, I_d the reference index, β the index exposure, and ΔW_d the residual stock-specific move. Cumulative ΔW_d isolates the profile after removing index co-movement. The intuition is attribution: do not charge the execution for a market move that the whole index experienced.

Caveat 7.4.39. *Autocorrelated Flow Bias*
Metaorder flow can remain signed and autocorrelated for days. If later same-direction executions are ignored, their impact may be falsely attributed to the original order.

7.4.6 Trade Duration and Event Time

Definition 7.4.40. *Trade Duration*
Trade duration is the elapsed time between consecutive events, usually trades, quotes, or order-book messages. Short durations indicate high activity; long durations indicate sparse information arrival or weak liquidity.

Taxonomy 7.4.41. *Market Time Segmentation*

i. *clock time*: calendar seconds or milliseconds;
ii. *trade time*: one step per print;
iii. *quote time*: one step per quote or book update;
iv. *volume time*: one step per fixed traded quantity or notional amount;
v. *business time*: activity-adjusted time after removing intraday seasonality.

Calendar risk, event response, and liquidity consumption are different in different times.

Model 7.4.42. *Duration Notation*
If event times are $0 < \tau_1 < \tau_2 < \cdots$, duration is $\Delta\tau_i = \tau_i - \tau_{i-1}$. A basic intensity view treats high event intensity as short expected duration and low intensity as long expected duration.

Principle 7.4.43. *Event Time for Microstructure Laws*
Many order-book relations are cleaner in event time because the book changes when messages arrive, not when the wall clock advances. Clock time is still needed for latency, risk limits, news, and benchmark windows.

Caveat 7.4.44. *Activity Time Can Hide Stress*
Business-time transformations stabilise model residuals, but they can also remove exactly the stress state a trader cares about. Keep both clock-time and event-time diagnostics.

7.4.7 Statistical Models for Order Flow

Definition 7.4.45. *Point-Process View*
Order-book events can be modelled as point processes: random arrival times for insertions, cancellations, and transactions. The intensity λ is the expected event rate over a short interval.

Definition 7.4.46. *Counting Process and Duration*
For event times $(\tau_i)_{i\geq 1}$, the counting process and duration process are $N_t = \sum_{i\geq 1} \mathbf{1}_{\{\tau_i \leq t\}}$, $\Delta\tau_i = \tau_i - \tau_{i-1}$. Here N_t counts events up to time t, τ_i is the ith event time, and $\Delta\tau_i$ is the waiting time between consecutive events.

Definition 7.4.47. *Conditional Intensity*
For filtration $\mathcal{F}_t$, the event intensity is $\lambda(t \mid \mathcal{F}_t) = \lim_{h\downarrow 0} \frac{1}{h}\mathbb{P}(N_{t+h} - N_t > 0 \mid \mathcal{F}_t)$. Here $\mathcal{F}_t$ is the information available before t. The intensity is the instantaneous event rate conditional on that information.

Taxonomy 7.4.48. *Order-Book Event Segmentation*

i. λ^+: limit-order insertions that add liquidity;
ii. λ^c: cancellations that remove resting liquidity;
iii. λ^x: transactions that consume resting liquidity.

Model 7.4.49. *Homogeneous Poisson Baseline*
The simplest model treats λ^+, λ^c, and λ^x as constant and independent of the book state. It is a useful null model, but it cannot reproduce realistic queue-size distributions because event rates are strongly state-dependent. Its value is diagnostic: any predictive model should beat this memoryless arrival clock.

Definition 7.4.50. *Integrated Intensity*
The integrated intensity over an inter-event interval is $\Lambda(\tau_{i-1}, \tau_i) = \int_{\tau_{i-1}}^{\tau_i} \lambda(s)\, ds$. Here Λ measures event time in intensity units. Under the time-change view, a correctly specified point process becomes a unit-rate Poisson process after this clock transformation.

Model 7.4.51. *Linear Hawkes Process*
A one-dimensional Hawkes intensity is $\lambda(t) = \lambda_0(t) + \int_{-\infty}^{t} \nu(t-s) dN_s = \lambda_0(t) + \sum_{\tau_i < t} \nu(t - \tau_i)$. Here $\lambda_0(t)$ is baseline intensity, $\nu(\cdot)$ is the excitation kernel, and past events raise current intensity. Hawkes models fit clustered liquidity because trades or messages make near-future events more likely.

Model 7.4.52. *Exponential Hawkes Kernel*
With $\nu(u) = \alpha e^{-\beta u}\mathbf{1}_{\{u\geq 0\}}$ and constant λ_0, the intensity is $\lambda(t) = \lambda_0 + \sum_{\tau_i < t} \alpha e^{-\beta(t-\tau_i)}$. Here α is jump size, β is decay speed, and α/β is the branching ratio. Stationarity requires $\alpha/\beta < 1$, and then $\mathbb{E}[\lambda(t)] = \lambda_0/(1-\alpha/\beta)$.

Mechanism 7.4.53. *Conditioning Improves Queue Dynamics*
Queue-reactive models condition event intensities on current and neighbouring queue sizes. Small best queues are easier to consume; large queues can trigger nearby cancellations. The model becomes more realistic because the book state changes trader incentives before the next event arrives.

Taxonomy 7.4.54. *Conditioning Segmentation*

i. *unconditional*: event type only;
ii. *own queue*: event type and current queue size;
iii. *neighbour queues*: own queue plus adjacent bid and ask queues;
iv. *event time*: count events rather than seconds when clock time is less informative.

Taxonomy 7.4.55. *Marked Flow Segmentation*

i. *event type*: insertions, cancellations, market orders, and trades;
ii. *side*: buy-initiated and sell-initiated flow;
iii. *agent class*: broker, HFT, investor, dealer, or internaliser when identifiers exist;
iv. *venue*: lit, dark, auction, and internalised pools.

A marked Hawkes model lets one class of event excite or inhibit another, useful when liquidity arrives in clusters.

7.4.8 Feature Engineering for Microstructure Alpha

Taxonomy 7.4.56. *Relative-Value Feature Segmentation*

i. *basis*: raw spread, carry-adjusted spread, z-score, percentile, and distance to arbitrage bounds;

ii. *hedge quality*: rolling beta, cointegration residual, factor residual, and hedge-ratio instability;

iii. *convergence clock*: time to delivery, expiry, fixing, rebalance, event date, or funding reset;

iv. *carry*: funding rate, borrow fee, dividend, storage, roll yield, and collateral return;

v. *execution synchrony*: leg liquidity, queue priority, fill correlation, and latency between venues;

vi. *stress capacity*: margin usage, drawdown to liquidation, short recall risk, and borrow availability.

These features separate cheapness from tradability.

Model 7.4.57. *Factor-Residual Arbitrage View*
For instruments i and j, write $r_i = \beta_i^\top f + \epsilon_i$ and $r_j = \beta_j^\top f + \epsilon_j$. A relative-value signal should explain whether the spread move came from common factors f, hedge-ratio error $\beta_i - \beta_j$, or residual shocks $\epsilon_i - \epsilon_j$.

Principle 7.4.58. *Alpha Needs Comparative Advantage*
A feature is useful only if it captures information or execution capacity that competitors do not fully exploit. A high in-sample score without an economic reason is usually a selection result, not an edge.

Taxonomy 7.4.59. *Microstructure Alpha Evaluation Segmentation*

i. *forecast*: relation to future midprice, spread, fill, or volatility;

ii. *tradability*: spread, queue loss, borrow, fees, and venue access;

iii. *capacity*: decay of performance with size and participation;

iv. *stability*: sensitivity to regime, tick size, latency, and participant mix;

v. *crowding*: whether many agents can observe and monetize the same state.

The feature is only the first filter; the executable strategy is the object being evaluated.

Principle 7.4.60. *Arbitrage Labels Need Path Information*
A spread that later converges is not automatically a good label.
The trade may still be infeasible if the interim drawdown breaches margin, borrow, inventory, or risk limits. Label both terminal convergence and maximum adverse excursion.

Taxonomy 7.4.61. *Arbitrage Event Label Segmentation*

i. *entry*: spread crosses estimated arbitrage bound after costs;

ii. *confirmation*: flow suggests uninformed pressure or slow common-factor adjustment;

iii. *failure*: specific news, borrow loss, deal break, or hedge-ratio regime change;

iv. *exit*: convergence, stop-out, forced liquidation, expiry, or capital reallocation.

7.4.9 Microstructure Data Engineering

Taxonomy 7.4.62. *Market-Data Depth Segmentation*

i. *L1*: best bid, best ask, sizes at the touch, and last trade;

ii. *L2*: visible depth by price level, usually aggregated by side and venue;

iii. *L3*: order-level messages, allowing queue reconstruction and cancellation analysis;

iv. *trades*: prints with price, size, time, venue, and often trade condition codes.

The empirical question should choose the shallowest feed that identifies the event cleanly.

Taxonomy 7.4.63. *Trading-Message Segmentation*

i. *order entry*: new, cancel, replace, reject, and execution-report messages;

ii. *market data*: public trades, quotes, status, and depth updates;

iii. *drop copy*: private execution and routing records for audit and reconciliation;

iv. *clearing feed*: allocations, positions, margin, fees, and settlement status.

FIX standardises much of order-entry layer. Venues still differ in native gateways, tags, throttles, recovery rules.

Caveat 7.4.64. *Ticker Is Not an Instrument Key*
Ticker symbols are display mnemonics. Robust microstructure data keys need venue, currency, instrument type, expiry, strike, option right, corporate-action history, and identifier validity dates.

Definition 7.4.65. *Microstructure Proportion*
A microstructure proportion is the share of events satisfying a binary condition: trade at the bid, execution on a venue, hidden-liquidity fill, HFT-tagged message, or cancellation before queue execution.

Model 7.4.66. *Bernoulli Proportion Estimator*
For independent indicators $X_1, \dots, X_n \in \{0, 1\}$ with success probability p,
$\hat{p} = \frac{1}{n} \sum_{\ell=1}^{n} X_\ell$, $\mathbb{E}[\hat{p}] = p$, $\mathrm{Var}(\hat{p}) = \frac{p(1-p)}{n}$. The usual large-sample interval is $\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.
Here X_ℓ is the indicator for event ℓ, n is the number of observations, p is the true success probability, $\hat{p}$ is the sample proportion, α is the error probability, and $z_{1-\alpha/2}$ is the corresponding standard-normal quantile.
For a 95% interval, $z_{1-\alpha/2} \approx 1.96$.

Caveat 7.4.67. *Normal Approximation Boundary*
The normal interval is fragile when the sample is small or the estimated proportion is close to zero or one. A practical check is $n\hat{p} > 5$ and $n(1 - \hat{p}) > 5$; otherwise use an exact or adjusted binomial interval.

Taxonomy 7.4.68. *Sampling Unit Segmentation*

i. *trade count*: estimates the probability that the next print has a property;
ii. *traded value*: estimates where economic turnover is concentrated;
iii. *message count*: estimates quoting, cancellation, or latency behaviour;
iv. *time bucket*: estimates realised activity inside a comparable clock segment;
v. *security bucket*: estimates whether the result is driven by large caps, small caps, sectors, or venues.

Changing the sampling unit changes the estimand, even when the same symbol $\hat{p}$ is used.

Model 7.4.69. *Simple Linear Regression Baseline*
For observations (x_i, y_i), a simple linear regression writes $y_i = a_0 + a_1 x_i + \epsilon_i$, $1 \le i \le n$.
Here y_i is the dependent variable, x_i is the explanatory variable, a_0 is the intercept, a_1 is the slope, ϵ_i is the residual, and n is the sample size.
Ordinary least squares estimates $\hat{a}_1 = \left(\sum_i (x_i - \bar{x})(y_i - \bar{y}) \right) / \left(\sum_i (x_i - \bar{x})^2 \right)$, $\hat{a}_0 = \bar{y} - \hat{a}_1 \bar{x}$. Here $\bar{x}$ and $\bar{y}$ are sample means, and $\hat{a}_0, \hat{a}_1$ are the estimated intercept and slope.

Definition 7.4.70. *Regression Fit Quality*
For fitted values $\hat{y}_i$ and residuals $\hat{\epsilon}_i = y_i - \hat{y}_i$, $R^2 = 1 - \frac{\sum_i \hat{\epsilon}_i^2}{\sum_i (y_i - \bar{y})^2}$.
Here $\hat{y}_i$ is the fitted value, $\hat{\epsilon}_i$ is the residual, and R^2 is explained variability relative to total variability. It measures fit, not causality or trading usefulness.

Model 7.4.71. *Mean Excess-Return Test*
For excess observations $x_1, \dots, x_n$, sample mean $\bar{x}$, and sample standard deviation s_x, the statistic is $t = \frac{\bar{x}}{s_x / \sqrt{n}}$.
It asks whether realised average excess performance is large relative to the noise in the same sample. In trading data, the hard part is defining x_i so it is comparable, independent enough, and not selected after the fact.

Proposition 7.4.72. *Power Is a Signal-to-Noise Problem*
For independent observations with true mean μ and standard deviation σ, the expected t-statistic is approximately $\mu\sqrt{n}/\sigma$. Detecting a small edge therefore requires n of order $(\sigma/\mu)^2$.

Proof. The standard error of the sample mean is $\sigma/\sqrt{n}$.
Dividing the true mean by this standard error gives the signal-to-noise ratio $\mu\sqrt{n}/\sigma$. □

Caveat 7.4.73. *More Ticks Are Not More History*
Sampling more frequently can estimate intraday mechanics more accurately, but does not create new independent market regimes. Serial correlation, order splitting, common shocks, and stale prices reduce effective sample size.

Caveat 7.4.74. *Regression Is an Empirical Approximation*
Linearity is often a local summary for spread, volatility, turnover, and impact relations. Check residuals, non-stationarity, heteroskedasticity, and whether a log or event-time transformation gives a more stable relation.

Taxonomy 7.4.75. *Performance-Inference Bias Segmentation*

i. *survivorship*: failed strategies, funds, venues, or symbols disappear from the dataset;

ii. *selection*: the sample is chosen because it already looked good;

iii. *multiple testing*: the best result among many trials is treated as a single unbiased test;

iv. *look-ahead*: timestamps, corporate actions, or benchmark constituents use future information;

v. *fillability*: simulated fills assume liquidity that was not actually reachable.

Most backtest errors are sampling errors before they are modelling errors.

Caveat 7.4.76. *Regression to the Mean*
Selecting the best recent performer raises the starting estimate. If the original outperformance mixed skill and luck, subsequent performance usually falls toward the population mean even when the strategy has not changed.

Caveat 7.4.77. *Distributional Shape Matters*
Fat tails, asymmetric payoff profiles, return smoothing, rare devaluations, and fraud-like reporting can make normal-based tests overconfident. Inspect drawdowns, skew, jump exposure, stale marks, and the economic source of returns before trusting a high t-statistic.

Caveat 7.4.78. *Averaging Effect*
An average intraday profile is not a typical realised path. Averaging volatility, volume, or spread across days can hide event days, close auctions, and isolated liquidity shocks.

Model 7.4.79. *Average of Regressions vs Regression of Averages*
For an index $I_t = \sum_{k=1}^{N} w_k S_t^k$, spreads aggregate linearly as $\psi^I = \sum_{k=1}^{N} w_k \psi^k$, but volatility generally does not satisfy $\sigma^I = \sum_{k=1}^{N} w_k \sigma^k$. Here w_k is the weight of stock k, S_t^k its price, ψ^k its spread, and σ^k its volatility. Regression coefficients estimated on index quantities therefore need not equal weighted averages of single-stock regression coefficients.

7.5 Market Making and Liquidity Provision

7.5.1 Economics of Market Making

Definition 7.5.1. *Dealer*
A dealer trades as principal: buying from clients that want to sell and selling to clients that want to buy. The service sold is immediacy; the compensation is spread, relationship value, or order-flow value.

Taxonomy 7.5.2. *Dealer Problem Segmentation*

i. *quote*: set bid, ask, and sizes;
ii. *order flow*: attract clients without buying toxic flow;
iii. *inventory*: keep positions near target levels;
iv. *information*: infer value from trades, quotes, news, and client behaviour.

Taxonomy 7.5.3. *Designated Market Maker Obligation Segmentation*

i. *affirmative*: quote meaningful two-sided markets and trade when public liquidity is absent;
ii. *negative*: yield to public orders and avoid trading ahead of customer liquidity;
iii. *auction*: organise openings or calls so orders receive orderly representation;
iv. *capital*: maintain enough balance sheet to supply liquidity in stressed conditions.

Obligations buy price continuity; privileges pay for the obligation.

Taxonomy 7.5.4. *Specialist Privilege Segmentation*

i. *order-flow information*: see routed orders, floor interest, or book imbalance earlier;
ii. *quote control*: set the disseminated quote within exposure rules;
iii. *participation option*: step in when public orders are thin or unattractive;
iv. *brokerage access*: earn fees or relationship value from represented flow.

Each privilege must be evaluated against the liquidity supplied when no one else wants to trade.

Taxonomy 7.5.5. *Dealer Quote Segmentation*

i. *two-sided*: quote both bid and ask; *one-sided*: quote only the desired side;
ii. *firm*: executable up to a stated size; *soft*: indication that may be revised;
iii. *displayed*: visible to the market; *request-only*: revealed after an RFQ or negotiation;
iv. *size*: maximum quantity available at the quoted price.

Taxonomy 7.5.6. *Dealer Spread vs Outside Spread*

i. *valuation target*: dealers quote around current market value; value traders trade on fundamental value;
ii. *holding period*: dealers try to turn inventory quickly; value traders wait for convergence;
iii. *risk source*: dealers bear inventory, client-selection risk.
Value traders bear valuation, financing, winner's-curse risk;
iv. *scale*: dealers can earn small spreads repeatedly; value traders need fewer, larger opportunities;
v. *profit capture*: dealers may earn a full round-trip spread.
Value trader often earns only the move from entry back toward value.

This is why dealer spreads is tight even when fundamental investors require a wide outside spread before trading.

Taxonomy 7.5.7. *Order-Flow Acquisition Segmentation*

i. *price*: quote aggressively enough to win trades;
ii. *size*: show capacity for larger trades;
iii. *service*: provide research, execution support, credit, and settlement reliability;
iv. *relationship*: cultivate clients and brokers;
v. *inducement*: pay or reciprocate for routed order flow.

Caveat 7.5.8. *Preferencing and Payment for Order Flow*
Preferencing can lower search costs when dealers compete for relationships, but it can also route orders toward economic arrangements rather than best all-in execution.

Practitioner Note 7.5.9. *Trading with Dealers*
Experienced clients often ask for a two-sided quote before revealing direction. A wide quote is information: the dealer may suspect informed flow, inventory pressure, or poor unwind conditions.

7.5.2 Optimal Quoting and Inventory Control

Model 7.5.10. *Specialist Participation Option*
Let V_{step} be the expected value of stepping ahead or trading with the book, $C_{\text{obligation}}$ the expected loss from required liquidity provision, and K the compliance and capital cost. A specialist assignment is privately attractive when $V_{\text{step}} > C_{\text{obligation}} + K$. Regulators should ask the public version: whether the price-continuity benefit exceeds the rents transferred through the privilege.

Caveat 7.5.11. *Tick Size and Participation*
Small ticks reduce the cost of stepping ahead of public orders, making specialist or DMM participation easier and public queue value weaker. Larger ticks protect queue rents but can widen quoted spreads.

7.5.3 Latency and Queue Competition

Principle 7.5.12. *Latency as Queue Option*
Speed buys queue priority. It is most valuable when queues are thin, stale, or information-sensitive.

Model 7.5.13. *HFMM and Investor Race Model*
Split participants into high-frequency market makers and longer-horizon investors.
A market maker earns spread surplus $s\Delta$ net of maker fee c_M; an investor earns value surplus Γ net of spread and taker fee c_T: $\pi_M = s\Delta - c_M$, $\pi_T = \Gamma - s\Delta - c_T$, $\bar{c} = c_M + c_T$.
Here s is spread in ticks, Δ is tick size, Γ is investor gross value gain, c_M, c_T are maker/taker fees, π_M, π_T are net per-trade gains, and $\bar{c}$ is venue revenue per trade. The model asks who should be subsidised to invest in speed, not whether speed is socially free.

Mechanism 7.5.14. *Race Intensity*
Let μ_m be market maker m's arrival intensity and τ_n investor n's arrival intensity, with averages $\bar{\mu}$ and $\bar{\tau}$.
If technology costs are quadratic, $\Pi_M(m) = \frac{\mu_m}{\bar{\mu}}\pi_M\mathcal{R} - \frac{1}{2}\beta\mu_m^2$, and $\Pi_T(n) = \frac{\tau_n}{\bar{\tau}}\pi_T\mathcal{R} - \frac{1}{2}\gamma\tau_n^2$.
Here $\mathcal{R}$ is trade rate, Π_M, Π_T are total expected gains, and β, γ are speed-cost coefficients. The ratios $\mu_m/\bar{\mu}$ and $\tau_n/\bar{\tau}$ act like race shares; lower cost raises optimal speed because winning one more race becomes cheaper.

Taxonomy 7.5.15. *Latency-Cost Segmentation*

i. *market-maker cost*: co-location, feed handling, quoting engines, and hedge automation;
ii. *investor cost*: broker connectivity, routing latency, algo responsiveness, and monitoring;
iii. *venue fee split*: maker fee or rebate versus taker fee;
iv. *surplus split*: spread capture for makers versus value capture for investors;
v. *competition level*: number of fast makers relative to investor flow.

Definition 7.5.16. *Efficient Maker Fee*
In the stylised fee-choice model, the efficient maker fee satisfies $c_M^\star = s\Delta - \frac{\Gamma - \bar{c}}{1 + \left(\frac{M}{N}\frac{\gamma}{\beta}\right)^{1/3}}$, $\bar{c} = c_T^\star + c_M^\star$.
Here M, N are the numbers of market makers and investors.
Negative $c_M^\star$ is a maker rebate; negative $c_T^\star$ is a taker rebate.

Caveat 7.5.17. *Fee Rebates Are Comparative*
Maker rebates fit scarce or costly market-making; taker rebates fit crowded makers and costly investor access.

Practitioner Note 7.5.18. *Queue Competition Diagnostic*
Do not classify a fee schedule as good or bad in isolation. Segment by spread size, maker count, investor urgency, queue stability, latency cost, and whether rebates improve displayed liquidity or merely pay for fleeting quotes.

7.5.4 Cross-Asset and Cross-Venue Market Making

Principle 7.5.19. *Arbitrageurs as Cross-Sectional Dealers*
Dealers use inventory to connect buyers and sellers across time. Arbitrageurs use hedge portfolios to connect buyers and sellers across instruments, venues, locations, maturities, and risk forms. Both sell immediacy, but arbitrageurs often sell a transformed exposure rather than the same instrument.

Taxonomy 7.5.20. *Cross-Asset Arbitrage Segmentation*

i. *cash-futures*: buy cash and sell futures, or reverse, after carry and delivery costs;
ii. *ETF-basket*: create or redeem ETF shares against the underlying basket;
iii. *ADR-local*: trade depositary receipts against local shares after FX, fees, and settlement frictions;

iv. *FX triangular*: compare direct exchange rates with synthetic cross rates;

v. *option conversion*: manufacture stock, cash, or volatility exposure from options and underlying hedges;

vi. *merger spread*: trade target and acquirer legs against deal terms and completion probability.

Each case answers: what synthetic instrument does hedge portfolio manufacture, and what costs bound basis?

Model 7.5.21. *Triangular FX No-Arbitrage*
For currencies A, B, C, no triangular arbitrage requires $S_{A/C} = S_{A/B}S_{B/C}$ after bid-ask spreads, funding, and settlement costs. A tradable deviation must exceed the cost of executing all three legs before quotes move.

Mechanism 7.5.22. *Derivative Conversion*
Derivatives let arbitrageurs repackage risk. A market maker who buys an option and delta-hedges with the underlying is converting option demand into stock trading, funding, and volatility exposure. The hedge is dynamic because delta, gamma, vega, and funding costs change with market state.

Caveat 7.5.23. *Legging and Synchronisation Risk*
Multi-leg arbitrage is most fragile while the hedge is incomplete. Market orders reduce legging risk but pay spread and impact; limit orders reduce cost but may leave one leg filled and the other missing. Execution logic therefore matters as much as fair-value logic.

Process 7.5.24. *Balanced Arbitrage Execution*

i. estimate fair basis, costs, and hedge ratios;

ii. choose leg order from liquidity, volatility, and unwind optionality;

iii. submit synchronised orders or hedge the most liquid leg last;

iv. monitor partial fills, basis movement, and residual exposure;

v. cancel, complete, or rebalance before residual risk dominates the spread.

Caveat 7.5.25. *Arbitrage Capital Constraint*
The widest spreads often appear when balance sheet is scarce.
If arbitrageurs cannot finance losses, meet margin, borrow securities, or carry inventory, forced liquidation can widen the basis precisely when the forward-looking trade is most attractive.

Chapter 8

Execution, trading costs and market structure

8.1 Execution and Algorithmic Trading

8.1.1 Optimal Execution Problem

Principle 8.1.1. *Impact-Risk Trade-Off*
Optimal execution balances two forces: market impact rewards slower trading, while price risk rewards faster completion. A schedule is a compromise between footprint and exposure.

Model 8.1.2. *Discrete Execution Schedule*
Plan a buy order of size $v^\star$ over N slices. During slice n, trade v_n shares, expected market volume is V_n, volatility is σ_n, and the fair price evolves as $S_{n+1} = S_n + \sigma_n \xi_{n+1}$, $\sum_{n=1}^{N} v_n = v^\star$.
Here ξ is a standard Gaussian shock. The schedule controls timing; the market supplies uncertainty.

Model 8.1.3. *Temporary Impact Cost*
A simple execution price for slice n is $\widetilde{S}_n(v_n) = S_n + \kappa\sigma_n \left(\frac{v_n}{V_n}\right)^\gamma$, $W = \sum_{n=1}^{N} v_n \widetilde{S}_n(v_n)$.
Here W is total execution notional, and κ and γ are stock-dependent impact parameters.
Participation v_n/V_n is the key lever: trading more of the available volume costs more.

Definition 8.1.4. *Expectation-Minimising Schedule*
Ignoring risk, the minimiser allocates more to high-volume and low-volatility slices: $v_n^{\exp} = \frac{V_n/\sigma_n^{1/\gamma}}{\sum_{\ell=1}^{N} V_\ell/\sigma_\ell^{1/\gamma}} v^\star$.
If V_n and σ_n are constant, the schedule is uniform: $v_n = v^\star/N$.

Mechanism 8.1.5. *Risk-Only Schedule*
If impact is ignored and only price-risk variance matters, the optimal action is to trade immediately. Waiting leaves remaining inventory exposed to random price moves.

Model 8.1.6. *Mean-Variance Execution Objective*
Let x_n be the remaining quantity after slice n, with $x_0 = v^\star$ and $x_N = 0$.
A compact objective is $\mathcal{C} = v^\star S_0 + \kappa \sum_{n=1}^{N} \sigma_n \frac{v_n^{\gamma+1}}{V_n^\gamma} + \lambda \sum_{n=1}^{N} x_n^2 \sigma_n^2$.
The middle term is impact cost; the last term is inventory risk. Higher risk aversion λ front-loads the schedule, while higher impact sensitivity back-loads it.

Taxonomy 8.1.7. *Execution Schedule Segmentation*

i. *risk-neutral*: follows volume and volatility to reduce expected impact;
ii. *risk-only*: finishes immediately to remove exposure;
iii. *mean-variance*: trades along a convex remaining-quantity path;
iv. *tactical overlay*: lets dark pools, SORs, alpha, and limits modify child orders inside the schedule.

Definition 8.1.8. *Block Order*
A block order is too large to complete easily through normal displayed liquidity.
Fixed-share thresholds are venue conventions; for execution research, a block is better measured as a fraction of average daily volume, typical depth, or expected participation capacity.

Taxonomy 8.1.9. *Block Execution Problem Segmentation*

i. *latent demand*: natural contra interest exists but has not posted orders;

ii. *order exposure*: revealing the parent attracts anticipators and liquidity withdrawal;

iii. *price discrimination*: liquidity suppliers wait because they fear selling too cheaply or buying too richly;

iv. *asymmetric information*: suppliers demand concessions when the initiator may be informed.

Block execution buys liquidity with information: disclose enough to find size, but not enough to be exploited.

8.1.2 Execution Benchmarks

Principle 8.1.10. *Benchmark Matches Trading Need*
Each execution benchmark expresses a different risk trade-off: avoid detection, follow market volume, capture price opportunity, or avoid missing liquidity. Benchmark choice should depend on order size, market depth, region, and investment style.

Taxonomy 8.1.11. *Regional Benchmark Preference*

i. *US*: liquidity seeking is natural when HFT is pervasive and stealth matters;

ii. *Asia*: VWAP and PoV help compare local market efficiency and fragmented market conventions;

iii. *Europe*: IS and routing discipline matter because orderbooks and venue interaction are fragmented.

Benchmark	Region	Order characteristics	Market context	Risk hedged
PoV	Asia	Large order size; more than 10% of ADV, using average daily consolidated volume	Possible negative news	Rapid news repricing
VWAP/TWAP	Asia and Europe	Medium size; 5–15% of ADV	Unusual volume is negligible	Slow information drift
Implementation Shortfall (IS)	Europe and US	Small size; 0–6% of ADV	Possible price opportunities	Vanishing price opportunity
Liquidity Seeker	US	Any size	Stock expected to oscillate around fair value	Liquidity bursts; relative-price dislocations

Table 8.1. Typical uses of benchmarks for trading algorithms.

Caveat 8.1.12. *Benchmark Is Not an Objective Alone*
The benchmark justifies the comparison, not the whole execution policy. Market conditions, order size, liquidity, and control freedom still determine the usable algorithm.

8.1.3 Execution Algorithms

Principle 8.1.13. *Algo as Layered Decision System*
An execution algorithm is not only a schedule. It is a layered system that turns a parent order into child orders, adapts to market state, and records enough context to explain the realised cost afterwards.

Taxonomy 8.1.14. *Algorithm Layer Segmentation*

i. *execution parameters*: side, size, benchmark, timing, limits, participation, urgency, and venues;

ii. *risk-control layer*: sets the curve from impact, timing risk, alpha, and context;

iii. *tactical layer*: uses SORs, liquidity seeking, pegging, and dark probes to realise it.

Risk control sets the envelope; tactics work inside it.

Principle 8.1.15. *Benchmark Customisation*
Execution styles can be combined: a VWAP can carry minimum participation, limit-price, or would-level constraints; a PoV can be capped by urgency; an IS schedule can still use an SOR and dark-pool probes. The benchmark defines the reference, while constraints define acceptable behaviour around it.

Caveat 8.1.16. *Hard Mixtures Are Fragile*
Binary benchmark switches are brittle in noisy markets. A trading envelope gives the schedule room to adapt without abandoning risk control.

Definition 8.1.17. *Trading Envelope*
A trading envelope is a band of acceptable execution trajectories around the target schedule. Being too fast implies pausing or posting passively; being too slow implies crossing the spread or increasing participation.

Taxonomy 8.1.18. *Envelope Construction Segmentation*

i. *benchmark path*: VWAP, TWAP, PoV, IS, or minimum-impact curve;
ii. *risk width*: tolerance for timing risk, price drift, and completion uncertainty;
iii. *market inputs*: volume curve, volatility, spread, depth, and impact estimates;
iv. *constraints*: limits, participation caps, venue restrictions, and common portfolio end times;
v. *tactics*: liquidity capture, dark probes, pegging, and routing choices used inside the band.

Process 8.1.19. *Customisation Workflow*

i. choose the desired risk profile and inject hard constraints into the envelope;
ii. choose liquidity tactics consistent with investment style and observed market state;
iii. monitor whether the realised path remains inside the envelope.

Process 8.1.20. *Execution Decision Loop*

i. initialise the parent order and benchmark;
ii. compute a target trajectory from historical profiles and model parameters;
iii. observe quotes, trades, fills, rejects, cancellations, and venue responses;
iv. revise child-order size, price, urgency, and venue split;
v. publish monitoring and performance indicators for the trader.

Taxonomy 8.1.21. *Pre-Trade Segmentation*

i. *expected cost*: spread, impact, volatility, and typical quantity on the book;
ii. *profile*: expected volume curve, duration, and participation bounds;
iii. *context*: news, volatility regime, sector, country, currency, and event risk;
iv. *portfolio coupling*: common end times, exposure constraints, and cross-order synchronisation.

Taxonomy 8.1.22. *Order-Submission Control Segmentation*

i. *urgency*: cross, post, peg, or wait;
ii. *price*: limit level, discretion range, and protection price;
iii. *quantity*: child size, minimum fill, reserve display, and parent slicing;
iv. *exposure*: lit, dark, brokered, indicated, or anonymous;
v. *intermediation*: single broker for control or multiple brokers for concealment;
vi. *venue*: one book, smart-routed books, crossing networks, or upstairs negotiation.

The execution algorithm controls transaction cost through these levers, not only through the target schedule.

Principle 8.1.23. *Reactive vs Proactive Execution*
Reactive execution waits for liquidity to appear and usually lowers leakage. Proactive execution advertises interest or crosses the spread to create a trade. The right point on this continuum depends on urgency, search cost, and how damaging exposure would be.

Taxonomy 8.1.24. *Portfolio Trading Segmentation*

i. *program trade*: simultaneous orders across many names, often index or factor related;
ii. *basket trade*: trade a list as one risk package;
iii. *package trade*: dealer quotes a firm bid or offer for the whole portfolio;
iv. *portfolio transition*: move from one target portfolio to another under tracking-error and cost limits.

Portfolio trading reduces leg count and search cost, but transfers basket characterisation and residual-risk pricing to the dealer or algorithm.

Mechanism 8.1.25. *Portfolio Insurance as Dynamic Execution*
Portfolio insurance synthetically buys downside protection by dynamically reducing risky exposure as the portfolio falls and increasing exposure as it rises. It transfers risk only if the market can absorb the hedge trades; when many investors run the same rule, the execution demand itself becomes market impact.

Proposition 8.1.26. *Dynamic Hedging Feedback*
Let a rule target risky holdings $q(S)$ as a function of market level S, with $q'(S) > 0$. A price fall $dS < 0$ forces $dq = q'(S)dS < 0$, so the rule sells into the decline. If temporary impact is λdq with $\lambda > 0$, the hedge flow has the same sign as the initial price move.

Proof. Since $q'(S) > 0$, $dS < 0$ implies $dq < 0$. A sell order has negative signed flow and negative temporary price impact, so the induced impact reinforces the initial decline. □

Taxonomy 8.1.27. *Basket Characterisation Inputs*

i. *risk*: beta, factor loadings, volatility, sector, currency, and country exposures;
ii. *liquidity*: ADV, spreads, depth, index membership, and short-sale or borrow constraints;
iii. *execution*: names, weights, limits, urgency, venue eligibility, and close/open constraints.

For index-like baskets, diversified residual risk lets dealers quote tighter than for unrelated single-stock blocks.

Taxonomy 8.1.28. *Index Rebalance Controls*

i. *timing*: trade before, during, or after the effective window subject to tracking error;
ii. *substitution*: use futures, ETFs, or correlated names when constituents are expensive;
iii. *liquidity supply*: post or cross opportunistically when forced flow pays a premium.

Index discipline constrains the portfolio, but execution still chooses how much immediacy to buy.

Benchmark	Stock type	Trade type	Main feature
PoV	Medium to large market depth	Long-duration position trade	Flow-following; reactive within participation bounds
VWAP/TWAP	Any market depth	Hedging order; long-duration position; tracking-error unwind	Volume-curve schedule; passive when possible
IS (Implementation Shortfall)	Medium liquidity depth	Alpha extraction; non-linear hedge, typically gamma; inventory-driven trade	Fast on opportunity; cuts losses on adverse moves
Liquidity seeker	Poor fragmented market depth	Alpha extraction; opportunistic position mounting; already split or scheduled order	Pool search; dispersed liquidity; low signalling

Table 8.2. Typical features of trading algorithms.

Practitioner Note 8.1.29. *What-If Controls*
Useful tools support live counterfactuals: finish now, slow down, switch benchmark, seek dark liquidity, or cap participation. The aim is disciplined comparison before changing trajectory.

8.1.4 Transaction Cost Analysis (TCA)

Definition 8.1.30. *TCA*
Transaction cost analysis compares realised execution prices with relevant benchmarks and explains the gap by market context, order design, broker behaviour, and trading style.

Taxonomy 8.1.31. *Transaction Cost Component Segmentation*

i. *explicit costs*: commissions, exchange fees, taxes, clearing, settlement, and borrow charges;
ii. *implicit costs*: bid-ask spread, market impact, delay cost, and adverse selection relative to a benchmark;
iii. *missed trade opportunity*: value lost when order is not filled, or filled too slowly to capture intended price.

Explicit costs are accounting entries. Implicit and missed-opportunity costs require a counterfactual benchmark.

Model 8.1.32. *Signed Benchmark Cost*
Let $d = +1$ for a buy and $d = -1$ for a sale, q be filled quantity, $\bar{p}$ the average fill price, and b the benchmark price. The signed benchmark cost is $C = dq(\bar{p} - b)$.
The sign convention makes high buy prices and low sale prices costly.

Model 8.1.33. *Implementation Shortfall with Missed Quantity*
For parent size Q, filled quantity q, average fill $\bar{p}$, decision benchmark p_0, and later evaluation price p_T,
$IS = dq(\bar{p} - p_0) + d(Q - q)(p_T - p_0) +$ explicit costs.
The first term is realised execution cost. The second term is missed trade opportunity, and it is meaningful only when the unfilled order was economically feasible.

Principle 8.1.34. *Marginal Aggressiveness Rule*
Trade more aggressively when the marginal missed-opportunity cost saved exceeds the marginal transaction cost paid. Trade less aggressively when the marginal transaction cost saved exceeds the additional missed-opportunity cost. Average cost is the wrong control variable for the last slice.

Taxonomy 8.1.35. *Benchmark Price Segmentation*

i. *contemporaneous midpoint*: clean for effective spread, weak for timing skill;
ii. *post-trade midpoint*: measures realised spread and adverse selection;
iii. *decision price*: natural for implementation shortfall and paper-portfolio comparison;
iv. *VWAP*: compares to average traded market price, but can hide own impact and split-order bias;
v. *open or close*: useful for sponsors with coarse timestamps, but noisy and gameable.

No benchmark is neutral; it defines which cost is visible.

Definition 8.1.36. *VWAP Benchmark*
For trades i in the benchmark window, VWAP $= \frac{\sum_i q_i p_i}{\sum_i q_i} = \sum_i w_i p_i$, where $w_i = q_i / \sum_j q_j$. It is an average market execution price, not a no-impact counterfactual.

Taxonomy 8.1.37. *Performance Measurement Segmentation*

i. *absolute*: total return or all-in execution price without a comparison asset;
ii. *relative*: result versus benchmark, peer, arrival price, market volume, or factor model;
iii. *holding-period*: investor-level return after cash flows and distributions;
iv. *money-weighted*: IRR, useful when capital flows are controlled by the investor;
v. *trade-weighted*: fill-quality measures, useful when the desk controls execution.

Relative measures more useful as most price movement is not caused by trader or manager being evaluated.

Model 8.1.38. *Risk-Adjusted Excess Return*
For portfolio/strategy return r_t^p, benchmark return r_t^m, a one-factor evaluation is $r_t^p - r_t^f = \alpha + \beta(r_t^m - r_t^f) + u_t$. Here α is realised selection value after market exposure, not proof of skill. Microstructure TCA uses the same logic at shorter horizons: remove common price movement before attributing residual cost to execution.

Taxonomy 8.1.39. *Benchmark Failure Modes*

i. *wrong exposure*: benchmark lacks the risk actually carried;
ii. *wrong clock*: benchmark window does not match the decision or execution horizon;
iii. *own-impact contamination*: the order helps set the benchmark it is judged against;
iv. *gaming*: agents can change behaviour to improve the measured score;
v. *selection*: only favourable orders, venues, or managers enter the sample.

Principle 8.1.40. *Pre-, Live-, and Post-Trade Consistency*
The same indicator should have a pre-trade expectation, a live monitored value, and a post-trade assessment. A bid-ask-spread indicator, for example, can move from expected average, to updated live likelihood, to realised influence on performance.

Taxonomy 8.1.41. *Performance Attribution Segmentation*

i. *immediate price*: the reference available when the order decision is made;
ii. *planned price*: the price expected from the risk-control trajectory;
iii. *unexpected market move*: exogenous price variation during execution;
iv. *trading efficiency*: quality of liquidity-seeking tactics relative to the plan;
v. *market impact*: price movement caused by the order itself.

This segmentation is useful only when the benchmark, horizon, and side are explicit.

Model 8.1.42. *Selection and Timing Decomposition*
Let r_t^p be realised return, r_t^m market return, and β_t the exposure chosen during the period.
A useful attribution separates: $r_t^p = r_t^m + (r_t^p - \beta_t r_t^m) + (\beta_t - 1) r_t^m$.
The three terms are market return, selection residual, and timing payoff from changing exposure. In execution, the analogue is market move, trading residual, and urgency or participation timing.

Caveat 8.1.43. *TCA Uses Wider Samples Than Post-Trade Review*
Post-trade review explains one execution in detail. TCA usually aggregates months of comparable trades to identify contexts where an algorithm, broker, or parameter set is systematically strong or weak.

Taxonomy 8.1.44. *TCA Estimator Quality Segmentation*

i. *data*: daily benchmarks are cheap; quote methods need synchronised trades and quotes;
ii. *benchmark*: closer references reduce noise but shift timing attribution;
iii. *inference*: splitting, momentum, informed flow, gaming, and small samples bias costs.

Taxonomy 8.1.45. *TCA Context Segmentation*

i. *liquidity*: spread, queue size, turnover, trade size, free float, and trading rate;
ii. *risk*: volatility, jumps, intraday trend, and index or sector co-movement;
iii. *order design*: size, duration, participation rate, benchmark, and limit constraints;
iv. *market state*: news, auctions, rebalances, corporate events, and venue outages.

Taxonomy 8.1.46. *Pre-Trade Cost Prediction Variables*

i. *order variables*: parent size, participation target, limit price, urgency, and placement;
ii. *current market variables*: spread, displayed depth, volatility, recent returns, and signed volume;
iii. *general market variables*: ADV, market capitalisation, risk regime, sector liquidity, and event calendar;
iv. *venue variables*: hidden fill rates, fee tier, queue loss rate, and routing latency.

Prediction is a conditioning problem: yesterday's cost matters only after adjusting for today's liquidity state.

Practitioner Note 8.1.47. *Broker and Algo Selection*
The practical output of TCA is a routing table: for a given market context and trading objective, send the next order to the broker or algorithm whose historical performance is best after adjusting for risk and liquidity.

Principle 8.1.48. *Best Execution as Auditability*
Best execution is not the cheapest commission or the fastest fill in isolation. It is the best auditable outcome given the client's objective, limit, urgency, available venues, and information risk.

Taxonomy 8.1.49. *Retail Execution Quality Segmentation*

i. *price*: execution relative to NBBO, midpoint, or effective spread;
ii. *speed*: time to fill, cancel, or reject;
iii. *certainty*: fill probability and partial-fill risk;
iv. *commission*: explicit brokerage charge net of rebates;
v. *adverse selection*: post-fill price movement and opportunity cost.

Retail traders observe commissions easily; execution price quality requires quote-synchronised data.

Taxonomy 8.1.50. *Broker-Conflict Segmentation*

i. *routing inducements*: rebates, payments for order flow, reciprocal flow, and affiliate venues;
ii. *information misuse*: front-running, inappropriate exposure, and prearranged trades;
iii. *allocation*: assigning favourable fills to preferred accounts;
iv. *activity incentives*: commissions, churning, soft dollars, and commission recapture;
v. *principal conflict*: internalisation, dual trading, and dealer inventory.

TCA and audit trails convert these conflicts from stories into measurable routing and fill-quality checks.

Caveat 8.1.51. *Gaming the Measure*
Any benchmark that brokers can influence can be gamed. Delaying a trade, matching the benchmark mechanically, splitting fills around measurement windows, or selecting only favourable venues can improve reported TCA while worsening the client's economics.

Caveat 8.1.52. *Smooth Returns Are Not Safer Returns*
Illiquid valuations, stale marks, and discretion in marking can reduce reported volatility and raise apparent risk-adjusted performance. The risk has not disappeared; it has moved into valuation lag and jump risk.

Process 8.1.53. *Block Motive Audit*

i. identify economic reason for trade: funding, diversification, hedging, benchmark change, or speculation;

ii. test whether the requested size and timing are consistent with that reason;

iii. compare the initiator's alternatives, constraints, and past behaviour;

iv. price the trade only after separating utilitarian urgency from possible information.

The audit is not moral judgement; it is adverse-selection control.

Caveat 8.1.54. *Block Trade Reporting Trade-Off*
Fast public reporting improves transparency and protects the regular market, but it can hurt dealers that still need to unwind facilitated inventory. Delayed reporting protects facilitation, but gives informed block traders more room to exploit stale prices.

8.1.5 Smart Order Routing

Definition 8.1.55. *Execution-Side SOR*
In execution, a smart order router chooses where each child order should go after the parent algorithm has chosen when and how much to trade. It is the venue-optimisation layer inside the execution stack.

Definition 8.1.56. *Marketable Price Across Venues*
With several books, marketability is venue-specific. A buy order can be marketable on one venue because its limit reaches the local ask, while not marketable elsewhere if the same instrument is offered at a higher price.

Taxonomy 8.1.57. *Routing Logic Segmentation*

i. *observable liquidity*: quotes, depth, queue position, hidden-liquidity signals, and auction state;

ii. *fill probability*: venue hit rate, cancellation speed, adverse selection, and reject history;

iii. *net economics*: explicit fees, maker-taker rebates, clearing, and broker-specific schedules;

iv. *latency*: feed delay, gateway delay, matching-engine delay, and stale-quote risk;

v. *information cost*: signalling from sweeping, probing, or repeatedly visiting the same venue.

Process 8.1.58. *Child-Order Routing Loop*

i. receive the child quantity, side, limit, and urgency from the execution algorithm;

ii. score venues by expected net cost and completion probability;

iii. split or sequence messages across venues;

iv. update venue scores from fills, partial fills, rejects, and missed liquidity.

Principle 8.1.59. *Best Execution Differs by Regime*
US routing leans on protected consolidated quotes. European brokers choose the consolidation layer, so routing quality includes venue access, data quality, and auditability.

Caveat 8.1.60. *Displayed Best Price Is Not Enough*
A venue with the best quote can still be a poor destination if the queue is unstable, the quote is stale, fees are high, or fills are followed by adverse price moves.

Mechanism 8.1.61. *Active Sweep Segmentation*

i. *local best price*: cheapest visible venue first;

ii. *consolidated price*: split across venues to minimise average price;

iii. *policy coverage*: restrict destinations to venues covered by the broker mandate;

iv. *residual choice*: when venues tie at a level, choose by fee, latency, and fill reliability.

Caveat 8.1.62. *Fee-Adjusted Price Priority*
Let P be the price on V, δ the relative price increase on V', Q the size, f the fee on V, and ϕ the fee saving on V'. For a buy order, V' is cheaper when $P(1+\delta)(1+f-\phi)Q < P(1+f)Q$.
Roughly, the price penalty must be below the fee saving.

Caveat 8.1.63. *Order Preferencing*
A broker may prefer a dealer, affiliate, or venue because of service, economics, or reciprocal flow rather than the displayed quote alone. Best-execution review must therefore compare all-in fills, not just routing intent.

Proposition 8.1.64. *Payment-for-Order-Flow Routing Conflict*
Let C_v be expected client execution cost on venue or dealer v, and let h_v be broker revenue from routing there. Client-optimal routing chooses $\arg\min_v C_v$; broker-private routing chooses $\arg\min_v(C_v - h_v)$. A conflict exists whenever some a, b satisfy $C_a < C_b$ but $C_a - h_a > C_b - h_b$.

Proof. The client pays execution cost C_v.
The broker internalises both client execution quality and routing revenue, so its private objective subtracts h_v.
The displayed inequalities state that a is better for the client while b is better for the broker. □

Caveat 8.1.65. *Latency Makes Liquidity Probabilistic*
Displayed depth is probabilistic: stale or duplicated quotes can vanish before the router arrives.

Principle 8.1.66. *Liquidity Seeking Objective*
A liquidity seeker scores the quantity likely to be available on arrival, using visible depth, hidden-liquidity signals, latency, duplicate quotes, fees, and fill rates.

Taxonomy 8.1.67. *Passive Posting Segmentation*

i. *queue rank*: post more where fewer shares stand ahead;
ii. *distance to touch*: prefer venues closer to becoming marketable;
iii. *trading rate*: prefer venues where opposing flow consumes the level faster;
iv. *resting-time target*: split to equalise expected completion time.

Definition 8.1.68. *Minimum Resting-Time Split*
Let D be total passive child size, Q_n queue ahead, r_n opposing-flow rate, f_n posted fraction, and $\tau^\star$ common completion time. For posted size x on venue n, $\tau_n(x) = \frac{x+Q_n}{r_n}$ is the expected completion time, so equal-time posting gives $\tau^\star = \frac{D+\sum_n Q_n}{\sum_n r_n}$ and $f_n = \frac{\tau^\star r_n - Q_n}{D}$. If $f_n < 0$, post nothing on venue n.

Criterion	1st venue	2nd venue	3rd venue
Rank in queue	better queue	next queue	deepest queue
Distance to best	nearest touch	next touch	farthest touch
Trading rate	fastest flow	middle flow	slowest flow
Minimum resting time	highest adjusted rate	next adjusted rate	residual venue

Table 8.3. Passive routing criteria can rank the same venues differently.

Process 8.1.69. *Liquidity-Seeker Build*

i. estimate hidden liquidity, mirrored orders, latency, fees, and adverse selection;
ii. remove liquidity when the probability-adjusted net price is attractive;
iii. post passively where queue, distance, and trading-rate criteria support fast completion;
iv. keep the parent order inside the execution envelope set by the benchmark layer.

8.1.6 High-Frequency Trading Strategies

Taxonomy 8.1.70. *HFT Strategy Segmentation*

i. *passive market making*: quote both sides, earn spread and rebates, manage inventory;
ii. *statistical arbitrage*: trade short-horizon relative value or lead-lag signals;
iii. *latency arbitrage*: react faster to stale quotes or delayed cross-venue price updates;
iv. *order anticipation*: infer large parent orders from child-order footprints;
v. *liquidity detection*: probe venues and dark pools for hidden or fleeting interest.

The same firm may run several of these, but the risk and social value differ by strategy.

Principle 8.1.71. *HFT Edge Decays With Diffusion*
Speed advantages are valuable only while competitors, venues, and clients cannot replicate them.
Once reaction speed and routing logic diffuse, durable profits must come from inventory control, better models, lower costs, or privileged order flow.

Caveat 8.1.72. *Message Profit Is Not Market Quality*
High message rates can improve displayed prices, but they can also increase cancellation risk and stale-book uncertainty. Evaluate HFT by spreads, depth, realised costs, adverse selection, and stress behaviour.

8.1.7 Reinforcement Learning and ML in Execution

Model 8.1.73. *Stochastic-Control Execution View*
Stochastic-control execution chooses a trading rate v_t from state variables such as price S_t, remaining inventory Q_t, cash X_t, and forecast market flow μ_t. A compact sell-side value function is

$$H_t(v) = \mathbb{E}_t\left[X_T^v + Q_T^v(S_T - AQ_T^v) - \phi\int_t^T (Q_s^v)^2 ds\right]$$

Here $H_t(v)$ is the time-t value of control v, $\mathbb{E}_t$ is conditional expectation at time t, T is the terminal time, and the superscript v indicates the state path generated by that control. A penalises terminal inventory and ϕ penalises carrying inventory through time.

Model 8.1.74. *Cartea-Jaimungal State Dynamics*
One useful specification lets exogenous market flow μ_t interact with the trader's own rate v_t:

$$dS_t = b_t(\mu_t - v_t)dt + \sigma dW_t, \qquad \widehat{S}_t = S_t - \left(\frac{\psi}{2} + \kappa v_t\right),$$
$$dQ_t^v = -v_t dt, \qquad dX_t^v = \widehat{S}_t v_t dt.$$

Here b_t is permanent-impact sensitivity, σ is price-diffusion volatility, W_t is a Brownian shock, $\widehat{S}_t$ is the execution price, ψ is the bid-ask spread, and κ is temporary-impact slope.

Principle 8.1.75. *Value Function Intuition*
The control trades off three terms: realised cash, leftover inventory value, and inventory-risk penalty. A larger ϕ or A pushes faster liquidation; stronger temporary impact κ pushes smoother trading.

Taxonomy 8.1.76. *Control vs Learning*

i. *stochastic control*: specify dynamics and solve for an optimal feedback policy;
ii. *reinforcement learning*: learn the policy or value function from simulated or historical interaction;
iii. *hybrid execution*: use control structure for constraints and ML for forecasts, fills, and regime detection.

8.1.8 Production Considerations

Principle 8.1.77. *Trading Architecture*
Automated trading needs a production architecture that synchronises market data, order gateways, historical databases, models, monitoring, and TCA. The hard part is not one model; it is keeping the decision chain coherent under fragmented venues and live latency.

Taxonomy 8.1.78. *Trading Tool Input Segmentation*

i. *market data*: trades, quotes, depth, timestamps, and book updates, preferably consolidated across venues;
ii. *venue connectivity*: private order-entry messages and public market-data feeds;
iii. *historical data*: raw events, real-time analytics, order messages, fills, and market context;
iv. *models*: impact, risk, alpha, volume, and context models with calibrated parameters;
v. *calendar*: holidays, auctions, expiries, index events, and expected halt days.

Taxonomy 8.1.79. *Broker-Dealer Operations Segmentation*

i. *front office*: client contact, order receipt, execution, sales, and research;
ii. *back office*: accounts, confirmations, clearing, settlement, custody, and statements;
iii. *risk and treasury*: margin, credit, stock loan, cash management, and firm inventory;
iv. *compliance*: suitability, best execution, surveillance, and regulatory reporting;
v. *technology*: order routing, market data, drop copies, books, and audit trails.

Mechanism 8.1.80. *Analytics Close to the Feed*
Spread, effective spread, volatility, book imbalance, and queue-size measures should be computed as soon as raw data arrive. Fast analytics become state variables for execution, monitoring, and later post-trade explanation.

Caveat 8.1.81. *Connections Are Two-Way*
An execution engine sends orders, cancellations, and modifications, then receives acknowledgements, rejects, fills, and public book updates. In dark venues, the private responses can also act as probes of hidden liquidity.

Taxonomy 8.1.82. *Model Component Segmentation*

i. *formula*: the functional relation between state variables and decisions;
ii. *estimation*: the procedure that calibrates parameters from historical data;
iii. *accuracy check*: the live or post-trade test that decides whether the model is still usable.

Process 8.1.83. *Indicator Publication*

i. models and algorithms publish their state as indicators;
ii. traders monitor exceptions, concentration, performance, and context shifts;
iii. post-trade systems store the same indicators for diagnosis and TCA.

This makes trading algorithms stateful agents in the production stack rather than black boxes.

Process 8.1.84. *Audit Trail*

i. record order receipt, route, modification, cancellation, execution, allocation, and timestamp;
ii. link private order messages to public market state and venue responses;
iii. preserve enough context to test best execution, front-running, allocation, and unauthorised trading.

Taxonomy 8.1.85. *Automated Trading Reliability Segmentation*

i. *availability*: redundant servers, gateways, power, and network paths;
ii. *integrity*: authenticated users, validated orders, and tamper-resistant logs;
iii. *recoverability*: backups, replayable feeds, drop copies, and disaster-recovery sites;
iv. *capacity*: headroom for message bursts, opens, closes, halts, and data-feed spikes;
v. *latency*: bounded delays for market data, decisions, order entry, and acknowledgements.

For a trading system, reliability is a market-quality variable, not only an IT variable.

Caveat 8.1.86. *Network Failure Creates Position Risk*
If order entry, market data, or drop copy fails, the trader may not know whether an order traded, whether a hedge executed, or whether the market moved through a stop.
Controls should prefer safe state reconciliation over blind resubmission.

Mechanism 8.1.87. *Automated System Cost Structure*
Electronic trading has high fixed costs for software, data networks, security, and recovery, but low marginal labour cost per order. Floor trading has higher human and physical operating costs, but can still be valuable when negotiation, discretion, and soft information dominate raw matching speed.

8.2 Asset-Class-Specific Microstructure

8.2.1 Equity Market Microstructure

Principle 8.2.1. *Equities as Microstructure Laboratory*
Equities are the cleanest empirical laboratory because trades, quotes, venue identifiers, and regulatory changes are comparatively well recorded. Lessons from equities transfer to other markets only after adjusting for contract standardisation, post-trade process, leverage, and dealer intermediation.

Caveat 8.2.2. *Asset-Class Transfer*
An equity-style CLOB is not the default architecture for every asset.
Futures, options, FX, fixed income, and crypto differ in tick size, margin, clearing, settlement, inventory risk, and whether liquidity is naturally multilateral or dealer-mediated.

Rule 8.2.3. *Equity Block Prints and Size Priority*
Equity block crosses may be negotiated away from the public book but printed through an exchange or trade-reporting facility. Size-priority exceptions let very large crosses avoid being broken by small resting orders, while price-priority rules protect displayed liquidity. The design trade-off is direct: protect the book too much and blocks leave; protect blocks too much and displayed liquidity loses precedence value.

Principle 8.2.4. *Index Markets Trade Common Risk*
Index products let traders buy or sell broad market risk without trading every constituent separately.
They are often cheaper because index flow is less security-specific, more active, easier to net, and easier to hedge with futures, ETFs, options, or baskets.

Taxonomy 8.2.5. *Index Product Segmentation*

i. *open-end index fund*: trades with the fund at NAV; low turnover but slower intraday access;
ii. *ETF*: exchange-traded claim with creation and redemption against a basket;
iii. *index future*: leveraged, cash-efficient exposure with basis and roll risk;
iv. *index option*: convex exposure to index level and volatility.

Product choice is mostly a choice among trading speed, funding, tracking error, tax, and operational constraints.

Mechanism 8.2.6. *ETF Basket Arbitrage*
Authorised participants create ETF shares by delivering the basket, or redeem ETF shares for the basket.
This links ETF price to constituent value, but the link weakens when creation costs, borrow, cash substitutions, taxes, or stressed constituent liquidity are material.

8.2.2 Futures Market Microstructure

Principle 8.2.7. *Centralised Futures Trading*
Liquid futures are highly electronic and usually trade on centralised limit order books. Fragmentation is limited because margining, clearing, delivery rules, and contract fungibility make post-trade duplication costly.

Caveat 8.2.8. *No Call Auction Norm*
Many futures markets do not rely on equity-style opening and closing call auctions. Instead, participants agree on liquid trading hours during which the contract can support reliable price discovery and hedging.

Mechanism 8.2.9. *Lead-Lag With Cash Markets*
Index futures and their cash baskets form prices jointly.
Futures may lead when they are cheaper to trade, more leveraged, or open during information arrival; cash markets may lead when constituent-level information dominates.

Principle 8.2.10. *Large Tick Futures*
Futures often have economically large ticks. A binding tick creates bid-ask bounce and price reversals, but it also supports large displayed quantities at the best limits relative to comparable equities.

Taxonomy 8.2.11. *Futures Architecture Drivers*

i. *standardisation*: contracts are fungible by expiry and specification;
ii. *central clearing*: margin and default management are built into the market;
iii. *leverage*: small price moves create large economic exposure;
iv. *hedging link*: futures connect to cash baskets, ETFs, options, and swaps.

Mechanism 8.2.12. *Variation Margin*
Futures gains and losses are settled through daily variation margin.
The broker or FCM manages client margin calls, liquidation rights, and cash balances; credit risk is therefore controlled continuously rather than only at final settlement.

Taxonomy 8.2.13. *Futures Block and EFP Segmentation*

i. *block trade*: negotiated futures size reported under exchange rules;
ii. *EFP*: exchange cash or physical exposure for offsetting futures;
iii. *transitory EFP*: short-lived EFP used when the book is closed or thin.

They move large risk off-book, subject to eligibility and reporting rules.

8.2.3 FX Market Microstructure

Principle 8.2.14. *FX Is Dealer-Centric and Fragmented*
Spot FX has no single consolidated exchange. Liquidity is split across dealer streams, ECNs, prime-brokered access, bank internalisers, and bilateral relationships.

Taxonomy 8.2.15. *FX Venue Segmentation*

i. *dealer streams*: bilateral executable prices from banks or non-bank market makers;
ii. *ECNs*: central electronic books with disclosed or anonymous participation;
iii. *RFQ*: request-for-quote negotiation for size or less liquid pairs;
iv. *internalisation*: dealer offsets client flow before external hedging;
v. *fixing windows*: benchmark-driven flow concentrated around reference-rate times.

Definition 8.2.16. *Last Look*
Last look gives a liquidity provider a short window to accept or reject an incoming trade request.
It reduces stale-quote risk for the provider, but creates execution uncertainty and potential information-leakage concerns for the client.

Caveat 8.2.17. *FX Cost Is Relationship-Dependent*
The displayed spread is only part of the cost. Credit, prime-broker limits, rejection rates, hold times, skewed streams, and information leakage can dominate fill quality.

8.2.4 Options Market Microstructure

Principle 8.2.18. *Options Are Surface Markets*
An option quote is linked to the underlying price, volatility, rates, dividends, strike, and maturity.
Liquidity therefore lives on a surface, not in one instrument, and market makers hedge across the underlying, futures, and related options.

Caveat 8.2.19. *Post-Trade Complexity*
Options fragmentation is constrained by clearing, exercise, assignment, margin, and position-risk management.
The trading interface may look electronic, but the economics are dominated by cross-contract risk.

Mechanism 8.2.20. *Market-Maker-Centric Liquidity*
Options markets rely heavily on professional market makers posting two-sided liquidity. Buy-side traders typically consume liquidity because maintaining coherent quotes across many strikes and maturities requires specialised models and hedging infrastructure.

Taxonomy 8.2.21. *Options Liquidity Segmentation*

i. *moneyness*: at-the-money contracts are usually deepest;
ii. *expiry*: near expiries trade differently from long-dated optionality;
iii. *underlying liquidity*: hedging cost transfers into option spreads;
iv. *volatility regime*: high uncertainty widens quotes and reduces size;
v. *complex orders*: spreads and multi-leg trades may be more relevant than outrights.

8.2.5 Fixed Income Microstructure

Taxonomy 8.2.22. *Fixed-Income Instrument Range*

i. *government bonds*: most standardised and electronically tradable;

ii. *corporate bonds*: fragmented by issuer, maturity, covenant, and liquidity;

iii. *credit derivatives*: standard indices trade more centrally than bespoke risk;

iv. *swaps and structured products*: often remain dealer-mediated because contracts differ.

Principle 8.2.23. *RFQ Dominance*
Many fixed-income products trade through RFQ because each bond or swap line is less standardised and less continuously active than an equity. Dealers compete by quote, balance sheet, inventory, and relationship rather than only queue position.

Mechanism 8.2.24. *Dealer Balance Sheet as Venue*
For less liquid bonds, the dealer's inventory and willingness to warehouse risk are part of the market structure. Liquidity is therefore partly bilateral and may disappear when balance-sheet cost rises.

Rule 8.2.25. *TRACE and Post-Trade Transparency*
TRACE reports U.S. corporate bond trades after execution, improving post-trade transparency even when pre-trade negotiation remains bilateral. Similar transparency pushes in Europe aim to make fixed-income liquidity easier to measure without forcing all products into a CLOB.

Caveat 8.2.26. *Pre-Trade Transparency Limits*
Publishing too much pre-trade interest in illiquid bonds can reveal inventory needs and increase dealer risk. Fixed-income transparency policy must balance market monitoring against the cost of exposing fragile liquidity.

Practitioner Note 8.2.27. *Data Changes the Market*
Richer order, quote, and trade records let asset managers measure transaction costs, compare dealers, and link investment decisions to liquidity. Electronification changes execution quality and governance of trading desks.

8.2.6 Securities Lending and Synthetic Shorts

Definition 8.2.28. *Street Name*
Street-name securities are registered to a broker or depository while the client remains beneficial owner. This eases transfer, custody, margin, and lending.

Mechanism 8.2.29. *Securities Lending*
Short selling requires borrowable shares. Brokers source inventory from street-name holdings, stock-loan desks, brokers, or institutions, earning lending fees or rebate spreads.

Taxonomy 8.2.30. *Borrow Availability Segmentation*

i. *general collateral*: easy-to-borrow securities with low lending fees;

ii. *specials*: scarce borrow with high fees and low or negative rebates;

iii. *recall risk*: lender can demand the shares back, forcing replacement borrow or buy-in;

iv. *settlement risk*: a failed locate or delivery can break the short-sale trade.

Definition 8.2.31. *Short Interest Rebate*
The short seller posts cash collateral against borrowed shares. The lender may rebate part of the collateral interest; the harder the borrow, the lower the rebate and the higher the effective shorting cost.

Principle 8.2.32. *Synthetic Short*
A long put and short call with the same strike and maturity replicate a short forward. It can replace scarce borrow, subject to funding, dividends, exercise, and option-liquidity costs.

Caveat 8.2.33. *Voting and Beneficial Ownership*
Stock lending can separate economic exposure from voting rights. Brokers and custodians need entitlement records because more investors may believe they own voting shares than there are record shares available.

8.2.7 Crypto Market Microstructure

Principle 8.2.34. *Crypto Liquidity Is Exchange-Fragmented*
Crypto trades across many centralised and decentralised venues with different custody, margin, liquidation, and fee rules. There is no universal consolidated tape, so fair value and best execution are venue-dependent.

Taxonomy 8.2.35. *Crypto Instrument Segmentation*

i. *spot*: direct token exchange with custody and transfer risk;
ii. *perpetual futures*: leveraged swaps kept near spot through funding;
iii. *dated futures*: basis trades tied to expiry and collateral terms;
iv. *options*: volatility exposure concentrated in fewer venues and strikes;
v. *on-chain pools*: automated market makers with inventory curves and gas costs.

Definition 8.2.36. *Perpetual Funding*
Funding is the periodic payment that links perpetual futures to spot. Positive funding means longs pay shorts; negative funding means shorts pay longs. It is both a carry term and a crowding signal.

Taxonomy 8.2.37. *Crypto Microstructure Risk Segmentation*

i. *venue risk*: outage, withdrawal halt, auto-deleveraging, or insolvency;
ii. *liquidation risk*: forced trades from leveraged accounts;
iii. *stablecoin risk*: collateral, redemption, and depeg exposure;
iv. *bridge and settlement risk*: chain congestion, reorgs, and transfer delay;
v. *fragmentation*: basis and price gaps across venues when capital cannot move quickly.

Sources outside the catalogue

`ane-geman-2000` Thierry Ané and Hélyette Geman, "Order Flow, Transaction Clock, and Normality of Asset Returns", *Journal of Finance* 55(5), 2259–2284, 2000. doi:10.1111/0022-1082.00286.

`crypto-cartea-errata` Álvaro Cartea, Sebastian Jaimungal and José Penalva, errata to *Algorithmic and High-Frequency Trading* (`crypto-cartea`), authors' web page, undated, retrieved 2026-10-15. sebastian.statistics.utoronto.ca.

`demsetz-1968` Harold Demsetz, "The Cost of Transacting", *Quarterly Journal of Economics* 82(1), 33–53, 1968. doi:10.2307/1882244.

`garman-klass-1980` Mark B. Garman and Michael J. Klass, "On the Estimation of Security Price Volatilities from Historical Data", *Journal of Business* 53(1), 67–78, 1980. doi:10.1086/296072.

`glosten-milgrom-1983` Lawrence R. Glosten and Paul R. Milgrom, "Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders", Discussion Paper 570, Center for Mathematical Studies in Economics and Management Science, Northwestern University, August 1983; the working-paper version of `s6-glosten`, whose pages and numbering may differ. RePEc:nwu:cmsems:570.

`grossman-miller-1988` Sanford J. Grossman and Merton H. Miller, "Liquidity and Market Structure", *Journal of Finance* 43(3), 617–633, 1988. doi:10.1111/j.1540-6261.1988.tb04594.x.

`grossman-miller-1988-nber` Sanford J. Grossman and Merton H. Miller, "Liquidity and Market Structure", NBER Working Paper 2641, July 1988; the working-paper version of `grossman-miller-1988`, whose pages and numbering may differ. nber.org/papers/w2641.

`ho-stoll-1981` Thomas Ho and Hans R. Stoll, "Optimal Dealer Pricing under Transactions and Return Uncertainty", *Journal of Financial Economics* 9(1), 47–73, 1981. doi:10.1016/0304-405X(81)90020-9.

`madhavan-2000` Ananth Madhavan, "Market Microstructure: A Survey", *Journal of Financial Markets* 3(3), 205–258, 2000. doi:10.1016/S1386-4181(00)00007-0.